

(12) **United States Patent**
Tekolste et al.

(10) **Patent No.:** **US 12,399,320 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **METHOD AND SYSTEM FOR VARIABLE OPTICAL THICKNESS WAVEGUIDES FOR AUGMENTED REALITY DEVICES**

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(72) Inventors: **Robert D. Tekolste**, Fort Lauderdale, FL (US); **Ryan Jason Ong**, Austin, TX (US); **Victor Kai Liu**, Mountain View, CA (US); **Samarth Bhargava**, Saratoga, CA (US); **Christophe Peroz**, Tokyo (JP); **Vikramjit Singh**, Pflugerville, TX (US); **Marlon Edward Menezes**, Austin, TX (US); **Shuqiang Yang**, Austin, TX (US); **Frank Y. Xu**, Austin, TX (US)

(73) Assignee: **Magic Leap, Inc.**, Plantation, FL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 166 days.

(21) Appl. No.: **17/705,202**

(22) Filed: **Mar. 25, 2022**

(65) **Prior Publication Data**

US 2022/0283371 A1 Sep. 8, 2022

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/308,407, filed on May 5, 2021, now Pat. No. 11,487,061, (Continued)

(51) **Int. Cl.**
G02B 6/12 (2006.01)
F21V 8/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G02B 6/13** (2013.01); **G02B 6/0046** (2013.01); **G02B 6/122** (2013.01); **G02B 6/1221** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC G02B 6/0046; G02B 26/0891
See application file for complete search history.

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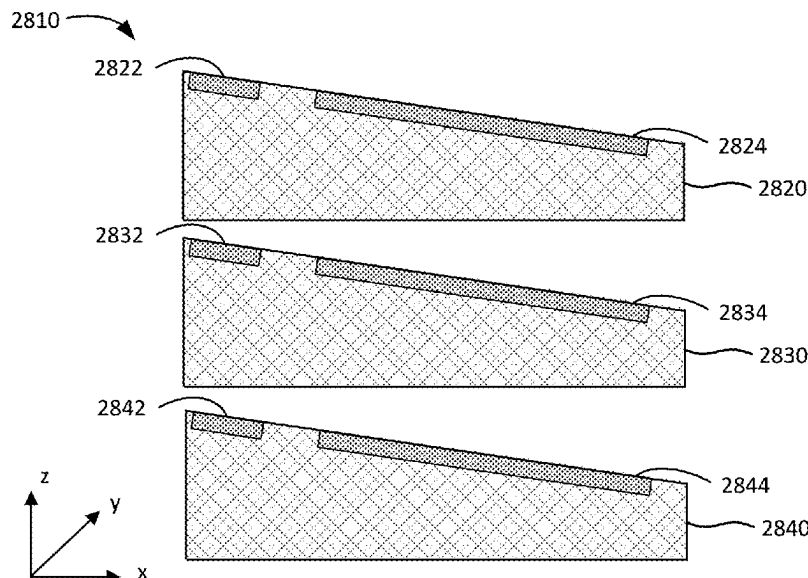
Primary Examiner — Charlie Y Peng

(74) *Attorney, Agent, or Firm* — KILPATRICK TOWNSEND & STOCKTON LLP

(57) **ABSTRACT**

An augmented reality device includes a projector, projector optics optically coupled to the projector, and an eyepiece optically coupled to the projector optics. The eyepiece includes an eyepiece waveguide characterized by lateral dimensions and an optical path length difference as a function of one or more of the lateral dimensions.

44 Claims, 65 Drawing Sheets



Related U.S. Application Data

which is a continuation of application No. 16/792,083, filed on Feb. 14, 2020, now Pat. No. 11,022,753.

- (60) Provisional application No. 62/820,769, filed on Mar. 19, 2019, provisional application No. 62/805,832, filed on Feb. 14, 2019.

(51) **Int. Cl.**

G02B 6/122 (2006.01)

G02B 6/13 (2006.01)

G02B 26/08 (2006.01)

(52) **U.S. Cl.**

CPC **G02B 26/0891** (2013.01); **G02B 2006/12038** (2013.01); **G02B 2006/12061** (2013.01); **G02B 2006/12069** (2013.01); **G02B 2006/12197** (2013.01)

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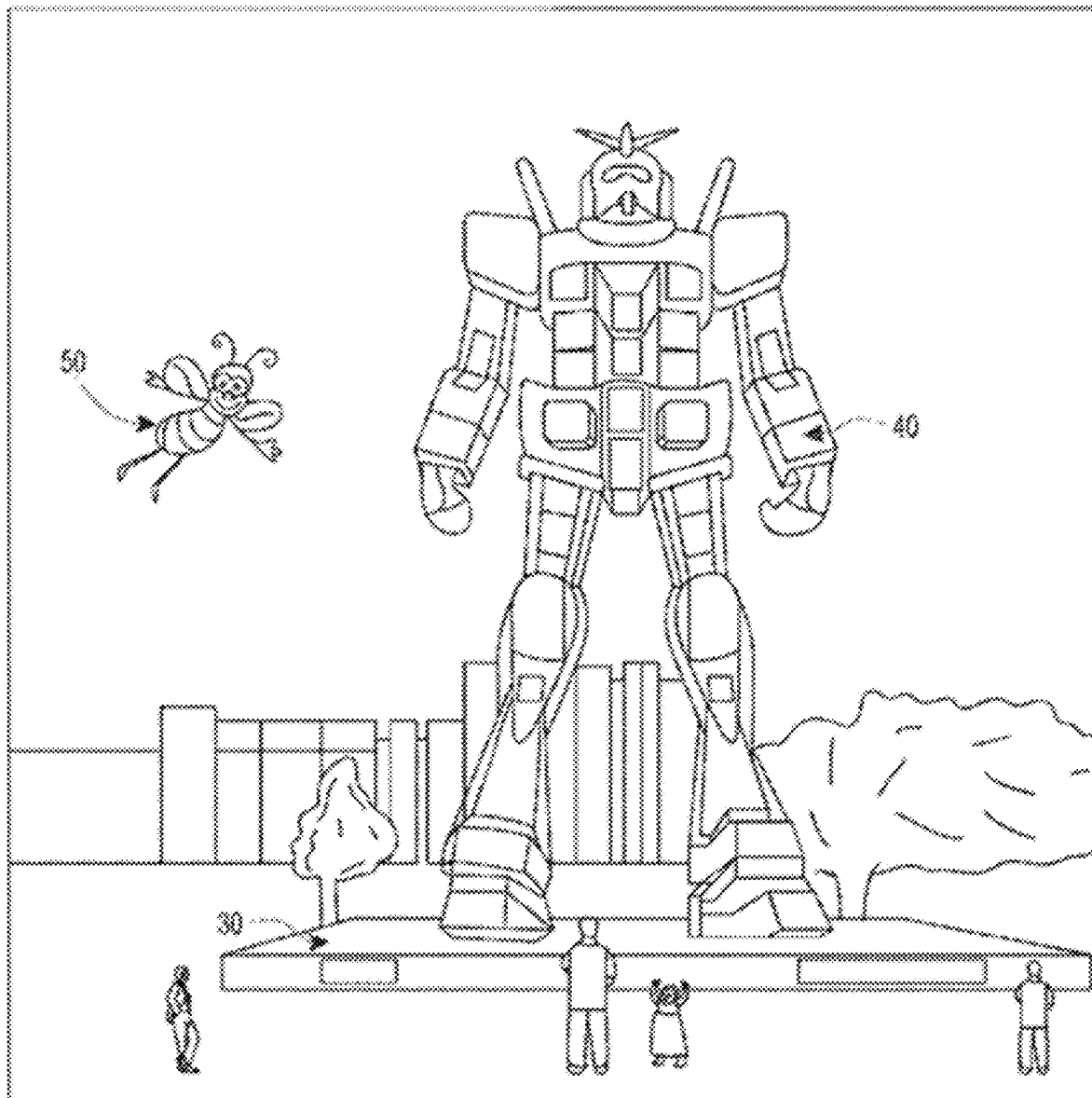
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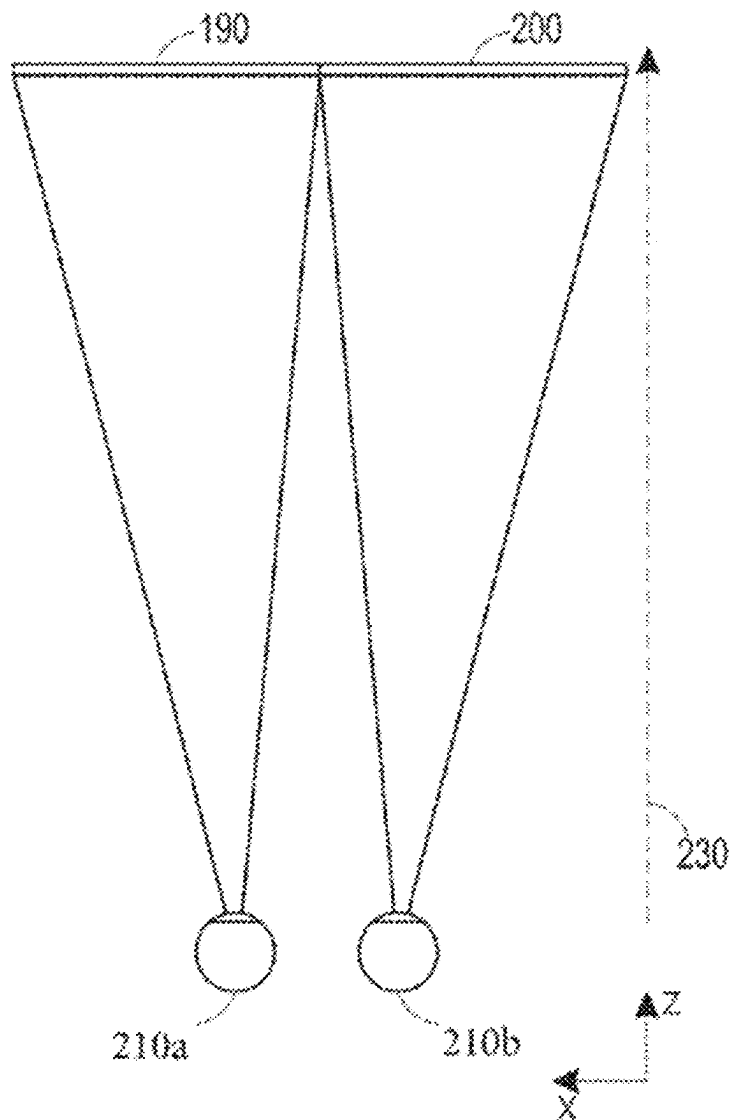
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**FIG. 1**

**FIG. 2**

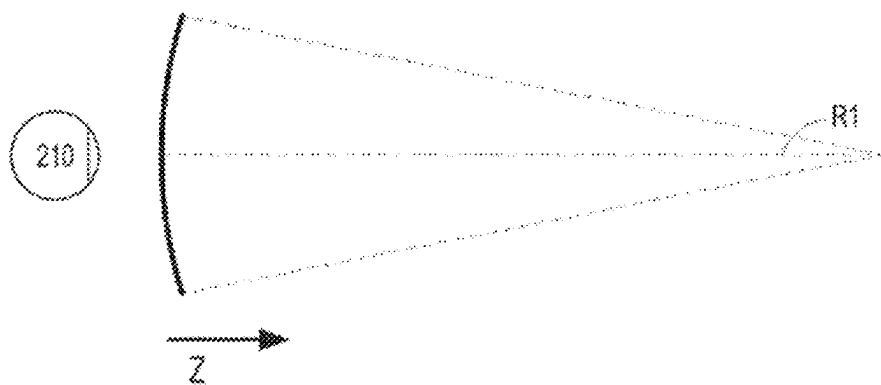


FIG. 3A

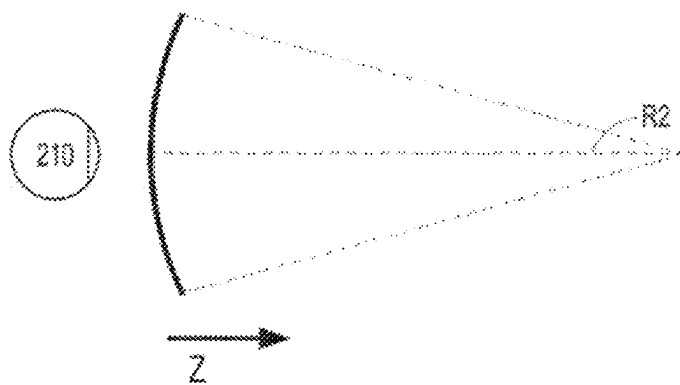


FIG. 3B

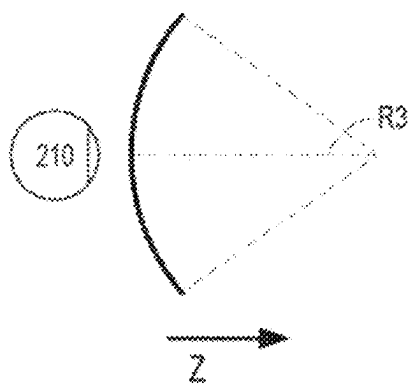


FIG. 3C

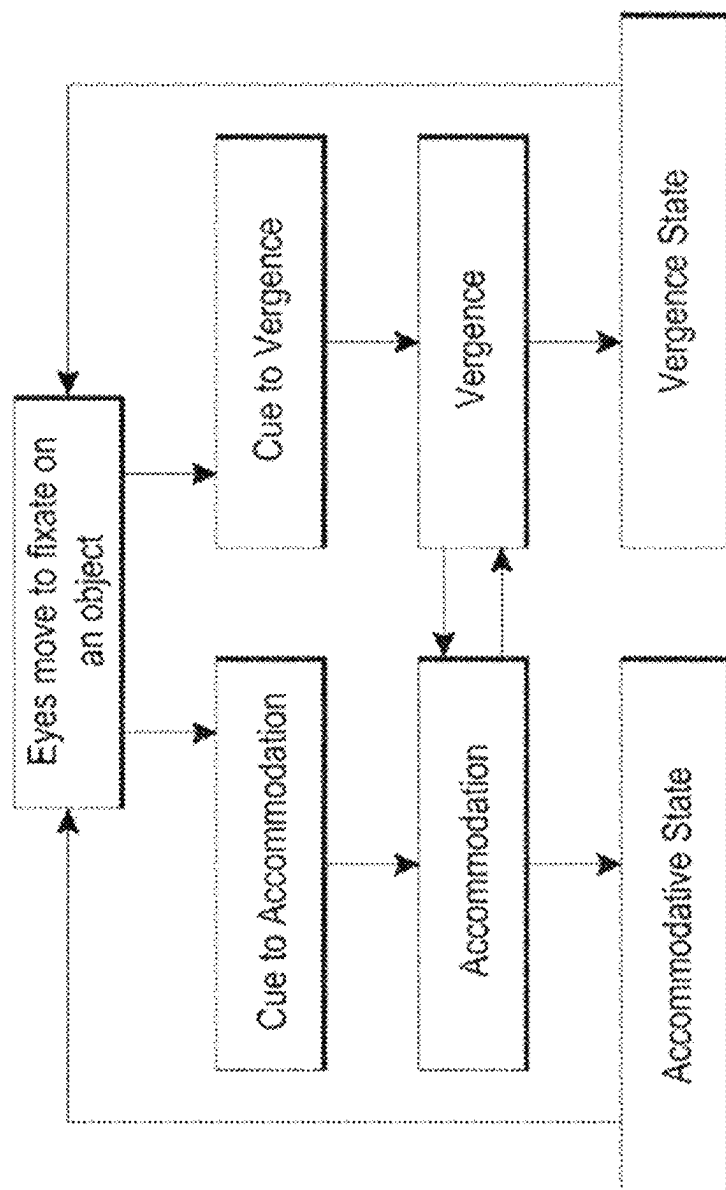


FIG. 4A

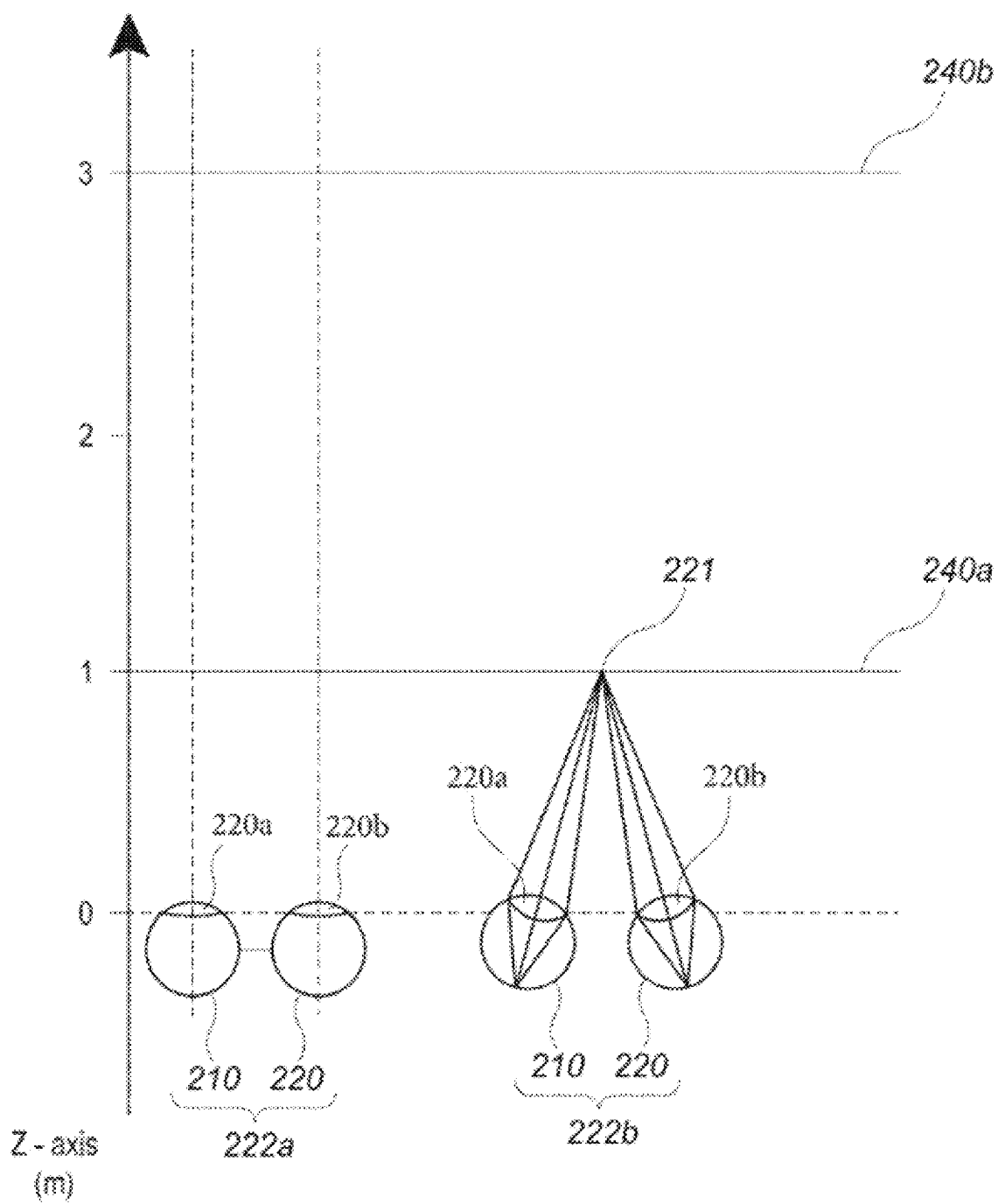


FIG. 4B

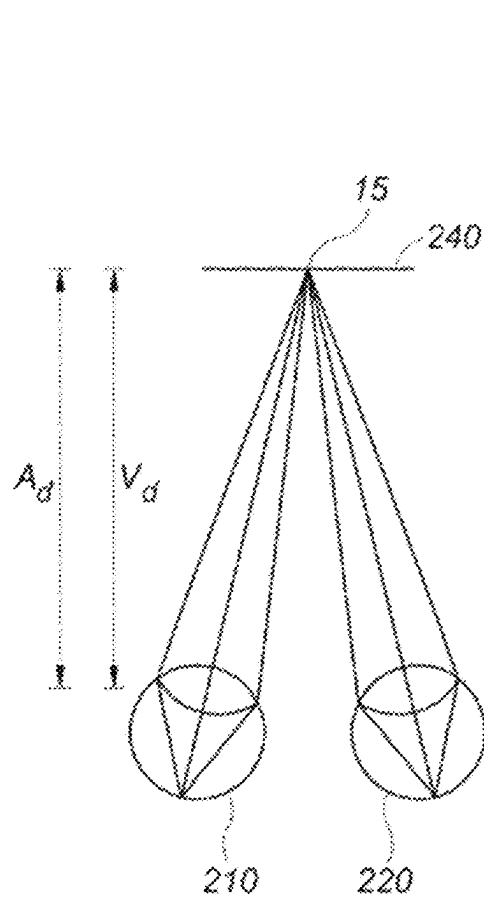


FIG. 4C

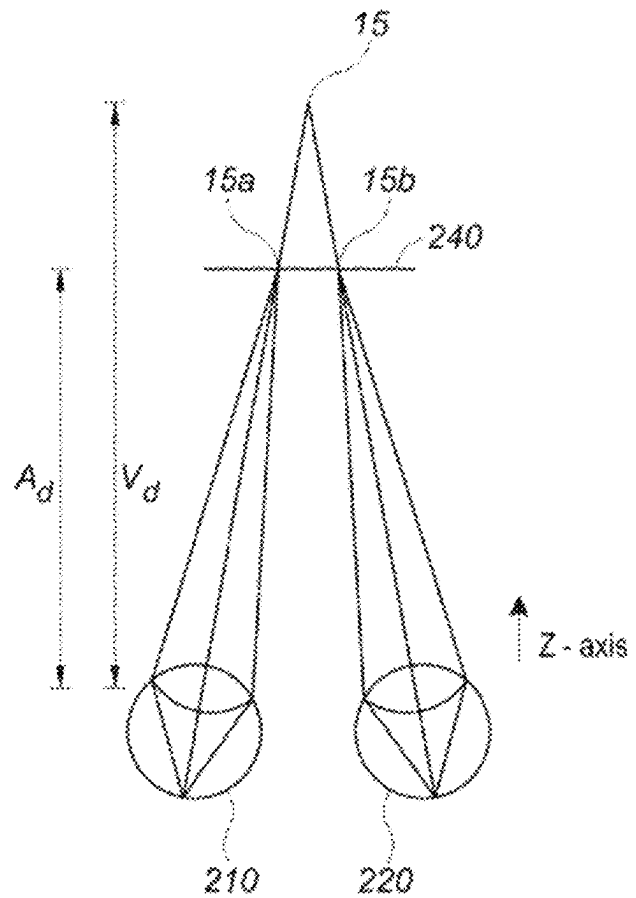


FIG. 4D

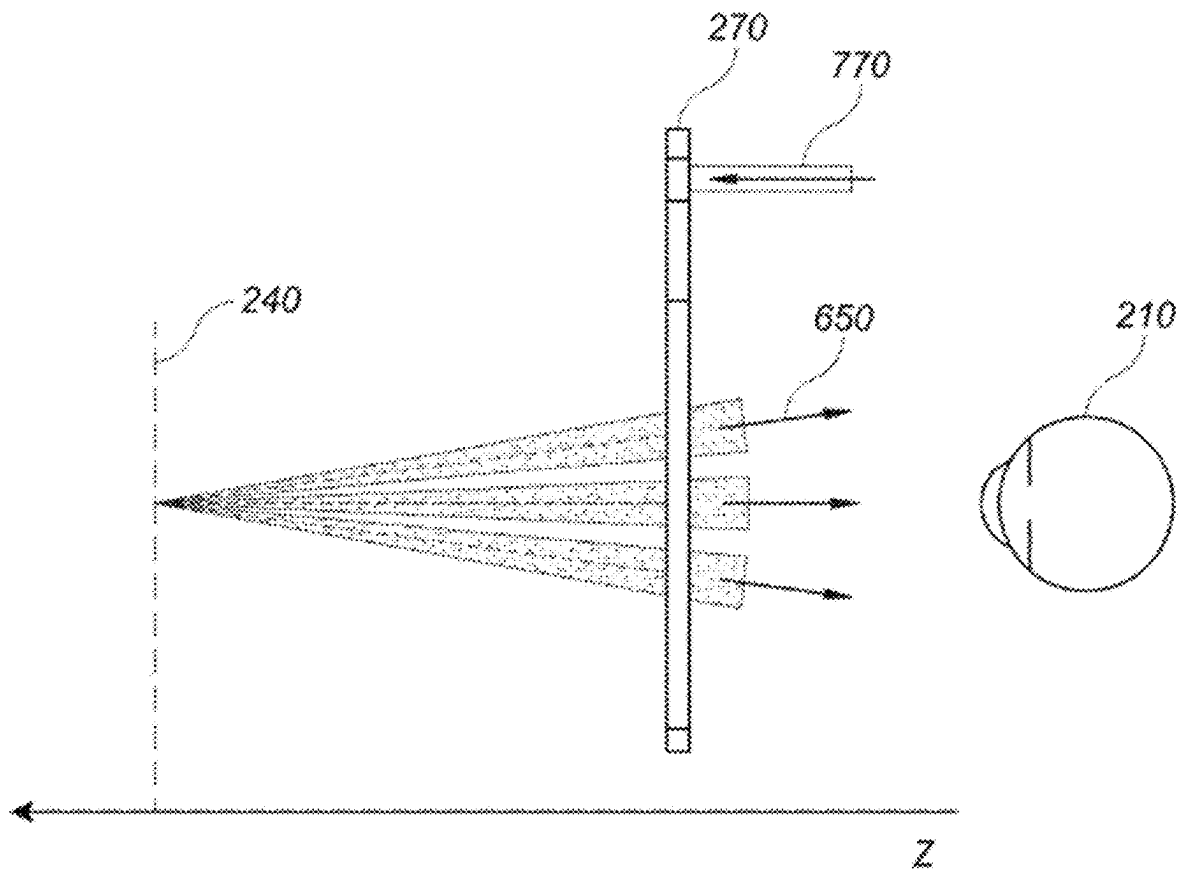


FIG. 5

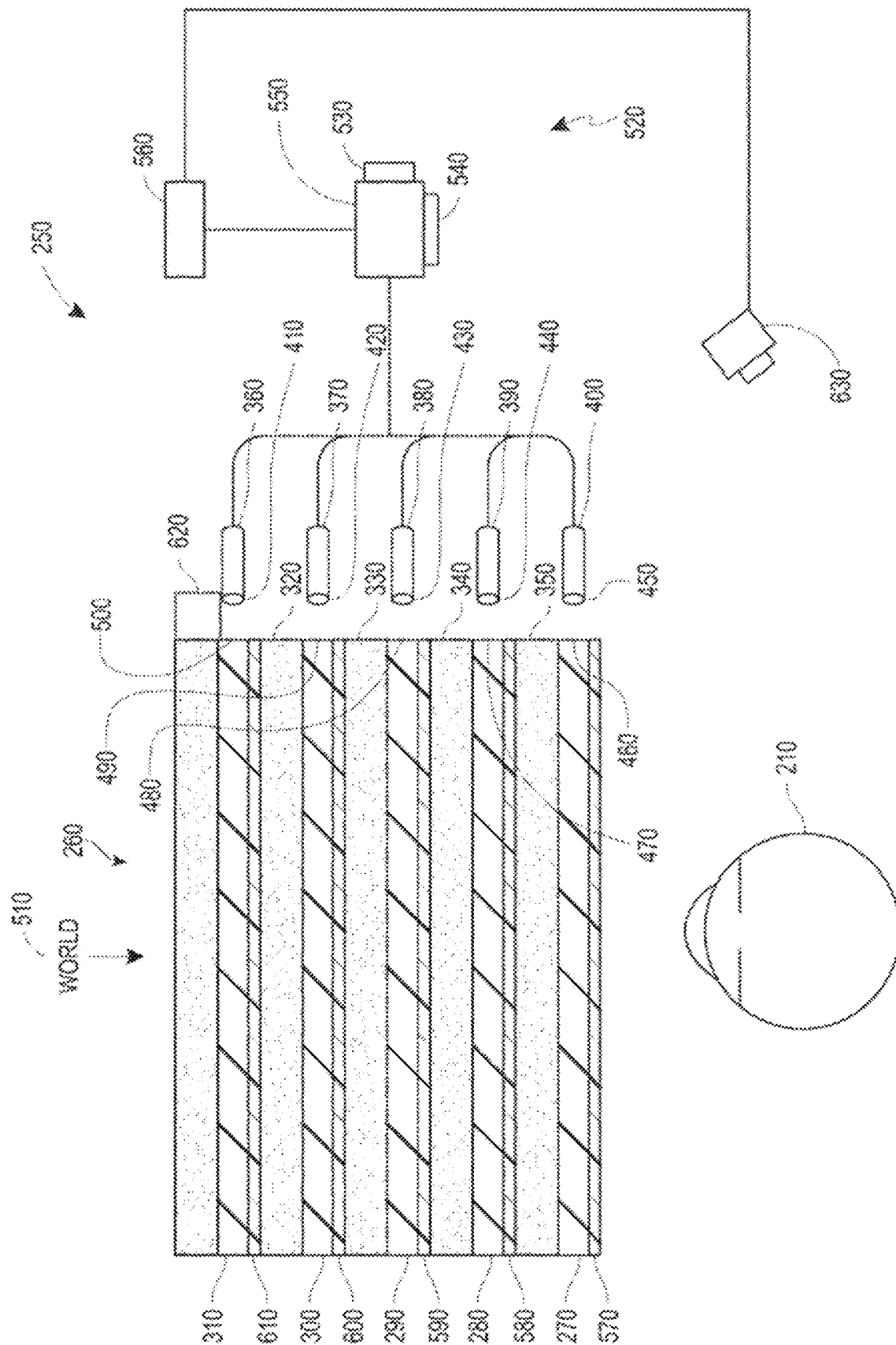


FIG. 6

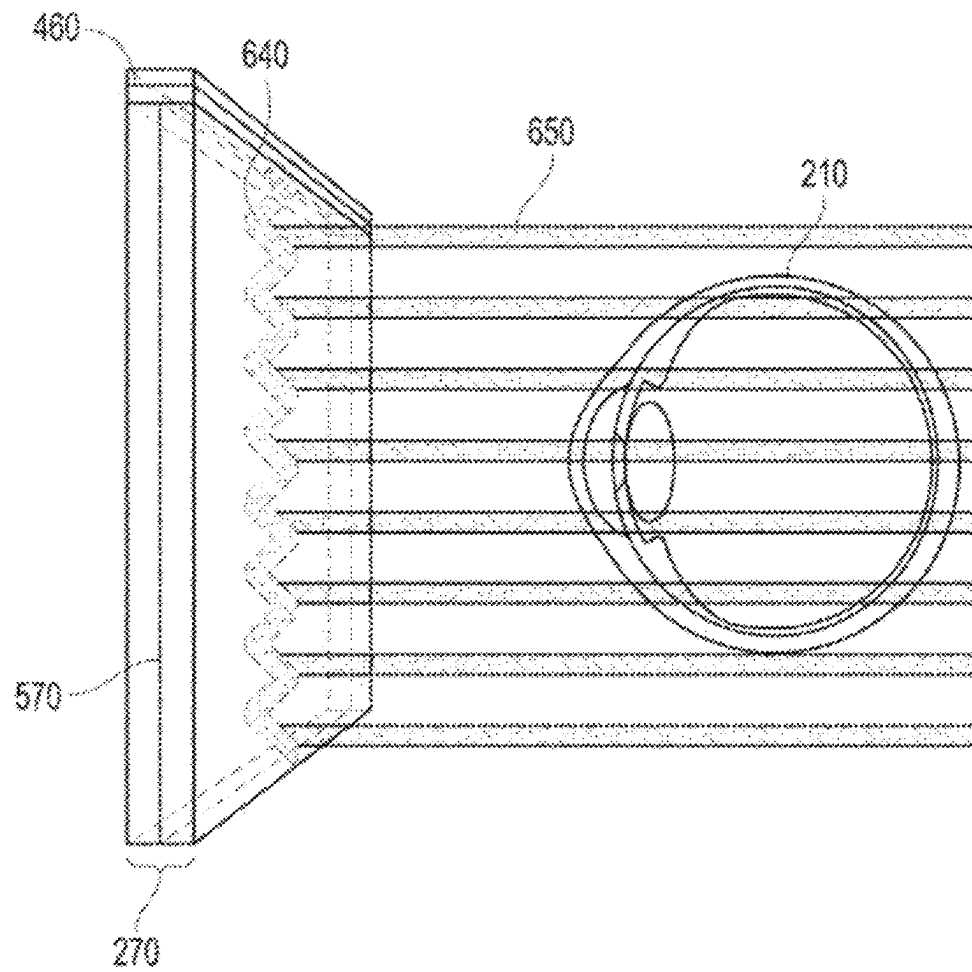


FIG. 7

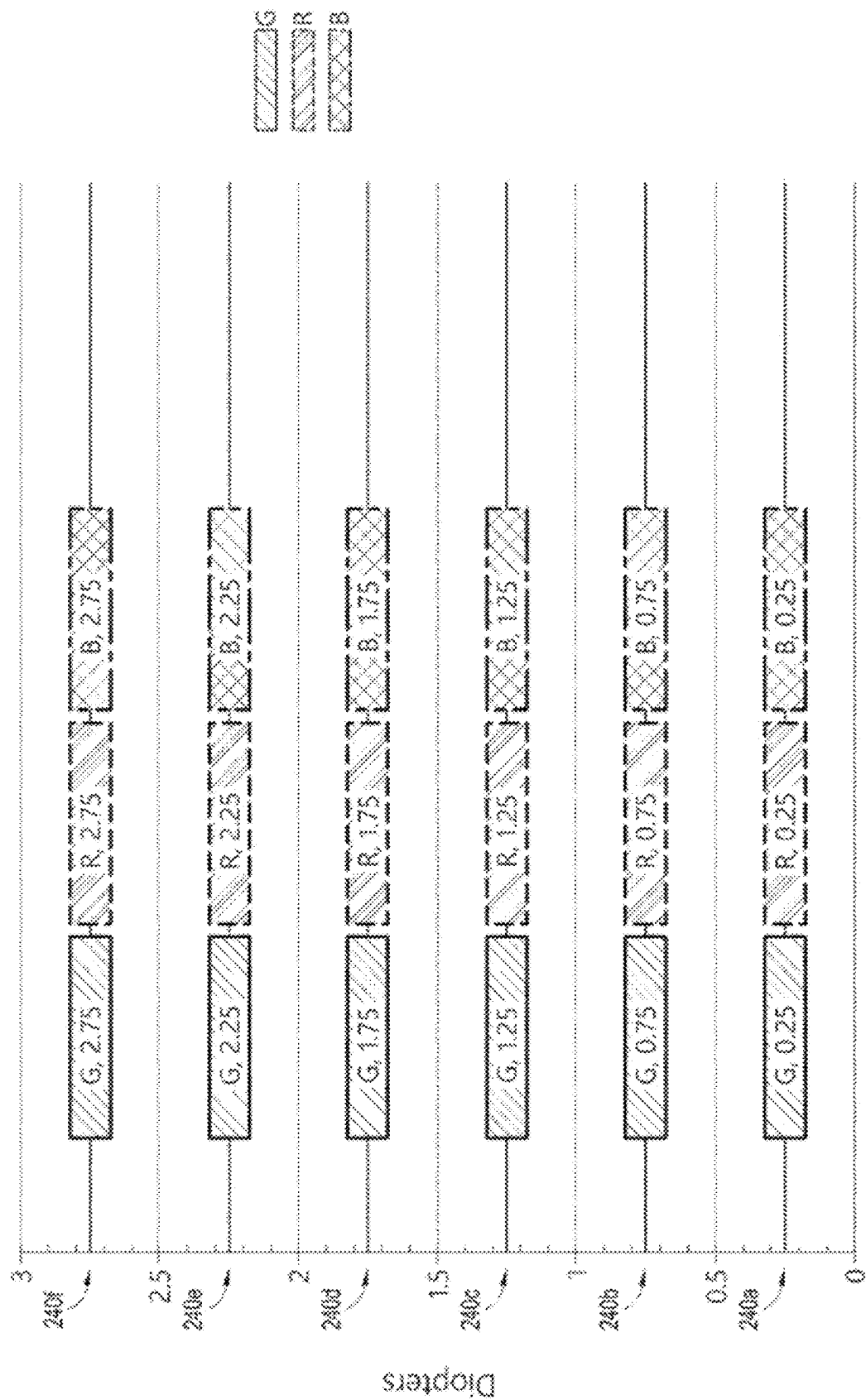


FIG. 8

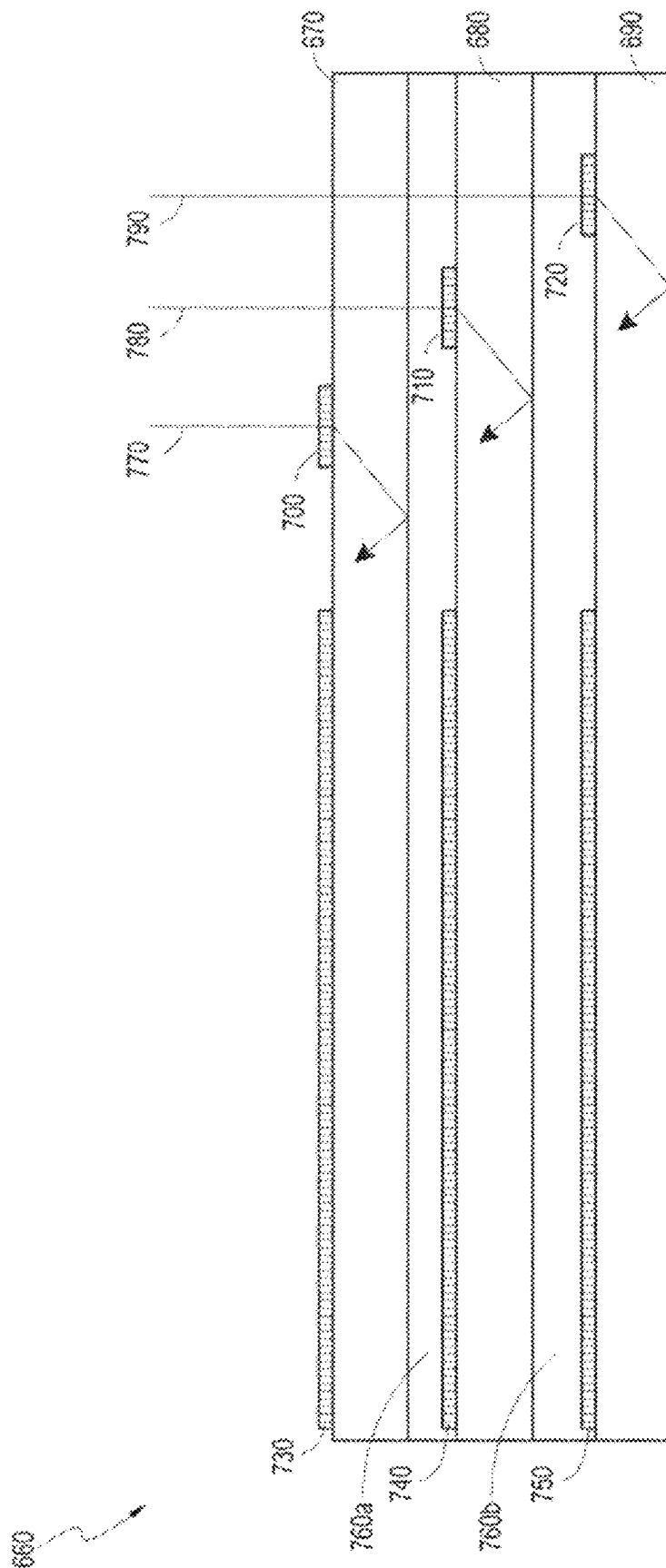


FIG. 9A

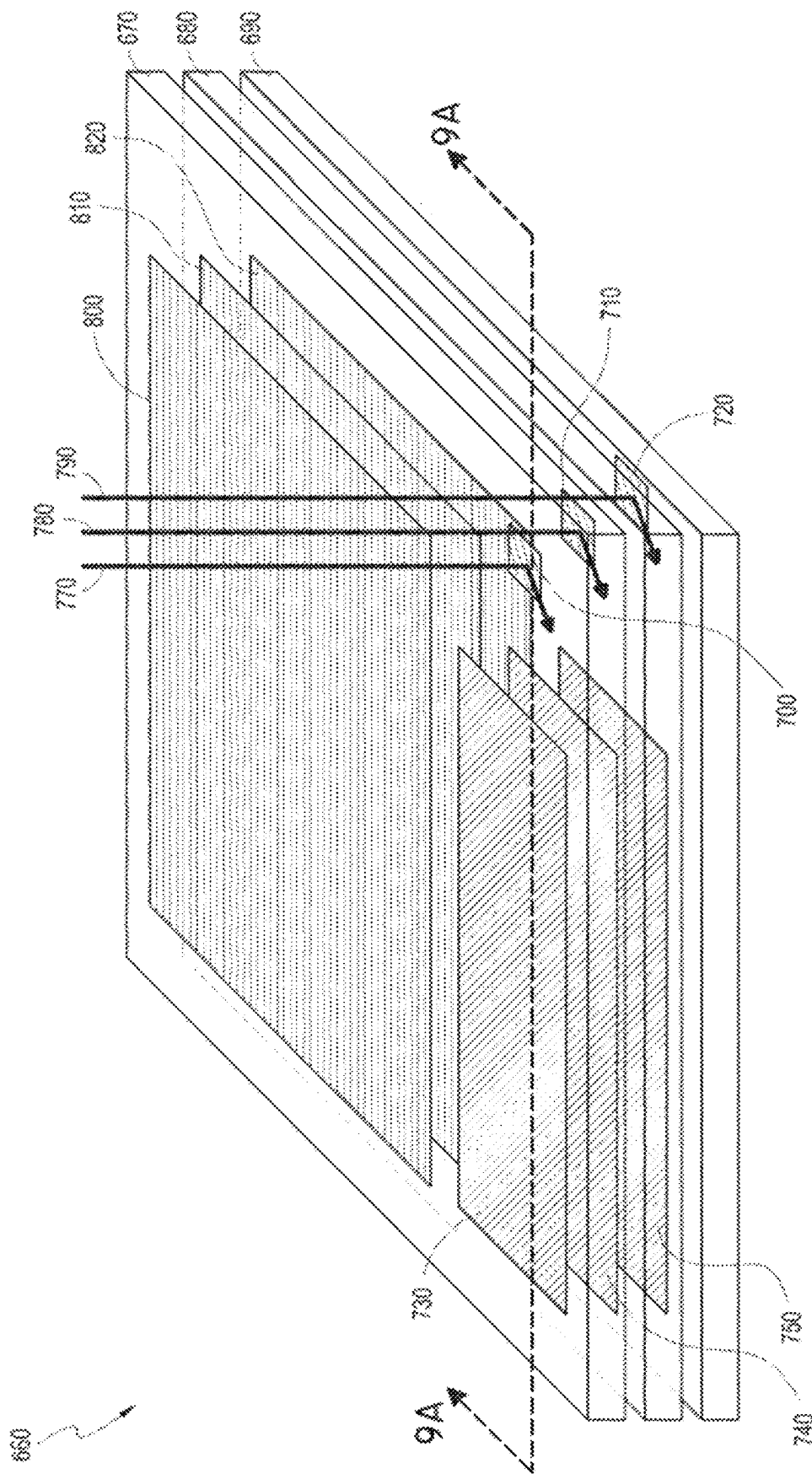


FIG. 9B

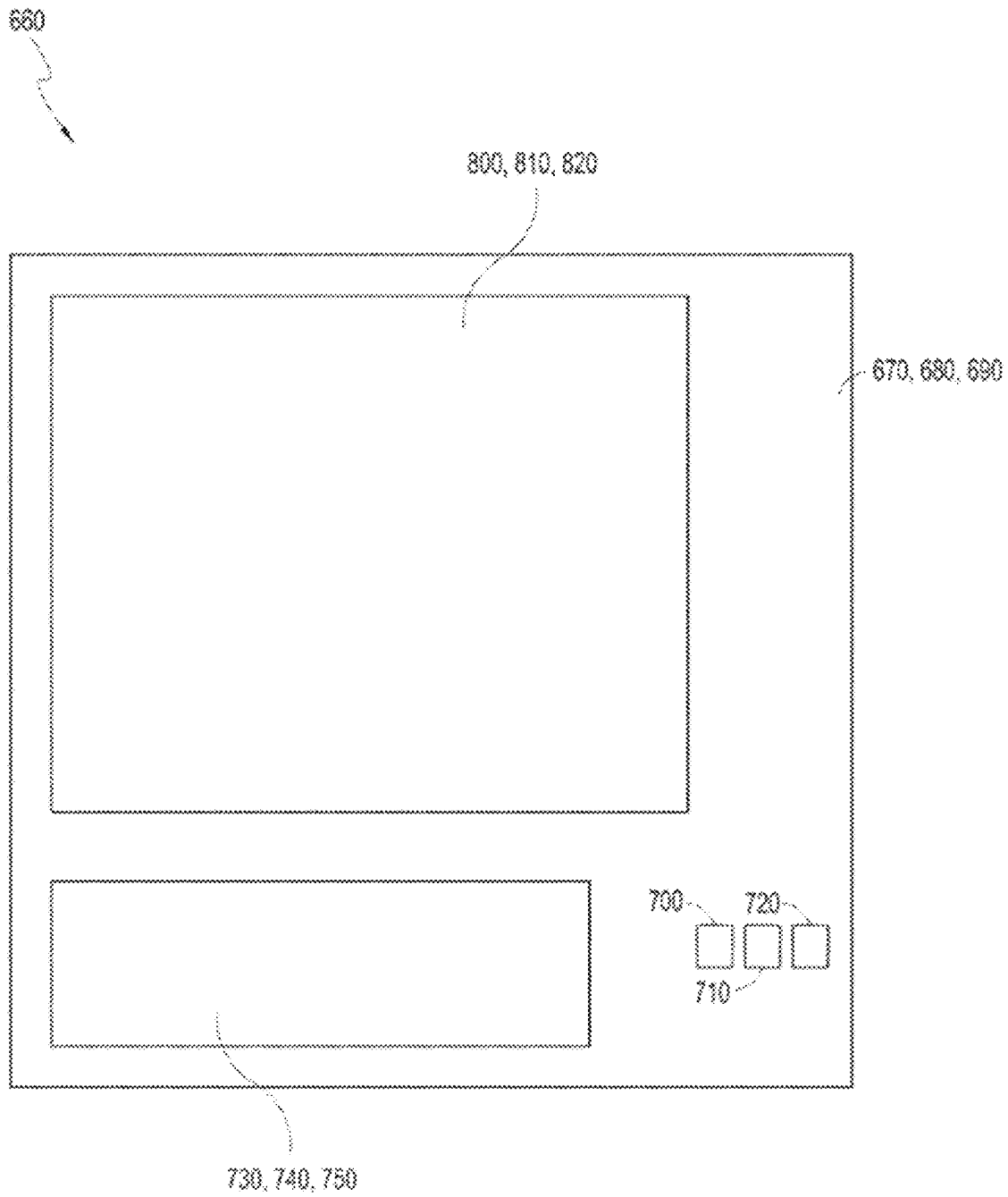


FIG 9C

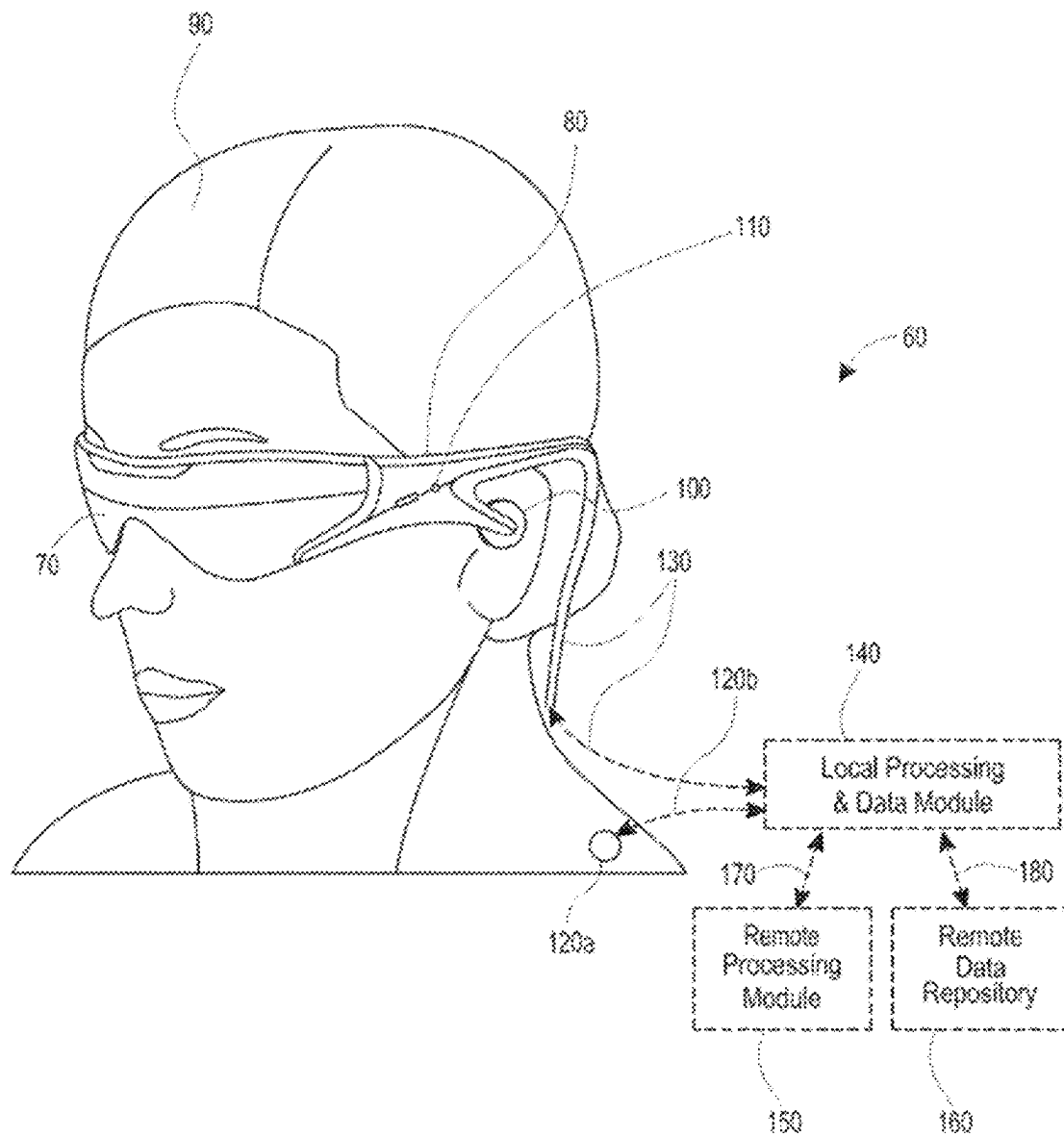


FIG 9D

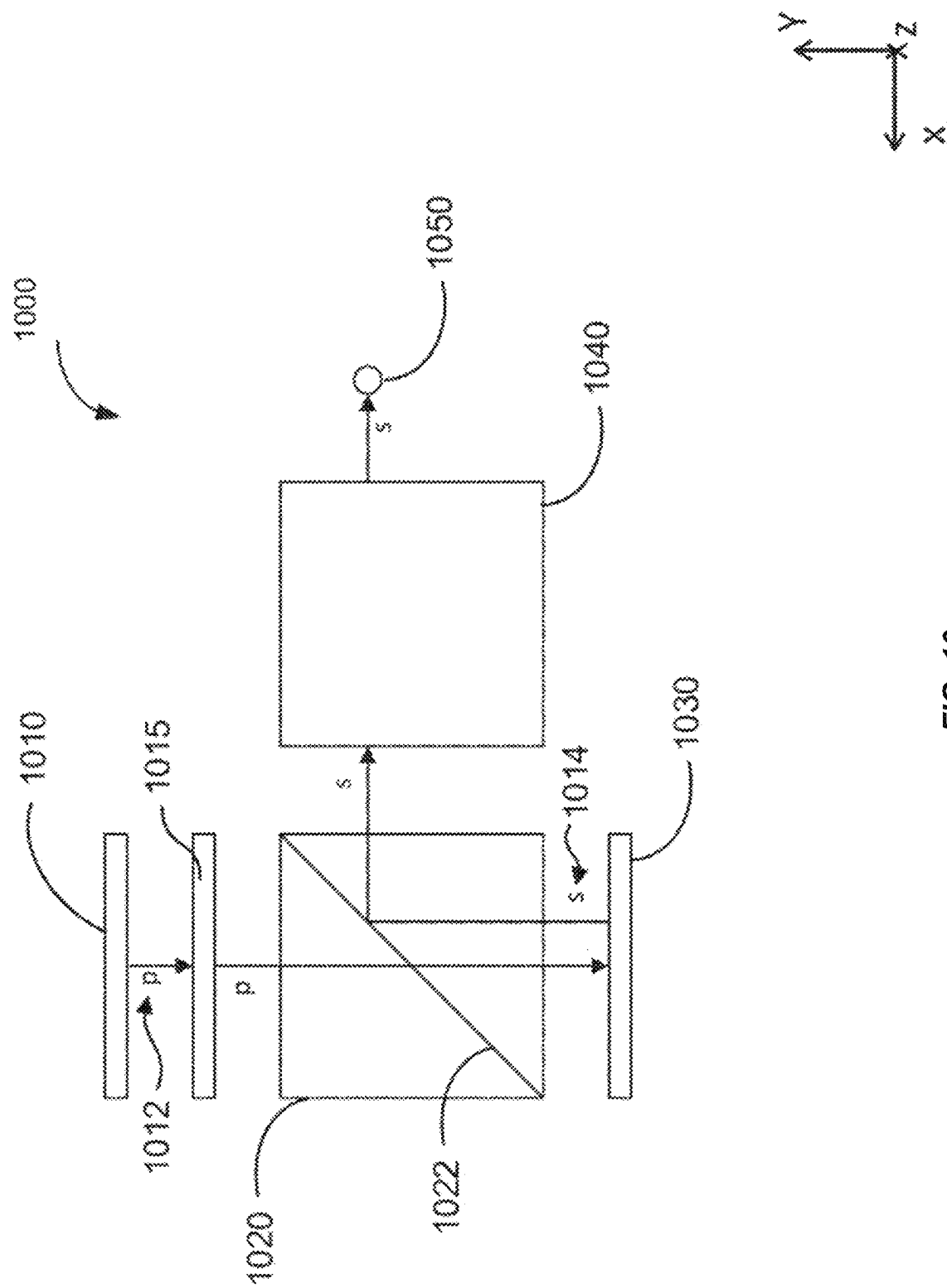


FIG. 10

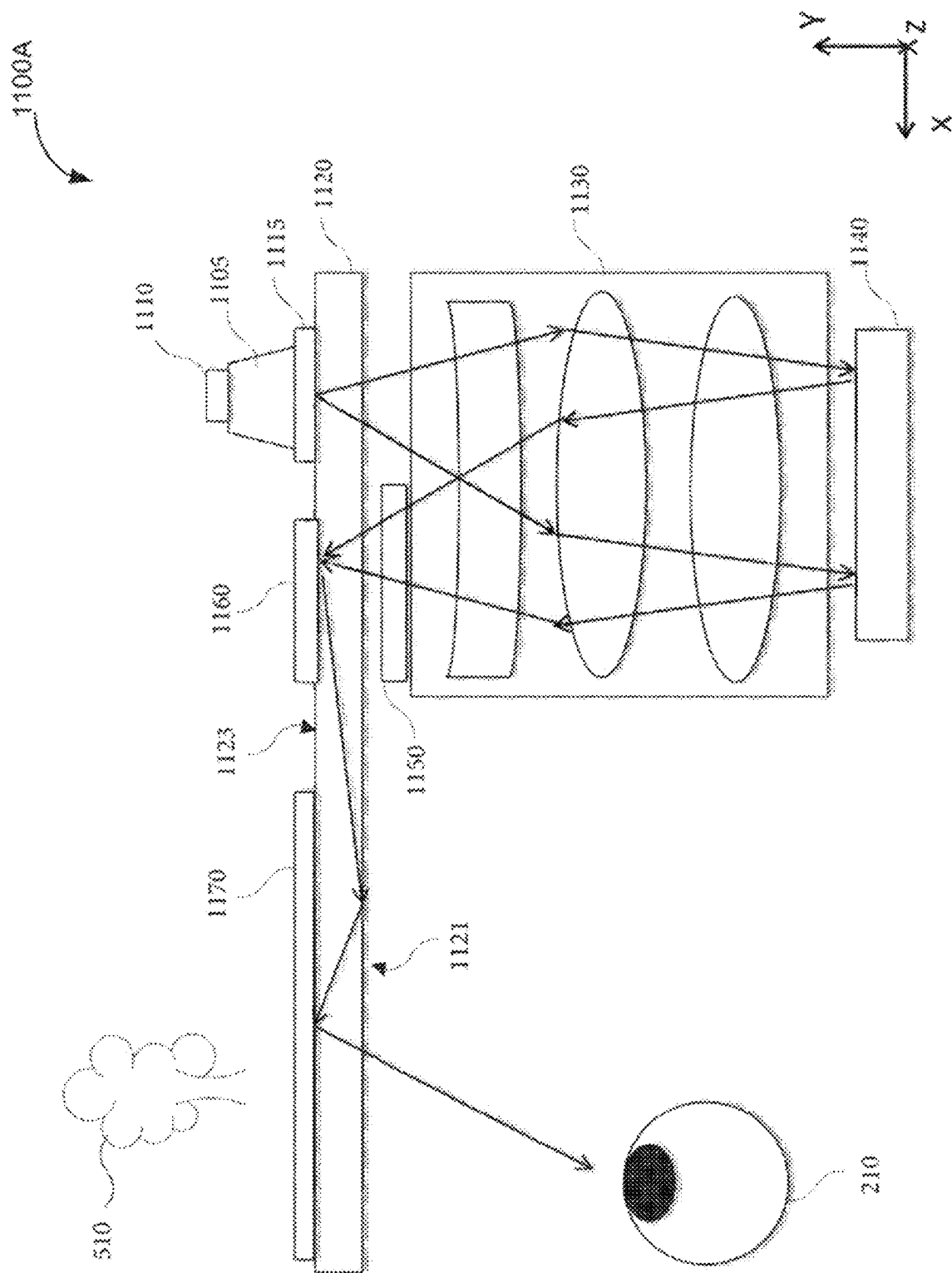


FIG. 11A

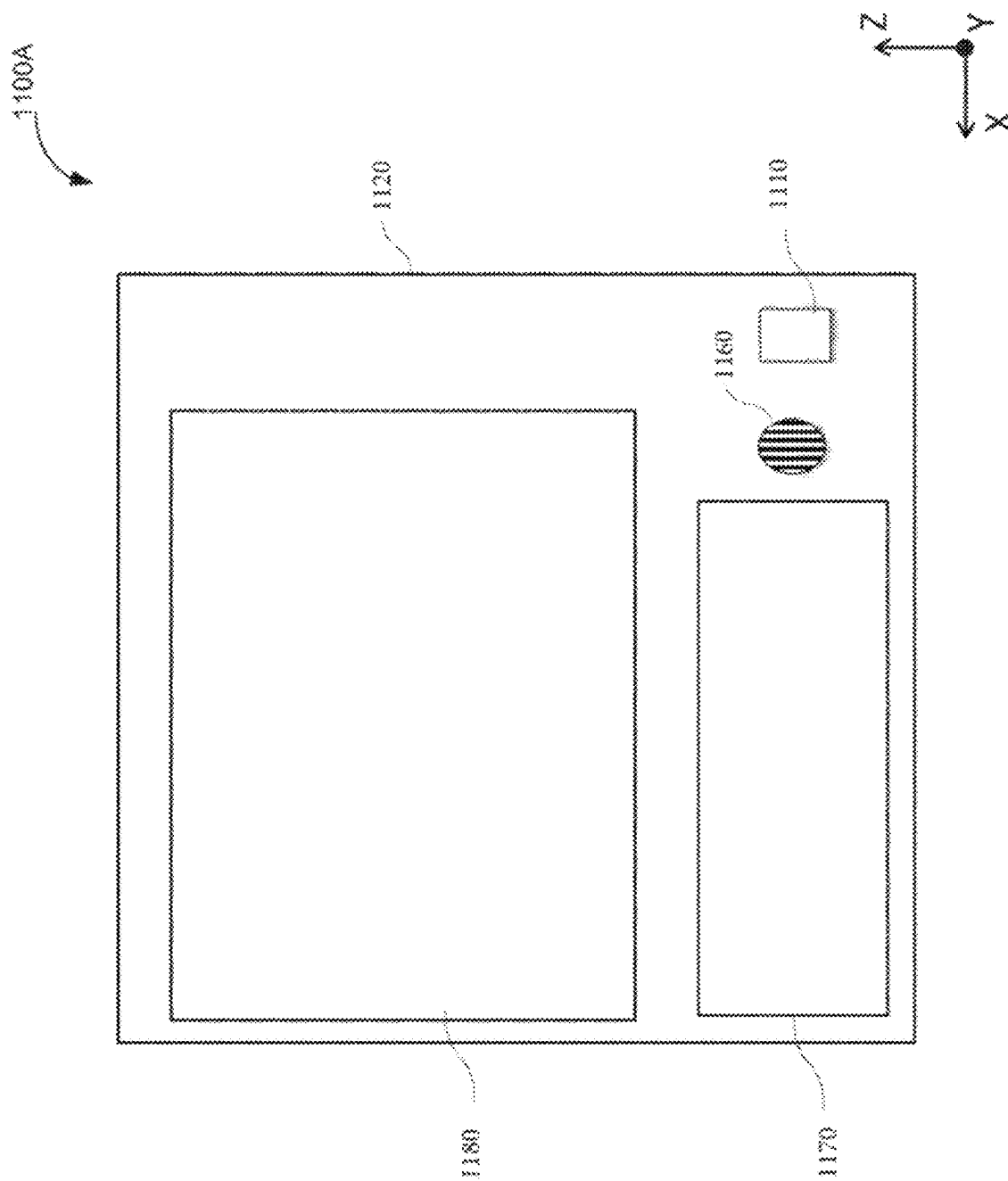


FIG. 11B

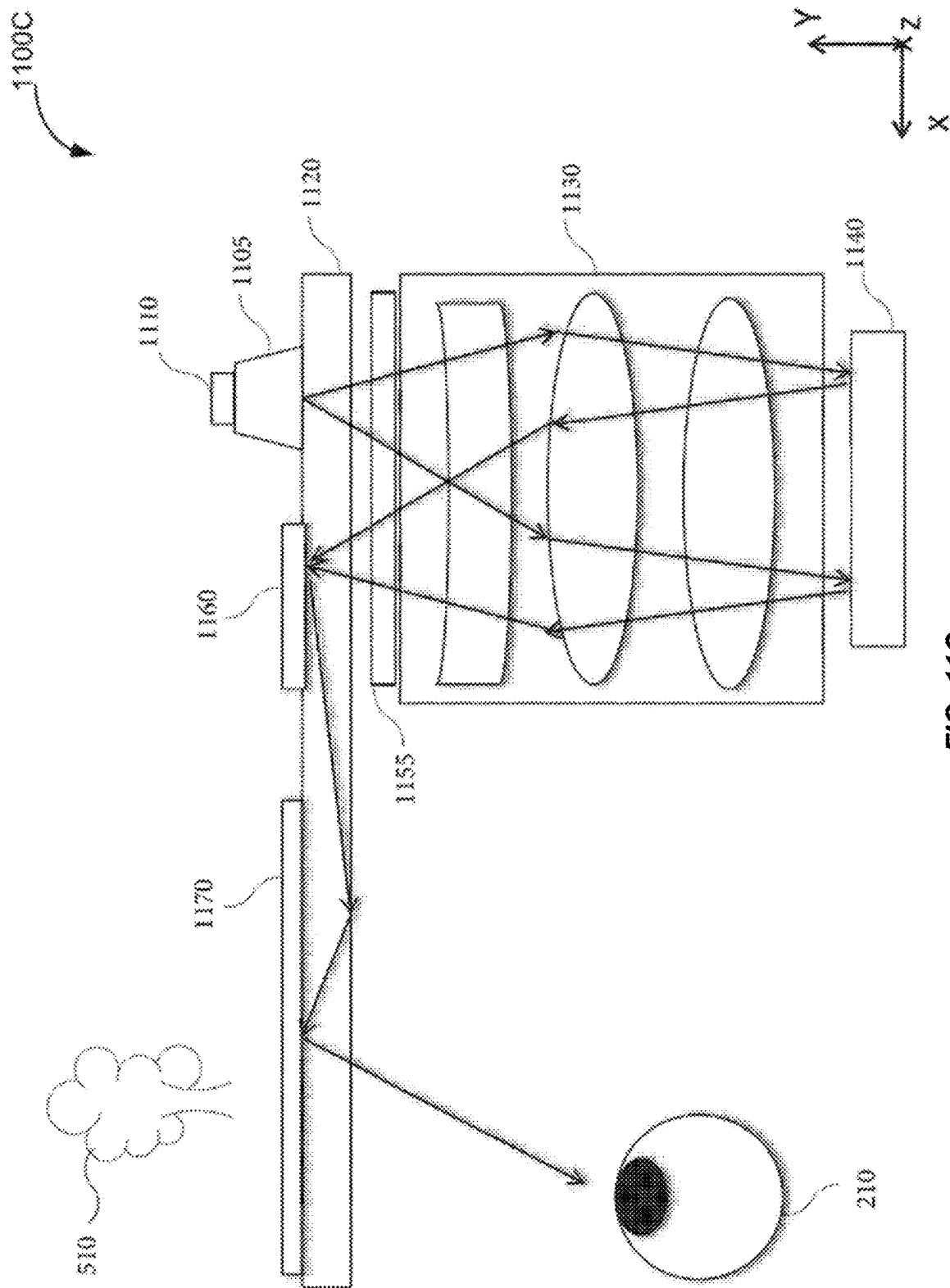


FIG. 11C

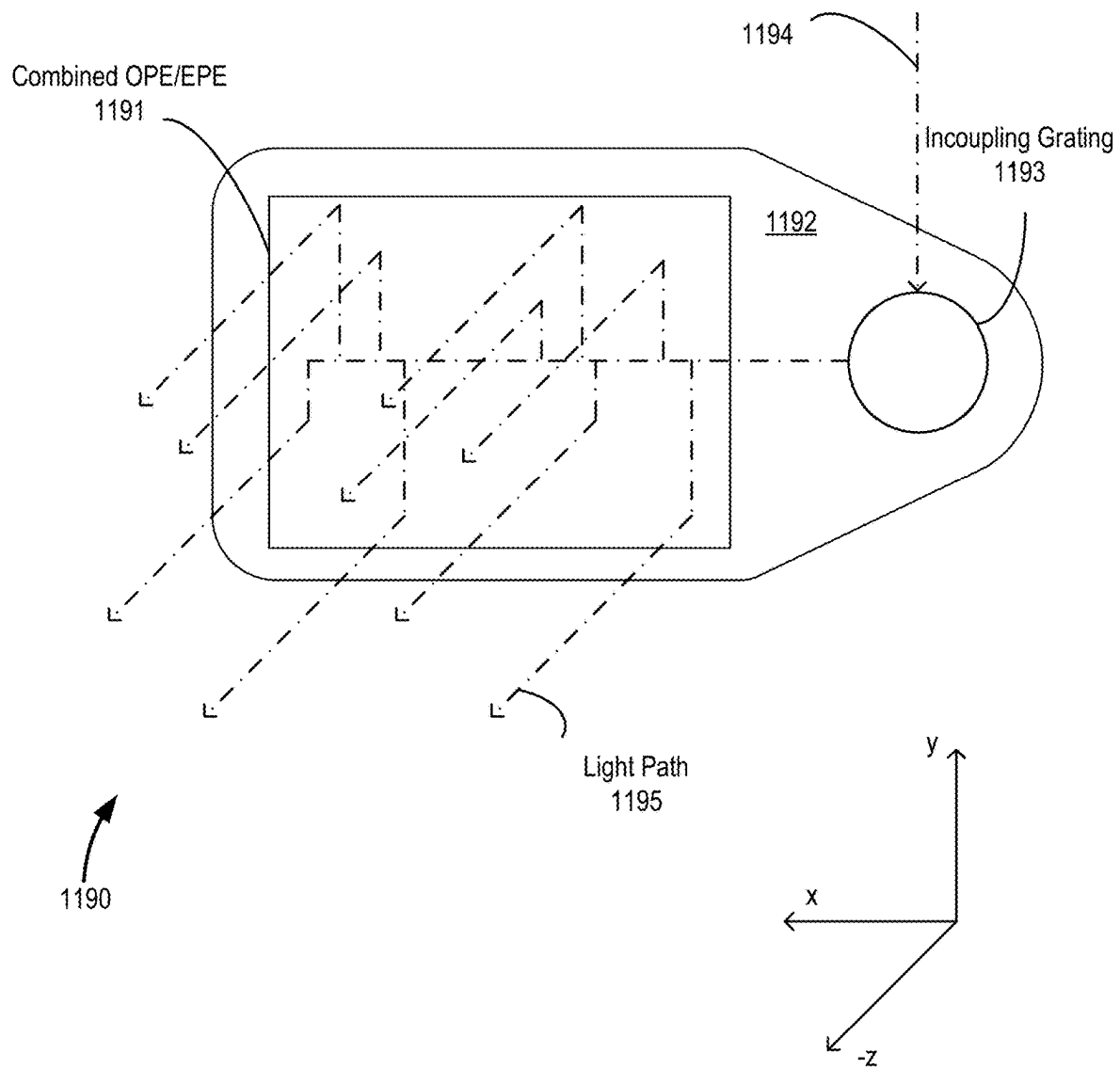
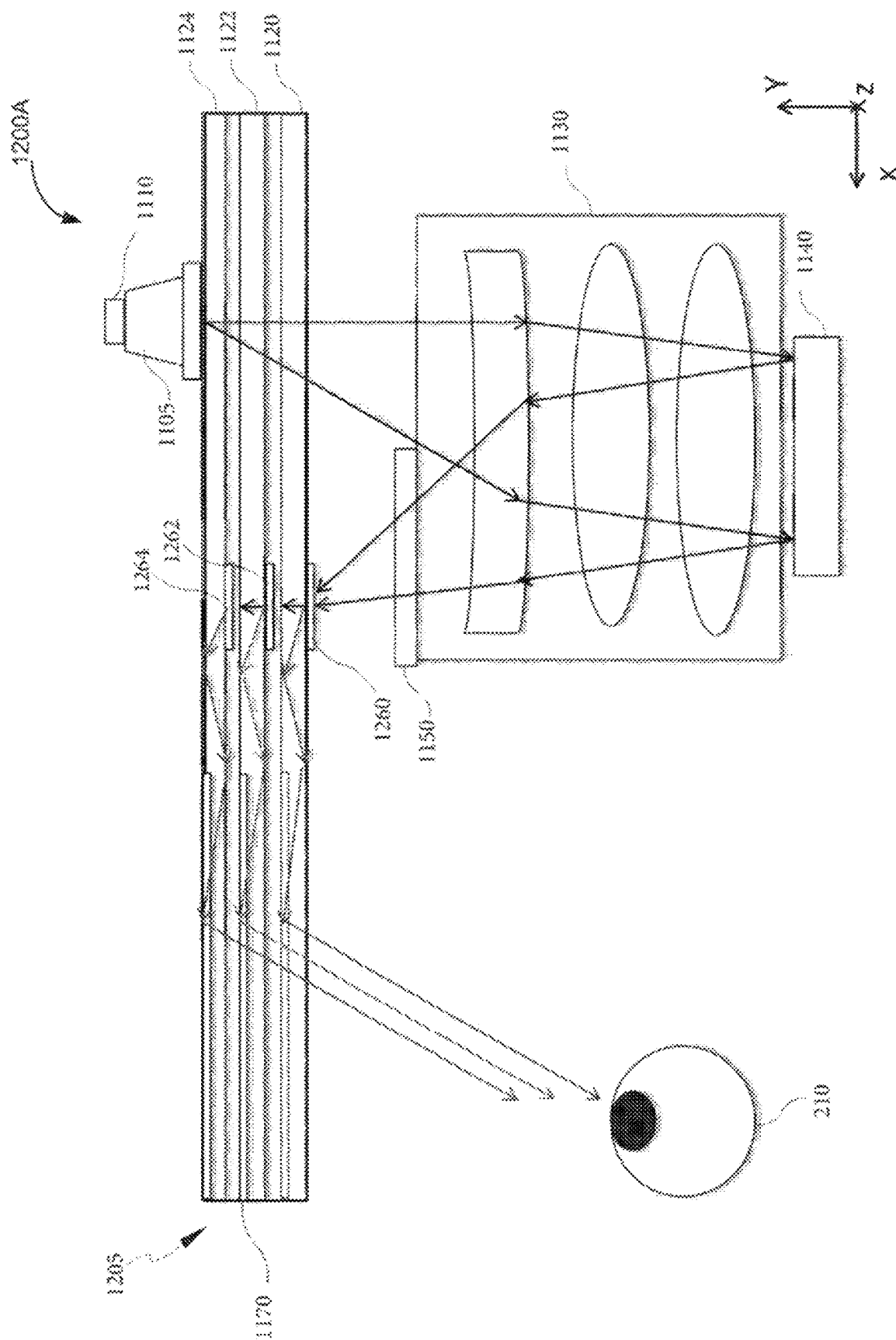
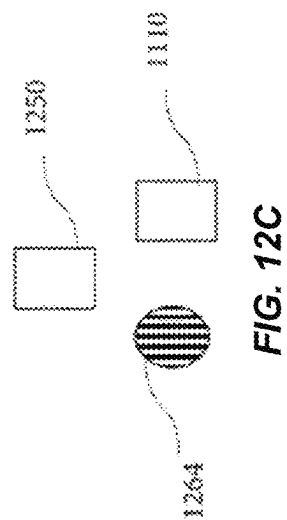
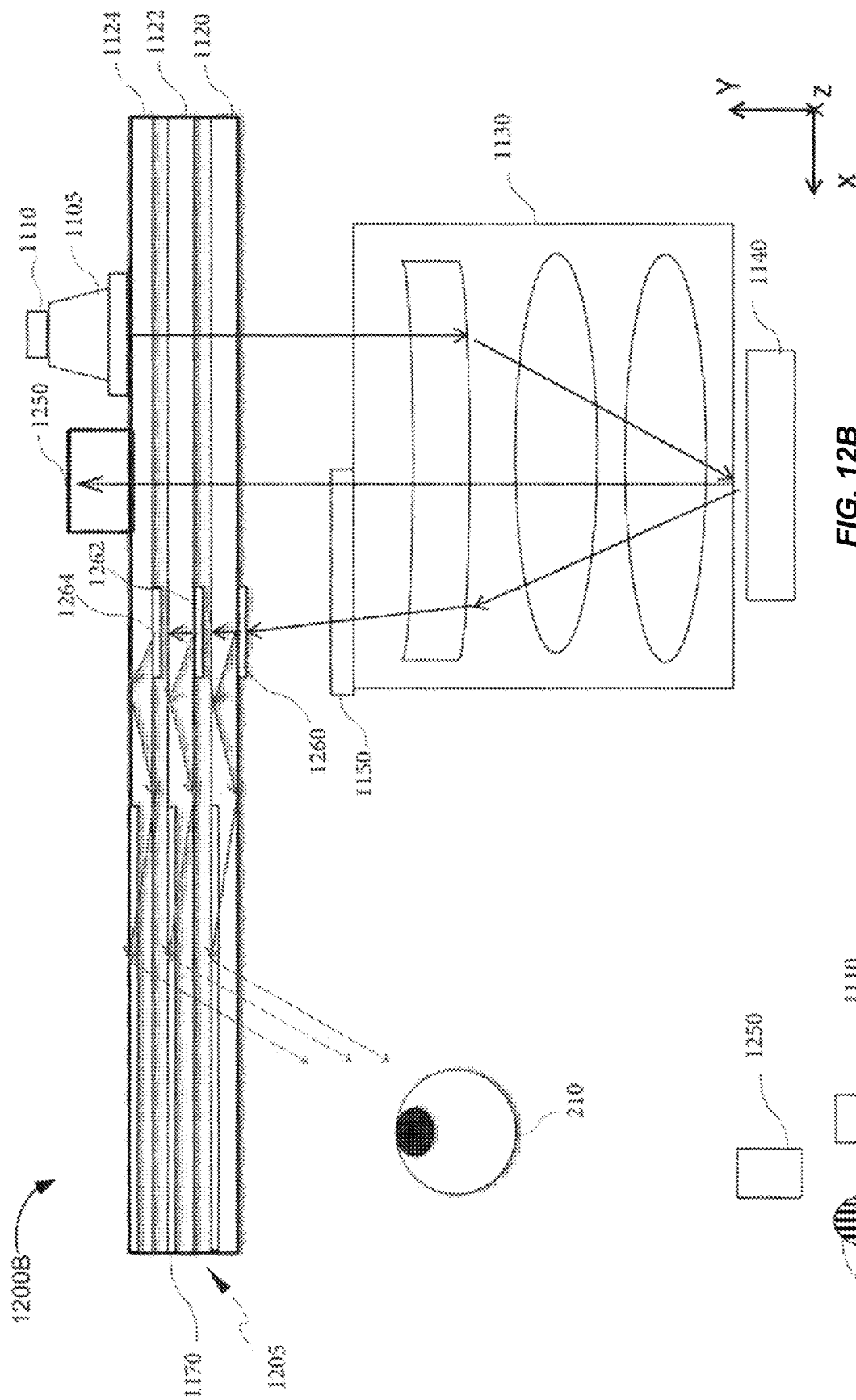


FIG. 11D





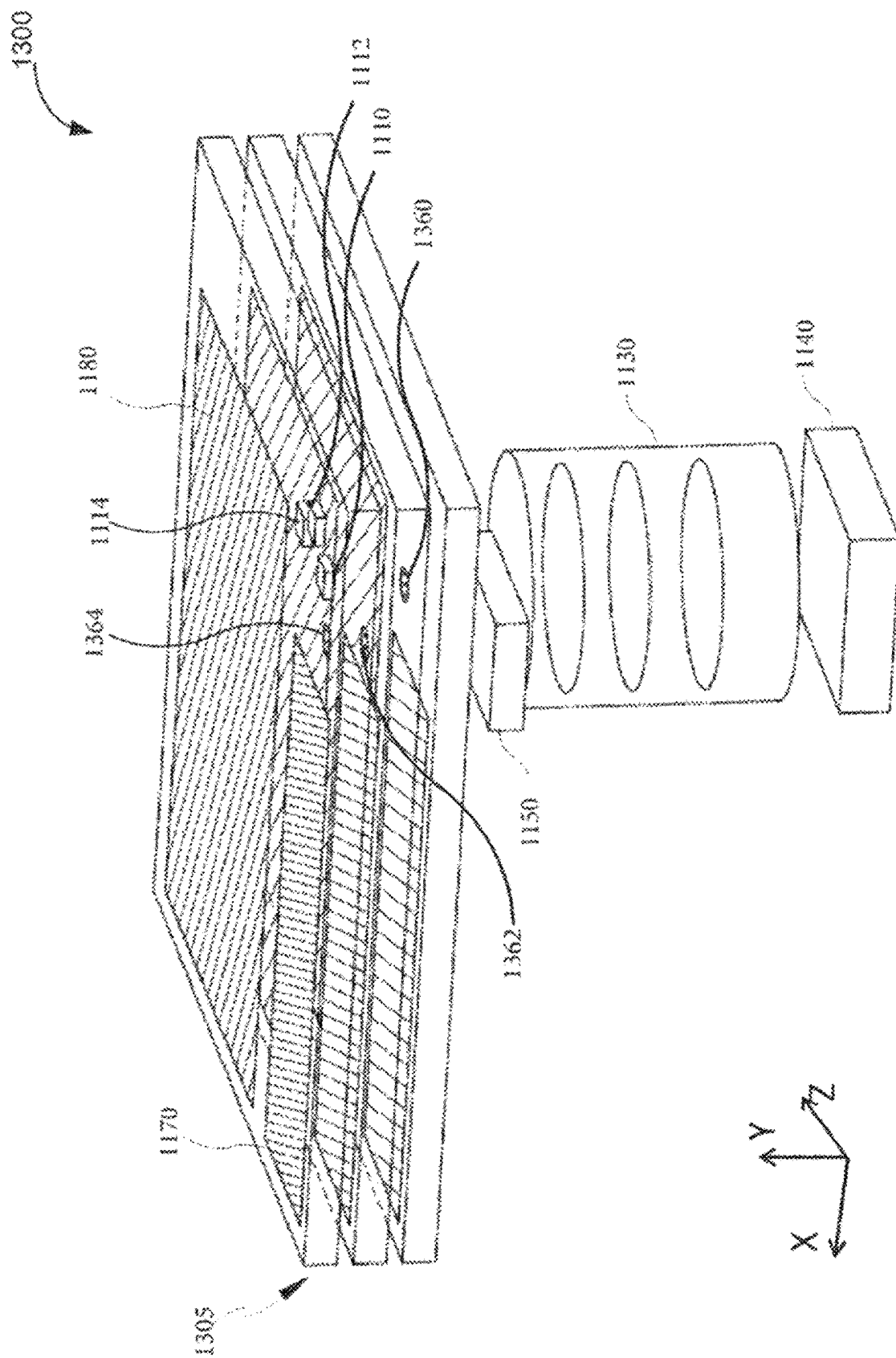


FIG. 13A

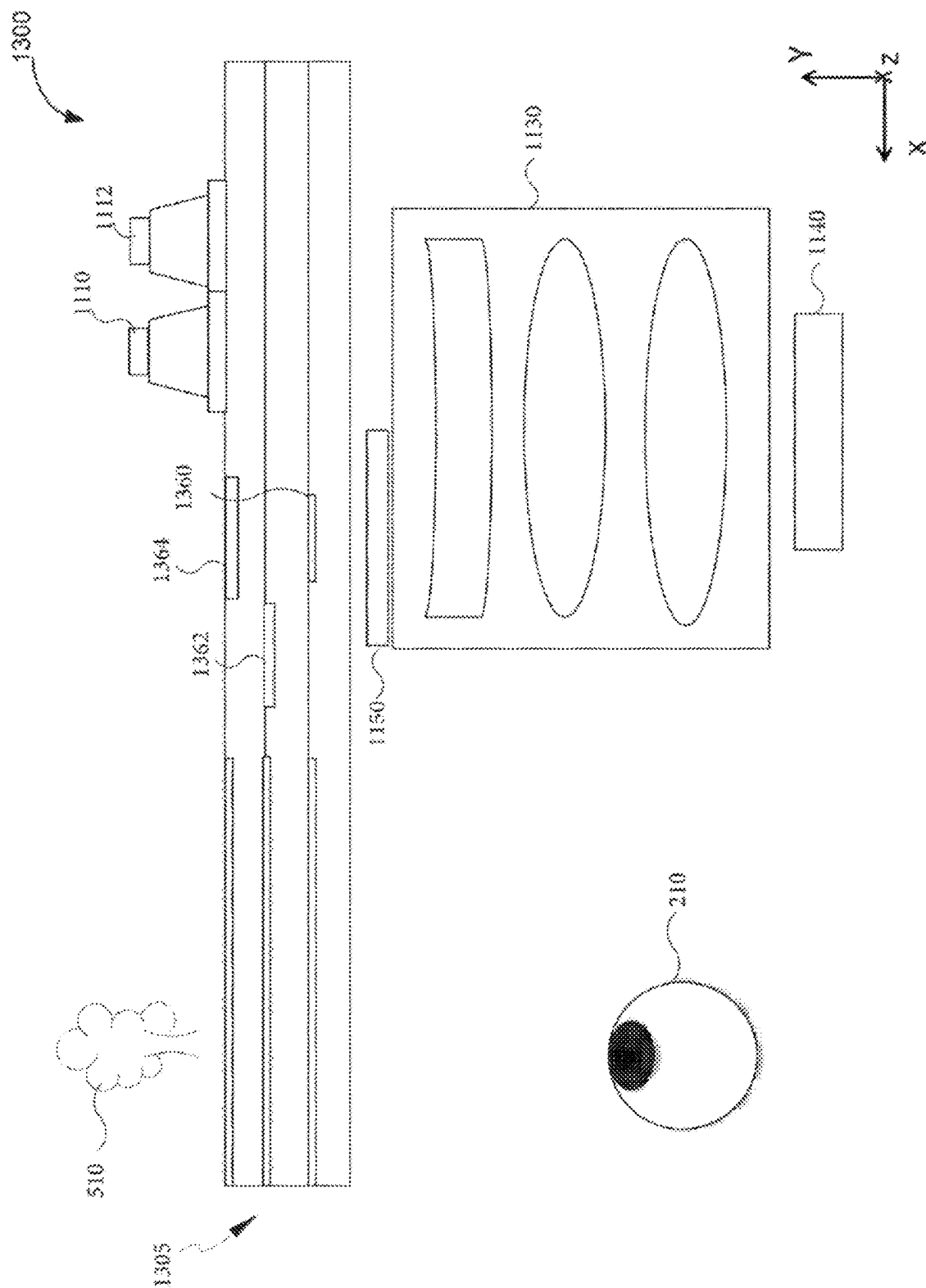


FIG. 13B

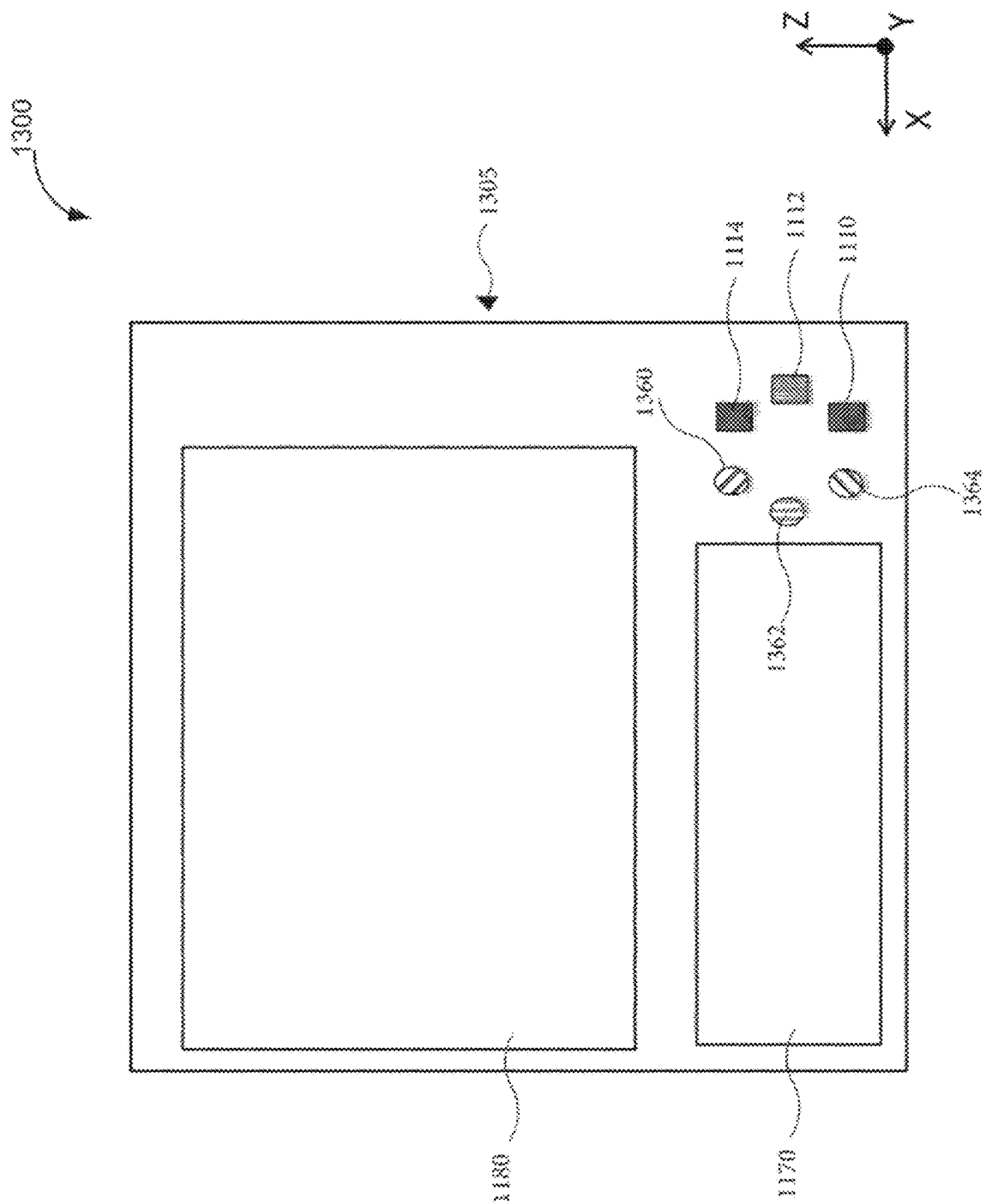


FIG. 13C

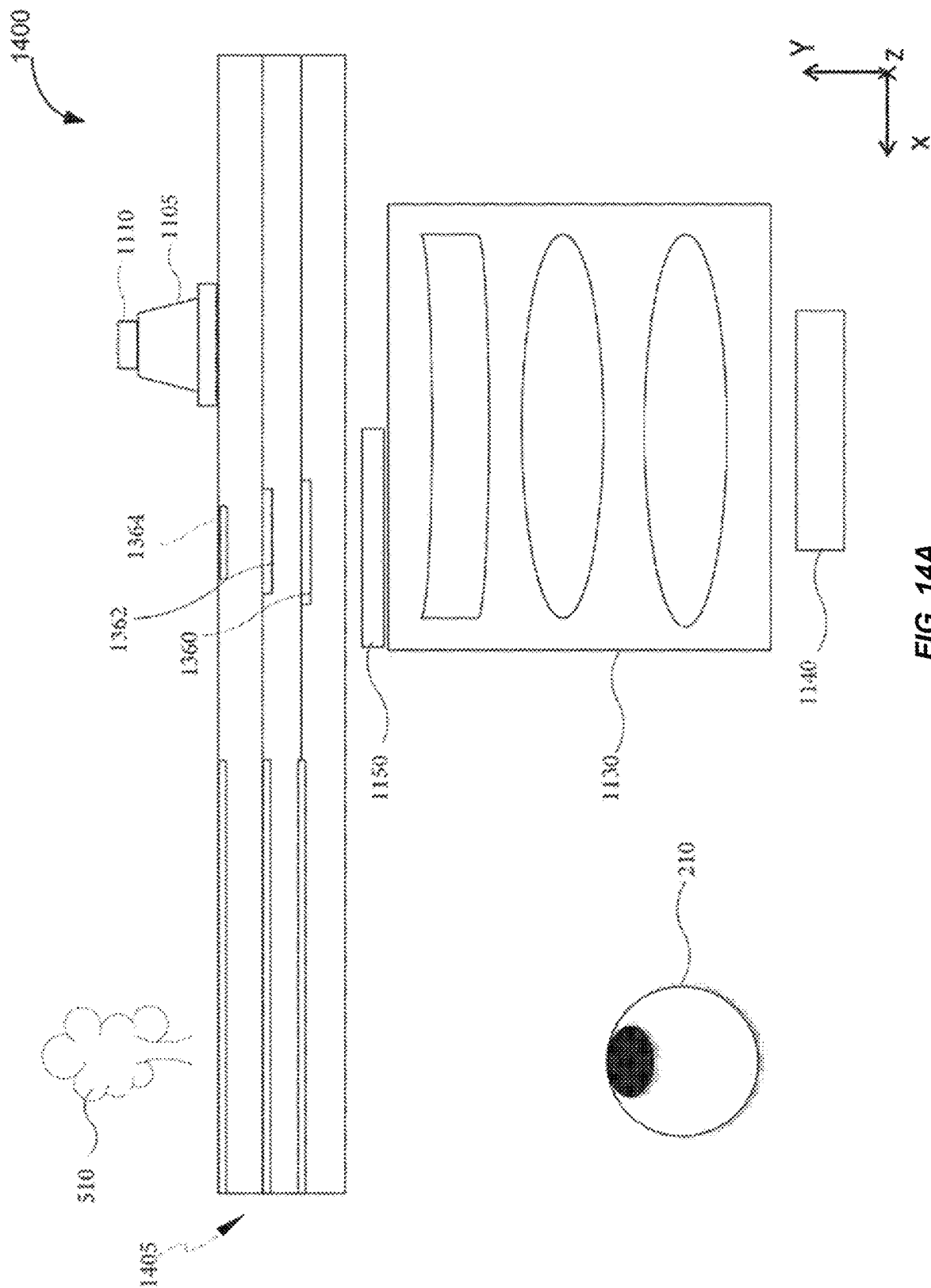


FIG. 14A

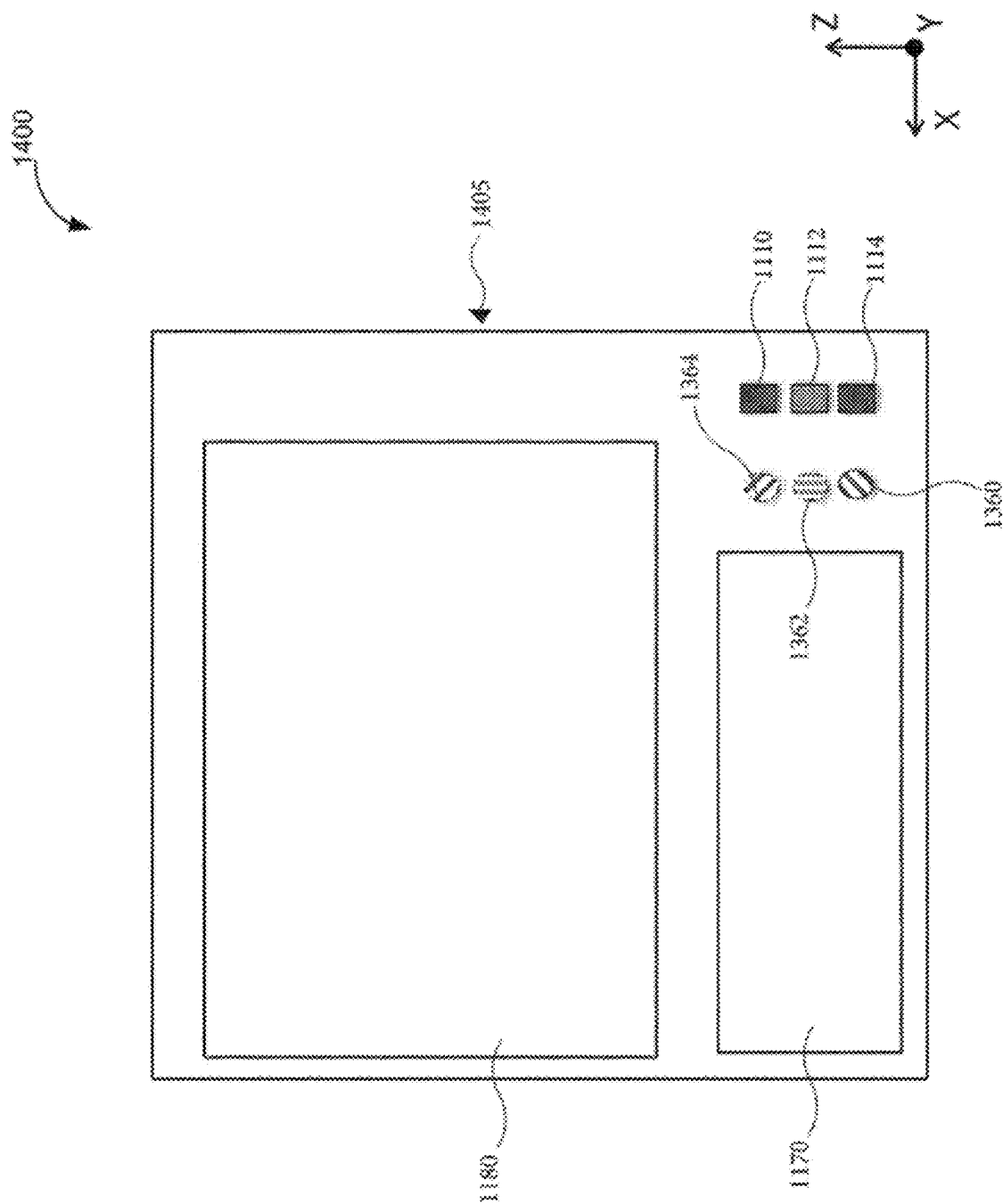


FIG. 14B

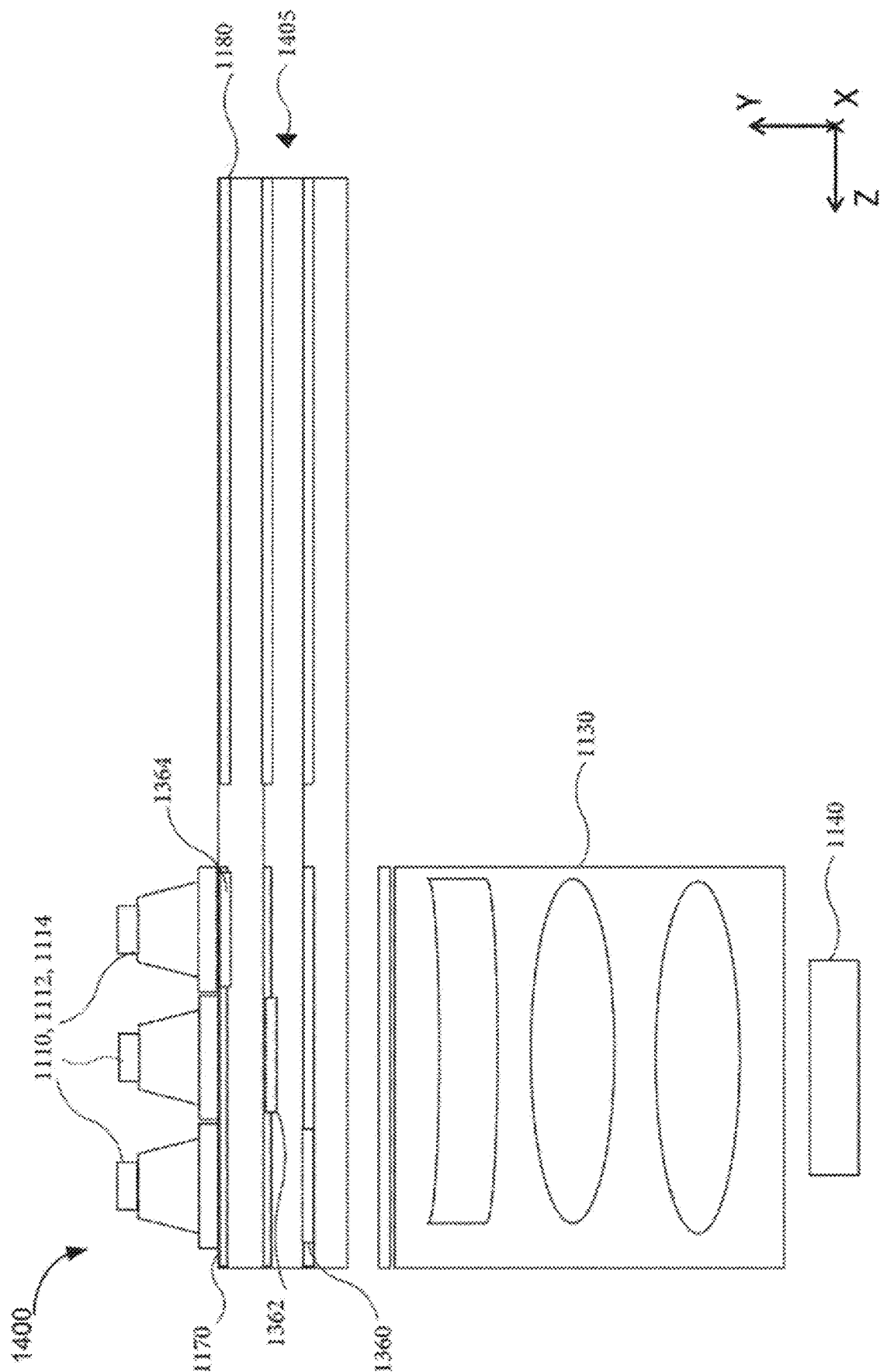


FIG. 14C

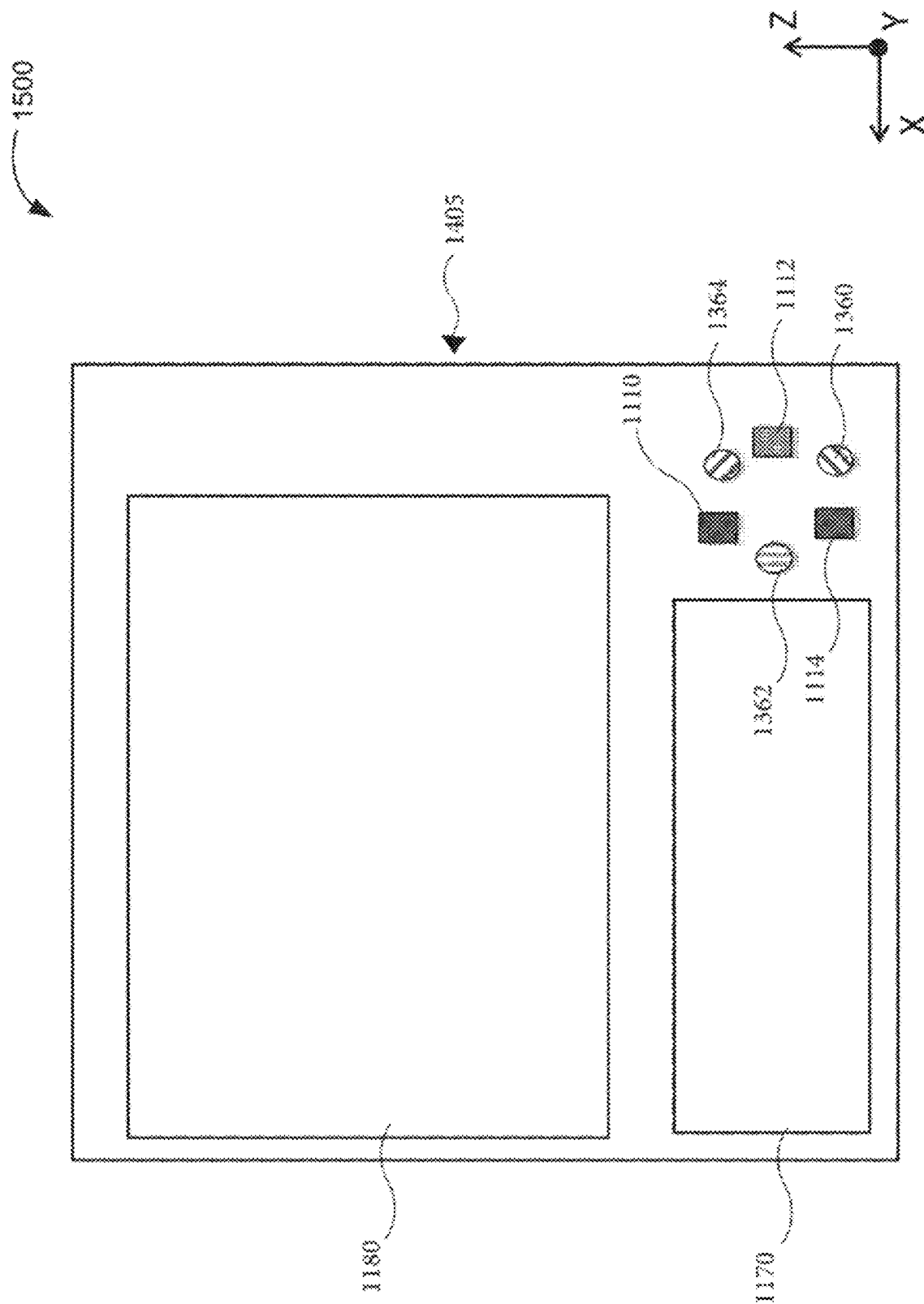
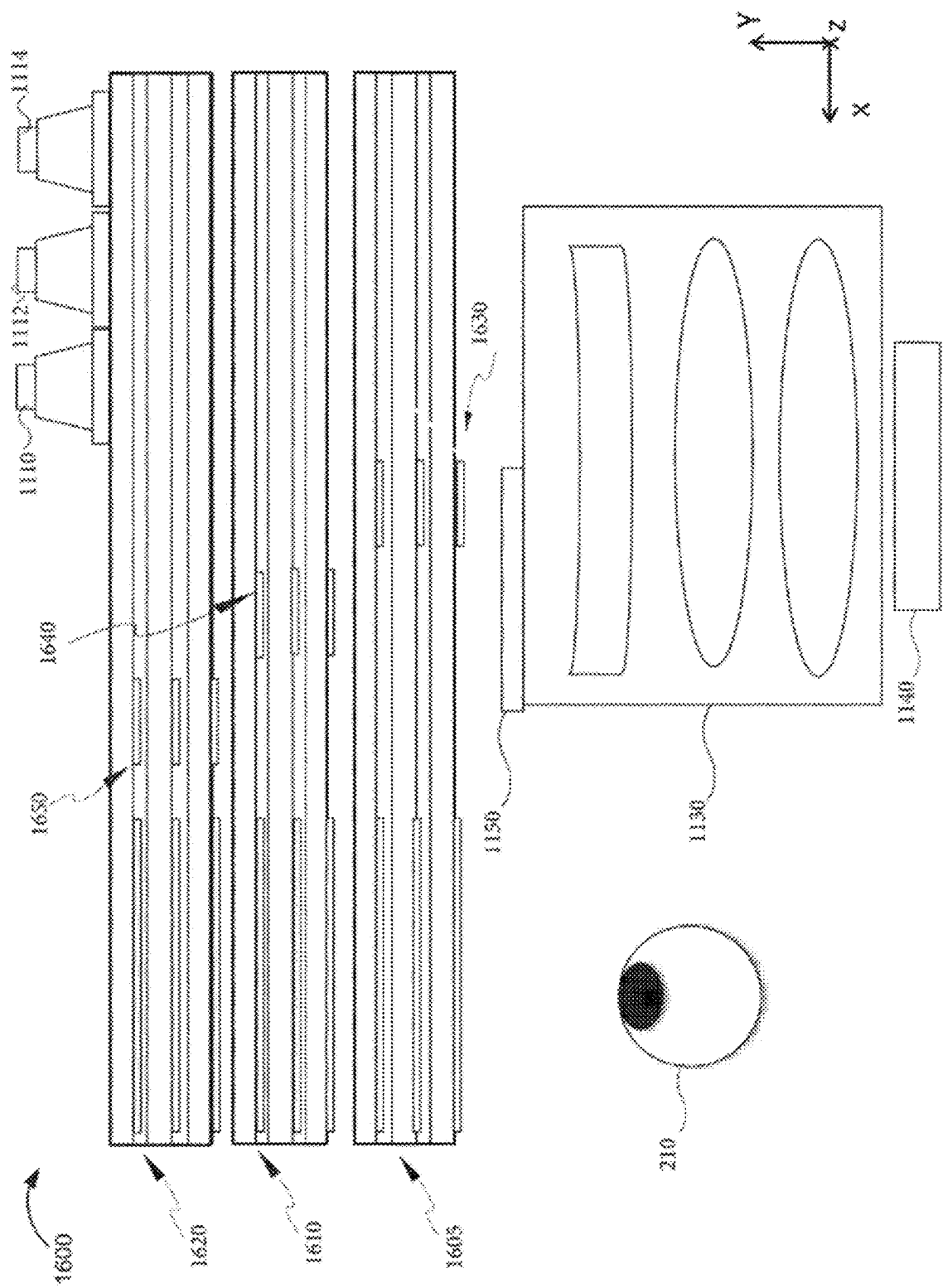


FIG. 15



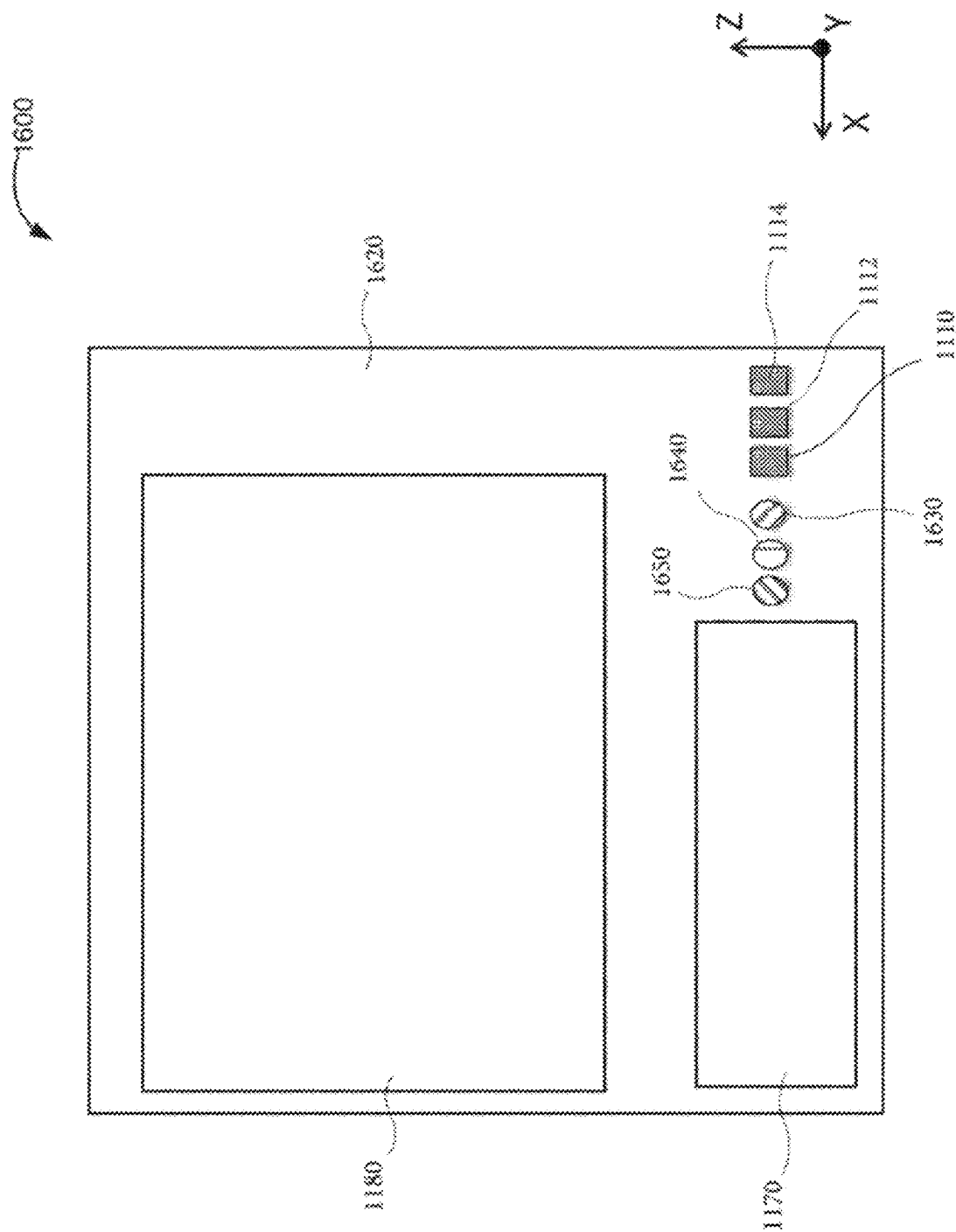


FIG. 16B

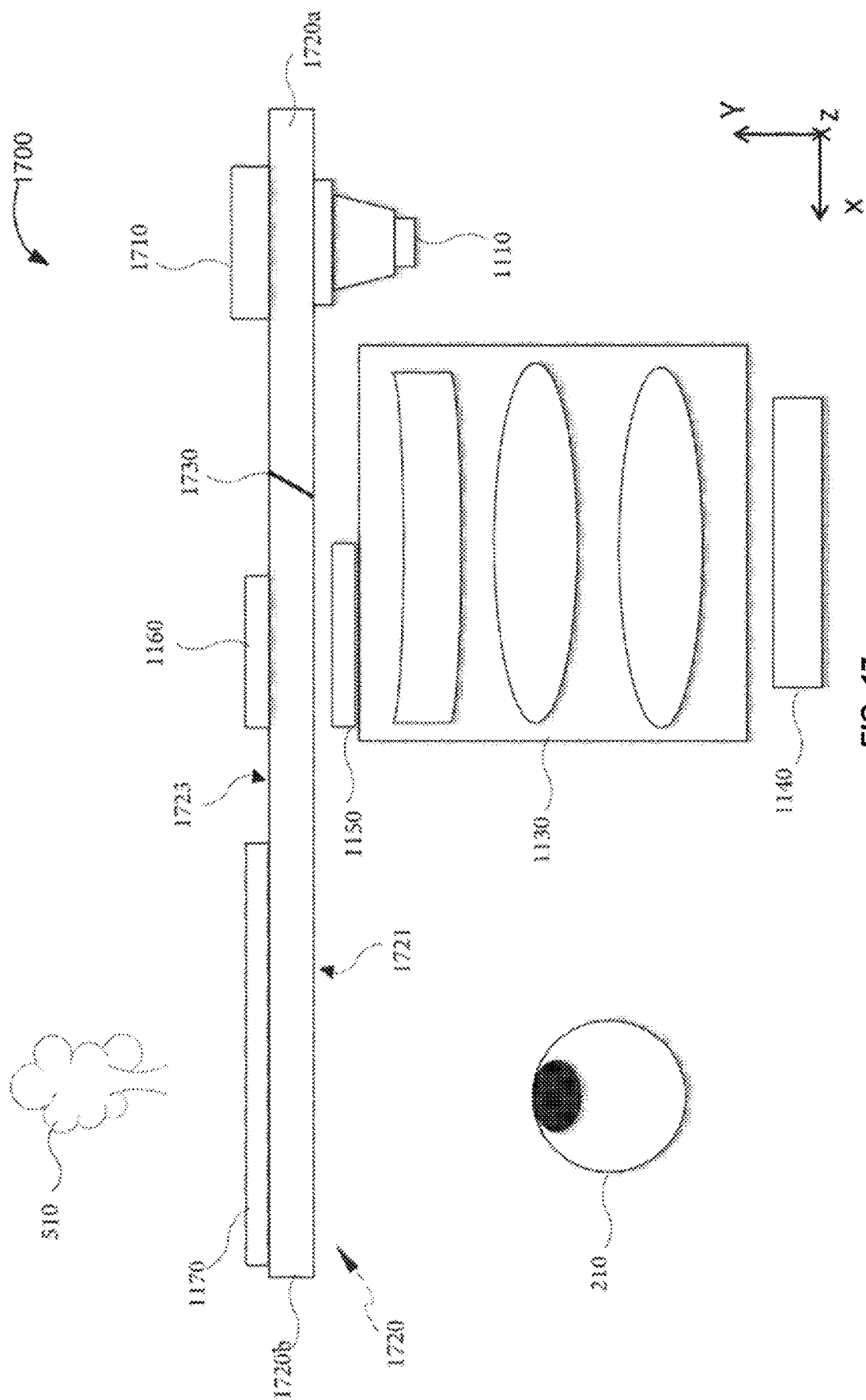


FIG. 17

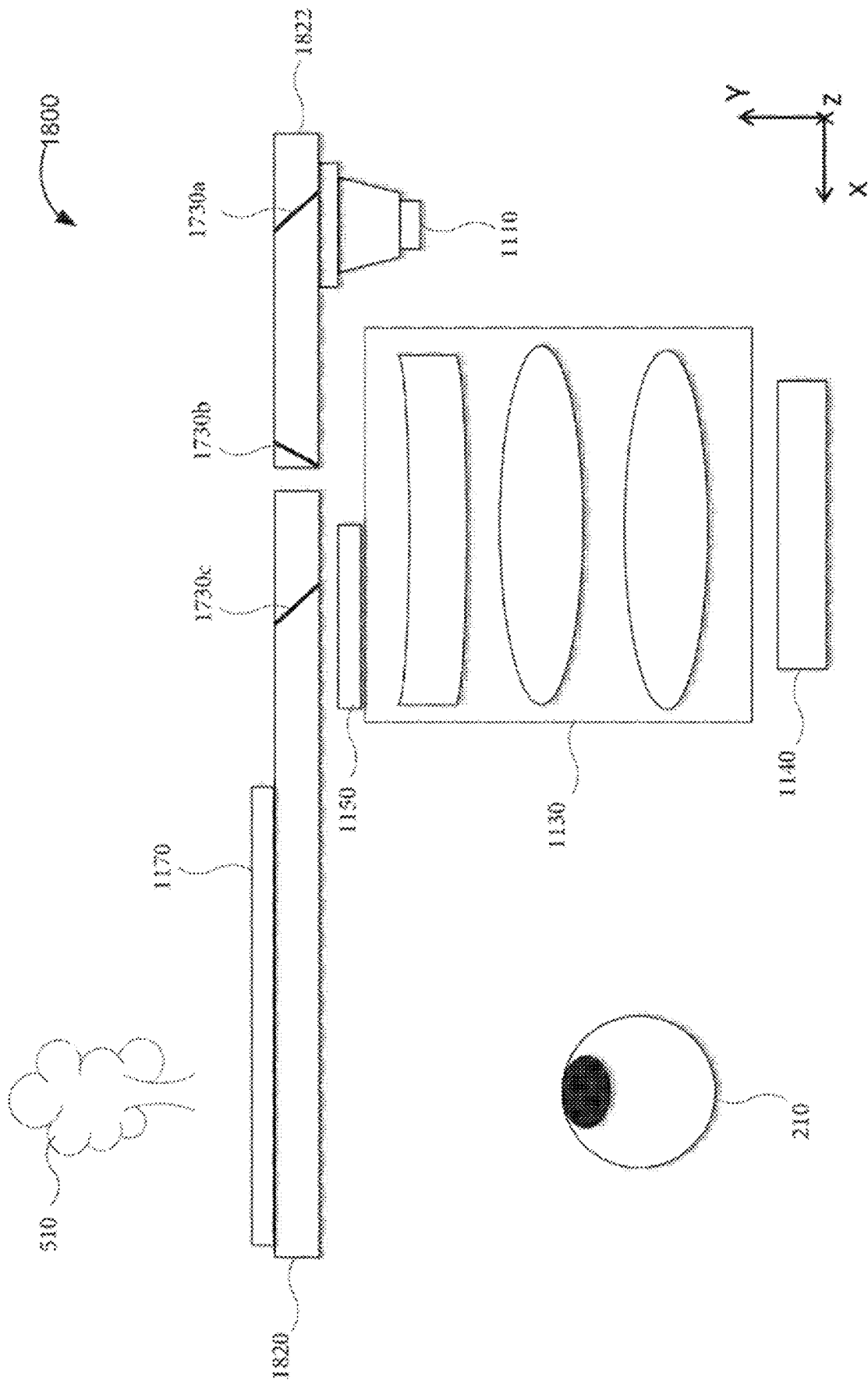


FIG. 18

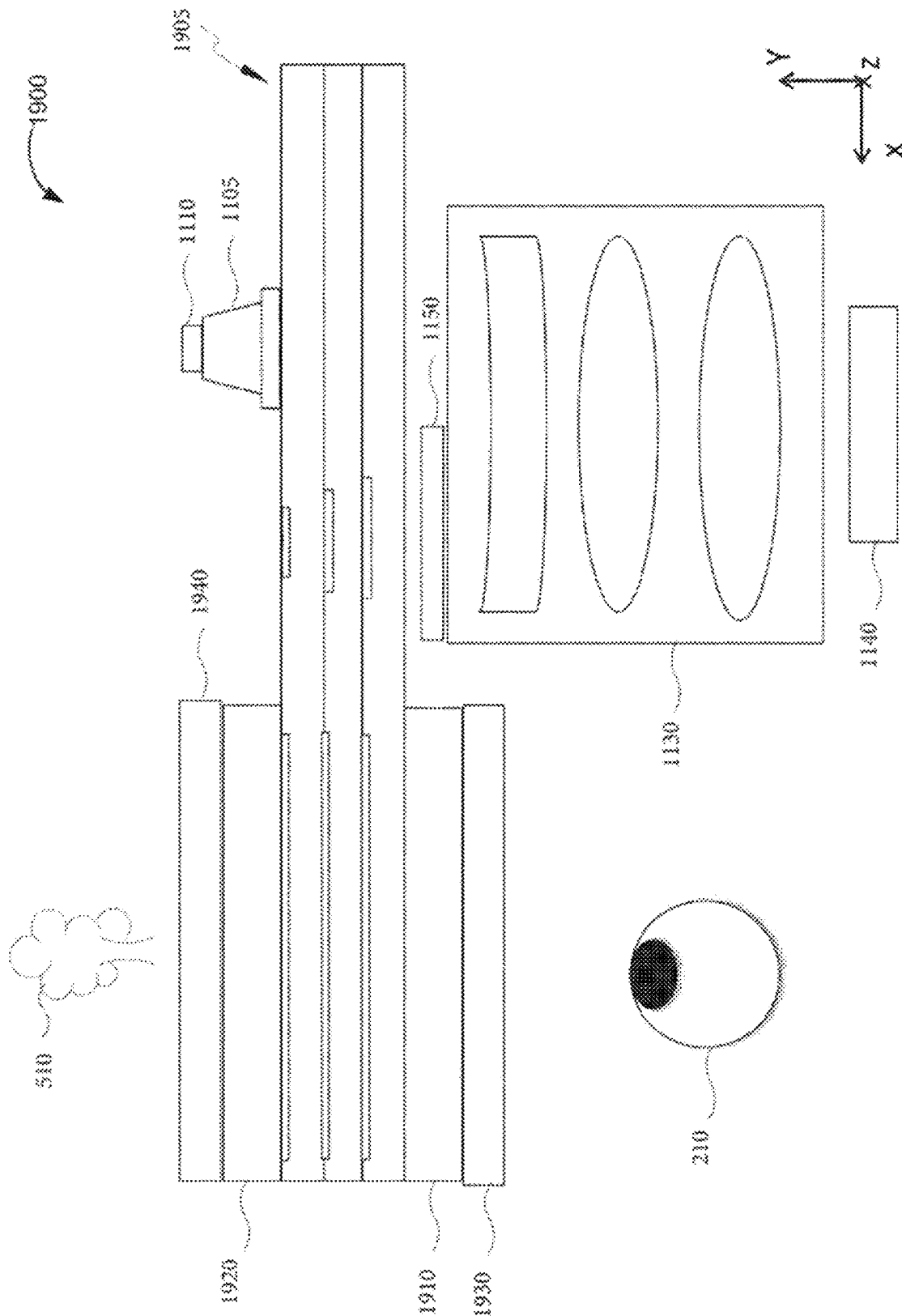
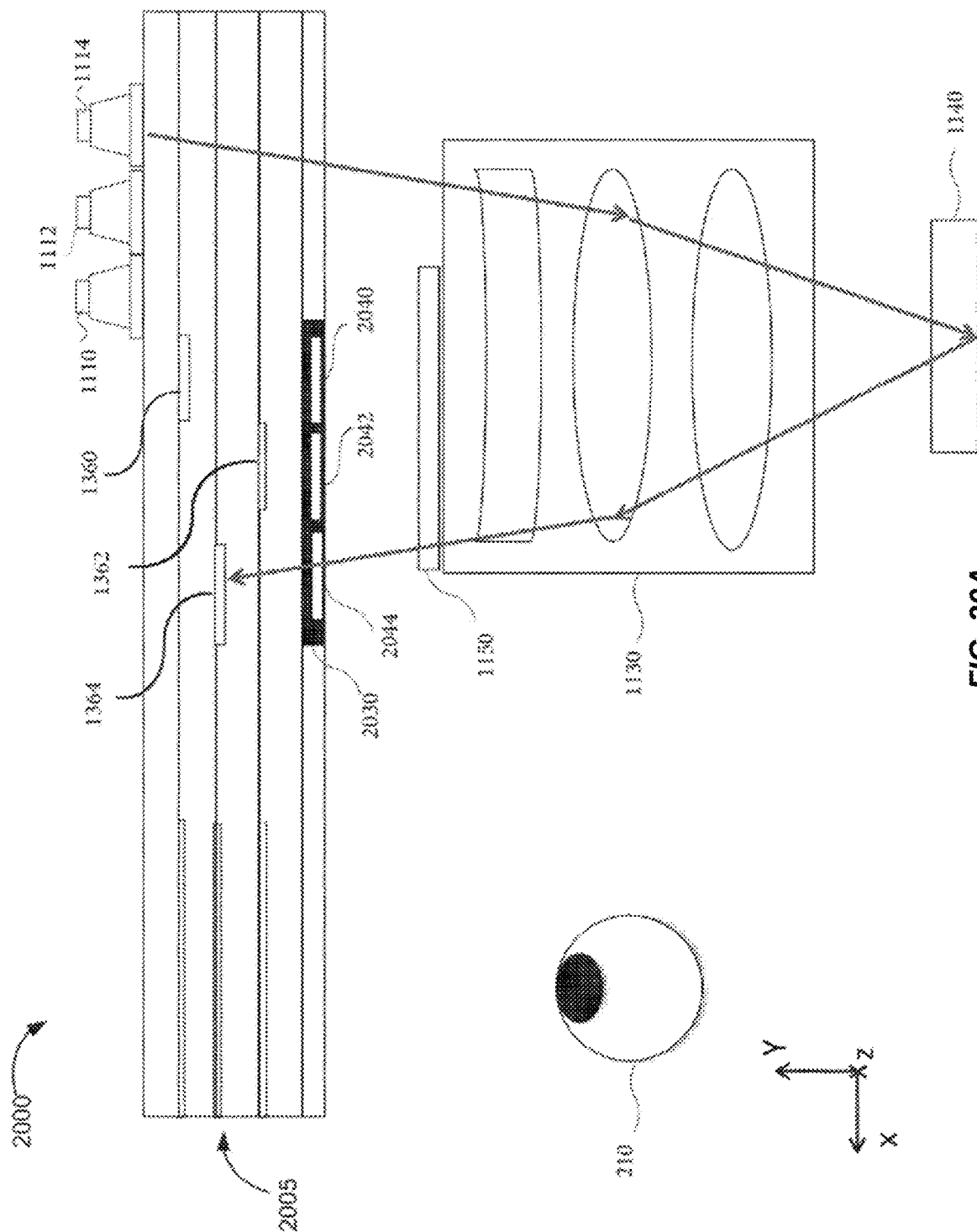
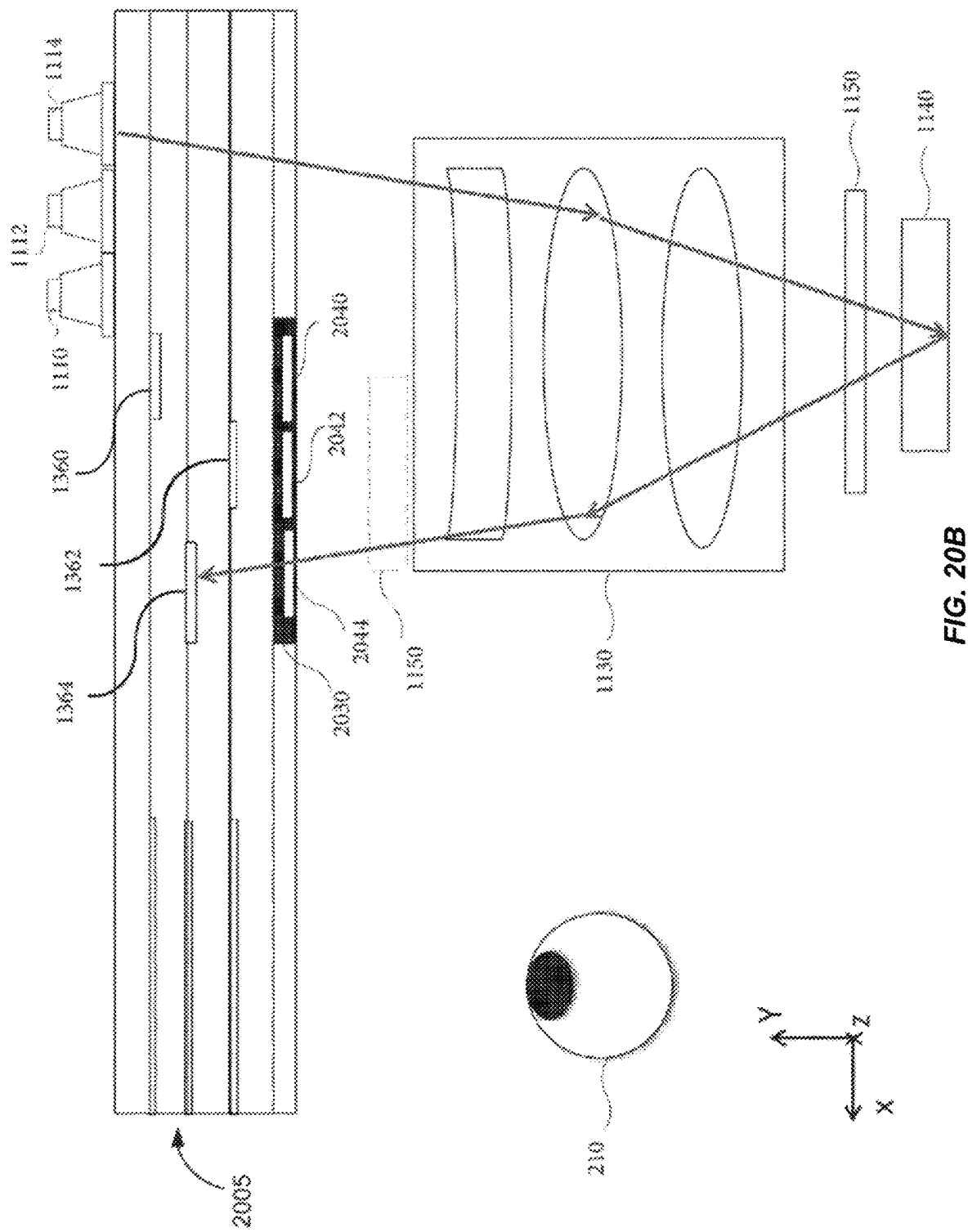
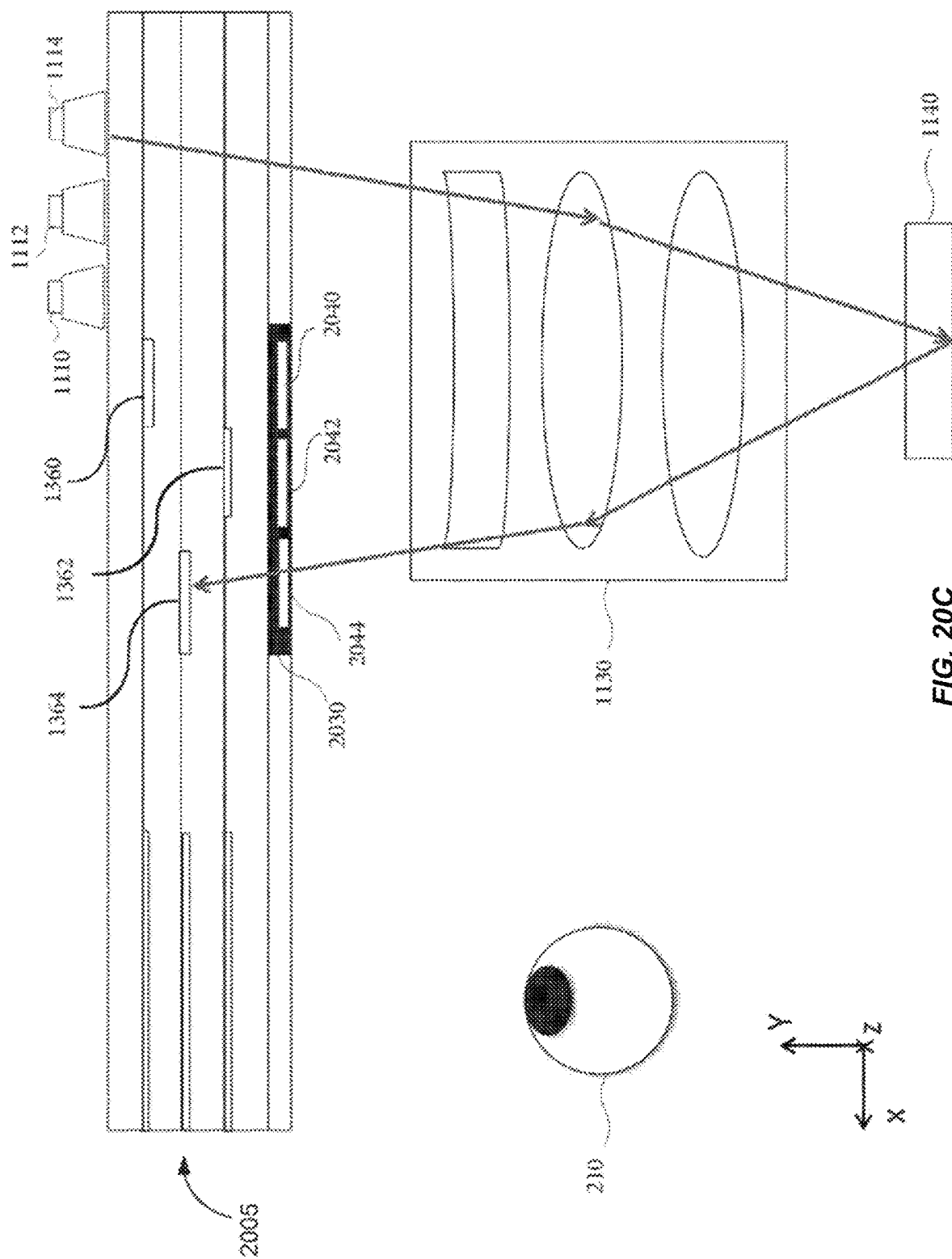


FIG. 19







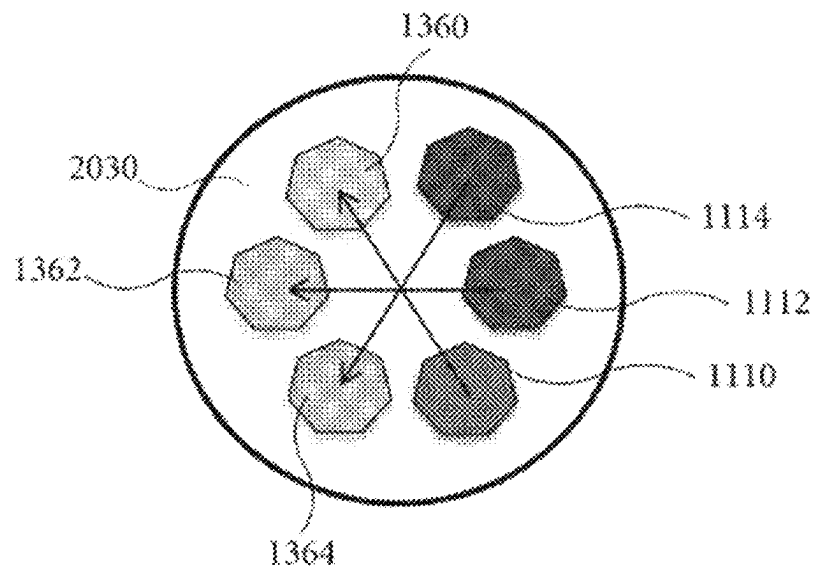


FIG 20D

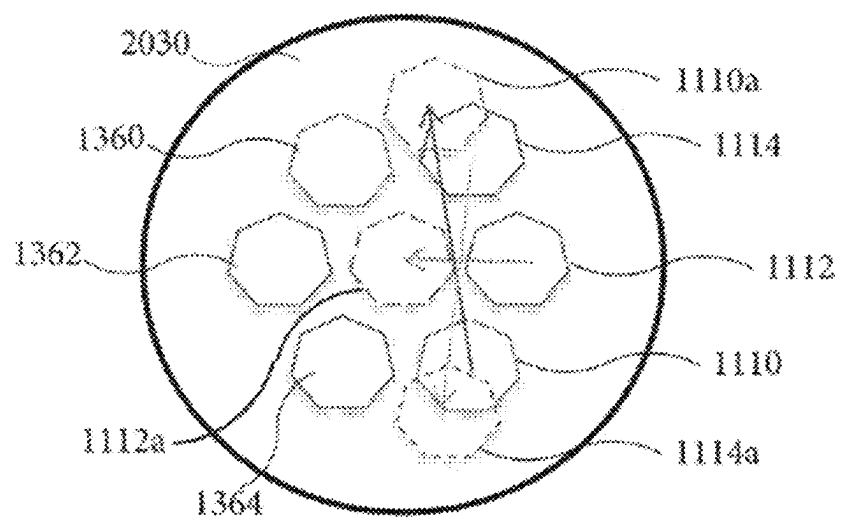
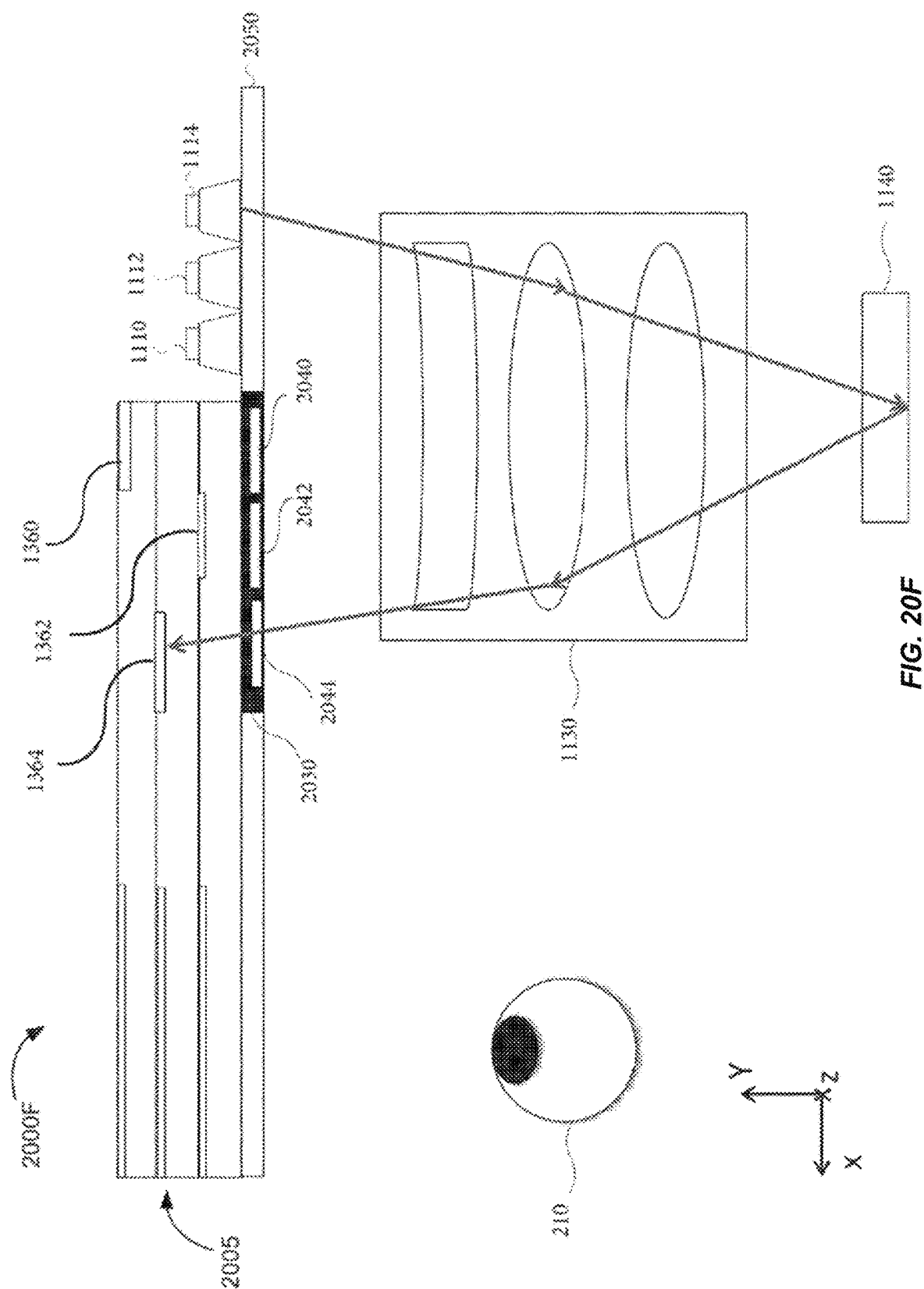
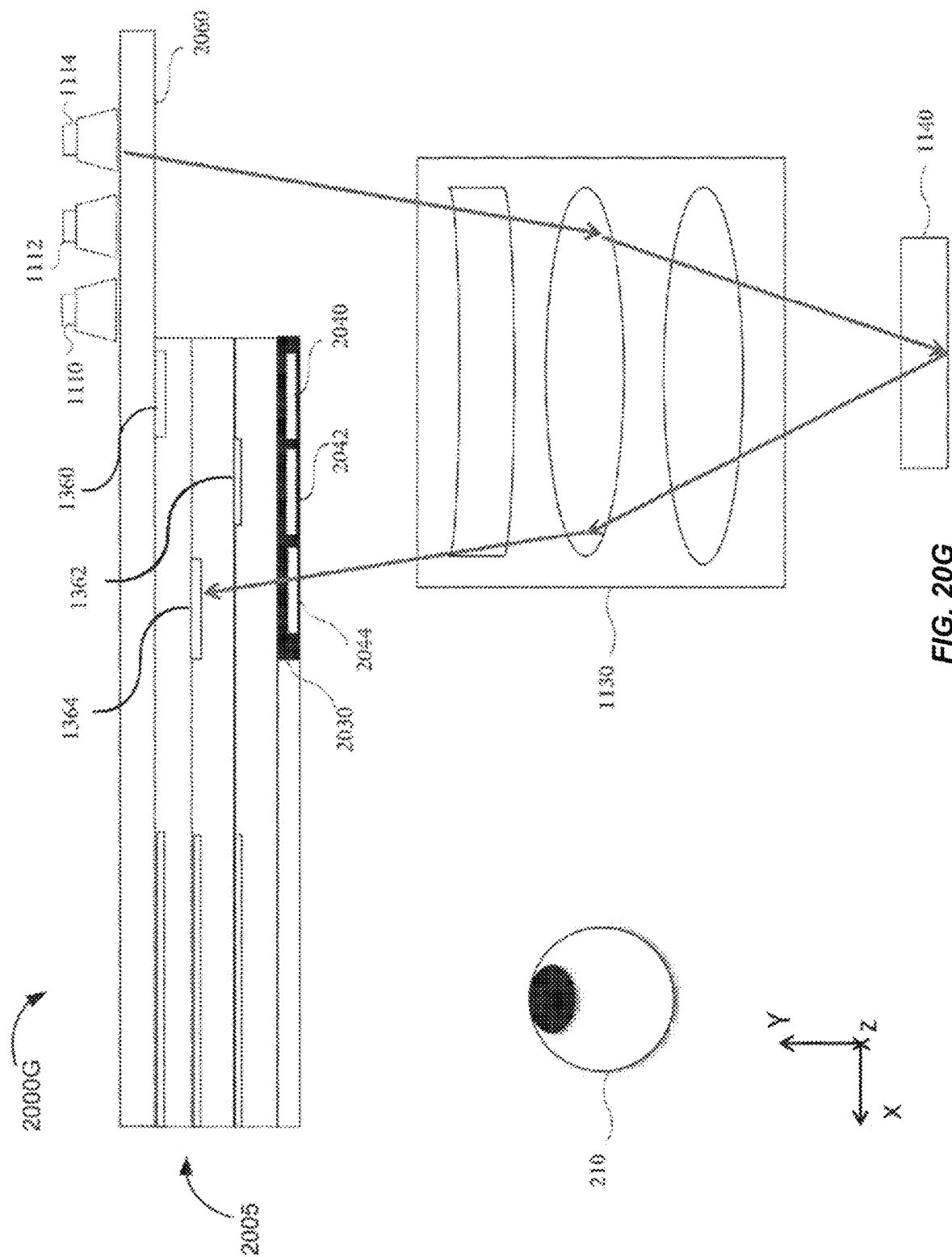


FIG 20E





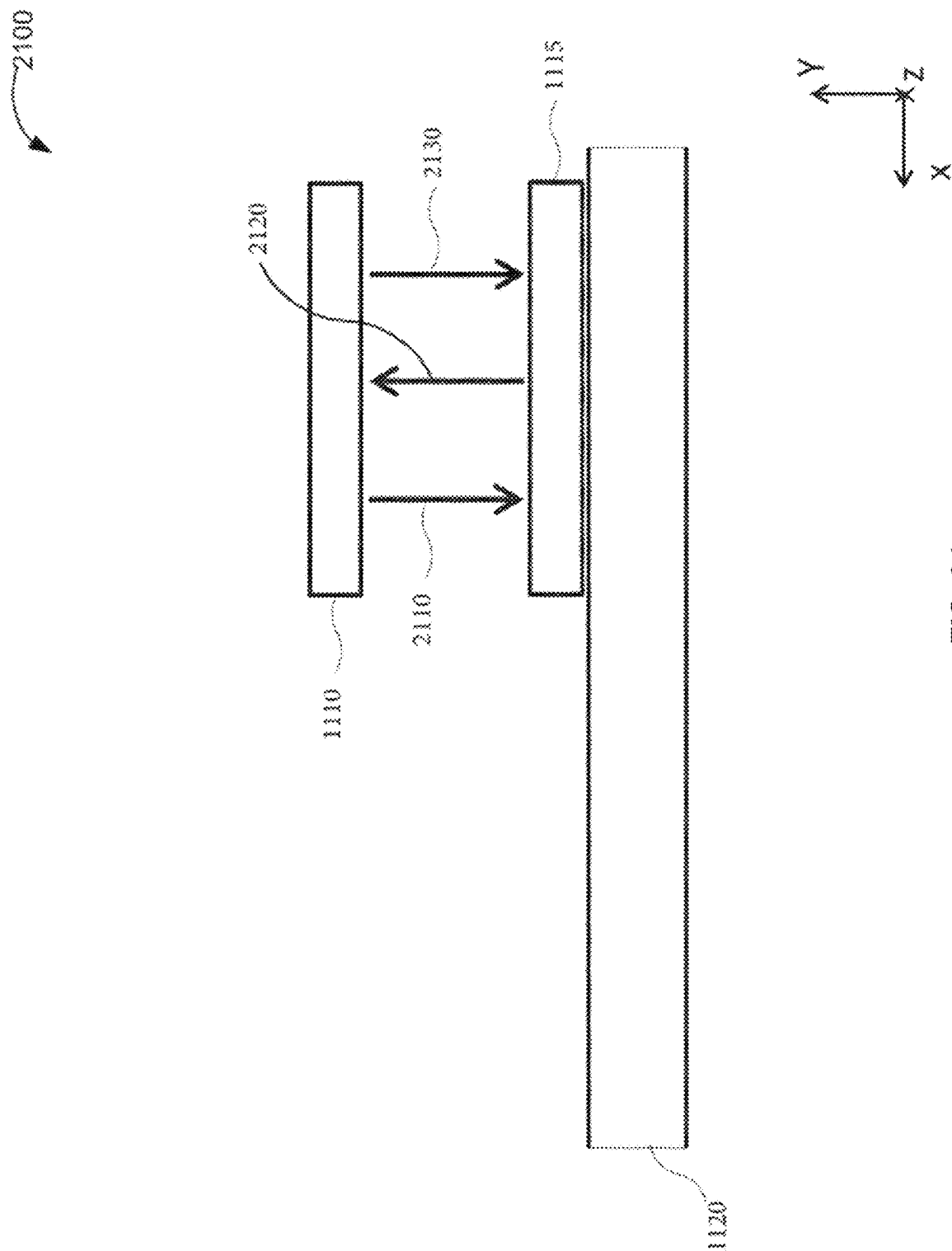


FIG. 21

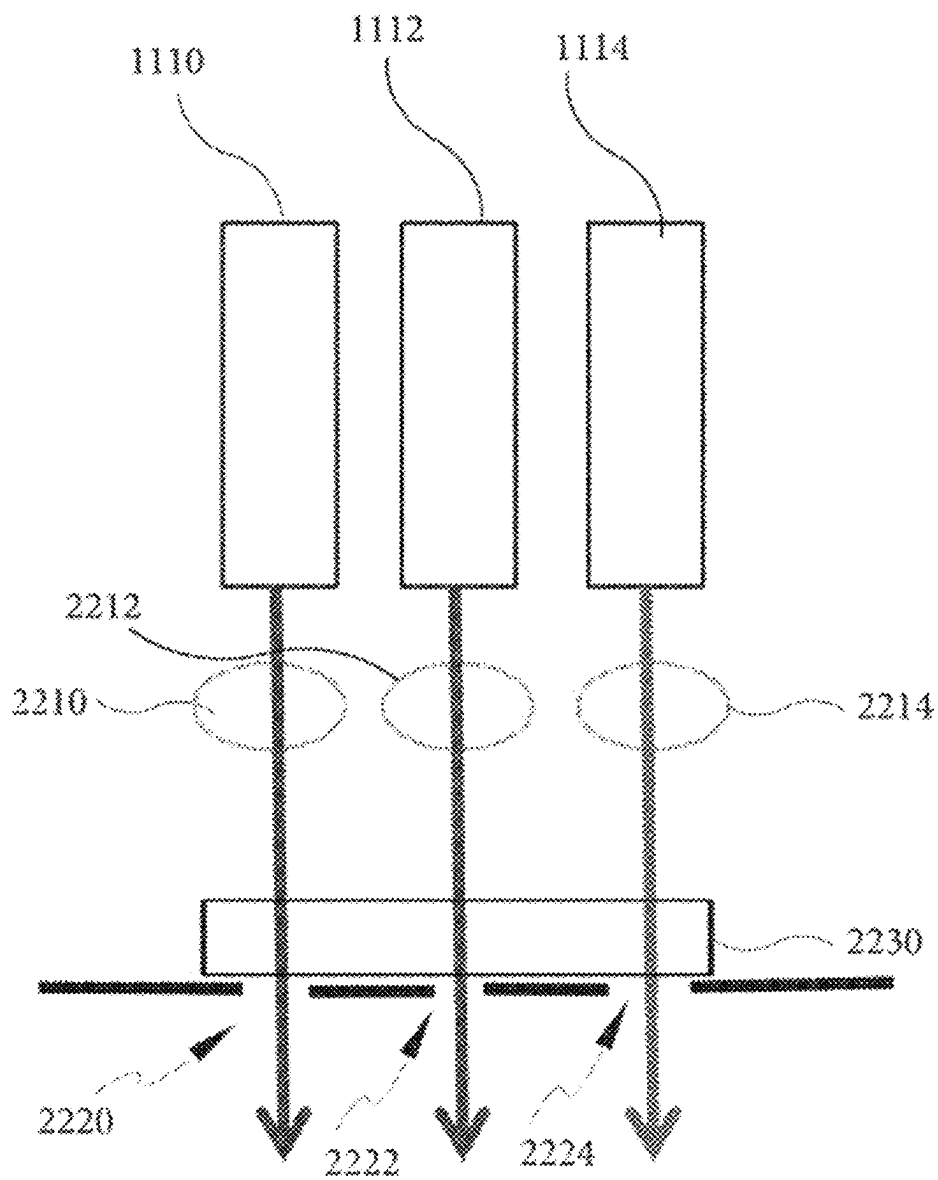


FIG 22

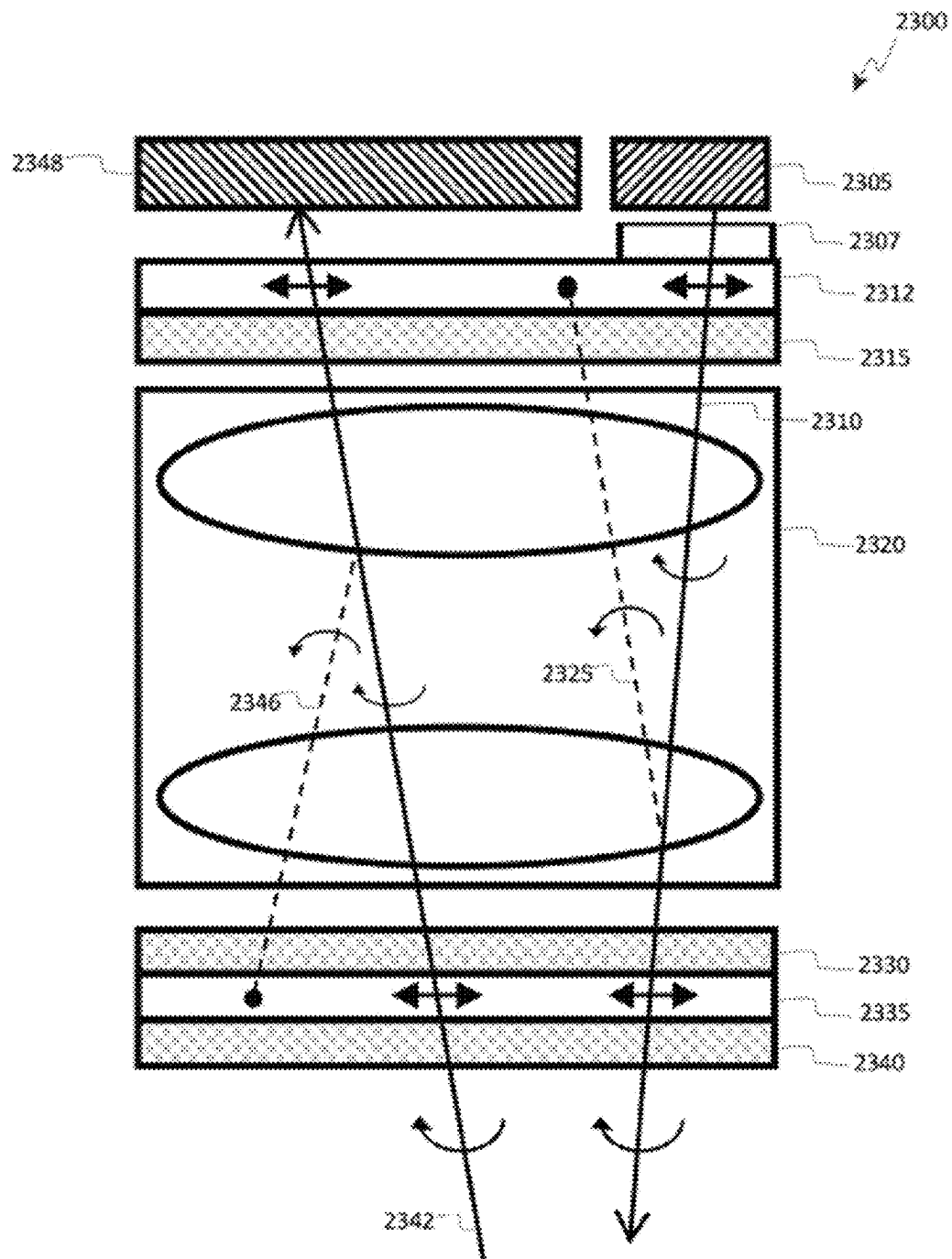


FIG 23A

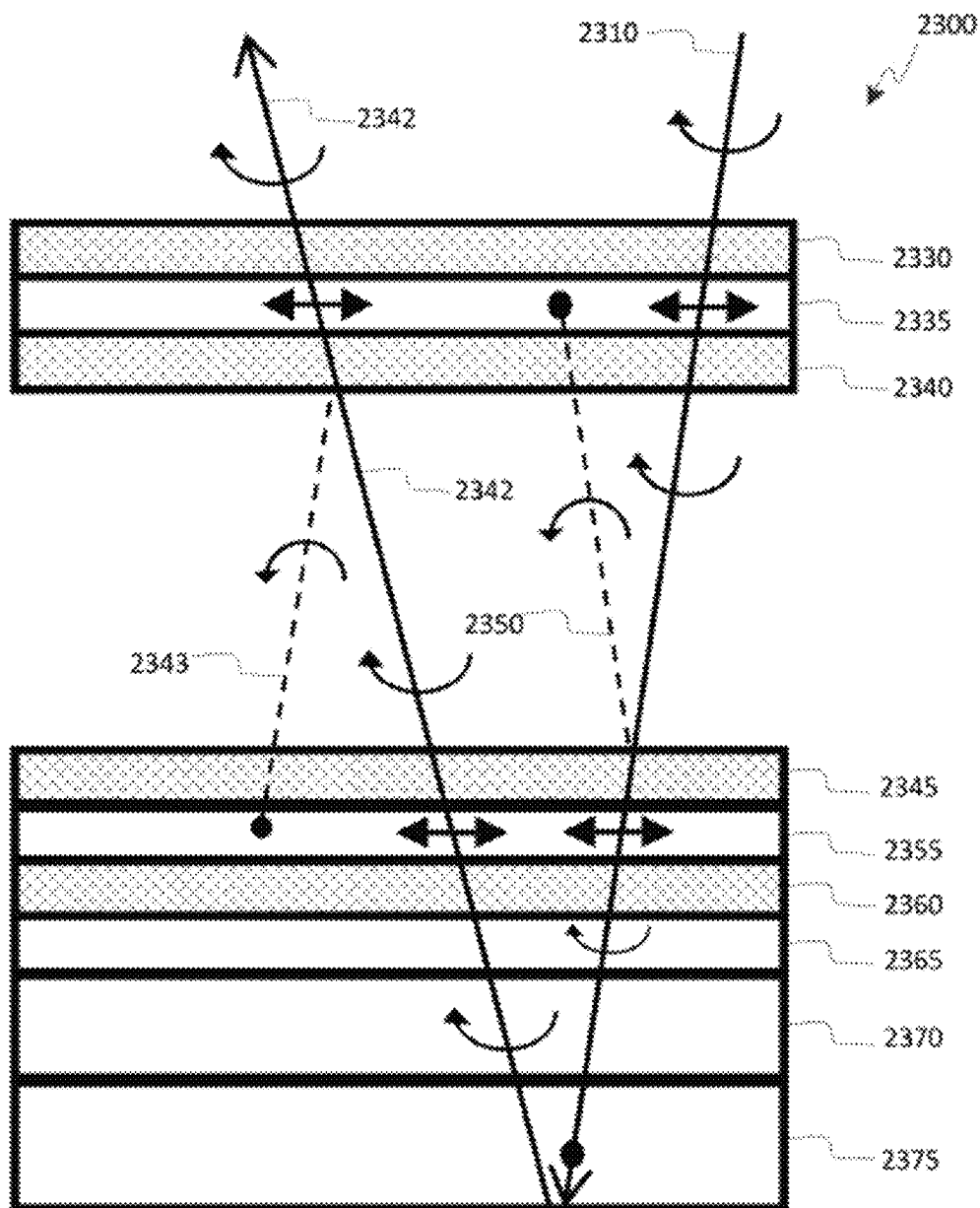


FIG 23B

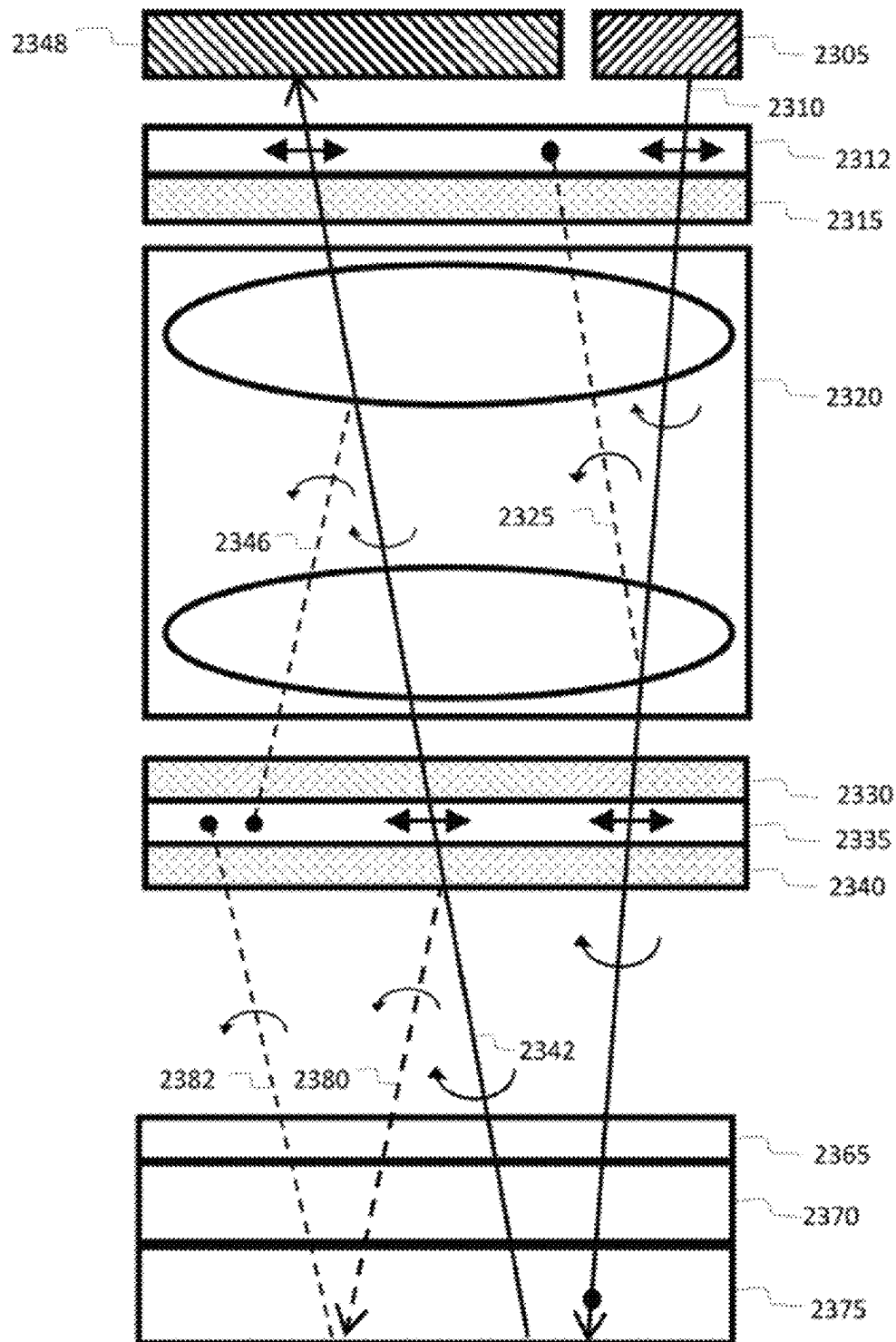


FIG 23C

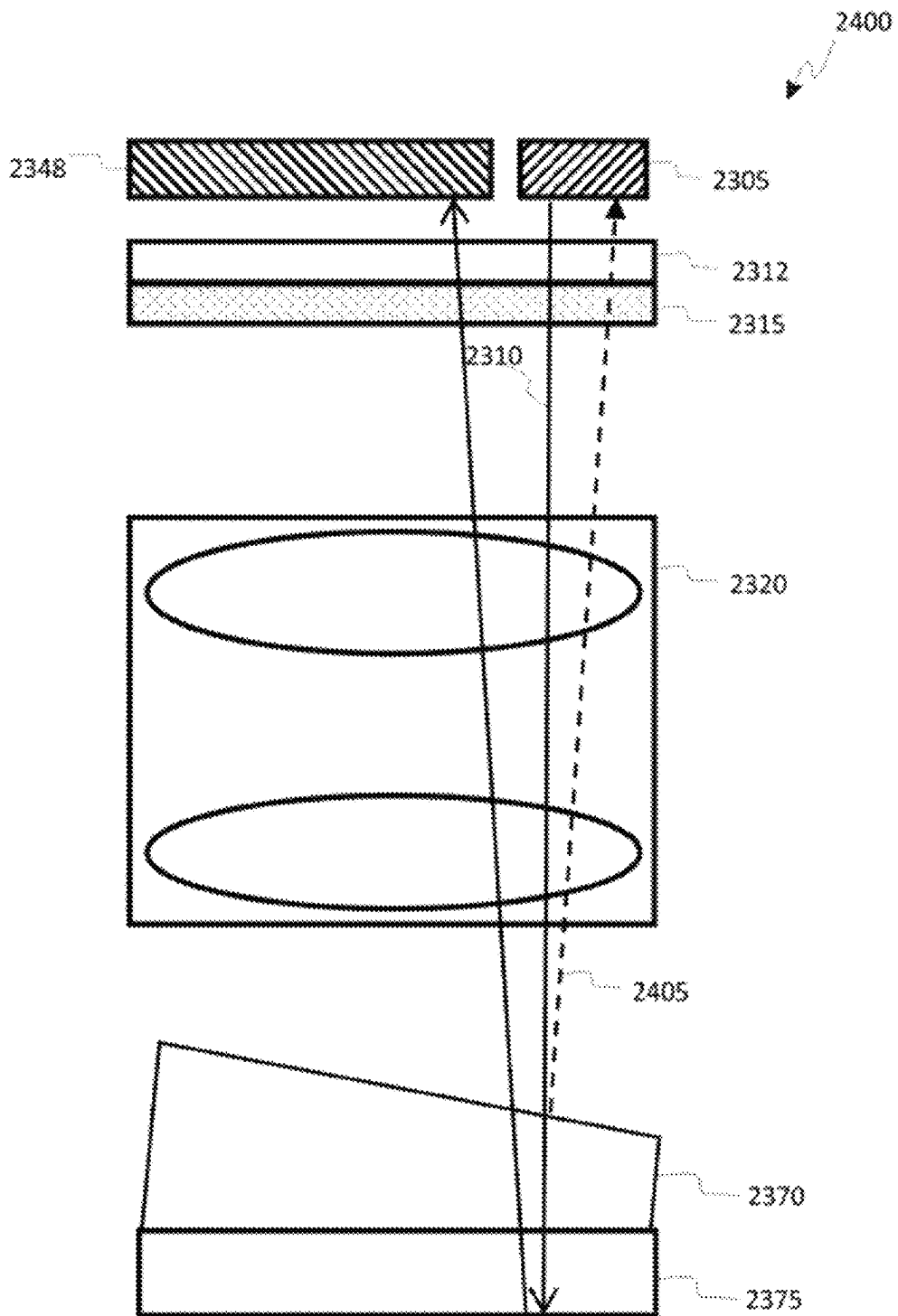


FIG 24

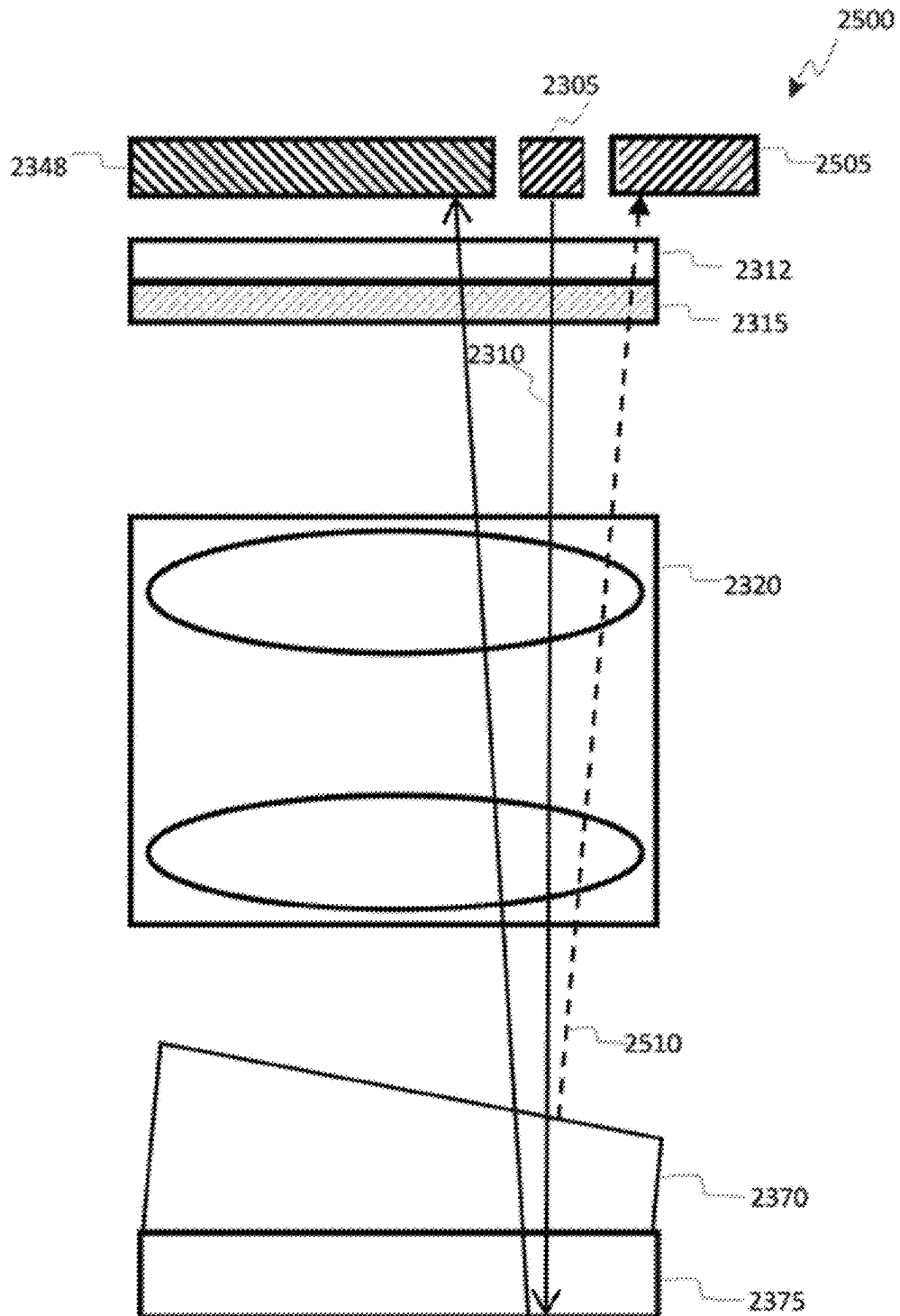


FIG 25

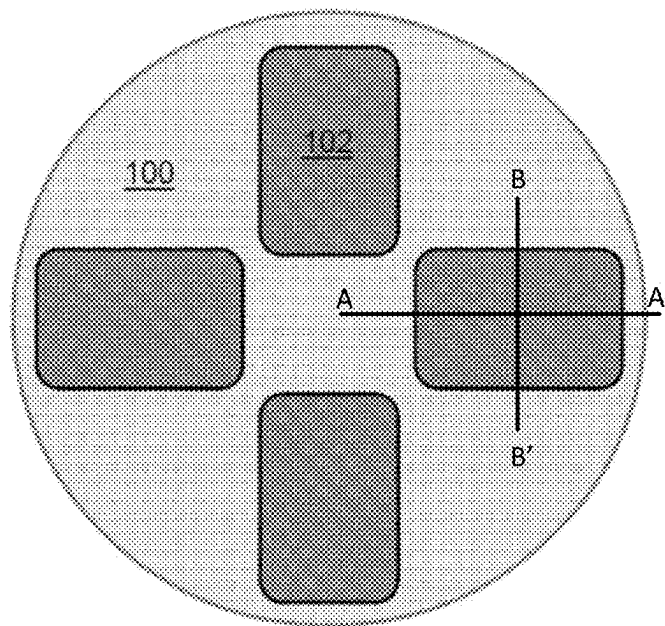


FIG. 26A

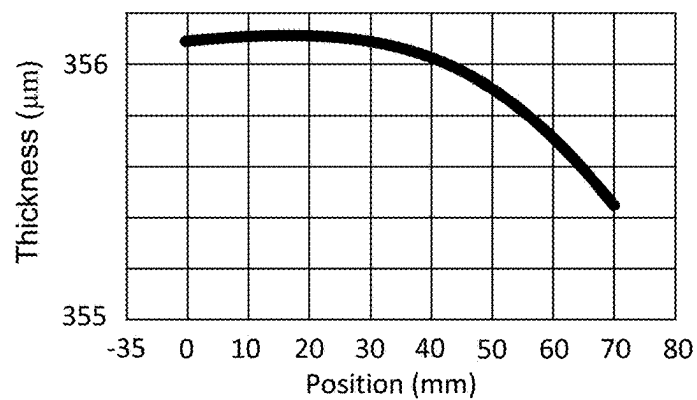


FIG. 26B

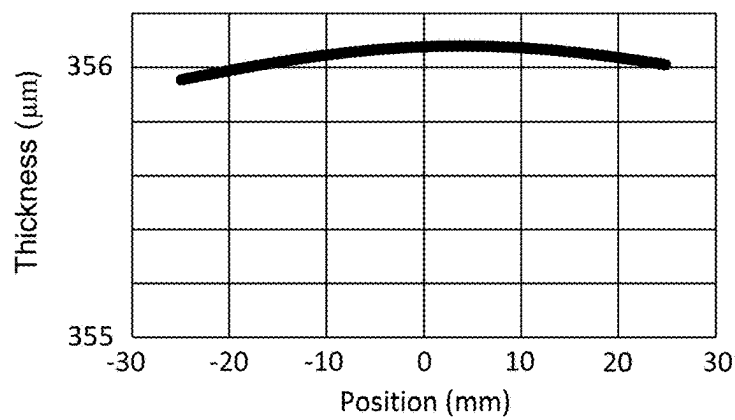


FIG. 26C

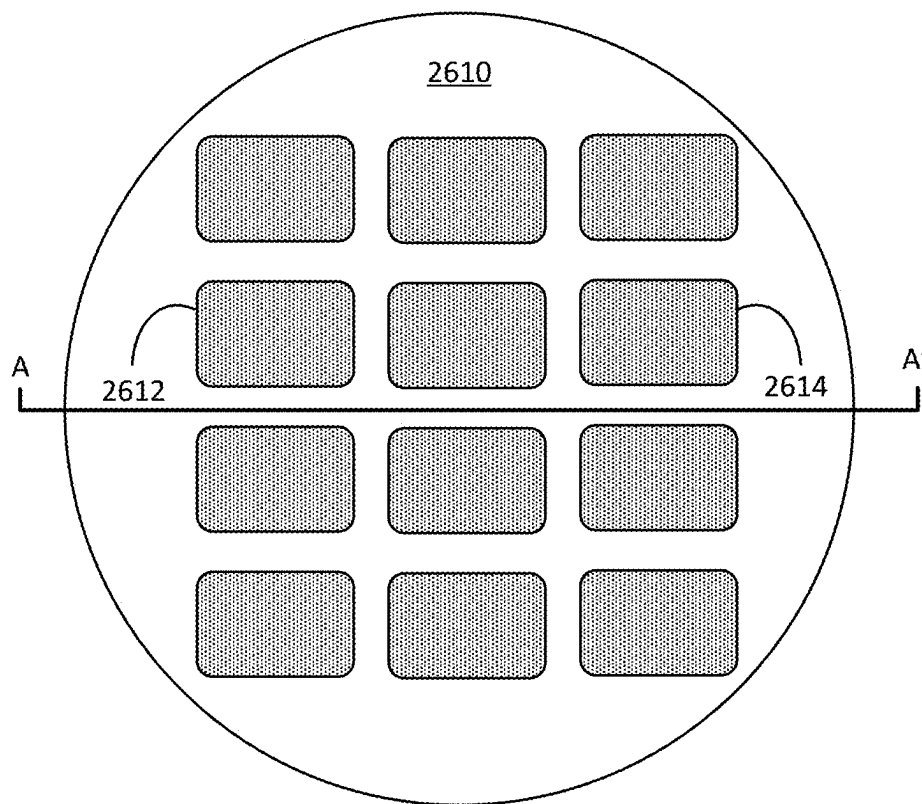


FIG. 26D

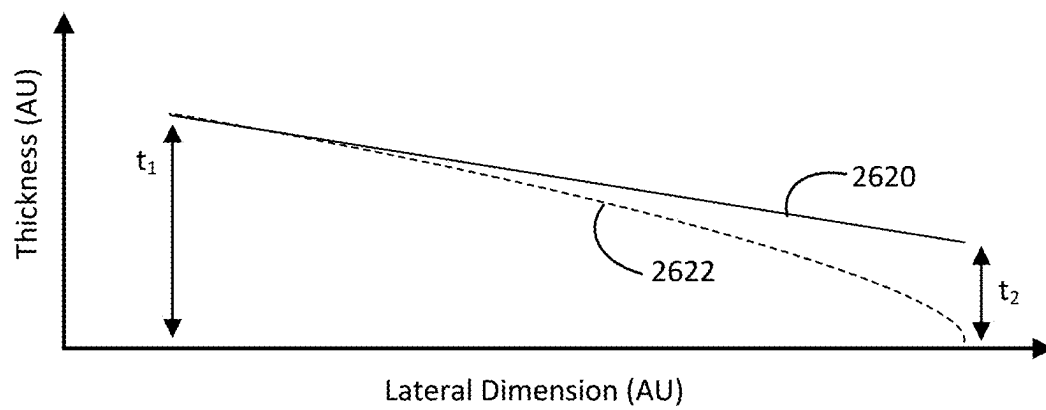
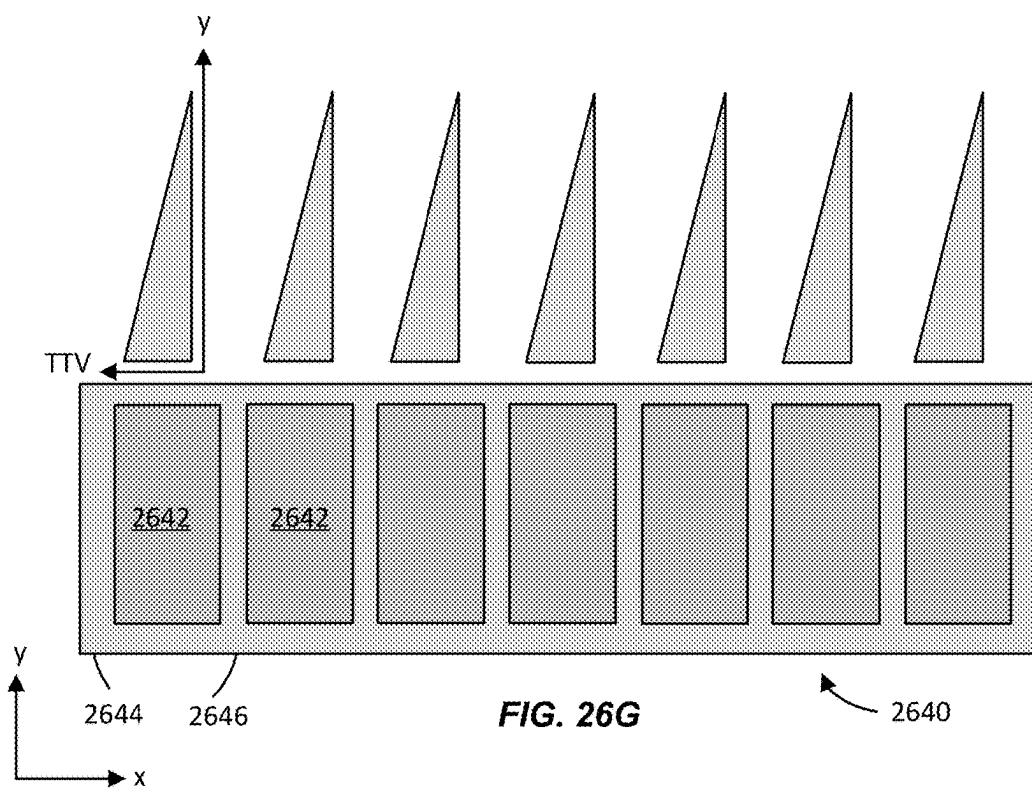
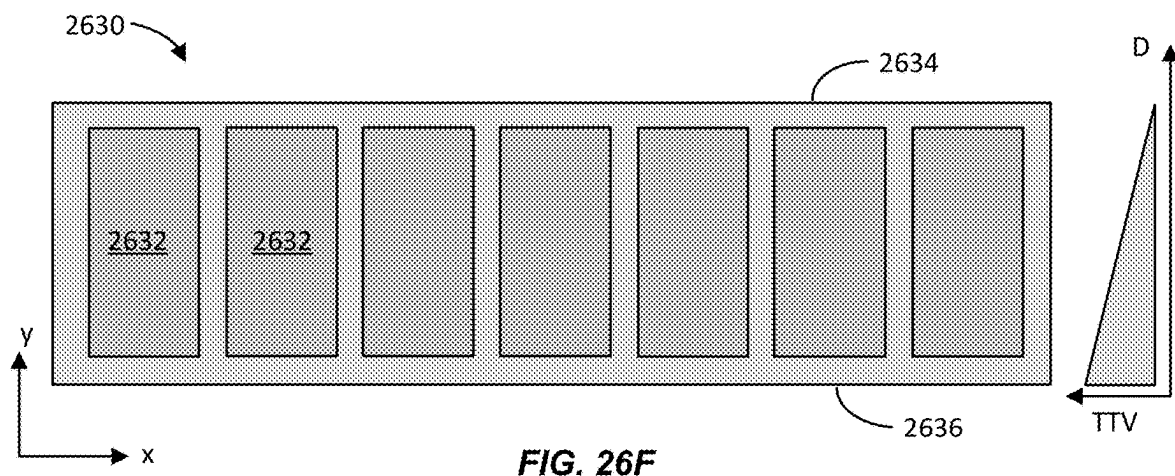


FIG. 26E



200



FIG. 27A

202



FIG. 27B

204



FIG. 27C

2706

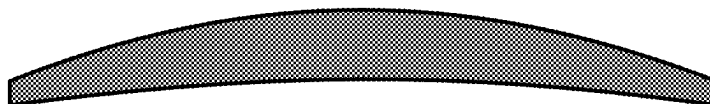


FIG. 27D

2708



FIG. 27E

2710



FIG. 27F

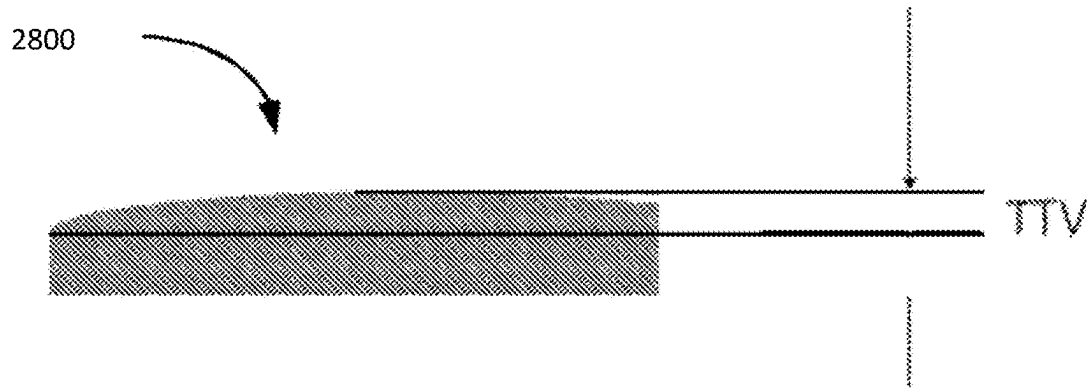


FIG. 28A

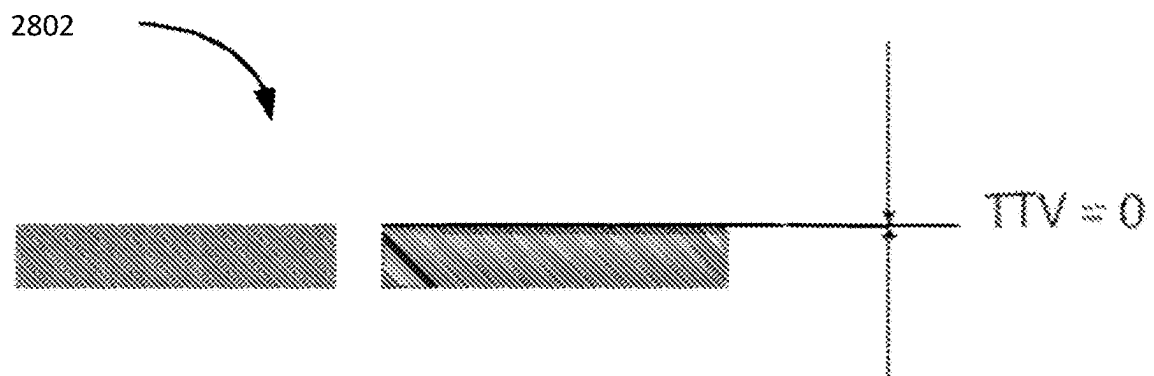
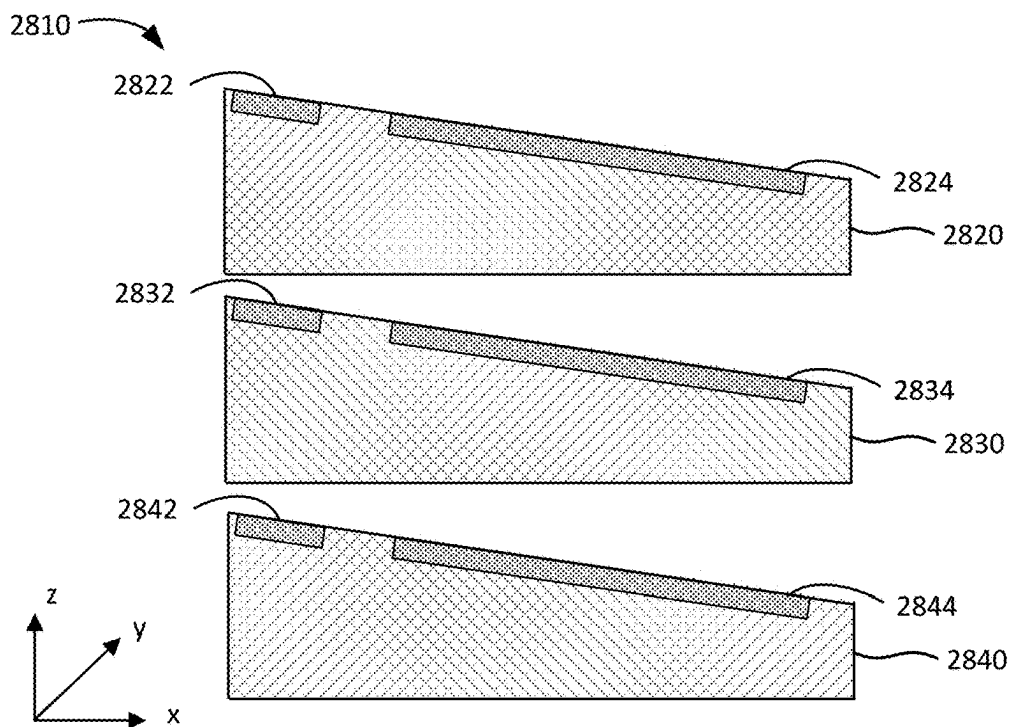
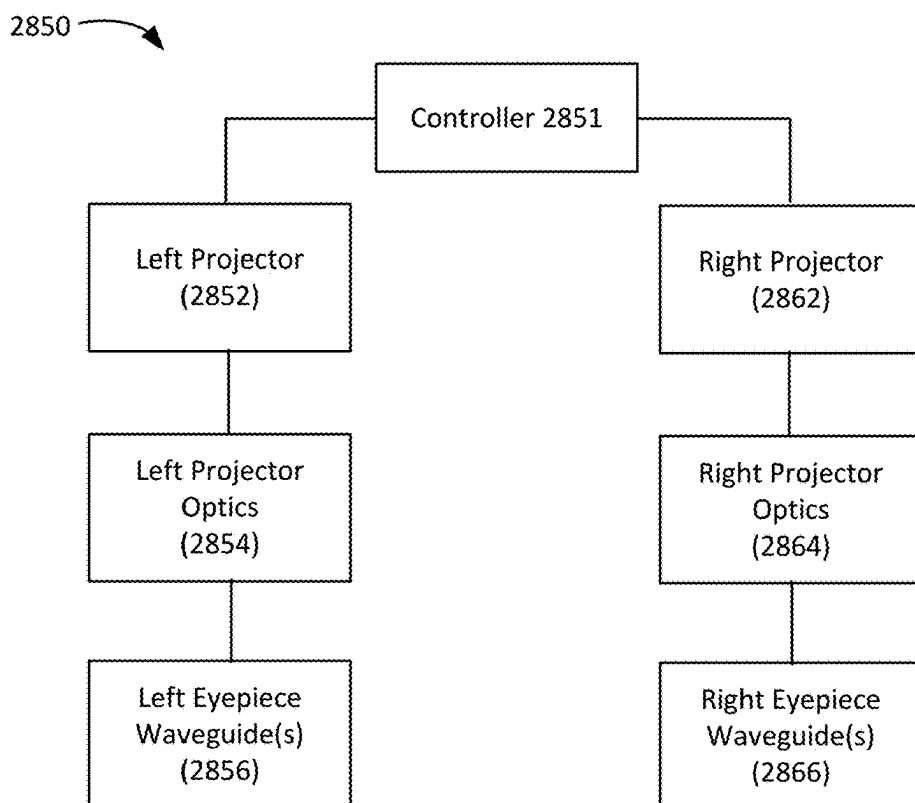


FIG. 28B

**FIG. 28C****FIG. 28D**

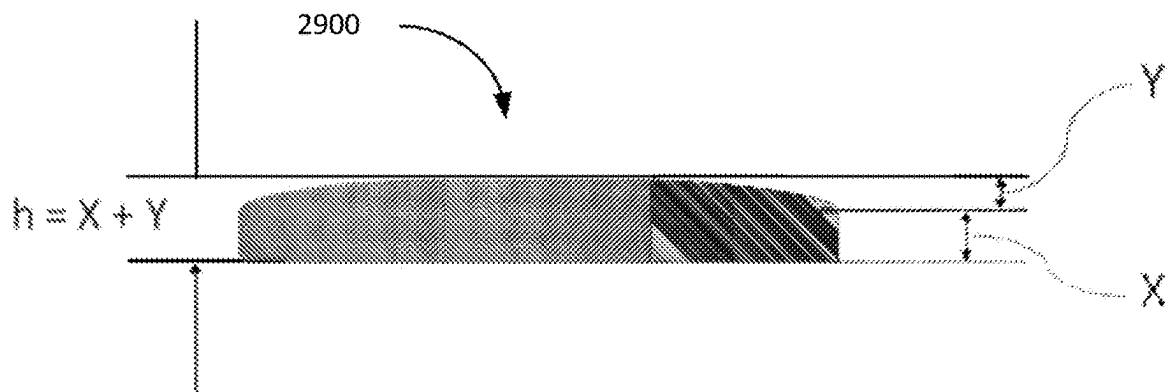


FIG. 29

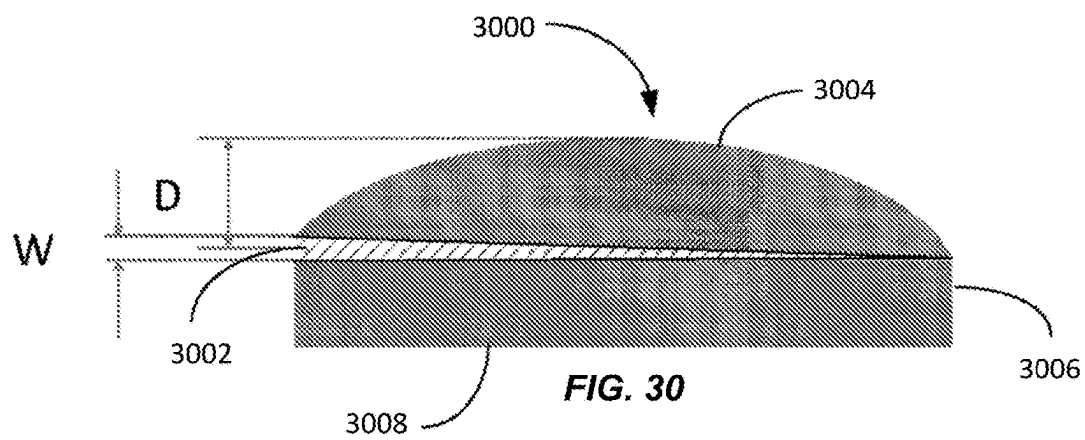


FIG. 30

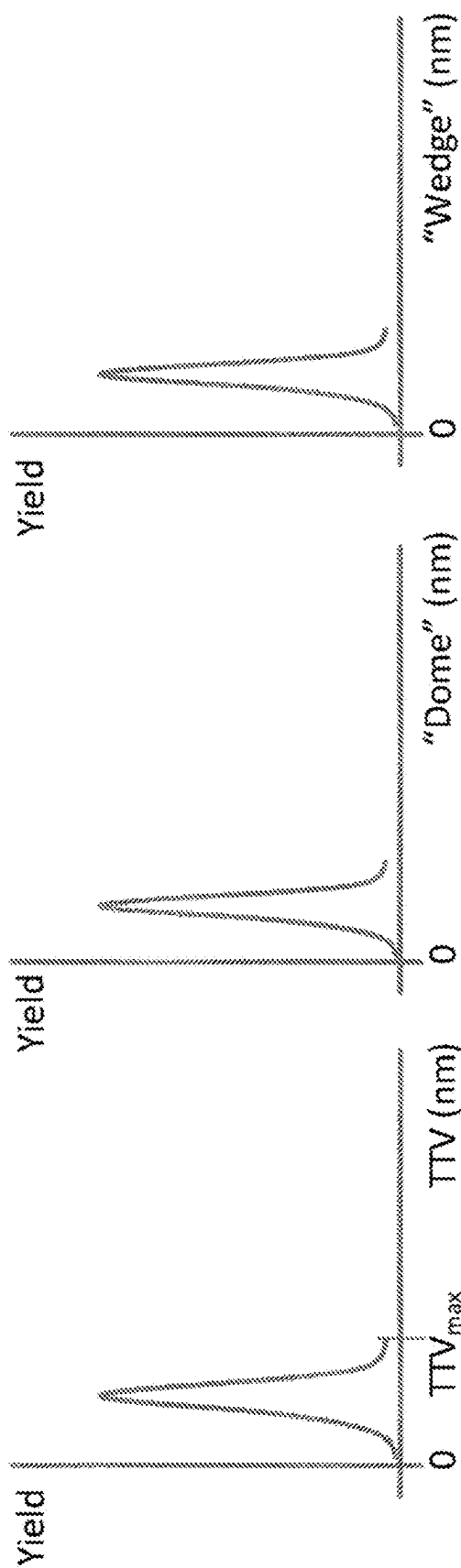


FIG. 31A

FIG. 31B

FIG. 31C

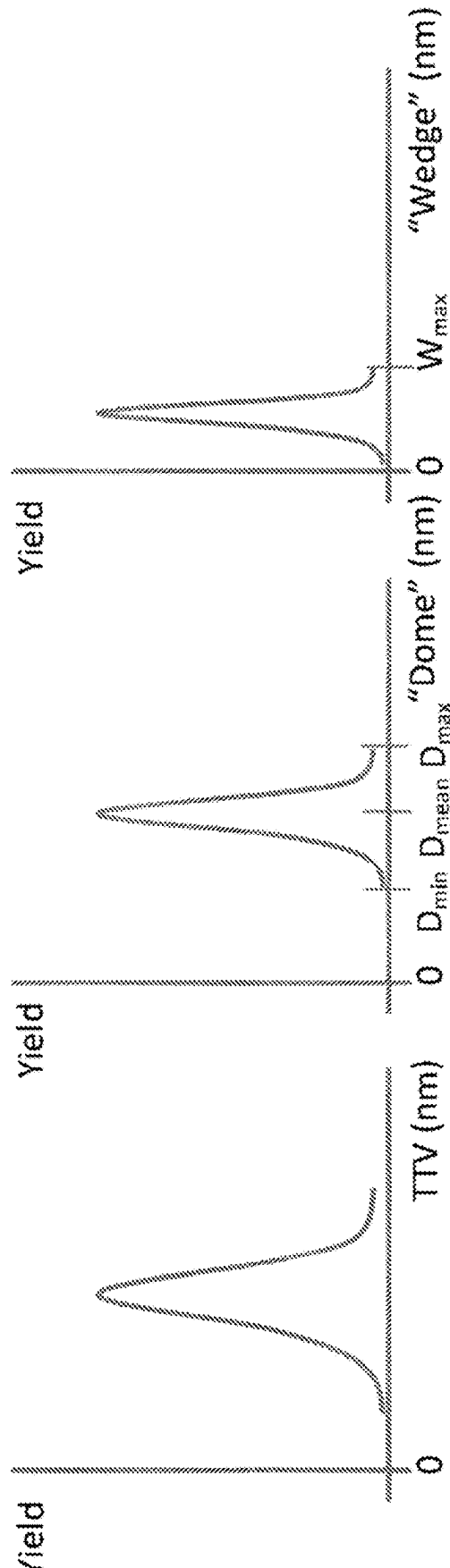
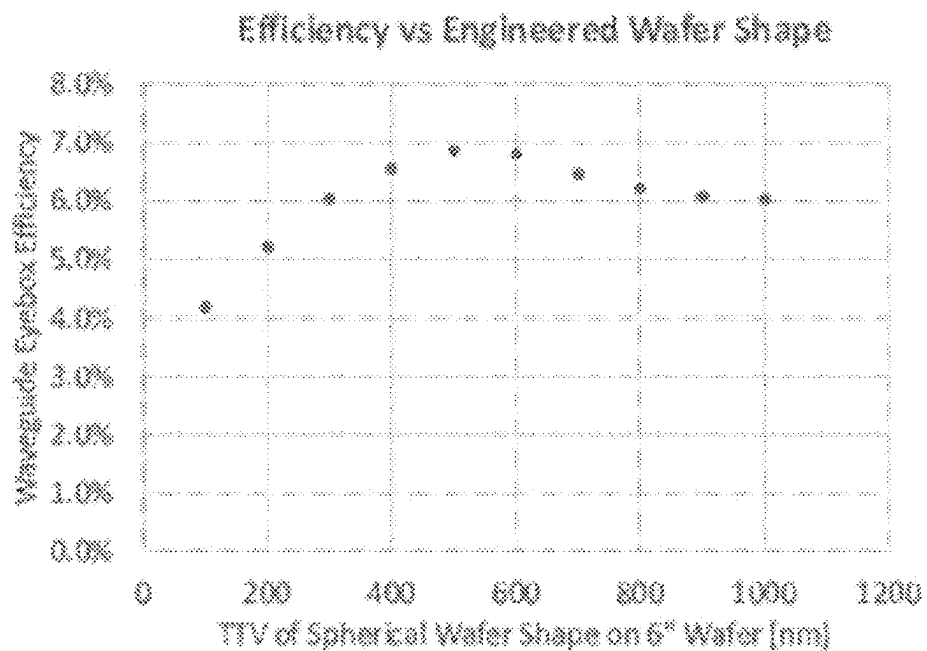
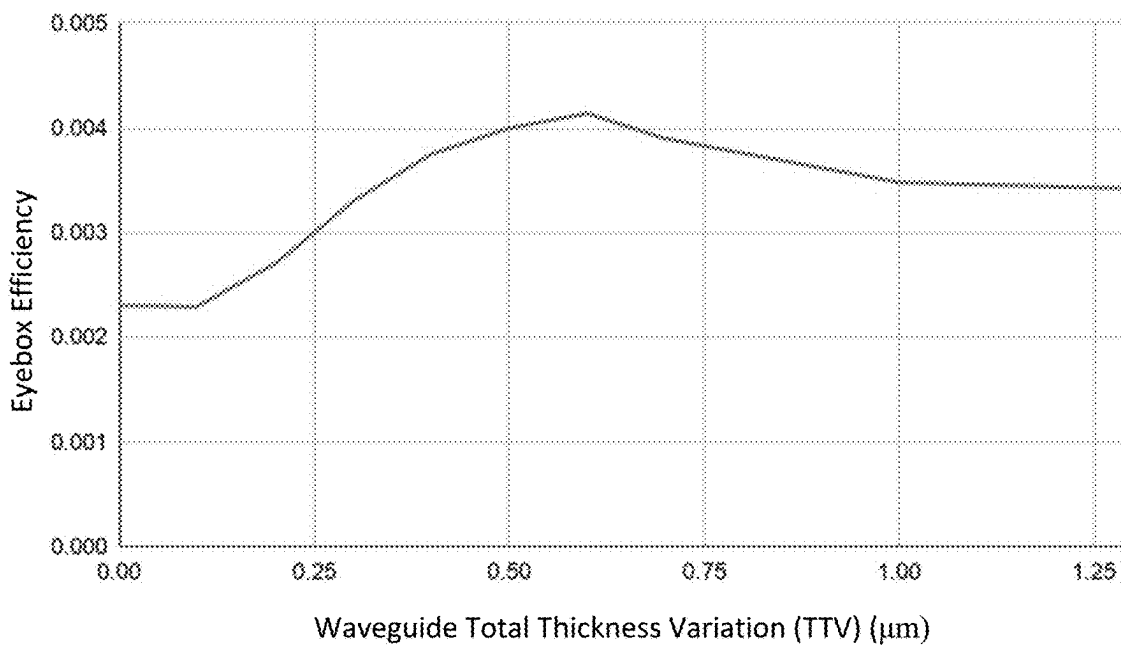
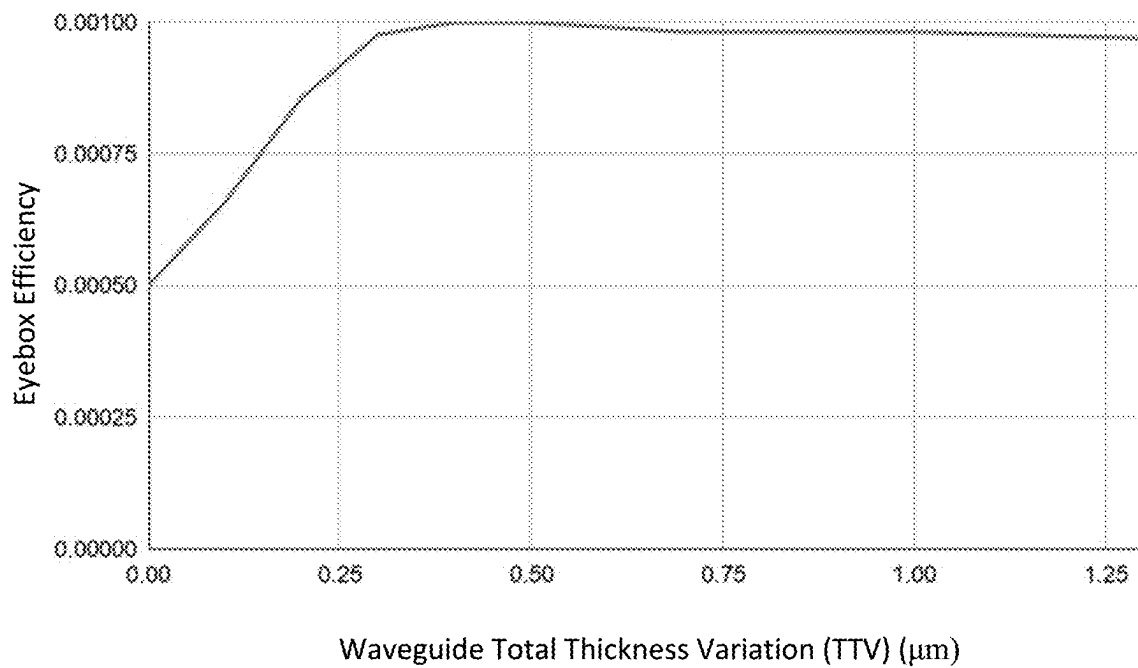


FIG. 31D

FIG. 31E

FIG. 31F

**FIG. 32A****FIG. 32B**

**FIG. 32C**

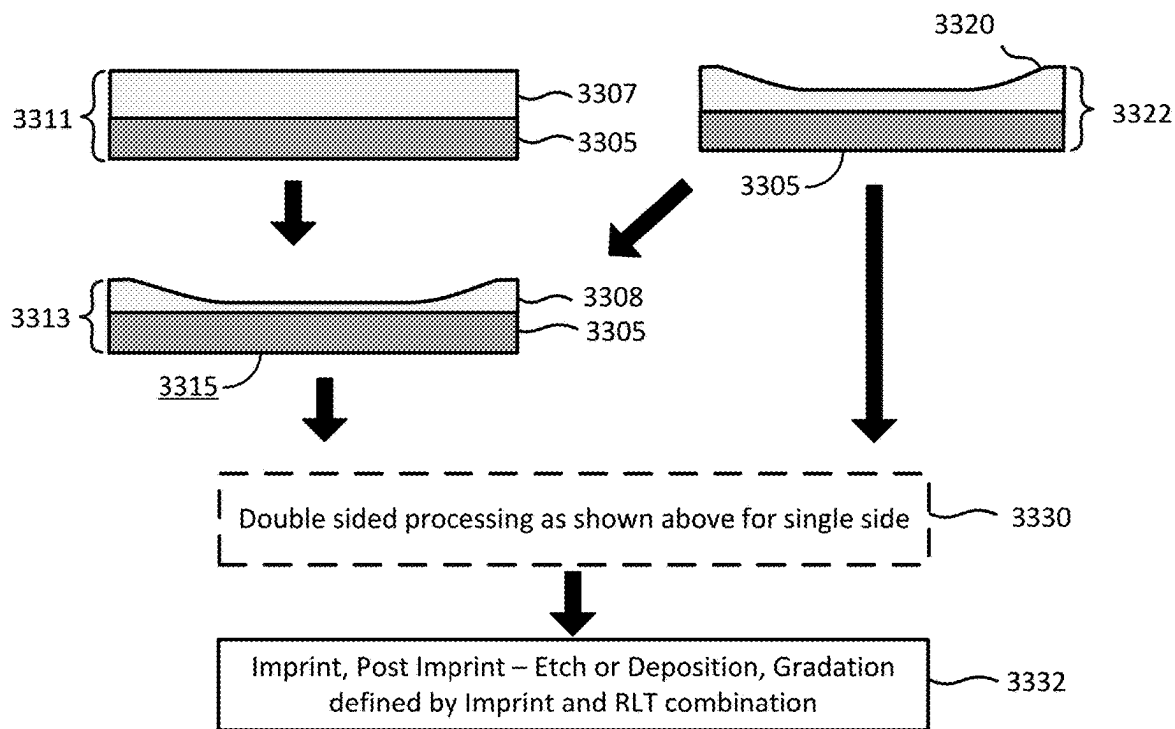


FIG. 33

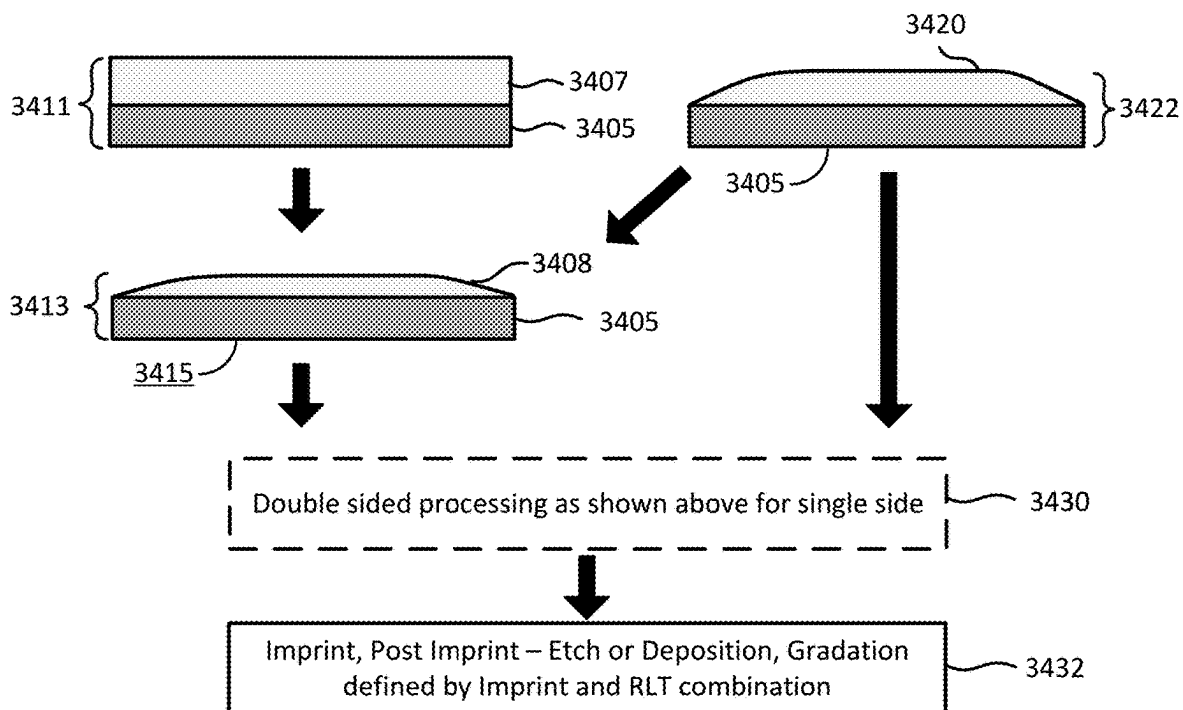


FIG. 34

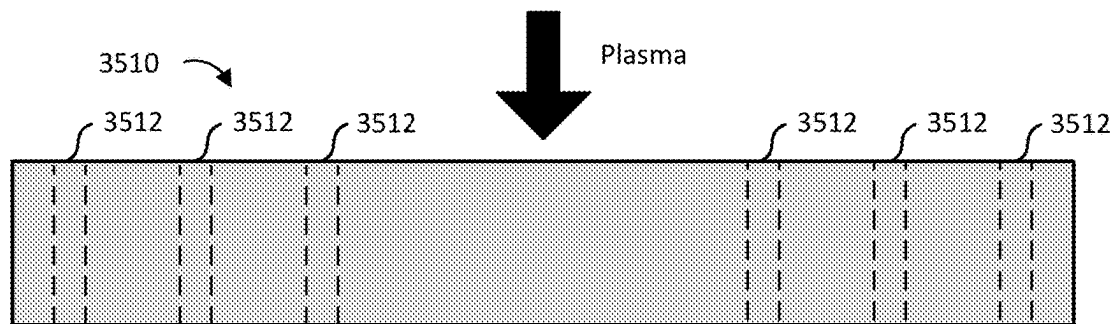


FIG. 35A

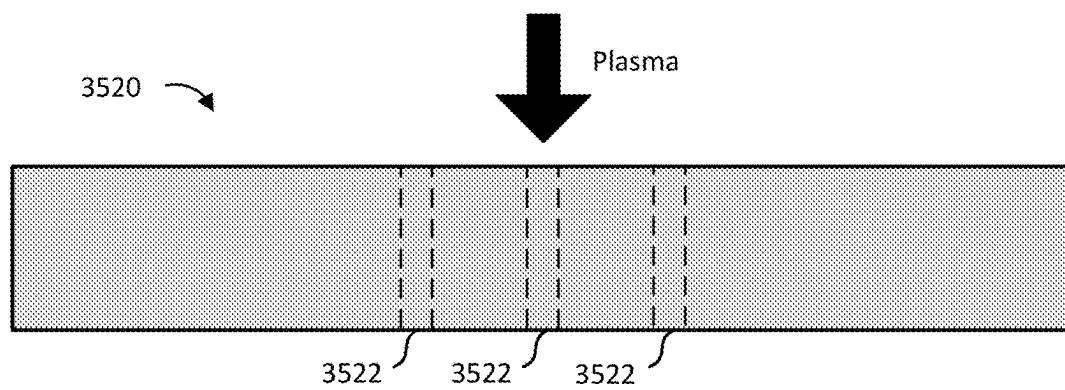


FIG. 35B

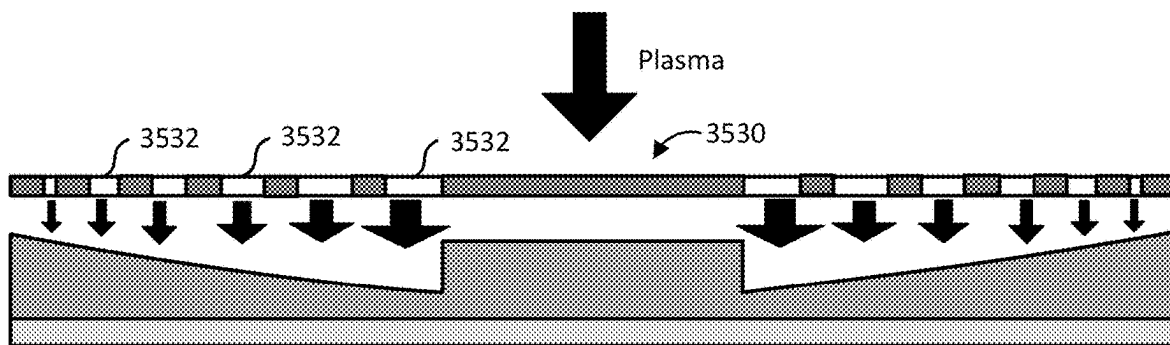


FIG. 35C

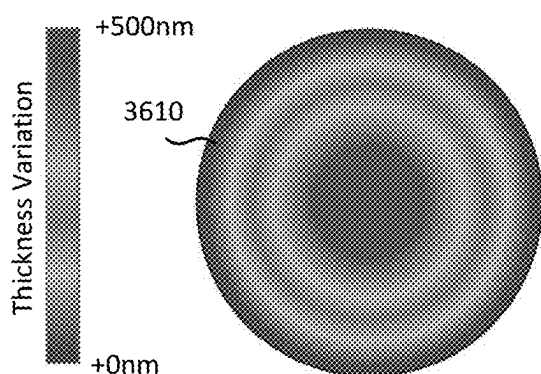


FIG. 36A

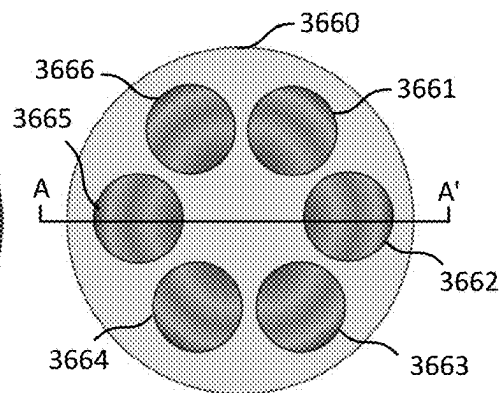


FIG. 36F

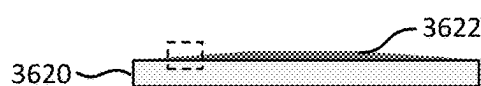


FIG. 36B



FIG. 36G

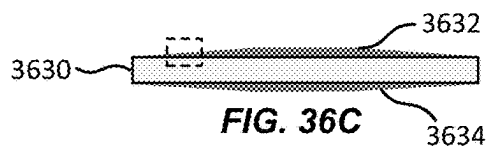


FIG. 36C



FIG. 36H

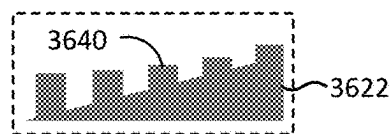


FIG. 36D



FIG. 36E

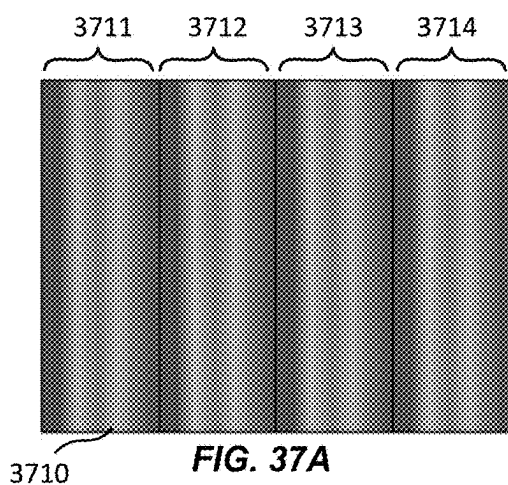


FIG. 37A

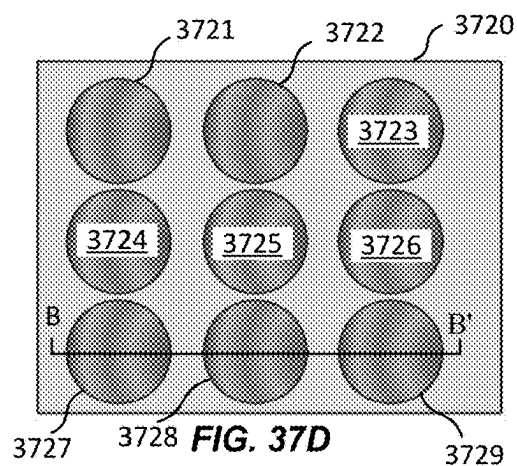


FIG. 37D



FIG. 37B



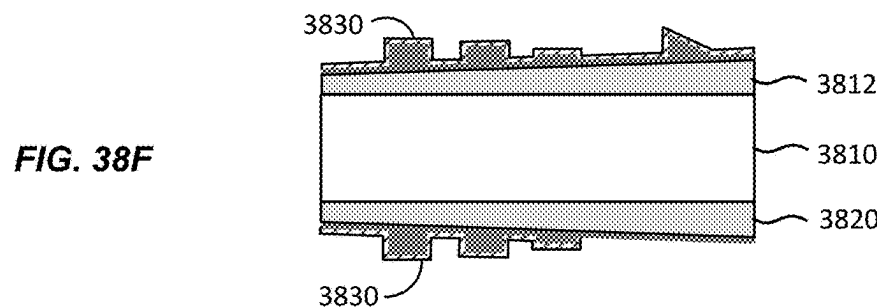
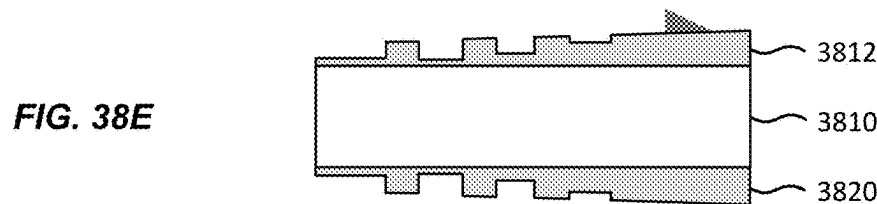
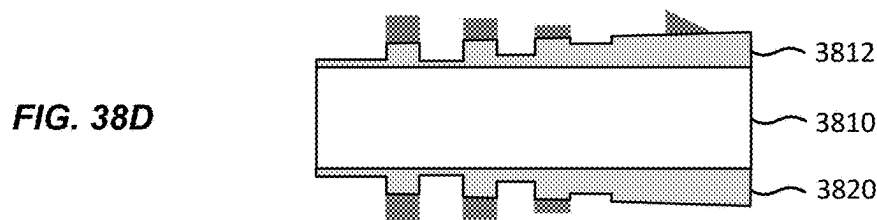
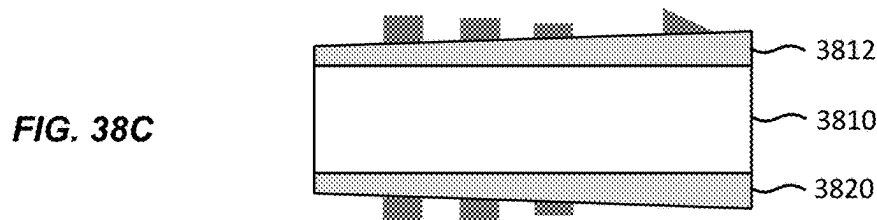
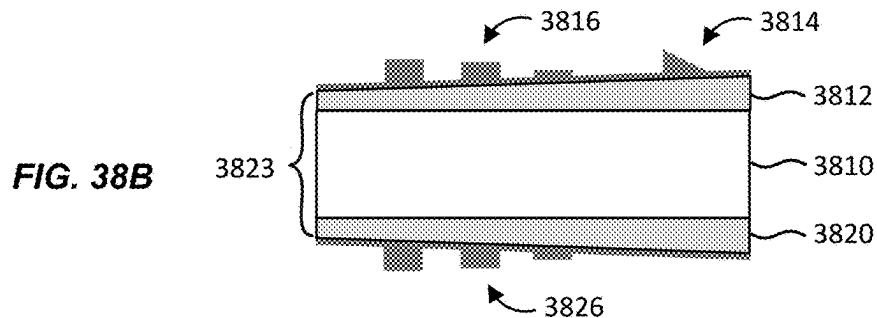
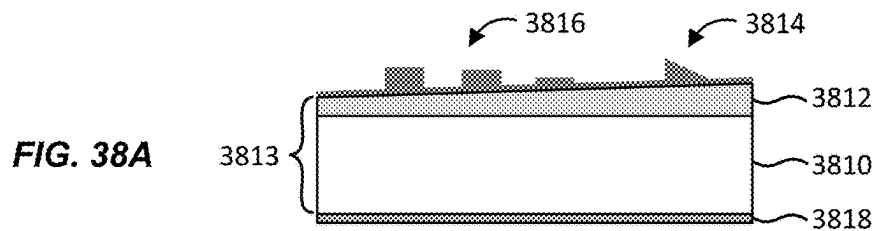
FIG. 37E

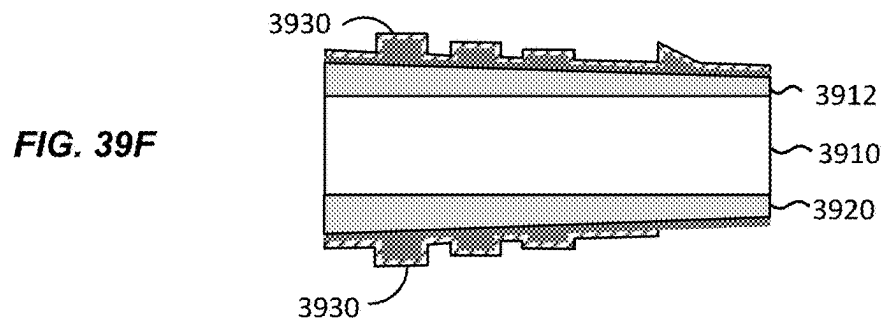
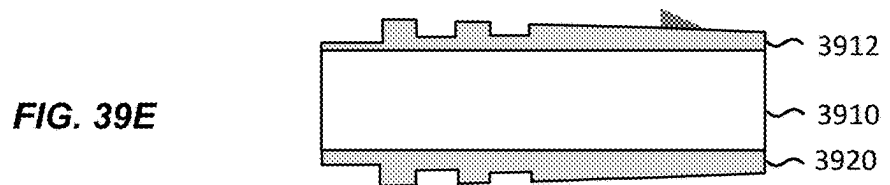
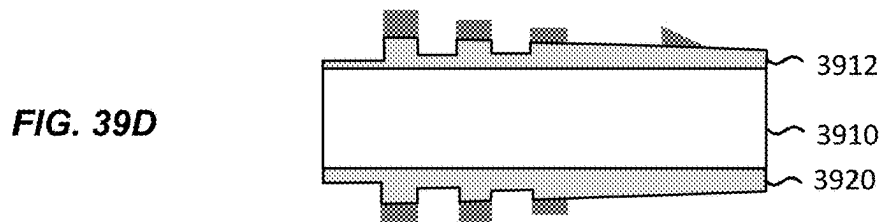
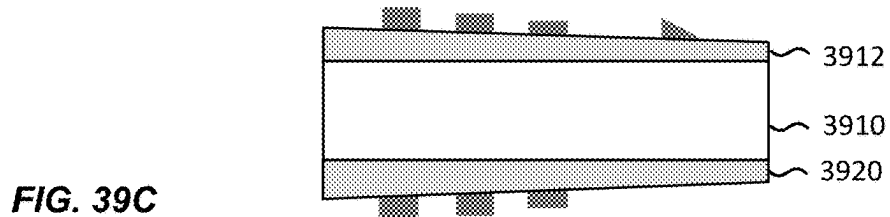
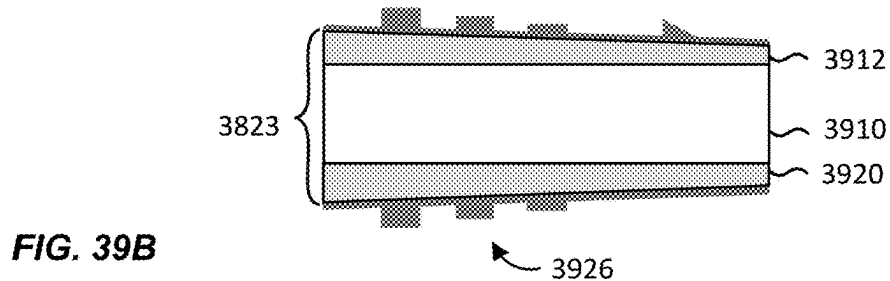
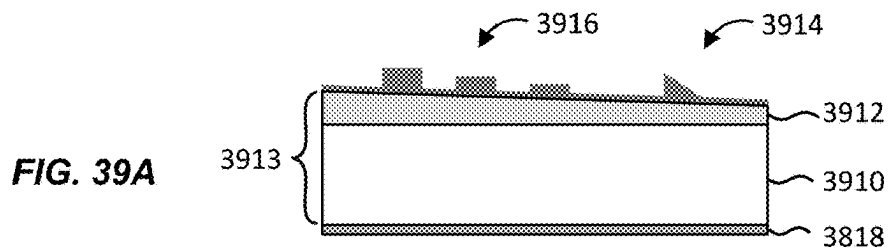


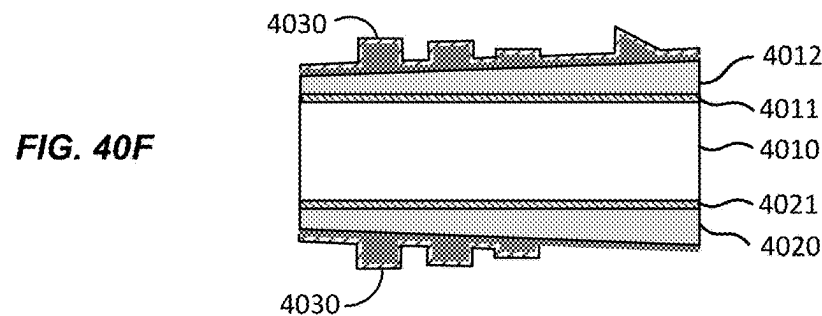
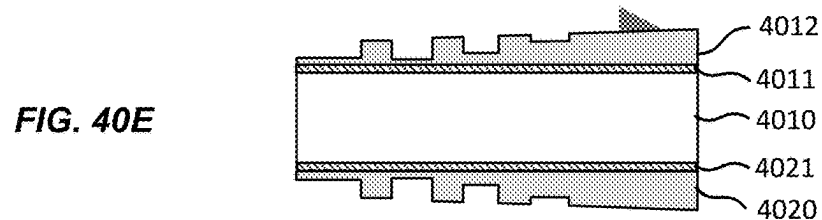
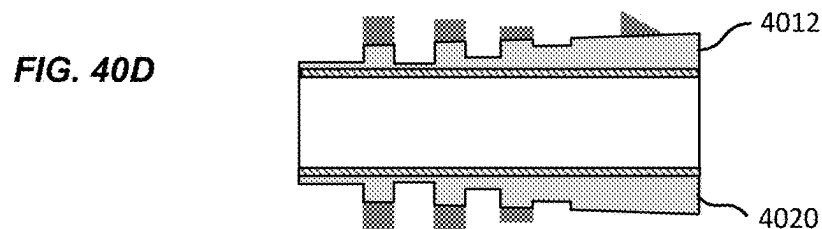
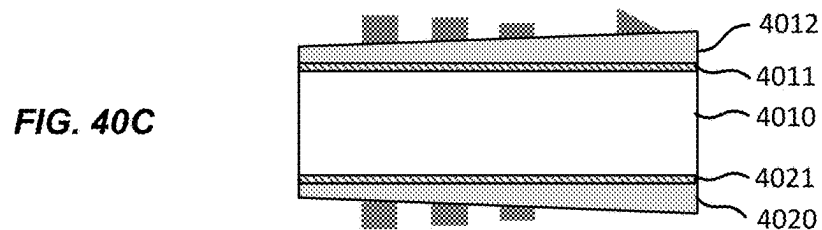
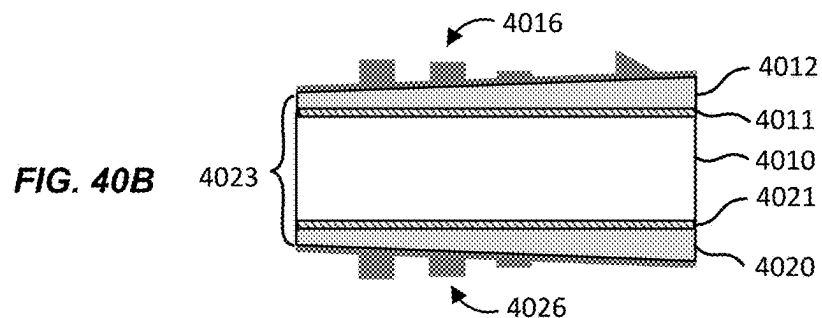
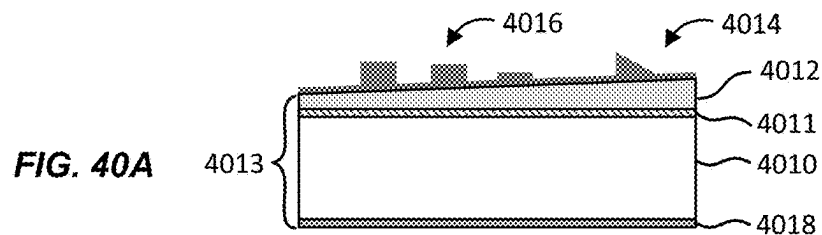
FIG. 37C



FIG. 37F







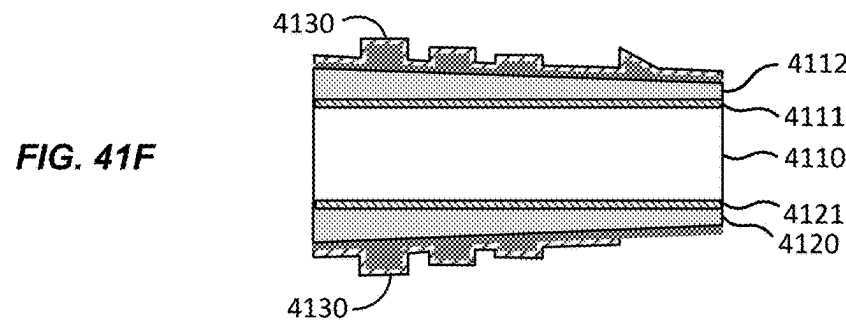
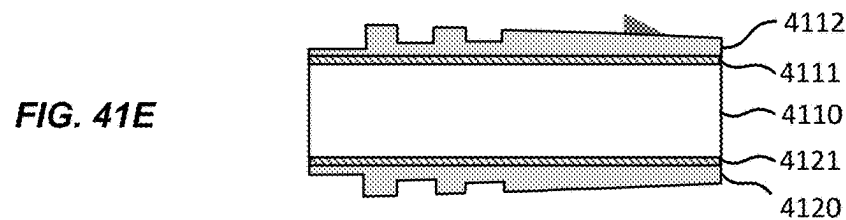
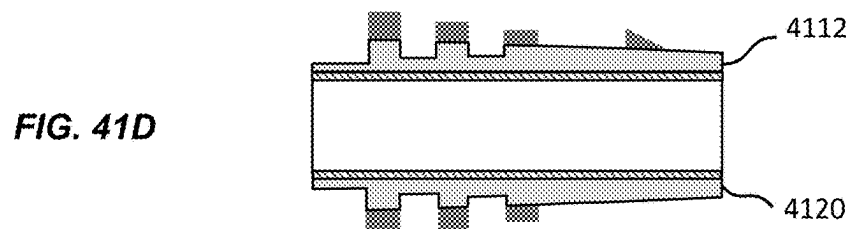
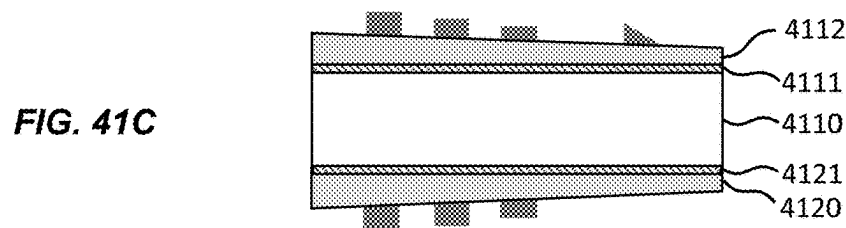
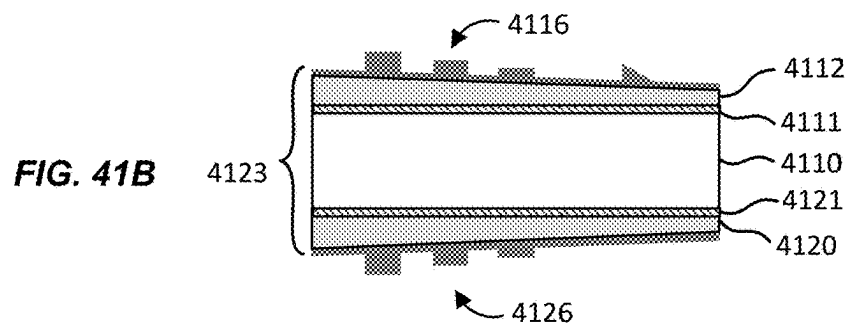
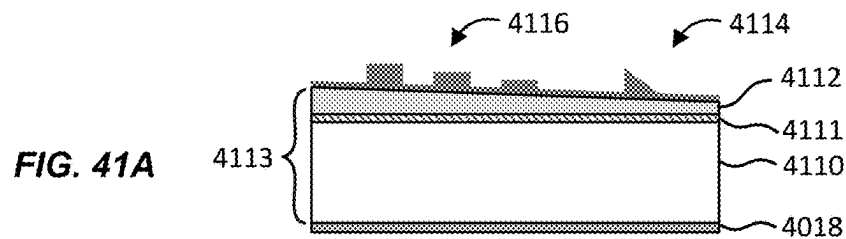


FIG. 42A

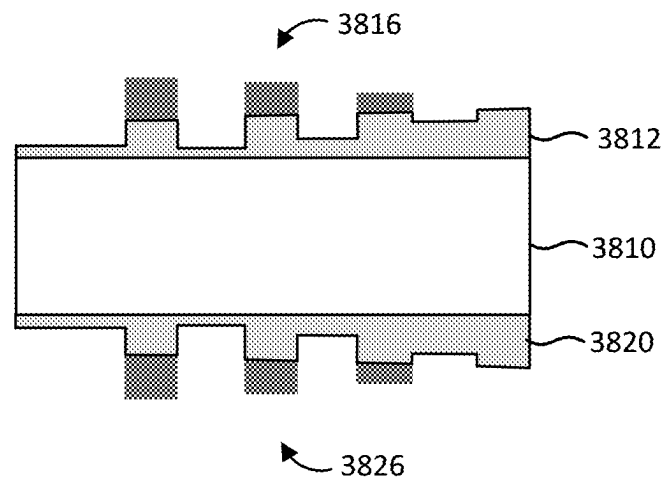


FIG. 42B

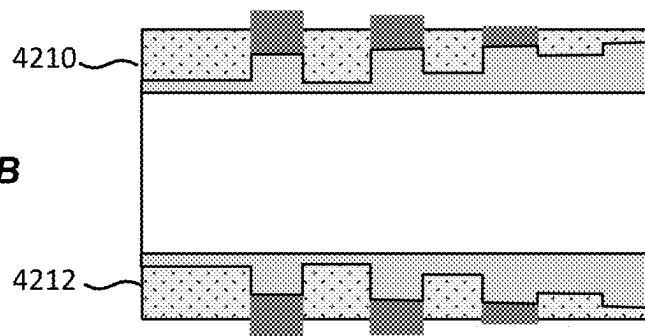


FIG. 42C

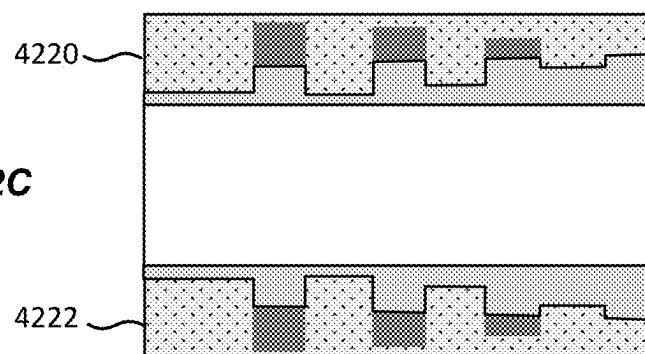


FIG. 43A

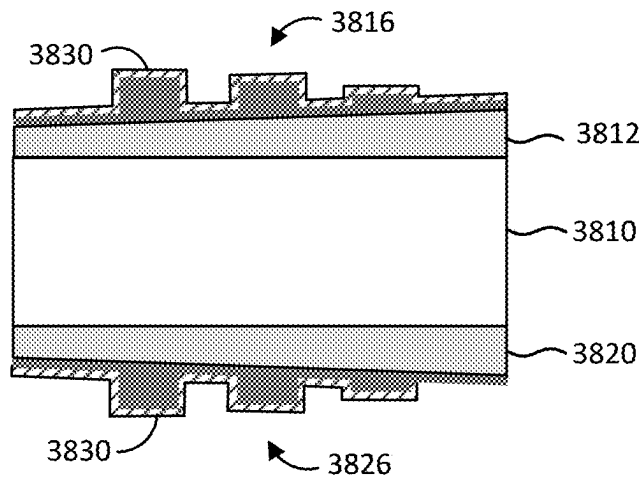


FIG. 43B

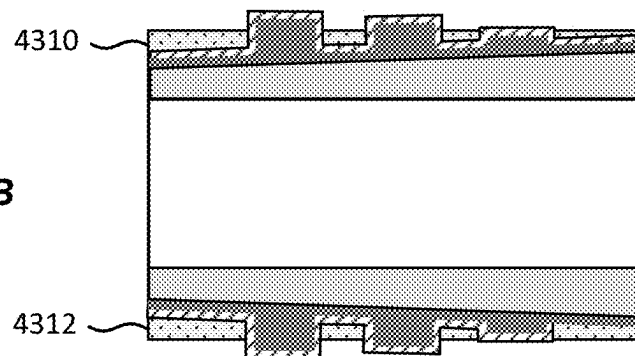
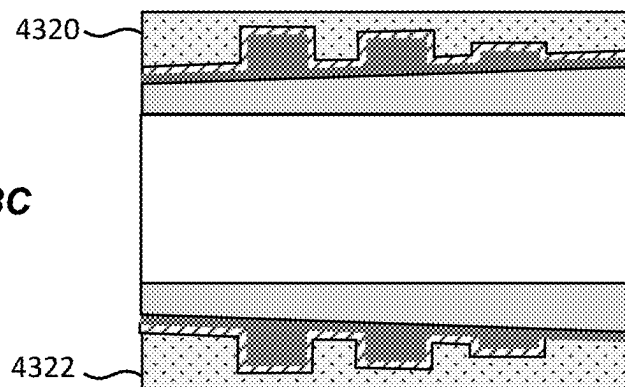


FIG. 43C



METHOD AND SYSTEM FOR VARIABLE OPTICAL THICKNESS WAVEGUIDES FOR AUGMENTED REALITY DEVICES

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation in part of U.S. patent application Ser. No. 17/308,407, filed on May 5, 2021, now U.S. Pat. No. 11,487,061, issued on Nov. 1, 2022, entitled “BIASED TOTAL THICKNESS VARIATIONS IN WAVEGUIDE DISPLAY SUBSTRATES,” which is a continuation of U.S. patent application Ser. No. 16/792,083, filed on Feb. 14, 2020, entitled “BIASED TOTAL THICKNESS VARIATIONS IN WAVEGUIDE DISPLAY SUBSTRATES,” now U.S. Pat. No. 11,022,753, issued on Jun. 6, 2021, which claims priority to U.S. Provisional Patent Application No. 62/820,769, filed on Mar. 19, 2019, and U.S. Provisional Patent Application No. 62/805,832, filed on Feb. 14, 2019, the disclosures of which are hereby incorporated by reference in their entirety for all purposes.

BACKGROUND OF THE INVENTION

Modern computing and display technologies have facilitated the development of systems for so called “virtual reality” or “augmented reality” experiences, wherein digitally reproduced images or portions thereof are presented to a viewer in a manner wherein they seem to be, or may be perceived as, real. A virtual reality, or “VR,” scenario typically involves presentation of digital or virtual image information without transparency to other actual real-world visual input; an augmented reality, or “AR,” scenario typically involves presentation of digital or virtual image information as an augmentation to visualization of the actual world around the viewer.

Referring to FIG. 1, an augmented reality scene 10 is depicted. The user of an AR technology sees a real-world park-like setting 20 featuring people, trees, buildings in the background, and a concrete platform 30. The user also perceives that he/she “sees” “virtual content” such as a robot statue 40 standing upon the real-world platform 30, and a flying cartoon-like avatar character 50 which seems to be a personification of a bumble bee. These elements 50, 40 are “virtual” in that they do not exist in the real world. Because the human visual perception system is complex, it is challenging to produce AR technology that facilitates a comfortable, natural-feeling, rich presentation of virtual image elements amongst other virtual or real-world imagery elements.

Despite the progress made in these display technologies, there is a need in the art for improved methods and systems related to augmented reality systems, particularly, display systems.

SUMMARY OF THE INVENTION

The present invention relates generally to methods and systems related to projection display systems including wearable displays. More particularly, embodiments of the present invention provide methods and systems including eyepiece waveguide layers with variable optical thickness utilized to perform optical imaging in augmented reality systems. The invention is applicable to a variety of applications in computer vision and image display systems.

According to an embodiment of the present invention, an augmented reality device is provided. The augmented reality

device includes a projector, projector optics optically coupled to the projector, and an eyepiece optically coupled to the projector optics. The eyepiece comprises an eyepiece waveguide characterized by lateral dimensions and an optical path length difference as a function of one or more of the lateral dimensions. The eyepiece can further include a second eyepiece waveguide characterized by second lateral dimensions and a second optical path length difference as a function of one or more of the second lateral dimensions and a third eyepiece waveguide characterized by third lateral dimensions and a third optical path length difference as a function of one or more of the third lateral dimensions. The eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide can form the eyepiece. The eyepiece can be a laminated structure including the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide. The augmented reality device can also include a second projector, second projector optics optically coupled to the projector, and a second eyepiece optically coupled to the second projector optics. The second eyepiece can include a fourth eyepiece waveguide characterized by fourth lateral dimensions and a fourth optical path length difference as a function of one or more of the fourth lateral dimensions. The second eyepiece can further include a fifth eyepiece waveguide characterized by fifth lateral dimensions and a fifth optical path length difference as a function of one or more of the fifth lateral dimensions and a sixth eyepiece waveguide characterized by sixth lateral dimensions and a sixth optical path length difference as a function of one or more of the sixth lateral dimensions. The projector, the projector optics, and the eyepiece can be mounted in an augmented reality headset.

In an embodiment, the eyepiece waveguide can include a combined pupil expander, the second eyepiece waveguide comprises a second combined pupil expander, and the third eyepiece waveguide comprises a third combined pupil expander. The eyepiece waveguide thickness varies across the combined pupil expander, the second eyepiece waveguide thickness varies across the second combined pupil expander, and the third eyepiece waveguide thickness varies across the third combined pupil expander. The eyepiece waveguide can be characterized by a thickness variation as a function of the one or more lateral dimensions. The eyepiece waveguide can be characterized by an index of refraction variation as a function of the one or more lateral dimensions. The eyepiece can include a three eyepiece waveguides. The optical path length difference can vary from a first value at a first region of each of the three eyepiece waveguides to a second value at a second region of each of the three eyepiece waveguides. The first region can correspond to an input coupling grating, the second region can correspond to a combined pupil expander, and the optical path length difference can decrease from the first value to the second value. The optical path length difference of each of the three eyepiece waveguides can extend along a common direction. The optical path length difference of each of the three eyepiece waveguides can decrease along a lateral dimension of each of the three eyepiece waveguides, wherein the lateral dimensions of each of the three eyepiece waveguides are parallel.

According to another embodiment of the present invention, an augmented reality device is provided. The augmented reality device includes a projector, projector optics optically coupled to the projector, and an eyepiece optically coupled to the projector optics. The eyepiece includes a first eyepiece waveguide characterized by lateral dimensions and a first optical path length difference gradient and a second

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eyepiece waveguide characterized by the lateral dimensions and a second optical path length difference gradient aligned with the first optical path length difference gradient. The eyepiece can further include a third eyepiece waveguide characterized by the lateral dimensions and a third optical path length difference gradient aligned with the first optical path length difference gradient and the third optical path length difference gradient. The first eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide can be laminated together. The projector, the projector optics, and the eyepiece can be mounted in an augmented reality headset. The first eyepiece waveguide can include a first combined pupil expander, wherein a thickness of the first eyepiece waveguide varies from a first portion of the first combined pupil expander to a second portion of the first combined pupil expander. The second eyepiece waveguide can include a second combined pupil expander, wherein a thickness of the second eyepiece waveguide varies from a first portion of the second combined pupil expander to a second portion of the second combined pupil expander. The first optical path length difference gradient can be aligned with the second optical path length difference gradient. The first optical path length difference gradient can correspond to a direction and the second optical path length difference gradient corresponds to the direction.

According to a specific embodiment of the present invention, an augmented reality device is provided. The augmented reality device includes a projector, projector optics optically coupled to the projector, and an eyepiece optically coupled to the projector optics. The eyepiece comprises an eyepiece waveguide characterized by lateral dimensions and one or more layers. At least one of the one or more layers is characterized by an optical path length difference as a function of one or more of the lateral dimensions. The eyepiece can include a substrate and a variable thickness layer coupled to the substrate. An index of refraction of the substrate can be substantially equal to an index of refraction of the variable thickness layer. The eyepiece can further include a second eyepiece waveguide characterized by second lateral dimensions and one or more second layers and a third eyepiece waveguide characterized by third lateral dimensions and a one or more third layers. At least one of the one or more second layers is characterized by a second optical path length difference as a function of one or more of the lateral dimensions and at least one of the one or more third layers is characterized by a third optical path length difference as a function of one or more of the lateral dimensions. The eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide can form the eyepiece. The eyepiece can be a laminated structure including the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide. The one or more layers, the one or more second layers, and the one or more third layers can be characterized by a thickness variation as a function of the one or more lateral dimensions and the thickness variation of each of the one or more layers, the one or more second layers, and the one or more third layers can decrease along a lateral dimension of each of the three eyepiece waveguides, wherein the lateral dimensions of each of the three eyepiece waveguides are parallel. The augmented reality device can further include a second projector, second projector optics optically coupled to the projector, and a second eyepiece optically coupled to the projector optics. The second eyepiece comprises a fourth eyepiece waveguide characterized by fourth lateral dimensions and one or more fourth layers, wherein at least one of the one or more fourth layers is characterized by a fourth

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optical path length difference as a function of one or more of the fourth lateral dimensions. The second eyepiece can further include a fifth eyepiece waveguide characterized by fifth lateral dimensions and one or more fifth layers, wherein at least one of the one or more fifth layers is characterized by a fifth optical path length difference as a function of one or more of the fifth lateral dimensions, and a sixth eyepiece waveguide characterized by sixth lateral dimensions and one or more sixth layers, wherein at least one of the one or more sixth layers is characterized by a sixth optical path length difference as a function of one or more of the sixth lateral dimensions.

In an embodiment, the projector, the projector optics, and the eyepiece are mounted in an augmented reality headset. The eyepiece waveguide can include a combined pupil expander, the second eyepiece waveguide can include a second combined pupil expander, and the third eyepiece waveguide can include a third combined pupil expander, wherein the one or more layers are characterized by a thickness that varies across the combined pupil expander, the one or more second layers are characterized by a thickness that varies across the second combined pupil expander, and the one or more third layers are characterized by a thickness that varies across the third combined pupil expander. The one or more layers can be characterized by a thickness variation as a function of the one or more of the lateral dimensions. The thickness variation can vary from a first value at a first region of the eyepiece waveguide to a second value at a second region of the eyepiece waveguide. The first region can correspond to an input coupling grating, the second region can correspond to a combined pupil expander, and the thickness variation can decrease from the first value to the second value. The one or more layers can be characterized by an index of refraction variation as a function of the one or more of the lateral dimensions. The eyepiece waveguide can include a nanopattern. The nanopattern can include a polymer grating having a polymer index of refraction less than an index of refraction of the eyepiece. The one or more layers can have a thickness less than 50 nm, less than 30 nm, or less than 10 nm. The nanopattern can include a conformally deposited structure or a directionally deposited structure formed in the one or more layers or the eyepiece. The nanopattern can include a polymer grating having a polymer index of refraction higher than an index of refraction of the eyepiece. The one or more layers can have a thickness less than 100 nm and include at least one of ZrO_2 or TiO_2 . The nanopattern can include an etched structure formed in the one or more layers or the eyepiece. The nanopattern can include binary gratings, blazed saw-tooth gratings, blazed multi-step grating, blazed slanted gratings, holes, or pillars.

According to another specific embodiment of the present invention, an augmented reality device includes a projector, projector optics optically coupled to the projector, and a substrate structure including a substrate having an incident surface and an opposing exit surface and a first variable thickness film coupled to the incident surface. The substrate structure can further include a first combined pupil expander coupled to the first variable thickness film, a second variable thickness film coupled to the exit surface, an incoupling grating coupled to the exit surface, and a second combined pupil expander coupled to the exit surface. The second variable thickness film can be thicker adjacent the incoupling grating than adjacent the second combined pupil expander. A variation in thickness of the substrate can be less than a variation in thickness of the first variable thickness film or the second variable thickness film. The augmented

reality device of can further include a first intermediate film disposed between the substrate and the first variable thickness film and a second intermediate film disposed between the substrate and the second variable thickness film. The first intermediate film can have an index of refraction less than an index of refraction of the substrate and an index of refraction of the first variable thickness film and the second intermediate film can have an index of refraction less than the index of refraction of the substrate and an index of refraction of the second variable thickness film.

According to a particular embodiment of the present invention, an augmented reality device includes a projector, projector optics optically coupled to the projector, and an eyepiece optically coupled to the projector optics. The eyepiece includes a first eyepiece waveguide characterized by lateral dimensions and a first variable thickness film having a first thickness gradient and a second eyepiece waveguide characterized by the lateral dimensions and a second variable thickness film having a second thickness gradient aligned with the first thickness gradient. The eyepiece can further include a third eyepiece waveguide characterized by the lateral dimensions and a third variable thickness film having a third thickness gradient aligned with the first thickness gradient and the third thickness gradient. The first eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide can be laminated together. The projector, the projector optics, and the eyepiece can be mounted in an augmented reality headset. The first eyepiece waveguide can include a first combined pupil expander, wherein a thickness of the first variable thickness film varies from a first portion of the first combined pupil expander to a second portion of the first combined pupil expander, and the second eyepiece waveguide can include a second combined pupil expander, wherein a thickness of the second variable thickness film varies from a first portion of the second combined pupil expander to a second portion of the second combined pupil expander. The first thickness gradient can be aligned with the second thickness gradient. The first thickness gradient can correspond to a direction and the second thickness gradient can correspond to the direction.

Numerous benefits are achieved by way of the present invention over conventional techniques. For example, embodiments of the present invention provide methods and systems that can improve the quality of virtual content, including image sharpness. These and other embodiments of the invention along with many of its advantages and features are described in more detail in conjunction with the text below and attached figures.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a user's view of augmented reality (AR) through an AR device.

FIG. 2 illustrates a conventional display system for simulating three-dimensional imagery for a user.

FIGS. 3A-3C illustrate relationships between radius of curvature and focal radius.

FIG. 4A illustrates a representation of the accommodation-vergence response of the human visual system.

FIG. 4B illustrates examples of different accommodative states and vergence states of a pair of eyes of the user.

FIG. 4C illustrates an example of a representation of a top-down view of a user viewing content via a display system.

FIG. 4D illustrates another example of a representation of a top-down view of a user viewing content via a display system.

FIG. 5 illustrates aspects of an approach for simulating three-dimensional imagery by modifying wavefront divergence.

FIG. 6 illustrates an example of a waveguide stack for outputting image information to a user.

FIG. 7 illustrates an example of exit beams outputted by a waveguide.

FIG. 8 illustrates an example of a stacked waveguide assembly in which each depth plane includes images formed using multiple different component colors.

FIG. 9A illustrates a cross-sectional side view of an example of a set of stacked waveguides that each includes an in-coupling optical element.

FIG. 9B illustrates a perspective view of an example of the one or more stacked waveguides of FIG. 9A.

FIG. 9C illustrates a top-down plan view of an example of the one or more stacked waveguides of FIGS. 9A and 9B.

FIG. 9D illustrates an example of wearable display system.

FIG. 10 is a side view of a projector assembly including a polarizing beam splitter with a light source injecting light into one side of the beamsplitter and projection optics receiving light from another side of the beamsplitter.

FIG. 11A is a side view of an augmented reality display system including a light source, a spatial light modulator, optics for illuminating the spatial light modulator and projecting an image of the spatial light modulator (SLM), and a waveguide for outputting image information to a user. The system includes an in-coupling optical element for coupling light from the optics into the waveguide as well as an out-coupling optical element for coupling light out of the waveguide to the eye.

FIG. 11B is a top view of the augmented reality display system illustrated in FIG. 11A showing the waveguide with the in-coupling optical element and the outcoupling optical elements as well as the light source disposed thereon. The top view also shows an orthogonal pupil expander.

FIG. 11C is a side view of the augmented reality display system of FIG. 11A with a shared polarizer/analyzer and polarization based spatial light modulator (e.g., a liquid crystal on silicon SLM).

FIG. 11D illustrates an example of a waveguide having a combined OPE/EPE according to an embodiment of the present invention.

FIG. 12A is a side view of an augmented reality display system including a multi-color light source (e.g., time multiplexed RGB LEDs or laser diodes), a spatial light modulator, optics for illuminating the spatial light modulator and projecting an image of the spatial light modulator to the eye, and a stack of waveguides, different waveguides including different color-selective in-coupling optical elements as well as out-coupling optical elements.

FIG. 12B is a side view of the augmented reality display system of FIG. 12A further including a MEMS (micro-electro-mechanical) based SLM such as an array of movable mirrors (e.g., Digital Light Processing (DLP™) technology) and a light dump.

FIG. 12C is a top view of a portion of the augmented reality display system of FIG. 12B schematically illustrating the lateral arrangement of one of the in-coupling optical elements and the light dump as well as the light source.

FIG. 13A is a perspective view of an augmented reality display system including a stack of waveguides, different waveguides including different in-coupling optical ele-

ments, wherein the in-coupling optical elements are displaced laterally with respect to each other. One or more light sources, also laterally displaced with respect to each other are disposed to direct light to respective in-coupling optical elements by passing light through optics, reflecting light off a spatial light modulator and passing the reflected light again through the optics.

FIG. 13B is a side view of the example illustrated in FIG. 13A showing the lateral displaced in-coupling optical elements and light sources as well as the optics and the spatial light modulator.

FIG. 13C is a top view of the augmented reality display system illustrated in FIGS. 13A and 13B showing one or more laterally displaced in-coupling optical elements and the associated one or more laterally displaced light sources.

FIG. 14A is a side view of an augmented reality display system including a waveguide stack, different waveguides including different in-coupling optical elements, where the in-coupling optical elements are laterally displaced with respect to each other (the lateral displacement occurring in the z direction in this example).

FIG. 14B is a top view of the display system illustrated in FIG. 14A showing the laterally displaced in-coupling optical elements and light sources.

FIG. 14C is an orthogonal-side view of the display system illustrated in FIGS. 14A and 14B.

FIG. 15 is a top view of an augmented reality display system including a set of stacked waveguides, different waveguides including different in-coupling optical elements. The light sources and in-coupling optical elements are arranged in an alternative configuration than that shown in FIG. 14A-14C.

FIG. 16A is a side view of an augmented reality display system including groups of in-coupling optical elements that are laterally displaced with respect to each other, each group including one or more color-selective in-optical coupling optical elements

FIG. 16B is a top view of the display system in FIG. 16A.

FIG. 17 is a side view of an augmented reality display system including a waveguide that is divided with a reflective surface that can couple light guided in a portion of the waveguide proximal to a light source out of that portion of the waveguide and into optics toward a spatial light modulator. In this example, the optics and a light source are shown disposed on a same side of the waveguide.

FIG. 18 is a side view of an augmented reality display system that includes a waveguide for receiving light from a light source and directing the light guided in the waveguide into optics and toward a spatial light modulator. The display system additionally includes a waveguide that receives light from the spatial light modulator that passes again through the optics. The waveguide includes a reflective surface to out-couple light. The waveguide also includes a reflective surface to in-couple light therein. In this example, the optics and the light source are shown disposed on the same side of the waveguide.

FIG. 19 is a side view of an augmented reality display system including adaptive optical elements or variable focus optical elements. A first variable optical element between the stack of waveguides and the eye can vary the divergence and collimation of light coupled out from the waveguides and directed to the eye to vary the depth at which the objects appear to be located. A second variable optical element on the opposite side of the stack of waveguides can compensate for the effect of the first optical element on light received from the environment in front of the augmented reality display system and the user. The augmented reality display

system further includes a prescription lens to provide ophthalmic correction such refractive correction for a user who has myopia, hyperopia, astigmatism, etc.

FIG. 20A is a side view of an augmented reality display system including color filter array. One or more laterally displaced in-coupling optical elements are located on different waveguides and laterally displaced color filters are aligned with respective in-coupling optical elements.

FIG. 20B shows the augmented reality display system of FIG. 20A with the analyzer located between the optics and the spatial light modulator.

FIG. 20C shows the augmented reality display system similar to that shown in FIGS. 20A and 20B however using a deflection-based spatial light modulator such as a movable micro-mirror based spatial light modulator.

FIG. 20D is a top view of a portion of an augmented reality display system such as shown in FIG. 20C schematically illustrating the laterally displaced light sources and corresponding laterally displaced in-coupling optical elements above a color filter array.

FIG. 20E illustrates how the deflection-based spatial light modulator directs the light away from the corresponding in-coupling optical elements and onto the mask surrounding the filters in the filter array for the augmented reality display system of FIG. 20D.

FIG. 20F is a side view of an augmented reality display system including a cover glass disposed on a user side of a stack of waveguides and a light source disposed on a world side of the cover glass.

FIG. 20G is a side view of an augmented reality display system including a cover glass disposed on a world side of a stack of waveguides and a light source disposed on a world side of the cover glass.

FIG. 21 is a side view of an augmented reality display system including a light source outfitted with a light recycler configured to recycling light such as light of one polarization.

FIG. 22 is a side view of one or more light sources propagating light through corresponding light collection optics and one or more apertures. The light may also propagate through a diffuser located proximal the one or more apertures.

FIG. 23A is a side view of a portion of an augmented reality display system including a light source, optics having optical power, a waveguide for receiving and outputting image information to a user's eye, wherein the system further includes one or more retarders and polarizers configured to reduce reflection from optical surfaces that may be input to the waveguide as a ghost image.

FIG. 23B is a side view of a portion of an augmented reality display system such as shown in FIG. 23A with additional retarders and polarizers configured to reduce reflections that may produce ghost images.

FIG. 23C is a side view of an augmented reality display system such as shown in FIGS. 23A and 23B with reduced retarders and polarizers configured to reduce reflection that may produce ghost images.

FIG. 24 is a side view of an augmented reality display system that utilizes a tilted surface such as a tilted surface on a cover glass to direct reflections away from being directed into an eye of a user potentially reducing ghost reflections.

FIG. 25 is an embodiment of the system of FIG. 24 wherein the tilted surface on the cover glass is configured to direct reflections toward a light dump that absorbs the light.

FIG. 26A depicts a sample waveguide display substrate with waveguide areas.

FIG. 26B is a plot showing waveguide display substrate thickness as a function of position according to an embodiment of the present invention.

FIG. 26C is a plot showing waveguide display substrate thickness as a function of position according to an embodiment of the present invention.

FIG. 26D depicts a waveguide display substrate with a rectilinear arrangement of waveguides according to an embodiment of the present invention.

FIG. 26E is a simplified schematic diagram illustrating thickness variation for waveguide display substrate with a rectilinear layout according to an embodiment of the present invention.

FIG. 26F is a simplified schematic diagram illustrating a rectangular waveguide display substrate with a linear arrangement of waveguides according to an embodiment of the present invention.

FIG. 26G is a simplified schematic diagram illustrating a waveguide display substrate with localized TTV according to an embodiment of the present invention.

FIG. 27A depicts a polished flat waveguide display substrate.

FIGS. 27B and 27C depict a polished convex waveguide display substrate and a polished concave waveguide display substrate, respectively.

FIG. 27D depicts a polished meniscus waveguide display substrate with one convex surface and one concave surface.

FIG. 27E depicts a polished bi-convex waveguide display substrate.

FIG. 27F depicts a polished bi-concave waveguide display substrate.

FIGS. 28A and 28B depict total thickness variation (TTV) comparisons of waveguide display substrates.

FIG. 28C depicts TTV for a set of eyepiece waveguides according to an embodiment of the present invention.

FIG. 28D is a simplified schematic diagram illustrating an augmented reality system according to an embodiment of the present invention.

FIG. 29 depicts TTV of a polished waveguide display substrate with a convex surface.

FIG. 30 shows a cross section of a waveguide display substrate with a biased TTV having linear (wedge) and nonlinear (dome) components.

FIGS. 31A-31C show waveguide display substrate yield versus TTV, dome height, and wedge height for ultra-low TTV waveguide display substrates.

FIGS. 31D-31F show waveguide display substrate yield versus TTV, dome height, and wedge height for biased TTV waveguide display substrates.

FIG. 32A shows waveguide eyebox efficiency versus "dome" TTV (nm) of a spherical waveguide display substrate shape on a 6 inch wafer for a typical diffractive waveguide display with 300 μm average thickness and 0 nm "wedge" TTV.

FIG. 32B shows waveguide eyebox efficiency versus "dome" TTV (in μm) of a spherical waveguide display substrate shape on a 6 inch wafer for a typical diffractive waveguide display with 300 μm average thickness.

FIG. 32C shows waveguide eyebox efficiency versus "dome" TTV (in μm) of a spherical waveguide display substrate shape on a 6 inch wafer for another diffractive waveguide display with 300 μm average thickness.

FIG. 33 is a simplified schematic diagram illustrating process flows used in forming an inverse dome thickness variation according to an embodiment of the present invention.

FIG. 34 is a simplified schematic diagram illustrating process flows used in forming an inverse dome thickness variation according to an embodiment of the present invention.

FIGS. 35A-35B are simplified cross-sectional diagrams illustrating showerhead designs according to various embodiments of the present invention.

FIG. 35C is a simplified schematic diagram illustrating an etch mask according to an embodiment of the present invention.

FIG. 36A is a simplified plan view of a substrate illustrating thickness variation according to an embodiment of the present invention.

FIG. 36B is a simplified cross-sectional diagram illustrating a single sided thickness variation according to an embodiment of the present invention.

FIG. 36C is a simplified cross-sectional diagram illustrating a double sided thickness variation according to an embodiment of the present invention.

FIG. 36D is a simplified schematic diagram illustrating varying height nanofeatures according to an embodiment of the present invention.

FIG. 36E is a simplified schematic diagram illustrating single height nanofeatures according to an embodiment of the present invention.

FIG. 36F is a simplified plan view of a substrate illustrating thickness variation according to another embodiment of the present invention.

FIG. 36G is a simplified cross-sectional diagram illustrating a single sided thickness variation according to an embodiment of the present invention.

FIG. 36H is a simplified cross-sectional diagram illustrating a double sided thickness variation according to an embodiment of the present invention.

FIG. 37A is a simplified plan view of a substrate illustrating thickness variation according to an embodiment of the present invention.

FIG. 37B is a simplified schematic diagram illustrating a single sided thickness variation according to an embodiment of the present invention.

FIG. 37C is a simplified schematic diagram illustrating a double sided thickness variation according to an embodiment of the present invention.

FIG. 37D is a simplified plan view of a substrate illustrating thickness variation according to another embodiment of the present invention.

FIG. 37E is a simplified schematic diagram illustrating a single sided thickness variation according to an embodiment of the present invention.

FIG. 37F is a simplified schematic diagram illustrating a double sided thickness variation according to an embodiment of the present invention.

FIG. 38A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation according to an embodiment of the present invention.

FIG. 38B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an embodiment of the present invention.

FIG. 38C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to another embodiment of the present invention.

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FIG. 38D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to yet another embodiment of the present invention.

FIG. 38E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an alternative embodiment of the present invention.

FIG. 38F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and an overcoat according to an embodiment of the present invention.

FIG. 39A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation according to an embodiment of the present invention.

FIG. 39B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an embodiment of the present invention.

FIG. 39C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to another embodiment of the present invention.

FIG. 39D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to yet another embodiment of the present invention.

FIG. 39E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an alternative embodiment of the present invention.

FIG. 39F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and an overcoat according to an embodiment of the present invention.

FIG. 40A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation and a low index film according to an embodiment of the present invention.

FIG. 40B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an embodiment of the present invention.

FIG. 40C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to another embodiment of the present invention.

FIG. 40D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to yet another embodiment of the present invention.

FIG. 40E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an alternative embodiment of the present invention.

FIG. 40F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film and an overcoat according to an embodiment of the present invention.

FIG. 41A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation and a low index film according to an embodiment of the present invention.

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FIG. 41B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an embodiment of the present invention.

FIG. 41C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to another embodiment of the present invention.

FIG. 41D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to yet another embodiment of the present invention.

FIG. 41E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an alternative embodiment of the present invention.

FIG. 41F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film and an overcoat according to an embodiment of the present invention.

FIG. 42A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an embodiment of the present invention.

FIG. 42B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a first level of planarization according to an embodiment of the present invention.

FIG. 42C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a second level of planarization according to an embodiment of the present invention.

FIG. 43A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and an overcoat according to an embodiment of the present invention.

FIG. 43B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation, an overcoat, and a first level of planarization according to an embodiment of the present invention.

FIG. 43C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation, an overcoat, and a second level of planarization according to an embodiment of the present invention.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

Reference will now be made to the drawings, in which like reference numerals refer to like parts throughout. Unless indicated otherwise, the drawings are schematic not necessarily drawn to scale.

FIG. 2 illustrates a conventional display system for simulating three-dimensional imagery for a user. It will be appreciated that a user's eyes are spaced apart and that, when looking at a real object in space, each eye will have a slightly different view of the object and may form an image of the object at different locations on the retina of each eye. This may be referred to as binocular disparity and may be utilized by the human visual system to provide a perception of depth. Conventional display systems simulate binocular disparity by presenting two distinct images **190**, **200** with slightly different views of the same virtual object—one for each eye **210a**, **210b** corresponding to the views of the virtual object that would be seen by each eye were the virtual

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object a real object at a desired depth. These images provide binocular cues that the user's visual system may interpret to derive a perception of depth.

With continued reference to FIG. 2, the images **190**, **200** are spaced from the eyes **210a**, **210b** by a distance **230** on a z-axis. The z-axis is parallel to the optical axis of the viewer with their eyes fixated on an object at optical infinity directly ahead of the viewer. The images **190**, **200** are flat and at a fixed distance from the eyes **210a**, **210b**. Based on the slightly different views of a virtual object in the images presented to the eyes **210a**, **210b**, respectively, the eyes may naturally rotate such that an image of the object falls on corresponding points on the retinas of each of the eyes, to maintain single binocular vision. This rotation may cause the lines of sight of each of the eyes **210a**, **210b** to converge onto a point in space at which the virtual object is perceived to be present. As a result, providing three-dimensional imagery conventionally involves providing binocular cues that may manipulate the vergence of the eyes **210a**, **210b**, and that the human visual system interprets to provide a perception of depth.

Generating a realistic and comfortable perception of depth is challenging, however. It will be appreciated that light from objects at different distances from the eyes have wavefronts with different amounts of divergence. FIGS. 3A-3C illustrate relationships between distance and the divergence of light rays. The distance between the object and the eye **210** is represented by, in order of decreasing distance, **R1**, **R2**, and **R3**. As shown in FIGS. 3A-3C, the light rays become more divergent as distance to the object decreases. Conversely, as distance increases, the light rays become more collimated. Stated another way, it may be said that the light field produced by a point (the object or a part of the object) has a spherical wavefront curvature, which is a function of how far away the point is from the eye of the user. The curvature increases with decreasing distance between the object and the eye **210**. While only a single eye **210** is illustrated for clarity of illustration in FIGS. 3A-3C and other FIGS. herein, the discussions regarding eye **210** may be applied to both eyes **210a** and **210b**.

With continued reference to FIGS. 3A-3C, light from an object that the viewer's eyes are fixated on may have different degrees of wavefront divergence. Due to the different amounts of wavefront divergence, the light may be focused differently by the lens of the eye, which in turn may require the lens to assume different shapes to form a focused image on the retina of the eye. Where a focused image is not formed on the retina, the resulting retinal blur acts as a cue to accommodation that causes a change in the shape of the lens of the eye until a focused image is formed on the retina. For example, the cue to accommodation may trigger the ciliary muscles surrounding the lens of the eye to relax or contract, thereby modulating the force applied to the suspensory ligaments holding the lens, thus causing the shape of the lens of the eye to change until retinal blur of an object of fixation is eliminated or minimized, thereby forming a focused image of the object of fixation on the retina (e.g., fovea) of the eye. The process by which the lens of the eye changes shape may be referred to as accommodation, and the shape of the lens of the eye required to form a focused image of the object of fixation on the retina (e.g., fovea) of the eye may be referred to as an accommodative state.

With reference now to FIG. 4A, a representation of the accommodation-vergence response of the human visual system is illustrated. The movement of the eyes to fixate on an object causes the eyes to receive light from the object, with the light forming an image on each of the retinas of the eyes.

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The presence of retinal blur in the image formed on the retina may provide a cue to accommodation, and the relative locations of the image on the retinas may provide a cue to vergence. The cue to accommodation causes accommodation to occur, resulting in the lenses of the eyes each assuming a particular accommodative state that forms a focused image of the object on the retina (e.g., fovea) of the eye. On the other hand, the cue to vergence causes vergence movements (rotation of the eyes) to occur such that the images formed on each retina of each eye are at corresponding retinal points that maintain single binocular vision. In these positions, the eyes may be said to have assumed a particular vergence state. With continued reference to FIG. 4A, accommodation may be understood to be the process by which the eye achieves a particular accommodative state, and vergence may be understood to be the process by which the eye achieves a particular vergence state. As indicated in FIG. 4A, the accommodative and vergence states of the eyes may change if the user fixates on another object. For example, the accommodated state may change if the user fixates on a new object at a different depth on the z-axis.

Without being limited by theory, it is believed that viewers of an object may perceive the object as being "three-dimensional" due to a combination of vergence and accommodation. As noted above, vergence movements (e.g., rotation of the eyes so that the pupils move toward or away from each other to converge the lines of sight of the eyes to fixate upon an object) of the two eyes relative to each other are closely associated with accommodation of the lenses of the eyes. Under normal conditions, changing the shapes of the lenses of the eyes to change focus from one object to another object at a different distance will automatically cause a matching change in vergence to the same distance, under a relationship known as the "accommodation-vergence reflex." Likewise, a change in vergence will trigger a matching change in lens shape under normal conditions.

With reference now to FIG. 4B, examples of different accommodative and vergence states of the eyes are illustrated. The pair of eyes **222a** is fixated on an object at optical infinity, while the pair eyes **222b** are fixated on an object **221** at less than optical infinity. Notably, the vergence states of each pair of eyes is different, with the pair of eyes **222a** directed straight ahead, while the pair of eyes **222** converge on the object **221**. The accommodative states of the eyes forming each pair of eyes **222a** and **222b** are also different, as represented by the different shapes of the lenses **220a**, **220b**.

Undesirably, many users of conventional "3-D" display systems find such conventional systems to be uncomfortable or may not perceive a sense of depth at all due to a mismatch between accommodative and vergence states in these displays. As noted above, many stereoscopic or "3-D" display systems display a scene by providing slightly different images to each eye. Such systems are uncomfortable for many viewers, since they, among other things, simply provide different presentations of a scene and cause changes in the vergence states of the eyes, but without a corresponding change in the accommodative states of those eyes. Rather, the images are shown by a display at a fixed distance from the eyes, such that the eyes view all the image information at a single accommodative state. Such an arrangement works against the "accommodation-vergence reflex" by causing changes in the vergence state without a matching change in the accommodative state. This mismatch is believed to cause viewer discomfort. Display systems that provide a better

match between accommodation and vergence may form more realistic and comfortable simulations of three-dimensional imagery.

Without being limited by theory, it is believed that the human eye typically may interpret a finite number of depth planes to provide depth perception. Consequently, a highly believable simulation of perceived depth may be achieved by providing, to the eye, different presentations of an image corresponding to each of these limited numbers of depth planes. In some embodiments, the different presentations may provide both cues to vergence and matching cues to accommodation, thereby providing physiologically correct accommodation-vergence matching.

With continued reference to FIG. 4B, two depth planes **240**, corresponding to different distances in space from the eyes **210a**, **210b**, are illustrated. For a given depth plane **240**, vergence cues may be provided by the displaying of images of appropriately different perspectives for each eye **210a**, **210b**. In addition, for a given depth plane **240**, light forming the images provided to each eye **210a**, **210b** may have a wavefront divergence corresponding to a light field produced by a point at the distance of that depth plane **240**.

In the illustrated embodiment, the distance, along the z-axis, of the depth plane **240** containing the point **221** is 1 m. As used herein, distances or depths along the z-axis may be measured with a zero-point located at the exit pupils of the user's eyes. Thus, a depth plane **240** located at a depth of 1 m corresponds to a distance of 1 m away from the exit pupils of the user's eyes, on the optical axis of those eyes with the eyes directed towards optical infinity. As an approximation, the depth or distance along the z-axis may be measured from the display in front of the user's eyes (e.g., from the surface of a waveguide), plus a value for the distance between the device and the exit pupils of the user's eyes. That value may be called the eye relief and corresponds to the distance between the exit pupil of the user's eye and the display worn by the user in front of the eye. In practice, the value for the eye relief may be a normalized value used generally for all viewers. For example, the eye relief may be assumed to be 20 mm and a depth plane that is at a depth of 1 m may be at a distance of 980 mm in front of the display.

With reference now to FIGS. 4C and 4D, examples of matched accommodation-vergence distances and mismatched accommodation-vergence distances are illustrated, respectively. As illustrated in FIG. 4C, the display system may provide images of a virtual object to each eye **210a**, **210b**. The images may cause the eyes **210a**, **210b** to assume a vergence state in which the eyes converge on a point **15** on a depth plane **240**. In addition, the images may be formed by a light having a wavefront curvature corresponding to real objects at that depth plane **240**. As a result, the eyes **210a**, **210b** assume an accommodative state in which the images are in focus on the retinas of those eyes. Thus, the user may perceive the virtual object as being at the point **15** on the depth plane **240**.

It will be appreciated that each of the accommodative and vergence states of the eyes **210a**, **210b** are associated with a particular distance on the z-axis. For example, an object at a particular distance from the eyes **210a**, **210b** causes those eyes to assume particular accommodative states based upon the distances of the object. The distance associated with a particular accommodative state may be referred to as the accommodation distance, Ad. Similarly, there are particular vergence distances, Vd, associated with the eyes in particular vergence states, or positions relative to one another. Where the accommodation distance and the vergence dis-

tance match, the relationship between accommodation and vergence may be said to be physiologically correct. This is considered to be the most comfortable scenario for a viewer.

In stereoscopic displays, however, the accommodation distance and the vergence distance may not always match. For example, as illustrated in FIG. 4D, images displayed to the eyes **210a**, **210b** may be displayed with wavefront divergence corresponding to depth plane **240**, and the eyes **210a**, **210b** may assume a particular accommodative state in which the points **15a**, **15b** on that depth plane are in focus. However, the images displayed to the eyes **210a**, **210b** may provide cues for vergence that cause the eyes **210a**, **210b** to converge on a point **15** that is not located on the depth plane **240**. As a result, the accommodation distance corresponds to the distance from the exit pupils of the eyes **210a**, **210b** to the depth plane **240**, while the vergence distance corresponds to the larger distance from the exit pupils of the eyes **210a**, **210b** to the point **15**, in some embodiments. The accommodation distance is different from the vergence distance. Consequently, there is an accommodation-vergence mismatch. Such a mismatch is considered undesirable and may cause discomfort in the user. It will be appreciated that the mismatch corresponds to distance (e.g., VaAd) and may be characterized using diopters.

In some embodiments, it will be appreciated that a reference point other than exit pupils of the eyes **210a**, **210b** may be utilized for determining distance for determining accommodation-vergence mismatch, so long as the same reference point is utilized for the accommodation distance and the vergence distance. For example, the distances could be measured from the cornea to the depth plane, from the retina to the depth plane, from the eyepiece (e.g., a waveguide of the display device) to the depth plane, and so on.

Without being limited by theory, it is believed that users may still perceive accommodation-vergence mismatches of up to about 0.25 diopter, up to about 0.33 diopter, and up to about 0.5 diopter as being physiologically correct, without the mismatch itself causing significant discomfort. In some embodiments, display systems disclosed herein (e.g., the display system **250**, FIG. 6) present images to the viewer having accommodation-vergence mismatch of about 0.5 diopter or less. In some other embodiments, the accommodation-vergence mismatch of the images provided by the display system is about 0.33 diopter or less. In yet other embodiments, the accommodation-vergence mismatch of the images provided by the display system is about 0.25 diopter or less, including about 0.1 diopter or less.

FIG. 5 illustrates aspects of an approach for simulating three-dimensional imagery by modifying wavefront divergence. The display system includes a waveguide **270** that is configured to receive light **770** that is encoded with image information, and to output that light to the user's eye **210**. The waveguide **270** may output the light **650** with a defined amount of wavefront divergence corresponding to the wavefront divergence of a light field produced by a point on a desired depth plane **240**. In some embodiments, the same amount of wavefront divergence is provided for all objects presented on that depth plane. In addition, it will be illustrated that the other eye of the user may be provided with image information from a similar waveguide.

In some embodiments, a single waveguide may be configured to output light with a set amount of wavefront divergence corresponding to a single or limited number of depth planes and/or the waveguide may be configured to output light of a limited range of wavelengths. Consequently, in some embodiments, a stack of waveguides may be utilized to provide different amounts of wavefront diver-

gence for different depth planes and/or to output light of different ranges of wavelengths. As used herein, it will be appreciated at a depth plane may follow the contours of a flat or a curved surface. In some embodiments, advantageously for simplicity, the depth planes may follow the contours of flat surfaces.

FIG. 6 illustrates an example of a waveguide stack for outputting image information to a user. A display system 250 includes a stack of waveguides, or stacked waveguide assembly, 260 that may be utilized to provide three-dimensional perception to the eye/brain using waveguides 270, 280, 290, 300, 310. It will be appreciated that the display system 250 may be considered a light field display in some embodiments. In addition, the waveguide assembly 260 may also be referred to as an eyepiece.

In some embodiments, the display system 250 may be configured to provide substantially continuous cues to vergence and multiple discrete cues to accommodation. The cues to vergence may be provided by displaying different images to each of the eyes of the user, and the cues to accommodation may be provided by outputting the light that forms the images with selectable discrete amounts of wavefront divergence. Stated another way, the display system 250 may be configured to output light with variable levels of wavefront divergence. In some embodiments, each discrete level of wavefront divergence corresponds to a particular depth plane and may be provided by a particular one of the waveguides 270, 280, 290, 300, 310.

With continued reference to FIG. 6, the waveguide assembly 260 may also include features 320, 330, 340, 350 between the waveguides. In some embodiments, the features 320, 330, 340, 350 may be one or more lenses. The waveguides 270, 280, 290, 300, 310 and/or the features (e.g., lenses) 320, 330, 340, 350 may be configured to send image information to the eye with various levels of wavefront curvature or light ray divergence. Each waveguide level may be associated with a particular depth plane and may be configured to output image information corresponding to that depth plane. Image injection devices 360, 370, 380, 390, 400 may function as a source of light for the waveguides and may be utilized to inject image information into the waveguides 270, 280, 290, 300, 310, each of which may be configured, as described herein, to distribute incoming light across each respective waveguide, for output toward the eye 210. Light exits an output surface 410, 420, 430, 440, 450 of the image injection devices 360, 370, 380, 390, 400 and is injected into a corresponding input surface 460, 470, 480, 490, 500 of the waveguides 270, 280, 290, 300, 310. In some embodiments, each of the input surfaces 460, 470, 480, 490, 500 may be an edge of a corresponding waveguide, or may be part of a major surface of the corresponding waveguide (that is, one of the waveguide surfaces directly facing the world 510 or the viewer's eye 210). In some embodiments, a single beam of light (e.g. a collimated beam) may be injected into each waveguide to output an entire field of cloned collimated beams that are directed toward the eye 210 at particular angles (and amounts of divergence) corresponding to the depth plane associated with a particular waveguide. In some embodiments, a single one of the image injection devices 360, 370, 380, 390, 400 may be associated with and inject light into one or more (e.g., three) of the waveguides 270, 280, 290, 300, 310.

In some embodiments, the image injection devices 360, 370, 380, 390, 400 are discrete displays that each produce image information for injection into a corresponding waveguide 270, 280, 290, 300, 310, respectively. In some other embodiments, the image injection devices 360, 370, 380,

390, 400 are the output ends of a single multiplexed display which may, e.g., pipe image information via one or more optical conduits (such as fiber optic cables) to each of the image injection devices 360, 370, 380, 390, 400. It will be appreciated that the image information provided by the image injection devices 360, 370, 380, 390, 400 may include light of different wavelengths, or colors (e.g., different component colors, as discussed herein).

In some embodiments, the light injected into the waveguides 270, 280, 290, 300, 310 is provided by a light projector system 520, which includes a light module 530, which may include a light emitter, such as a light emitting diode (LED). The light from the light module 530 may be directed to and modified by a light modulator 540, e.g., a spatial light modulator, via a beam splitter 550. The light modulator 540 may be configured to change the perceived intensity of the light injected into the waveguides 270, 280, 290, 300, 310 to encode the light with image information. Examples of spatial light modulators include liquid crystal displays (LCD) including a liquid crystal on silicon (LCoS) displays. It will be appreciated that the image injection devices 360, 370, 380, 390, 400 are illustrated schematically and, in some embodiments, these image injection devices may represent different light paths and locations in a common projection system configured to output light into associated ones of the waveguides 270, 280, 290, 300, 310. In some embodiments, the waveguides of the waveguide assembly 260 may function as ideal lens while relaying light injected into the waveguides out to the user's eyes. In this conception, the object may be the spatial light modulator 540 and the image may be the image on the depth plane.

In some embodiments, the display system 250 may be a scanning fiber display comprising one or more scanning fibers configured to project light in various patterns (e.g., raster scan, spiral scan, Lissajous patterns, etc.) into one or more waveguides 270, 280, 290, 300, 310 and ultimately to the eye 210 of the viewer. In some embodiments, the illustrated image injection devices 360, 370, 380, 390, 400 may schematically represent a single scanning fiber or a bundle of scanning fibers configured to inject light into one or more waveguides of the waveguides 270, 280, 290, 300, 310. In some other embodiments, the illustrated image injection devices 360, 370, 380, 390, 400 may schematically represent one or more scanning fibers or one or more bundles of scanning fibers, each of which are configured to inject light into an associated one of the waveguides 270, 280, 290, 300, 310. It will be appreciated that one or more optical fibers may be configured to transmit light from the light module 530 to the one or more waveguides 270, 280, 290, 300, 310. It will be appreciated that one or more intervening optical structures may be provided between the scanning fiber, or fibers, and the one or more waveguides 270, 280, 290, 300, 310 to, e.g., redirect light exiting the scanning fiber into the one or more waveguides 270, 280, 290, 300, 310.

A controller 560 controls the operation of one or more of the stacked waveguide assembly 260, including operation of the image injection devices 360, 370, 380, 390, 400, the light source 530, and the light modulator 540. In some embodiments, the controller 560 is part of the local data processing module 140. The controller 560 includes programming (e.g., instructions in a non-transitory medium) that regulates the timing and provision of image information to the waveguides 270, 280, 290, 300, 310 according to, e.g., any of the various schemes disclosed herein. In some embodiments, the controller may be a single integral device, or a distributed system connected by wired or wireless communication

channels. The controller **560** may be part of the processing modules **140** or **150** (FIG. 9D) in some embodiments.

With continued reference to FIG. 6, the waveguides **270**, **280**, **290**, **300**, **310** may be configured to propagate light within each respective waveguide by total internal reflection (TIR). The waveguides **270**, **280**, **290**, **300**, **310** may each be planar or have another shape (e.g., curved), with major top and bottom surfaces and edges extending between those major top and bottom surfaces. In the illustrated configuration, the waveguides **270**, **280**, **290**, **300**, **310** may each include out-coupling optical elements **570**, **580**, **590**, **600**, **610** that are configured to extract light out of a waveguide by redirecting the light, propagating within each respective waveguide, out of the waveguide to output image information to the eye **210**. Although referred to as “out-coupling optical element” through the specification, the out-coupling optical element need not be an optical element and may be a non-optical element. Extracted light may also be referred to as out-coupled light and the outcoupling optical elements light may also be referred to light extracting optical elements. An extracted beam of light may be outputted by the waveguide at locations at which the light propagating in the waveguide strikes a light extracting optical element. The out-coupling optical elements **570**, **580**, **590**, **600**, **610** may, for example, be gratings, including diffractive optical features, as discussed further herein. While illustrated disposed at the bottom major surfaces of the waveguides **270**, **280**, **290**, **300**, **310**, for ease of description and drawing clarity, in some embodiments, the outcoupling optical elements **570**, **580**, **590**, **600**, **610** may be disposed at the top and/or bottom major surfaces, and/or may be disposed directly in the volume of the waveguides **270**, **280**, **290**, **300**, **310**, as discussed further herein. In some embodiments, the out-coupling optical elements **570**, **580**, **590**, **600**, **610** may be formed in a layer of material that is attached to a transparent substrate to form the waveguides **270**, **280**, **290**, **300**, **310**. In some other embodiments, the waveguides **270**, **280**, **290**, **300**, **310** may be a monolithic piece of material and the out-coupling optical elements **570**, **580**, **590**, **600**, **610** may be formed on a surface and/or in the interior of that piece of material.

With continued reference to FIG. 6, as discussed herein, each waveguide **270**, **280**, **290**, **300**, **310** is configured to output light to form an image corresponding to a particular depth plane. For example, the waveguide **270** nearest the eye may be configured to deliver collimated light (which was injected into such waveguide **270**), to the eye **210**. The collimated light may be representative of the optical infinity focal plane. The next waveguide up **280** may be configured to send out collimated light which passes through the first lens **350** (e.g., a negative lens) before it may reach the eye **210**; such first lens **350** may be configured to create a slight convex wavefront curvature so that the eye/brain interprets light coming from that next waveguide up **280** as coming from a first focal plane closer inward toward the eye **210** from optical infinity. Similarly, the third up waveguide **290** passes its output light through both the first **350** and second **340** lenses before reaching the eye **210**; the combined optical power of the first **350** and second **340** lenses may be configured to create another incremental amount of wavefront curvature so that the eye/brain interprets light coming from the third waveguide **290** as coming from a second focal plane that is even closer inward toward the person from optical infinity than was light from the next waveguide up **280**.

The other waveguide layers **300**, **310** and lenses **330**, **320** are similarly configured, with the highest waveguide **310** in

the stack sending its output through all of the lenses between it and the eye for an aggregate focal power representative of the closest focal plane to the person. To compensate for the stack of lenses **320**, **330**, **340**, **350** when viewing/interpreting light coming from the world **510** on the other side of the stacked waveguide assembly **260**, a compensating lens layer **620** may be disposed at the top of the stack to compensate for the aggregate power of the lens stack **320**, **330**, **340**, **350** below. Such a configuration provides as many perceived focal planes as there are available waveguide/lens pairings. Both the out-coupling optical elements of the waveguides and the focusing aspects of the lenses may be static (i.e., not dynamic or electro-active). In some alternative embodiments, either or both may be dynamic using electro-active features.

In some embodiments, two or more of the waveguides **270**, **280**, **290**, **300**, **310** may have the same associated depth plane. For example, multiple waveguides **270**, **280**, **290**, **300**, **310** may be configured to output images set to the same depth plane, or multiple subsets of the waveguides **270**, **280**, **290**, **300**, **310** may be configured to output images set to the same one or more depth planes, with one set for each depth plane. This may provide advantages for forming a tiled image to provide an expanded field of view at those depth planes.

With continued reference to FIG. 6, the out-coupling optical elements **570**, **580**, **590**, **600**, **610** may be configured to both redirect light out of their respective waveguides and to output this light with the appropriate amount of divergence or collimation for a particular depth plane associated with the waveguide. As a result, waveguides having different associated depth planes may have different configurations of out-coupling optical elements **570**, **580**, **590**, **600**, **610**, which output light with a different amount of divergence depending on the associated depth plane. In some embodiments, the light extracting optical elements **570**, **580**, **590**, **600**, **610** may be volumetric or surface features, which may be configured to output light at specific angles. For example, the light extracting optical elements **570**, **580**, **590**, **600**, **610** may be volume holograms, surface holograms, and/or diffraction gratings. In some embodiments, the features **320**, **330**, **340**, **350** may not be lenses; rather, they may simply be spacers (e.g., cladding layers and/or structures for forming air gaps).

In some embodiments, the out-coupling optical elements **570**, **580**, **590**, **600**, **610** are diffractive features that form a diffraction pattern, or “diffractive optical element” (also referred to herein as a “DOE”). Preferably, the DOEs have a sufficiently low diffraction efficiency so that only a portion of the light of the beam is deflected away toward the eye **210** with each intersection of the DOE, while the rest continues to move through a waveguide via TIR. The light carrying the image information is thus divided into a number of related exit beams that exit the waveguide at a multiplicity of locations and the result is a fairly uniform pattern of exit emission toward the eye **210** for this particular collimated beam bouncing around within a waveguide.

In some embodiments, one or more DOEs may be switchable between “on” states in which they actively diffract, and “off” states in which they do not significantly diffract. For instance, a switchable DOE may comprise a layer of polymer dispersed liquid crystal, in which microdroplets comprise a diffraction pattern in a host medium, and the refractive index of the microdroplets may be switched to substantially match the refractive index of the host material (in which case the pattern does not appreciably diffract incident light) or the microdroplet may be switched to an

index that does not match that of the host medium (in which case the pattern actively diffracts incident light).

In some embodiments, a camera assembly **630** (e.g., a digital camera, including visible light and infrared light cameras) may be provided to capture images of the eye **210** and/or tissue around the eye **210** to, e.g., detect user inputs and/or to monitor the physiological state of the user. As used herein, a camera may be any image capture device. In some embodiments, the camera assembly **630** may include an image capture device and a light source to project light (e.g., infrared light) to the eye, which may then be reflected by the eye and detected by the image capture device. In some embodiments, the camera assembly **630** may be attached to the frame **80** (FIG. 9D) and may be in electrical communication with the processing modules **140** and/or **150**, which may process image information from the camera assembly **630**. In some embodiments, one camera assembly **630** may be utilized for each eye, to separately monitor each eye.

With reference now to FIG. 7, an example of exit beams outputted by a waveguide is shown. One waveguide is illustrated, but it will be appreciated that other waveguides in the waveguide assembly **260** (FIG. 6) may function similarly, where the waveguide assembly **260** includes multiple waveguides. Light **640** is injected into the waveguide **270** at the input surface **460** of the waveguide **270** and propagates within the waveguide **270** by TIR. At points where the light **640** impinges on the DOE **570**, a portion of the light exits the waveguide as exit beams **650**. The exit beams **650** are illustrated as substantially parallel but, as discussed herein, they may also be redirected to propagate to the eye **210** at an angle (e.g., forming divergent exit beams), depending on the depth plane associated with the waveguide **270**. It will be appreciated that substantially parallel exit beams may be indicative of a waveguide with out-coupling optical elements that out-couple light to form images that appear to be set on a depth plane at a large distance (e.g., optical infinity) from the eye **210**. Other waveguides or other sets of out-coupling optical elements may output an exit beam pattern that is more divergent, which would require the eye **210** to accommodate to a closer distance to bring it into focus on the retina and would be interpreted by the brain as light from a distance closer to the eye **210** than optical infinity.

In some embodiments, a full color image may be formed at each depth plane by overlaying images in each of the component colors, e.g., three or more component colors.

FIG. 8 illustrates an example of a stacked waveguide assembly in which each depth plane includes images formed using multiple different component colors. The illustrated embodiment shows depth planes **240a-240f**, although more or fewer depths are also contemplated. Each depth plane may have three or more component color images associated with it, including: a first image of a first color, G; a second image of a second color, R; and a third image of a third color, B. Different depth planes are indicated in the FIG. by different numbers for diopters (dpt) following the letters G, R, and B. Just as examples, the numbers following each of these letters indicate diopters (1/m), or inverse distance of the depth plane from a viewer, and each box in the FIGS. represents an individual component color image. In some embodiments, to account for differences in the eye's focusing of light of different wavelengths, the exact placement of the depth planes for different component colors may vary. For example, different component color images for a given depth plane may be placed on depth planes corresponding to

different distances from the user. Such an arrangement may increase visual acuity and user comfort and/or may decrease chromatic aberrations.

In some embodiments, light of each component color may be outputted by a single dedicated waveguide and, consequently, each depth plane may have multiple waveguides associated with it. In such embodiments, each box in the FIGS. including the letters G, R, or B may be understood to represent an individual waveguide, and three waveguides may be provided per depth plane where three component color images are provided per depth plane. While the waveguides associated with each depth plane are shown adjacent to one another in this drawing for ease of description, it will be appreciated that, in a physical device, the waveguides may all be arranged in a stack with one waveguide per level. In some other embodiments, multiple component colors may be outputted by the same waveguide, such that, e.g., only a single waveguide may be provided per depth plane.

With continued reference to FIG. 8, in some embodiments, G is the color green, R is the color red, and B is the color blue. In some other embodiments, other colors associated with other wavelengths of light, including magenta and cyan, may be used in addition to or may replace one or more of red, green, or blue.

It will be appreciated that references to a given color of light throughout this disclosure will be understood to encompass light of one or more wavelengths within a range of wavelengths of light that are perceived by a viewer as being of that given color. For example, red light may include light of one or more wavelengths in the range of about 620-780 nm, green light may include light of one or more wavelengths in the range of about 492-577 nm, and blue light may include light of one or more wavelengths in the range of about 435-493 nm.

In some embodiments, the light source **530** (FIG. 6) may be configured to emit light of one or more wavelengths outside the visual perception range of the viewer, for example, infrared and/or ultraviolet wavelengths. In addition, the in-coupling, out-coupling, and other light redirecting structures of the waveguides of the display **250** may be configured to direct and emit this light out of the display towards the eye **210**, e.g., for imaging and/or user stimulation applications.

With reference now to FIG. 9A, in some embodiments, light impinging on a waveguide may need to be redirected to in-couple that light into the waveguide. An in-coupling optical element may be used to redirect and in-couple the light into its corresponding waveguide. Although referred to as "in-coupling optical element" through the specification, the in-coupling optical element need not be an optical element and may be a non-optical element. FIG. 9A illustrates a cross-sectional side view of an example of a set **660** of stacked waveguides that each includes an in-coupling optical element. The waveguides may each be configured to output light of one or more different wavelengths, or one or more different ranges of wavelengths. It will be appreciated that the stack **660** may correspond to the stack **260** (FIG. 6) and the illustrated waveguides of the stack **660** may correspond to part of the waveguides **270**, **280**, **290**, **300**, **310**, except that light from one or more of the image injection devices **360**, **370**, **380**, **390**, **400** is injected into the waveguides from a position that requires light to be redirected for in-coupling.

The illustrated set **660** of stacked waveguides includes waveguides **670**, **680**, and **690**. Each waveguide includes an associated in-coupling optical element (which may also be

referred to as a light input area on the waveguide), with, e.g., in-coupling optical element **700** disposed on a major surface (e.g., an upper major surface) of waveguide **670**, in-coupling optical element **710** disposed on a major surface (e.g., an upper major surface) of waveguide **680**, and in-coupling optical element **720** disposed on a major surface (e.g., an upper major surface) of waveguide **690**. In some embodiments, one or more of the in-coupling optical elements **700**, **710**, **720** may be disposed on the bottom major surface of the respective waveguide **670**, **680**, **690** (particularly where the one or more in-coupling optical elements are reflective, deflecting optical elements). As illustrated, the in-coupling optical elements **700**, **710**, **720** may be disposed on the upper major surface of their respective waveguide **670**, **680**, **690** (or the top of the next lower waveguide), particularly where those in-coupling optical elements are transmissive, deflecting optical elements. In some embodiments, the in-coupling optical elements **700**, **710**, **720** may be disposed in the body of the respective waveguide **670**, **680**, **690**. In some embodiments, as discussed herein, the in-coupling optical elements **700**, **710**, **720** are wavelength selective, such that they selectively redirect one or more wavelengths of light, while transmitting other wavelengths of light. While illustrated on one side or corner of their respective waveguide **670**, **680**, **690**, it will be appreciated that the in-coupling optical elements **700**, **710**, **720** may be disposed in other areas of their respective waveguide **670**, **680**, **690** in some embodiments.

As illustrated, the in-coupling optical elements **700**, **710**, **720** may be laterally offset from one another. In some embodiments, each in-coupling optical element may be offset such that it receives light without that light passing through another in-coupling optical element. For example, each in-coupling optical element **700**, **710**, **720** may be configured to receive light from a different image injection device **360**, **370**, **380**, **390**, and **400** as shown in FIG. 6, and may be separated (e.g., laterally spaced apart) from other in-coupling optical elements **700**, **710**, **720** such that it substantially does not receive light from the other ones of the in-coupling optical elements **700**, **710**, **720**.

Each waveguide also includes associated light distributing elements, with, e.g., light distributing elements **730** disposed on a major surface (e.g., a top major surface) of waveguide **670**, light distributing elements **740** disposed on a major surface (e.g., a top major surface) of waveguide **680**, and light distributing elements **750** disposed on a major surface (e.g., a top major surface) of waveguide **690**. In some other embodiments, the light distributing elements **730**, **740**, **750**, may be disposed on a bottom major surface of associated waveguides **670**, **680**, **690**, respectively. In some other embodiments, the light distributing elements **730**, **740**, **750**, may be disposed on both top and bottom major surface of associated waveguides **670**, **680**, **690**, respectively; or the light distributing elements **730**, **740**, **750**, may be disposed on different ones of the top and bottom major surfaces in different associated waveguides **670**, **680**, **690**, respectively.

The waveguides **670**, **680**, **690** may be spaced apart and separated by, e.g., gas, liquid, and/or solid layers of material. For example, as illustrated, layer **760a** may separate waveguides **670** and **680**; and layer **760b** may separate waveguides **680** and **690**. In some embodiments, the layers **760a** and **760b** are formed of low refractive index materials (that is, materials having a lower refractive index than the material forming the immediately adjacent one of waveguides **670**, **680**, **690**). Preferably, the refractive index of the material forming the layers **760a**, **760b** is 0.05 or more, or

0.10 or less than the refractive index of the material forming the waveguides **670**, **680**, **690**. Advantageously, the lower refractive index layers **760a**, **760b** may function as cladding layers that facilitate total internal reflection (TIR) of light through the waveguides **670**, **680**, **690** (e.g., TIR between the top and bottom major surfaces of each waveguide). In some embodiments, the layers **760a**, **760b** are formed of air. While not illustrated, it will be appreciated that the top and bottom of the illustrated set **660** of waveguides may include immediately neighboring cladding layers.

Preferably, for ease of manufacturing and other considerations, the material forming the waveguides **670**, **680**, **690** are similar or the same, and the material forming the layers **760a**, **760b** are similar or the same. In some embodiments, the material forming the waveguides **670**, **680**, **690** may be different between one or more waveguides, and/or the material forming the layers **760a**, **760b** may be different, while still holding to the various refractive index relationships noted above. A variety of materials can be utilized to form the waveguides. Although glass is one material that can be utilized to fabricate the waveguides, other materials, including LiNbO₃, SiC, ZnS, or the like can be utilized. These materials can be in the form of optical quality single crystal materials or materials that are of optical quality, but not single crystal. Additionally, multi-grain ceramics of similar compositions can be utilized to form the waveguides. As an example, nano-crystalline materials can be utilized in the fabrication of the waveguides.

With continued reference to FIG. 9A, light rays **770**, **780**, **790** are incident on the set **660** of waveguides. It will be appreciated that the light rays **770**, **780**, **790** may be injected into the waveguides **670**, **680**, **690** by one or more image injection devices **360**, **370**, **380**, **390**, **400** (FIG. 6).

In some embodiments, the light rays **770**, **780**, **790** have different properties, e.g., different wavelengths or different ranges of wavelengths, which may correspond to different colors. The in-coupling optical elements **700**, **710**, **720** each deflect the incident light such that the light propagates through a respective one of the waveguides **670**, **680**, **690** by TIR. In some embodiments, the in-coupling optical elements **700**, **710**, **720** each selectively deflect one or more particular wavelengths of light, while transmitting other wavelengths to an underlying waveguide and associated in-coupling optical element.

For example, in-coupling optical element **700** may be configured to deflect ray **770**, which has a first wavelength or range of wavelengths, while transmitting rays **780** and **790**, which have different second and third wavelengths or ranges of wavelengths, respectively. The transmitted ray **780** impinges on and is deflected by the in-coupling optical element **710**, which is configured to deflect light of a second wavelength or range of wavelengths. The ray **790** is deflected by the in-coupling optical element **720**, which is configured to selectively deflect light of third wavelength or range of wavelengths.

With continued reference to FIG. 9A, the deflected light rays **770**, **780**, **790** are deflected so that they propagate through a corresponding waveguide **670**, **680**, **690**; that is, the in-coupling optical elements **700**, **710**, **720** of each waveguide deflects light into that corresponding waveguide **670**, **680**, **690** to in-couple light into that corresponding waveguide. The light rays **770**, **780**, **790** are deflected at angles that cause the light to propagate through the respective waveguide **670**, **680**, **690** by TIR. The light rays **770**, **780**, **790** propagate through the respective waveguide **670**, **680**, **690** by TIR until impinging on the waveguide's corresponding light distributing elements **730**, **740**, **750**.

With reference now to FIG. 9B, a perspective view of an example of the stacked waveguides of FIG. 9A is illustrated. As noted above, the in-coupled light rays 770, 780, 790, are deflected by the in-coupling optical elements 700, 710, 720, respectively, and then propagate by TIR within the waveguides 670, 680, 690, respectively. The light rays 770, 780, 790 then impinge on the light distributing elements 730, 740, 750, respectively. The light distributing elements 730, 740, 750 deflect the light rays 770, 780, 790 so that they propagate towards the out-coupling optical elements 800, 810, 820, respectively.

In some embodiments, the light distributing elements 730, 740, 750 are orthogonal pupil expanders (OPEs). In some embodiments, the OPEs deflect or distribute light to the out-coupling optical elements 800, 810, 820 and, in some embodiments, may also increase the beam or spot size of this light as it propagates to the out-coupling optical elements. In some embodiments, the light distributing elements 730, 740, 750 may be omitted and the in-coupling optical elements 700, 710, 720 may be configured to deflect light directly to the out-coupling optical elements 800, 810, 820. For example, with reference to FIG. 9A, the light distributing elements 730, 740, 750 may be replaced with out-coupling optical elements 800, 810, 820, respectively. In some embodiments, the out-coupling optical elements 800, 810, 820 are exit pupils (EPs) or exit pupil expanders (EPEs) that direct light in the eye 210 (FIG. 7). It will be appreciated that the OPEs may be configured to increase the dimensions of the eye box in at least one axis and the EPEs may be to increase the eye box in an axis crossing, e.g., orthogonal to, the axis of the OPEs. For example, each OPE may be configured to redirect a portion of the light striking the OPE to an EPE of the same waveguide, while allowing the remaining portion of the light to continue to propagate down the waveguide. Upon impinging on the OPE again, another portion of the remaining light is redirected to the EPE, and the remaining portion of that portion continues to propagate further down the waveguide, and so on. Similarly, upon striking the EPE, a portion of the impinging light is directed out of the waveguide towards the user, and a remaining portion of that light continues to propagate through the waveguide until it strikes the EP again, at which time another portion of the impinging light is directed out of the waveguide, and so on. Consequently, a single beam of in-coupled light may be “replicated” each time a portion of that light is redirected by an OPE or EPE, thereby forming a field of cloned beams of light, as shown in FIG. 6. In some embodiments, the OPE and/or EPE may be configured to modify a size of the beams of light.

Accordingly, with reference to FIGS. 9A and 9B, in some embodiments, the set 660 of waveguides includes waveguides 670, 680, 690; in-coupling optical elements 700, 710, 720; light distributing elements (e.g., OPEs) 730, 740, 750; and out-coupling optical elements (e.g., EPs) 800, 810, 820 for each component color. The waveguides 670, 680, 690 may be stacked with an air gap/cladding layer between each one. The in-coupling optical elements 700, 710, 720 redirect or deflect incident light (with different in-coupling optical elements receiving light of different wavelengths) into its waveguide. The light then propagates at an angle which will result in TIR within the respective waveguide 670, 680, 690. In the example shown, light ray 770 (e.g., blue light) is deflected by the first in-coupling optical element 700, and then continues to bounce down the waveguide, interacting with the light distributing element (e.g., OPEs) 730 and then the out-coupling optical element (e.g., EPs) 800, in a manner described earlier. The light rays 780 and 790 (e.g., green and red light, respectively) will pass through the waveguide 670,

with light ray 780 impinging on and being deflected by in-coupling optical element 710. The light ray 780 then bounces down the waveguide 680 via TIR, proceeding on to its light distributing element (e.g., OPEs) 740 and then the out-coupling optical element (e.g., EPs) 810. Finally, light ray 790 (e.g., red light) passes through the waveguide 690 to impinge on the light in-coupling optical elements 720 of the waveguide 690. The light in-coupling optical elements 720 deflect the light ray 790 such that the light ray propagates to light distributing element (e.g., OPEs) 750 by TIR, and then to the out-coupling optical element (e.g., EPs) 820 by TIR. The out-coupling optical element 820 then finally out-couples the light ray 790 to the viewer, who also receives the out-coupled light from the other waveguides 670, 680.

FIG. 9C illustrates a top-down plan view of an example of the stacked waveguides of FIGS. 9A and 9B. As illustrated, the waveguides 670, 680, 690, along with each waveguide's associated light distributing element 730, 740, 750 and associated out-coupling optical element 800, 810, 820, may be vertically aligned. However, as discussed herein, the in-coupling optical elements 700, 710, 720 are not vertically aligned; rather, the in-coupling optical elements are preferably nonoverlapping (e.g., laterally spaced apart as seen in the top-down view). As discussed further herein, this nonoverlapping spatial arrangement facilitates the injection of light from different resources into different waveguides on a one-to-one basis, thereby allowing a specific light source to be uniquely coupled to a specific waveguide. In some embodiments, arrangements including nonoverlapping spatially-separated in-coupling optical elements may be referred to as a shifted pupil system, and the in-coupling optical elements within these arrangements may correspond to sub pupils.

FIG. 9D illustrates an example of wearable display system 60 into which the various waveguides and related systems disclosed herein may be integrated. In some embodiments, the display system 60 is the system 250 of FIG. 6, with FIG. 6 schematically showing some parts of that system 60 in greater detail. For example, the waveguide assembly 260 of FIG. 6 may be part of the display 70.

With continued reference to FIG. 9D, the display system 60 includes a display 70, and various mechanical and electronic modules and systems to support the functioning of that display 70. The display 70 may be coupled to a frame 80, which is wearable by a display system user or viewer 90 and which is configured to position the display 70 in front of the eyes of the user 90. The display 70 may be considered eyewear in some embodiments. In some embodiments, a speaker 100 is coupled to the frame 80 and configured to be positioned adjacent the ear canal of the user 90 (in some embodiments, another speaker, not shown, may optionally be positioned adjacent the other ear canal of the user to provide stereo/shapeable sound control). The display system 60 may also include one or more microphones 110 or other devices to detect sound. In some embodiments, the microphone is configured to allow the user to provide inputs or commands to the system 60 (e.g., the selection of voice menu commands, natural language questions, etc.), and/or may allow audio communication with other persons (e.g., with other users of similar display systems. The microphone may further be configured as a peripheral sensor to collect audio data (e.g., sounds from the user and/or environment). In some embodiments, the display system 60 may further include one or more outwardly-directed environmental sensors 112 configured to detect objects, stimuli, people, animals, locations, or other aspects of the world around the user. For example, environmental sensors 112 may include

one or more cameras, which may be located, for example, facing outward so as to capture images similar to at least a portion of an ordinary field of view of the user **90**. In some embodiments, the display system may also include a peripheral sensor **120a**, which may be separate from the frame **80** and attached to the body of the user **90** (e.g., on the head, torso, an extremity, etc. of the user **90**). The peripheral sensor **120a** may be configured to acquire data characterizing a physiological state of the user **90** in some embodiments. For example, the sensor **120a** may be an electrode.

With continued reference to FIG. 9D, the display **70** is operatively coupled by communications link **130**, such as by a wired lead or wireless connectivity, to a local data processing module **140** which may be mounted in a variety of configurations, such as fixedly attached to the frame **80**, fixedly attached to a helmet or hat worn by the user, embedded in headphones, or otherwise removably attached to the user **90** (e.g., in a backpack-style configuration, in a belt-coupling style configuration). Similarly, the sensor **120a** may be operatively coupled by communications link **120b**, e.g., a wired lead or wireless connectivity, to the local processor and data module **140**. The local processing and data module **140** may comprise a hardware processor, as well as digital memory, such as non-volatile memory (e.g., flash memory or hard disk drives), both of which may be utilized to assist in the processing, caching, and storage of data. Optionally, the local processor and data module **140** may include one or more central processing units (CPUs), graphics processing units (GPUs), dedicated processing hardware, and so on. The data may include data a) captured from sensors (which may be, e.g., operatively coupled to the frame **80** or otherwise attached to the user **90**), such as image capture devices (such as cameras), microphones, inertial measurement units, accelerometers, compasses, GPS units, radio devices, gyros, and/or other sensors disclosed herein; and/or b) acquired and/or processed using remote processing module **150** and/or remote data repository **160** (including data relating to virtual content), possibly for passage to the display **70** after such processing or retrieval. The local processing and data module **140** may be operatively coupled by communication links **170**, **180**, such as via a wired or wireless communication links, to the remote processing module **150** and remote data repository **160** such that these remote modules **150**, **160** are operatively coupled to each other and available as resources to the local processing and data module **140**. In some embodiments, the local processing and data module **140** may include one or more of the image capture devices, microphones, inertial measurement units, accelerometers, compasses, GPS units, radio devices, and/or gyros. In some other embodiments, one or more of these sensors may be attached to the frame **80**, or may be standalone structures that communicate with the local processing and data module **140** by wired or wireless communication pathways.

With continued reference to FIG. 9D, in some embodiments, the remote processing module **150** may comprise one or more processors configured to analyze and process data and/or image information, for instance including one or more central processing units (CPUs), graphics processing units (GPUs), dedicated processing hardware, and so on. In some embodiments, the remote data repository **160** may comprise a digital data storage facility, which may be available through the internet or other networking configuration in a "cloud" resource configuration. In some embodiments, the remote data repository **160** may include one or more remote servers, which provide information, e.g., information for generating augmented reality content, to the local

processing and data module **140** and/or the remote processing module **150**. In some embodiments, all data is stored and all computations are performed in the local processing and data module, allowing fully autonomous use from a remote module. Optionally, an outside system (e.g., a system of one or more processors, one or more computers) that includes CPUs, GPUs, and so on, may perform at least a portion of processing (e.g., generating image information, processing data) and provide information to, and receive information from, modules **140**, **150**, **160**, for instance via wireless or wired connections.

FIG. **10** is a schematic diagram illustrating a projector assembly **1000** that utilizes a polarization beamsplitter (PBS) **1020** to illuminate a spatial light modulator (SLM) **1030** and redirect the light from the SLM **1030** through projection optics **1040** to an eyepiece (not shown). The projector assembly **1000** includes an illumination source **1010**, which can include, for example, light emitting diodes (LEDs), lasers (e.g., laser diodes), or other type of light source. This light may be collimated by collimating optics. The illumination source **1010** can emit polarized, unpolarized, or partially polarized light. In the illustrated design, the illumination source **1010** may emit light **1012** polarized having a p-polarization. A first optical element **1015** (e.g., a pre-polarizer) is aligned to pass light with the first polarization (e.g., p-polarization).

This light is directed to the polarizing beam splitter **1020**. Initially, light passes through an interface **1022** (e.g., a polarizing interface) of the PBS **1020**, which is configured to transmit light of the first polarization (e.g., p-polarization). Accordingly, the light continues to and is incident on the spatial light modulator **1030**. As illustrated, the SLM **1030** is a reflective SLM configured to retro-reflect the light incident and selectively modulate the light. The SLM **1030**, for example, includes one or more pixels that can have different states. The light incident on respective pixels may be modulated based on the state of the pixel. Accordingly, the SLM **1030** can be driven to modulate the light so as to provide an image. In this example, the SLM **1030** may be a polarization based SLM that modulates the polarization of the light incident thereon. For example, in an on state, a pixel of the SLM **1030** changes input light from a first polarization state (e.g., p-polarization state) to a second polarization state (e.g., s-polarization state) such that a bright state (e.g., white pixel) is shown. The second polarization state may be the first polarization state modulated (e.g., rotated) by 90°. In the on state, the light having the second polarization state is reflected by the interface **1022** and propagates downstream to the projector optics **1040**. In an off state, the SLM **1030** does not change the polarization state of the light incident thereon, for example, does not rotate the input light from the first polarization state, thus a dark state (e.g., black pixel) is shown. In the off state, the light having the first polarization state is transmitted through the interface **1022** and propagates upstream back to the illumination source **1010** and not to a user's eye.

After reflection from the SLM **1030**, a portion of the light **1014** (e.g., the modulated light) is reflected from the interface **1022** and exits the PBS **1020** to be directed to the user's eye. The emitted light passes through the projector optics **1040** and is imaged onto an in-coupling grating (ICG) **1050** of an eyepiece (not shown).

FIG. **11A** illustrates a system (e.g., an augmented reality display system) **1100A** for presenting images to the user's eye **210** and for viewing the world **510** that has an alternative configuration to that shown in FIG. **10**. The system **1100** includes a light source **1110**, a spatial light modulator (SLM)

1140, and a waveguide 1120, also referred to as an eyepiece waveguide, arranged such that light from the light source 1110 illuminates the SLM 1140, and light reflected from the SLM 1140 is coupled into the waveguide 1120 to be directed to the eye 210. The system 1100A includes optics 1130 disposed to both illuminate the SLM 1140 and project an image of the SLM 1140. Light from the light source 1110, for example, propagates in a first direction through the optics 1130 onto the SLM 1140 thereby illuminating the SLM 1140. Light reflected from the SLM 1140 propagates again through the optics 1130 in a second direction opposite the first direction and is directed to the waveguide 1120 and coupled therein.

The light source 1110 may include light emitting diodes (LEDs), lasers (e.g., laser diodes), or other type of light source. The light source 1110 may be a polarized light source, however the light source 1110 need not be so limited. In some implementations, a polarizer 1115 may be positioned between the light source 1110 and the SLM 1140. As illustrated, the polarizer 1115 is between the light source 1110 and the waveguide 1120. This polarizer 1115 may also be a light recycler, transmitting light of a first polarization and reflecting light of a second polarization back to the light source 1110. Such a polarizer 1115 may be, for example, a wire grid polarizer. A coupling optic 1105, such as a non-imaging optical element (e.g., cone, compound parabolic collector (CPC, lenses)), may be disposed with respect to the light source 1110 to receive light output from the light source 1110. The coupling optic 1105 may collect the light from the light source 1110 and may, in some cases, reduce the divergence of light emitted from the light source 1110. The coupling optic 1105 may, for example, collimate the light output from the light source 1110. The coupling optic 1105 may collect light that matches the angular spectrum field of view of the system 1100A. Accordingly, the coupling optic 1105 may match an angular spectrum of the light output by the light source 1110 with the field of view of the system 1100A. The coupling optic 1105 may have an asymmetric profile to operate on the light emitted from the light source 1110 asymmetrically. For example, the coupling optic 1105 may reduce the divergence a different amount in orthogonal directions (e.g., x and z directions). Such asymmetry in the coupling optic 1105 may address asymmetry in the light emitted from the light source 1110 which may include, for example, a laser diode that emits a wider range of angles of light in one direction (e.g., x or z) as opposed to the orthogonal direction (e.g., z or x, respectively).

As discussed above, the system 1100A includes optics 1130 configured to illuminate the SLM 1140 that is disposed in an optical path between the light source 1110 and the SLM 1140. The optics 1130 may include transmissive optics that transmits light from the light source 1110 to the SLM 1140. The optics 1130 may also be configured to project an image of the SLM 1140 or formed by the SLM 1140 into the waveguide 1120. An image may be projected into the eye of the eye 210. In some designs, the optics 1130 may include one or more lenses or optical elements having optic power. The optic 1130 may, for example, have positive optical power. The optics 1130 may include one or more refractive optical elements such as refractive lenses. Other types of optical elements may also possibly be used.

The SLM 1140 may be reflective, modulating and reflecting light therefrom. The SLM 1140 may be a polarization based SLM configured to modulate polarization. The SLM 1140 may, for example, include a liquid crystal (LC) SLM (e.g., a liquid crystal on silicon (LCoS) SLM). The LC SLM may, for example, include twisted nematic (TN) liquid

crystal. The SLM 1140 may be substantially similar to the SLM 1030 with reference to FIG. 10. The SLM 1140 may, for example, include one or more pixels that are configured to selectively modulate light incident on the pixel depending on the state of the pixel. For some types of SLMs 1140, the pixel may, for example, modulate the beam incident thereon by altering the polarization state such as rotating the polarization (e.g., rotating the orientation of linearly polarized light).

As discussed above, the SLM 1140 may be a LCoS SLM 1140. In a cross-polarizer configuration, the LCoS SLM 1140 may be nominally white. When a pixel is off (e.g., 0 voltage), it has a bright state, and when the pixel is on (e.g., voltage above a threshold turn on voltage), it has a dark state. In this cross-polarization configuration, leakage is minimized when a pixel is on and it has a dark state.

In a parallel-polarizer configuration, the LCoS SLM 1140 is nominally black. When a pixel is off (e.g., 0 voltage), it has a dark state, and when the pixel is on (e.g., voltage above a threshold turn on voltage), it has a bright state. In this parallel-polarizer configuration, leakage is minimized when a pixel is off and it has a dark state. The dark state may be (re)optimized using rub direction and compensator angle. Compensator angle may refer to an angle of a compensator which may be between the optics 1130 and the SLM 1140, for example, as illustrated in FIG. 20B.

Dynamic range and throughput for parallel-polarizer configurations may be different than that of cross-polarizer configurations. Further, parallel-polarizer configurations may be optimized for contrast differently than cross-polarizer configurations.

The system 1100A includes the waveguide 1120 for outputting image information to the eye 210. The waveguide 1120 may be substantially similar to waveguides 270, 280, 290, 300, 310, 670, 680, and 690 discussed above. The waveguide 1120 may include substantially transparent material having a refractive index sufficient to guide light therein. As illustrated, the waveguide 1120 may include a first side 1121 and a second side 1123 opposite the first side 1121 and corresponding upper and lower major surfaces as well as edges there around. The first and second major 1121, 1123 surface may be sufficiently flat such that image information may be retained upon propagating light from the SLM 1140 to the eye 210 such than an image formed by the SLM 1140 may be injected into the eye. The optics 1130 and the SLM 1140 may be positioned on the first side 1121 of the waveguide 1120. The light source 1110 may be disposed on the second side 1123 such that light from the light source 1110 is incident on the second side 1123 prior to passing through the waveguide 1120 and through the optics 1130 to the SLM 1140. Accordingly, the waveguide 1120 may be disposed between the light source 1110 and the optics 1130. Additionally, at least a portion of the waveguide 1120 may extend between the light source 1110 and the optics 1130, whereby light passes through the portion of the waveguide 1120 to the optics 1130. Light emitted from the light source 1110 can therefore be directed through the waveguide 1120, into and through the optics 1130 and incident on the SLM 1140. The SLM 1140 reflects the light back through the optics 1130 and to the waveguide 1120.

The system 1100A also includes an in-coupling optical element 1160 for coupling light from the optics 1130 into the waveguide 1120. The in-coupling optical element 1160 may be disposed on a major surface (e.g., an upper major surface 1123) of the waveguide 1120. In some designs, the in-coupling optical element 1160 may be disposed on the lower major surface 1121 of the waveguide 1120. In some designs,

the in-coupling optical element **1160** may be disposed in the body of the waveguide **1120**. While illustrated on one side or corner of the waveguide **1120**, the in-coupling optical element **1160** may be disposed in/on other areas of the waveguide **1120**. The in-coupling optical element **1160** may be substantially similar to the in-coupling optical elements **700**, **710**, **720** described above with reference to FIGS. **9A**, **9B**, and **9C**. The in-coupling optical element **1160** may be a diffractive optical element or a reflector. Other structures may be used as the in-coupling optical element **1160**. The in-coupling optical element **1160** may be configured to direct the light incident thereon into the waveguide **1120** at a sufficiently large grazing angle (e.g., greater than the critical angle) with respect to the upper and lower major surfaces **1123**, **1121** of the waveguide **1120** to be guided therein by total internal reflection. Further, the in-coupling optical element **1160** may operate on a wide range of wavelengths and thus be configured to couple light of multiple colors into the waveguide **1120**. For instance, the in-coupling optical element **1160** may be configured to couple red light, green light, and blue light into the waveguide **1120**. The light source **1110** may emit red, green, and blue color light at different times.

The system **1100A** includes a light distributing element **1170** disposed on or in the waveguide **1120**. The light distributing element **1170** may be substantially similar to the light distributing elements **730**, **740**, and **750** described above with respect to FIG. **9B**. For instance, the light distributing element **1170** may be an orthogonal pupil expander (OPE). The light distributing element **1170** may be configured to spread the light within the waveguide **1120** by turning the light propagating in the x direction, for example, toward the z direction illustrated in the top view FIG. **11B**. The light distributing element **1170** may, thus, be configured to increase dimensions of the eyebox along the z-axis; see FIG. **11B**. The light distributing element **1170** may, for example, include one or more diffractive optical elements configured to diffract the light propagating within the waveguide **1120** incident the diffractive optical elements so as to redirect that light, for example, in a generally orthogonal direction. Other configurations are possible.

As shown in FIG. **11B**, the system **1100** may also include an out-coupling optical element **1180** for coupling light out of the waveguide **1120** to the eye **210**. The out-coupling optical element **1180** may be configured to redirect light propagating within the waveguide **1120** by total internal reflection (TIR) at an angle more normal to the upper and/or lower major surfaces **1123**, **1121** of the waveguide **1120** such that the light is not guided within the waveguide **1120**. Instead, this light is direct out of the waveguide **1120** through, for example, the lower major surface **1121**. The out-coupling optical element **1180** may, for example, include one or more diffractive optical elements configured to diffract the light propagating within the waveguide **1120** incident the diffractive optical element so as to redirect that light, for example, out of the waveguide **1120**. Other configurations are possible.

FIG. **11B** also shows the location of the in-coupling optical element **1160** laterally disposed with respect to the light distributing optical element (e.g., orthogonal pupil expander) **1170** and the out-coupling optical element **1180**. FIG. **11B** also shows the location of the light source **1110** laterally disposed with respect to the in-coupling optical element **1160**, the light distributing optical element (e.g., orthogonal pupil expander) **1170**, and the out-coupling optical element **1180**.

In operation, the light source **1110** of the system **1100A** emits light into the coupling optic **1105** and through the polarizer **1115**. This light may therefore be polarized, for example, linearly polarized in a first direction. This polarized light may be transmitted through the waveguide **1120**, entering the second major surface of the waveguide **1120** and exiting the first major surface of the waveguide **1120**. This light may propagate through the optics **1130** to the SLM **1140**. The optics **1130** quasi-collimates and/or selects the light from the light source **1110** to thereby illuminate the SLM **1140**, which may include a polarization based modulator that modulates the polarization of light incident thereon such as by selectively rotating the orientation of the modulator on a pixel by pixel basis depending on the state of the pixel. For example, a first pixel may be in a first state and rotate polarization while a second pixel may be in a second state and not rotate polarization. The light between the coupling optic **1105** and the optics **1130** may fairly uniformly illuminate the SLM **1140**. After being incident on the SLM **1140**, the light is reflected back through the optics **1130**. The optics **1130** may be configured to project images from the SLM **1140** into the waveguide **1120** and ultimately into the eye **210** so that the image is visible to the eye **210**. In some designs, the retina of the eye **210** is the optical conjugate to the SLM **1140** and/or images formed by and/or on the SLM **1140**. The power of the optics **1130** may facilitate the projection of the image on the SLM **1140** into the eye **210** and onto the retina of the eye **210**. In some implementations, optical power, for example, provided by the out-coupling optical element **1180** may assist in and/or affect the image ultimately formed in the eye **210**. The optics **1130** acts as a projection lens as light reflected from the SLM **1140** travels through the optics toward the waveguide **1120**. The optics may function roughly as a Fourier transform of the image on the SLM **1140** to a plane in the waveguide **1120** near the in-coupling optical elements **1160**. Together, both passes through the optics **1130** (a first from the light source **1110** to the SLM **1140**, and a second from the SLM **1140** to the waveguide **1120**) may act to roughly image pupils of the coupling optic **1105**. The alignment and orientation of the light source **1110** (possibly also coupling optic **1105** and/or the polarizer **1115**), the optics **1130**, the SLM **1140** are such that light from the light source **1110** that is reflected from the SLM **1140** is directed onto the in-coupling optical element **1160**. The pupil associated with the coupling optic **1105** may be aligned with the in-coupling optical element **1160**. The light may pass through the analyzer **1150** (e.g., a polarizer) in an optical path between the SLM **1140** and the eye **210**. As depicted in FIG. **11A**, an analyzer (e.g., polarizer) **1150** may be disposed in an optical path between the optics **1130** and the in-coupling optical element **1160**. The analyzer **1150** may, for example, be a linear polarizer having an orientation to transmit light of the first polarization (p-polarization) and block light of the second polarization (s-polarization) or vice versa. The analyzer **1150** may be a clean-up polarizer and further block light of a polarization that is blocked by another polarizer between the SLM **1140** and the analyzer **1150** or within the SLM **1140**. The analyzer **1150** may, for example, be a circular polarizer that acts as an isolator to mitigate reflections from the waveguide **1120**, specifically the in-coupling optical element **1160**, back toward the SLM **1140**. The analyzer **1150** may, as any of the polarizers disclosed herein, include wire grid polarizers such as an absorptive wire grid polarizer. Such polarizers may offer appreciable absorption of unwanted light and therefore increased contrast. Some such polarizers can be made to include one or more dielectric layers on top of the wires

and/or multilayer films. In some implementations the SLM 1140 may be a liquid crystal on silicon (LCoS) SLM and may include LC cells and a retarder (e.g., compensator). In some implementations, the analyzer 1150 may be a compensator intended to provide a more consistent polarization rotation (e.g., of 90°) of the SLM 1140 for different angles of incidence and different wavelengths. A compensator may be used to improve contrast of the display by improving the rotation polarization for rays that are incident across a spread of angles and wavelengths. The SLM 1140 may include, for example, a TN LCoS that is configured to rotate incident light of a first polarization (e.g., s-polarization) to a second polarization (e.g., p-polarization) for a first pixel to produce a bright pixel state as the light will pass through the analyzer 1150. Conversely, the SLM 1140 may be configured to not rotate incident light of the first polarization (e.g., s-polarization) to the second polarization (e.g., p-polarization) for a second pixel such that the reflected light remains the first polarization to produce a dark pixel state as the light will be attenuated or blocked by the analyzer 1150. In such a configuration, the polarizer 1115 closer along the optical path to the light source 1110 may be oriented different (e.g., orthogonal) to the analyzer 1150 farther along the optical path from the light source 1110. Other, for example, opposite, configurations are possible.

The light is then deflected, for example, turned by the in-coupling optical element 1160, so as to be guided in the waveguide 1120 where it propagates by TIR. The light then impinges on the light distributing element 1170 turning the light in another direction (e.g., more towards the z direction) causing an increase in dimensions of an eyepiece along the direction of the z-axis as shown in FIG. 11B. The light is thus deflected toward the out-coupling optical element 1180 which causes the light to be directed out of the waveguide 1120 toward the eye 210 (e.g., the user's eye as shown). Light being coupled out by different portions of the out-coupling optical element 1180 along the z direction causes an increase in dimensions of the eyepiece along at least the direction parallel to the z-axis as defined in FIG. 11B. Notably, in this configuration, the optics 1130 are used both for illuminating the SLM 1140 and projecting an image onto the in-coupling optical element 1160. Accordingly, the optics 1130 may act as projection optics distributing light from the light source 1110 (e.g., uniformly) as well as imaging optics providing an image of the SLM 1140 and/or of an image formed by the SLM 1140 into the eye. The system 1100A in FIGS. 11A/B may in some instances be more compact than the system 1000 in FIG. 10. In some cases, not employing the PBS 1020 shown in FIG. 10 can possibly reduce cost and/or size of the system. Additionally, without the PBS 1020, the system can be more symmetric and is easier to design by shortening the back focal length of the optics 1130.

As referred to above, alternative configurations are possible. With reference to FIG. 11C, for example, in some designs, a system 1100C may be configured to pass light having a polarization not rotated by the SLM 1140. In one implementation, for example, the SLM 1140 be a liquid crystal (LC) based SLM and may include vertically aligned (VA) LC on silicon (LCoS). The SLM 1140 may have a first pixel that is in a first state that does not rotate the polarization and a second pixel that is in a second state that rotates the polarization. In the configuration illustrated in FIG. 11C, a shared analyzer/polarizer 1155 is utilized. This analyzer 1155 may transmit light of a first polarization (e.g., s-polarization) and attenuate or reduce transmission of a second polarization (e.g., p-polarization). Accordingly, light (e.g., s-polarized light) incident on a first pixel in the first

state that does not rotate the polarization orientation is reflected from the SLM 1140 and passes through the analyzer 1155 to the waveguide 1120. Conversely, light (e.g., s-polarized light) incident on the second pixel in the second state that rotates the polarization orientation is reflected from the SLM 1140 and attenuated, reduced, or not passed through the analyzer 1155 to the waveguide 1120. This configuration, may thereby permit the polarizer 1115 and the analyzer 1150 shown in FIG. 11A to be incorporated into a shared optical element, the analyzer 1155 shown in FIG. 11C, thereby possibly simplifying the system 1100 of FIGS. 11A/B by reducing the number of optical components. The analyzer 1155 may be disposed between the waveguide 1120 and the optics 1130. In other implementations, a separate analyzer/polarizer and analyzer/polarizer may be used such as shown in system 1100 of FIGS. 11A/B. FIGS. 11A and 11B illustrate the polarizer 1115 between the light source 1110 and the waveguide 1120, and the analyzer 1140 between the optics 1130 and the waveguide 1120.

FIG. 11D illustrates an example of a waveguide having a combined OPE/EPE according to an embodiment of the present invention. Referring to FIG. 11D, the waveguide 1190 with the combined OPE/EPE region 1191 includes gratings corresponding to both an OPE and an EPE that spatially overlap in the x-direction and the y-direction. In some embodiments, the gratings corresponding to both the OPE and the EPE are located on the same side of a substrate such that either the OPE gratings are superimposed onto the EPE gratings or the EPE gratings are superimposed onto the OPE gratings (or both). In other embodiments, the OPE gratings are located on the opposite side of the substrate from the EPE gratings such that the gratings spatially overlap in the x-direction and the y-direction but are separated from each other in the z-direction (i.e., in different planes). Thus, the combined OPE/EPE region 1191 can be implemented in either a single-sided configuration or in a two-sided configuration.

The light path within the eyepiece waveguide 1190 includes an incident light 1194 that is coupled into the eyepiece waveguide 1190 at the ICG 1193. The incoupled light propagates in the substrate 1192 toward the combined OPE/EPE 1191 by total internal reflection. When these rays encounter combined OPE/EPE 1191, also referred to as a combined pupil expander (CPE), light is diffracted in the +y-direction and is subsequently diffracted in the -z-direction out of the waveguide toward the user's eye along light path 1195. Similarly, the incoupled light may alternatively encounter the combined OPE/EPE 1191 and be diffracted in the -y-direction and be subsequently diffracted out of the waveguide toward the user's eye along light path 1195.

As described more fully herein, embodiments, of the present invention utilize eyepiece waveguides that have differences in the optical path length, for example, the thickness of the eyepiece waveguide as a function of lateral position, that is, the position in the x-y plane. In some embodiments, the portion of the eyepiece waveguide in which the ICG is formed is thicker than the portion of the eyepiece waveguide in which the CPE is formed. Additionally, in some embodiments, the thickness of the CPE varies, with the thickness in the portion adjacent the ICG being greater than the thickness in the portion distal with respect to the ICG. In other embodiments, the physical thickness is uniform, but the index of refraction varies as a function of lateral position, resulting in an optical path length difference characterizing the eyepiece waveguide as a function of lateral position.

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A wide variety of other configurations may be employed that utilize the optics **1130** for both illumination of the SLM **1140** and imaging of the image formed by the SLM **1140**. For example, although FIGS. **11A-11D** show a single waveguide **1120**, one or more waveguides such as a stack of waveguide (possibly different waveguides for different color light) may be used.

FIG. **12A**, for example, illustrates a cross-sectional side view of an example system **1200A** including a stack **1205** including waveguides **1120**, **1122**, **1124** that each includes an in-coupling optical element **1260**, **1262**, **1264**. The waveguides **1120**, **1122**, **1124** may each be configured to output light of one or more different wavelengths, or one or more different ranges of wavelengths. The stack **1205** may be substantially similar to the stack **260** and **660** (FIGS. **6** and **9A**) and the illustrated waveguides **1120**, **1122**, **1124** of the stack **1205** may correspond to part of the waveguides **670**, **680**, **690**, however, the stack **1205** and waveguides **1120**, **1122**, **1124** need not be so limited. As illustrated in FIG. **12A**, the in-coupling optical elements **1260**, **1262**, **1264** may be, for example, associated with, included in or on the waveguides **1120**, **1122**, **1124**, respectively. The in-coupling optical elements **1260**, **1262**, **1264** may be color selective and may primarily divert or redirect certain wavelengths into the corresponding waveguides **1120**, **1122**, **1124** to be guided therein. As illustrated, because the in-coupling optical elements **1260**, **1262**, **1264** are color selective, the in-coupling optical elements **1260**, **1262**, **1264** need not be laterally displaced and may be stacked over each other. Wavelength multiplexing may be employed to couple the particular color into the corresponding waveguide. For example, the red in-coupling optical element may in-couple red light into the waveguide designated for propagating red light while not in-coupling blue or green light, which is coupled instead into the other waveguides by the other blue and green color selective waveguides, respectively.

In some implementations, the light source **1110** may be a multi-color light source capable of emitting different colored light at different times. For instance, the light source **1110** may emit red, green, and blue (RGB) light and may be configured to, at a first time period emit red and not more than negligible amounts of green and blue, at a second time period emit green and not more than negligible amounts of red and blue, and at a third time period emit blue and not more than negligible amounts of red and green. These cycles can be repeated and the SLM **1140** can be coordinated so as to produce the suitable pattern of pixel states for the particular color (red, green, or blue) to provide the proper image color component for a given image frame. The different waveguides **1120**, **1122**, **1124** of the stack **1205** may each be configured to output light with different respective colors. For example, as depicted in FIG. **12A**, the waveguides **1120**, **1122**, **1124** may be configured to output blue, green, and red color light respectively. Of course, other colors are possible, for example, the light source **1110** may emit other colors and the color selective in-coupling optical element **1260**, **1262**, **1264**, out-coupling optical element etc., can be configured for such other colors. Additionally, individual red, green, and blue emitters may be located close enough in proximity to effectively function as a single pupil light source. The red, green, and blue emitters may be combined with lenses and dichroic splitters to form a single red, green, and blue pupil source. The multiplexing of a single pupil may be extended beyond, or in addition to, color selectivity and may include the use of polarization sensitive gratings and polarization switching. These color or polarization gratings can also be

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used in combination with multiple display pupils to increase the number of layers that can be addressed.

The different in-coupling optical elements **1260**, **1262**, **1264** in the different waveguides **1120**, **1122**, **1124** may be disposed over and/or under and aligned laterally with respect to each other (e.g., in the x and z directions shown in FIG. **12A**) as opposed to being laterally displaced with each other and not aligned. Accordingly, in some implementations, for example, the different in-coupling optical elements **1260**, **1262**, **1264** can be so configured such that light of a first color can be coupled by the in-coupling optical element **1260** into waveguide **1120** to be guided therein and light of a second color different from the first color can pass through the in-coupling optical element **1260** to the next in-coupling optical element **1262** and can be coupled by the in-coupling optical element **1262** into the waveguide **1122** to be guided therein. Light of a third color different from the first color and the second color can pass through in-coupling optical elements **1260** and **1262** to the in-coupling optical element **1264** and can be coupled into the waveguide **1124** to be guided therein. Additionally, the in-coupling optical elements **1260**, **1262**, **1264** may be polarization selective. For example, the different in-coupling optical elements **1260**, **1262**, **1264** can be so configured such that light of a certain polarization either is coupled into the waveguide by a corresponding polarization selective in-coupling optical element **1260**, **1262**, **1264** or passes through the in-coupling optical element **1260**, **1262**, **1264**.

Depending on the configuration, the SLM **1140** may include a polarization based SLM that modulates the polarization. The system **1200A** can include polarizers and/or analyzers so as to modulate the light injected into the stack **1205** on a pixel by pixel basis, for example, depending on the state of the respective pixel (e.g., whether the pixel rotates the polarization orientation or not). Various aspects of such systems that employ polarization based SLMs are discussed above and any one of such features may be employed in combination with any other features described herein. Other designs, however, are still possible.

For example, a deflection-based SLM **1140** may be employed. For example, the SLM **1140** may include one or more moveable optical elements such as moveable mirror that can reflect and/or deflect light along different directions depending on the state of the optical element. The SLM **1140** may, for example, include one or more pixels including such optical elements such as micro-mirrors or reflectors. The SLM **1140** may incorporate, for example, Digital Light Processing (DLP™) technology which uses digital micro-mirror devices (DMD). An example of a system **1200B** that uses such a deflection-based SLM **1140** is shown in FIG. **12B**. The system **1200B** includes a deflection based SLM **1140** as well as a light dump **1250**. The light dump **1250** may include an absorbing material or structure that is configured to absorb light. The deflection-based SLM **1140** may include one or more micro moveable mirrors that can be selectively tilted to deflect light in different directions. For example, the deflection based SLM **1140** may be configured to deflect light from the light source **1110** incident thereon to the in-coupling optical elements **1260**, **1262**, **1264** when a given pixel is in a bright state. As discussed above, this light will thus be coupled by one of the in-coupling optical elements **1260**, **1262**, **1264**, for example, depending on the color of light, into one of the respective waveguides **1120**, **1122**, **1124** and directed to the eye **210**. Conversely, when a given pixel is in a dark state, light from the light source **1110** may be deflected to the light dump **1250** and the light is not coupled by one of the in-coupling optical elements **1260**,

1262, 1264 into one of the respective waveguides 1120, 1122, 1124 and directed to the eye 210. The light may instead be absorbed by absorbing material comprising the light dump 1250. In some implementations, the analyzer 1150 may be a polarizer (e.g., "clean-up" polarizer) used to eliminate undesired reflections from the in-coupling optical elements 1260, 1262, 1264. This polarizer may be useful as the optics 1130 may include plastic optical elements, which have birefringence and may alter polarization. A "clean-up" polarizer may attenuate or remove light (e.g., reflections) having unwanted polarization from being directed onto the waveguides 1120, 1122, 1124. Other types of light conditioning elements may be disposed between the SLM 1140 and the waveguides 1120, 1122, 1124 such as between the optics 1130 and the waveguides 1120, 1122, 1124. For example, such a light conditioning element may also include a circular polarizer (i.e., linear polarization and retarder such as a quarter waveplate). The circular polarizer may reduce the amount of reflection from the waveguides 1120, 1122, 1124 or in-coupling optical elements 1260, 1262, 1264 that are again incident on the waveguides 1120, 1122, 1124 and coupled therein. Reflected light may be circular polarized and may possess a circular polarization opposite to that of the incident light (e.g., right-handed circularly polarized light is converted to left-handed circularly polarized light, or vice versa, upon reflection). The retarder in the circular polarizer may convert the circular polarized light to linearly polarized light, such as of the orthogonal polarization of the polarizer, which is attenuated, e.g., absorbed, by the linear polarizer in the circular polarizer. The clean-up polarizer may be used with a polarization independent modulator such as a DMD. As mentioned above, the clean-up polarizer may be useful for suppressing reflections and/or improving coupling of light into the in-coupling optical elements 1260, 1262, 1264 with optimal polarization states.

FIG. 12B illustrates a side or cross-sectional view of such the system 1200B, while FIG. 12C shows a top view of the lateral arrangement of the in-coupling optical element 1264, the light dump 1250, and the light source 1110. The SLM 1140 would be configured, depending on the state of the particular pixel, to reflect, deflect, and/or direct the light from the light source 1110 to either the lateral location of the in-coupling optical element 1264 (as well as the other in-coupling optical elements 1260, 1262) or the light dump 1250.

In certain designs, the light dump 1250 may include an energy harvesting system. The light dump 1250 may, for example, include an optical energy conversion element that is configured to convert optical energy into electrical energy. The optical energy conversion element may include, for example, a solar cell. The optical energy conversion element may include, for example, a photovoltaic detector that produces electrical output when light is incident thereon. The optical energy conversion element may be electrically connected to electrical components, for example, conductive electrical lines to direct the electrical output so as to provide the power to the system 1200B and/or possibly charge one or more batteries.

Laterally displaced, non-color selective or broadband or multi-colored in-coupling optical elements may be used in certain designs. FIG. 13A, for example, is a perspective view of a system 1300 including a stack 1305 including waveguides. The stack 1305 may be substantially similar to the stack 1205 with reference to FIG. 12A. Each waveguide in the stack 1305 may include in-coupling optical elements 1360, 1362, 1364, however, in contrast to the design shown in FIG. 12A, the in-coupling optical elements 1360, 1362,

1364 are displaced laterally with respect to each other. As illustrated in FIGS. 13A, 13B, and 13C, light sources 1110, 1112, 1114, are also laterally displaced with respect to each other and may be disposed to direct light to respective in-coupling optical elements 1360, 1362, 1364 by passing light through optics 1130, reflecting light off the SLM 1140 and passing the reflected light again through the optics 1130. The system 1300 of FIG. 13B is depicted such that light source 1114 is located behind light source 1110 and therefore is not illustrated in FIG. 13B. The light sources 1110, 1112, 1114 may correspond to in-coupling optical elements 1360, 1362, 1364 respectively. In one design, for example, the light sources 1110, 1112, 1114 and corresponding in-coupling optical element 1360, 1362, 1364 are disposed roughly equidistant from (symmetrically about) a center of the optics 1130 along a common (optical) axis. The common (optical) axis may intersect the center of the optics 1130. In one design, for example, the light sources 1110, 1112, 1114 and corresponding in-coupling optical element 1360, 1362, 1364 are not disposed equidistant from (symmetrically about) the center of the optics 1130 along the common (optical) axis.

The in-coupling optical elements 1360, 1362, 1364 may be configured to couple light of multiple colors into their respective waveguides. Accordingly, these in-coupling optical elements 1360, 1362, 1364 may be referred to herein as broadband, multi-color, or non-color selective in-coupling optical elements 1360, 1362, 1364. For example, in some cases each one of these in-coupling optical elements 1360, 1362, 1364 is configured to in-couple red, green, and blue color light into the associated waveguide in which the in-coupling optical element 1360, 1362, 1364 is included and such that such colored light is guided within the waveguide by TIR. Such a broadband in-coupling optical element 1360, 1362, 1364 may, for example, operate across a wide range of wavelengths in, for example, the visible range or select wavelengths or wavelength regions spread across, for example, the visible range. Accordingly, such broadband or multi-color or non-color selective in-coupling optical elements 1360, 1362, 1364 may be configured to turn a variety of different colors (e.g., red, green, and blue) of light into a waveguide to be guided therein by TIR. Although red, green, blue colors (RGB) are referred to herein such as in connection with the light source, in-coupling optical elements, waveguides, etc., other colors or colors system could additionally or alternatively be used, such as for example but not limited to magenta, cyan, yellow (CMY).

As illustrated in FIG. 13A the light sources 1110, 1112, 1114 are shown above the uppermost waveguide and displaced with respect to each other (e.g., in the x and z direction). Similarly, three in-coupling optical elements 1360, 1362, 1364 are shown on three respective waveguides and are displaced with respect to each other (e.g., in the x, y, and z directions). FIG. 13B is a side view of the system 1300 illustrated in FIG. 13A showing the in-coupling optical elements 1360, 1362, 1364 laterally spatially displaced with respect to each other (e.g., in the x and z direction) as well as some of the light sources 1110, 1112, 1114 laterally displaced with respect to each other (e.g., in the x and z direction). FIG. 13B also shows the optics 1130 and the SLM 1140.

FIG. 13C is a top view of the augmented reality display system illustrated in FIGS. 13A and 13B showing the in-coupling optical elements 1360, 1362, 1364 and the associated light sources 1110, 1112, 1114. In this design, the in-coupling optical elements 1360, 1362, 1364 and the associated light sources 1110, 1112, 1114 are disposed in a ring-like pattern about a center point of a common (optical)

axis. As illustrated, the light sources **1110**, **1112**, **1114** and corresponding in-coupling optical elements **1360**, **1362**, **1364** are disposed roughly equidistant about the center point of the common (optical) axis, however, this need to be the case. In some designs, this center point may correspond to the center of the optics **1130** along a common (optical) axis that intersects the center of the optics **1130** and/or a location along an optical axis of the optics **1130**). Also as a result, the non-color selective in-coupling optical elements **1360**, **1362**, **1364** as well as the light sources **1110**, **1112**, **1114** are laterally displaced with respect to each other (e.g., in the x and z directions).

Other arrangements of lateral placements are possible. FIGS. **14A-14C** illustrates an alternative configuration of a system **1400** including a stack **1405** including waveguides where the in-coupling optical elements **1360**, **1362**, **1364** as well as the light sources **1110**, **1112**, **1114** are laterally displaced with respect to each other. FIG. **14A** is a side view while FIG. **14B** is a top view of the system **1400** illustrated in FIG. **14A** showing the laterally displaced in-coupling optical elements **1360**, **1362**, **1364** and light sources **1110**, **1112**, **1114**. FIG. **14C** is an orthogonal-side view of the system **1400** illustrated in FIGS. **14A** and **14B**.

The side views of FIGS. **14A** and **14C** show how the in-coupling optical elements **1360**, **1362**, **1364** are disposed on separate waveguides within the stack **1405** such that light can be coupled by the respective laterally displaced in-coupling optical element **1360**, **1362**, **1364** into the corresponding waveguide. The in-coupling optical elements **1360**, **1362**, **1364** are shown disposed in an upper major surface of the waveguides in FIGS. **14A** and **14C**. However, the in-coupling optical elements **1360**, **1362**, **1364** can alternatively be disposed on the lower major surface of the respective waveguides or in the bulk of the waveguides. A wide variety of configurations are possible.

As shown in the top view of FIG. **14B**, the in-coupling optical elements **1360**, **1362**, **1364** are disposed in a column, laterally displaced along with respect to each other along the z direction but not along the x direction. Similarly, the light sources **1110**, **1112**, **1114** are disposed in a column, also laterally displaced with respect to each other along the z direction but not along the x direction. The in-coupling optical elements **1360**, **1362**, **1364** are laterally displaced with respect to the light sources **1110**, **1112**, **1114** in the x direction.

Still other configurations are possible. FIG. **15** is a top view of a system **1500** showing an alternative configuration of the light sources **1110**, **1112**, **1114** and the in-coupling optical elements **1360**, **1362**, **1364**. In contrast to having all light sources **1110**, **1112**, **1114** generally on one side (for example of a ring like pattern) and all in-coupling optical elements **1360**, **1362**, **1364** generally on one side (i.e., an opposite side) as in FIG. **13C**, the light sources **1110**, **1112**, **1114** and in-coupling optical elements **1360**, **1362**, **1364** are interspersed or alternate along the circumference of the ring like pattern.

In some implementations, however, the in-coupling optical elements **1360**, **1362**, **1364** and the associated one or more light sources **1110**, **1112**, **1114** are also disposed in a ring-like pattern about a center point. As a result, the light source **1110**, **1112**, **1114** and corresponding in-coupling optical element **1360**, **1362**, **1364** may be disposed roughly about equidistant from a center. In some designs, this center may correspond to the center of the optics **1130** along a common central axis that intersects the center of the optics **1130** and/or a location along an optical axis of the optics). Accordingly, the light from the first light source **1110** may

be coupled via the optics **1130** into the in-coupling optical element **1360** across the center or central axis or optical axis of the optics **1130** (as seen from the top view of FIG. **15**). Similarly, the light from the second light source **1112** may be coupled via the optics **1130** into the in-coupling optical element **1362** across the center or central axis or optical axis of the optics **1130**. Likewise, the light from the third light source **1114** may be coupled via the optics **1130** into the in-coupling optical element **1364** across the center or central axis or optical axis of the optics **1130**. Also as a result, the non-color selective in-coupling optical elements **1360**, **1362**, **1364** as well as the light sources **1110**, **1112**, **1114** are laterally displaced with respect to each other (e.g., in the x and z directions). The optics **1130** may be designed such that the focus is more into the stack **1405** so that locations of sub-pupils and the in-coupling optical elements **1360**, **1362**, **1364** are closer in they-direction. In this configuration, the in-coupling optical elements **1360**, **1362**, **1364** may be smaller since they are closer to the focus of the optics **1130**. The light source **1110** may be on a user side of the stack **1405** (e.g., similar to FIGS. **17** and **18**) and thus decrease a distance or optical path between the light source **1110** and the optics **1130**.

In various implementations above such as shown in FIGS. **12A-15**, a stack (e.g., stack **1205**, **1305**, **1405**) including multiple waveguides (e.g., stack **1205** including waveguides **1120**, **1122**, **1124**, stack **1305** including waveguides (not labeled), and stack **1405** including waveguides (not labeled)) may be included to handle different colors, (e.g., red, green, and blue). Different waveguides may be for different colors. Similarly, multiple stacks can be included to provide different optical properties to the light out-coupled from the respective stack. For example, the waveguides **1120**, **1122**, **1124** of the stack **1205** of FIGS. **12A-12B** may be configured to output light having an optical property (e.g., optical power to provide a particular wavefront shape) possibly associated with the apparent depth from which the light appears to be emanating. For example, wavefronts having different amounts of divergence, convergence, or collimation may appear as if projected from different distances from the eye **210**. Accordingly, multiple stacks may be included with different stacks configured such that light out-coupled by out-coupling optical elements have different amounts convergence, divergence, or collimation and thus appear to originate from different depths. In some designs, the different stacks may include different lenses such as diffractive lenses or other diffractive optical elements to provide different amounts of optical power to the different stacks. Consequently, different stacks will produce different amounts of, convergence, divergence, or collimation and thus light from the different stacks will appear as if associated with different depth planes or objects at different distances from the eye **210**.

FIG. **16A** is a side view of a system **1600** including stacks **1605**, **1610**, **1620**. As illustrated in FIG. **16A**, the system **1600** includes three stacks **1605**, **1610**, **1620**, however, this need not be the case. A system may be devised with fewer or more stacks. Each of the stacks **1605**, **1610**, and **1620** includes one or more (e.g., three) waveguides. FIG. **16A** also shows groups **1630**, **1640**, **1650** of in-coupling optical elements. A first group **1630** is associated with a first stack **1605**, a second group **1640** is associated with a second stack **1610**, and a third group **1650** is associated with a third stack **1620**. The groups **1630**, **1640**, **1650** are laterally displaced with respect to each other. The groups **1630**, **1640**, **1650** each include color-selective in-coupling optical elements configured to in-couple different respective colors substan-

tially similar to in-coupling optical elements **1260**, **1262**, **1264** of FIG. **12A**. As illustrated in FIG. **16A**, the in-coupling optical elements within each of the groups **1630**, **1640**, **1650** are not laterally displaced with respect to each other, however, this need not be the case. A system may be devised in which in-coupling optical elements in a group are laterally displaced with respect to each other. The system **1600** may be configured such that light out-coupled from each of the stacks **1605**, **1610**, **1620** have different amounts of optical power. For example, waveguides in a stack may have out-coupling optical elements or diffractive lenses having a given optical power. The optical power for the different stacks **1605**, **1610**, **1615** may be different such that light from one stack may appear to be originating at a depth different from light from another stack. The optical power of one stack, for example, may cause the light from that stack to be collimated whereas the optical power of another stack may cause the light therefrom to be diverging. The diverging light may appear to originate from an object that is close distance from the eye **210** while the collimated light may appear to originate from an object that is at a far distance. Accordingly, light out-coupled from the first stack **1605**, the second stack **1610**, and the third stack **1620** may have different amounts of at least one of convergence, divergence, and collimation and thus appear to originate from different depths. In some implementations, the light out-coupled from one of the stacks may be collimated, while light out-coupled by a different stack may diverge. The light out-coupled from one of the other stacks might also diverge, but diverge a different amount.

As illustrated in FIG. **16A**, the light source **1110** may be disposed with respect to the optics **1130** and the SLM **1140** to direct light into the group **1630** of in-coupling optical elements, the light source **1112** may be disposed with respect to the optics **1130** and the SLM **1140** to direct light into the group **1640** of in-coupling optical elements, and the light source **1114** may be disposed with respect to the optics **1130** and the SLM **1140** to direct light into the group **1650** of in-coupling optical elements. The light sources **1110**, **1112**, **1114** may be configured to emit different color light at different times. Likewise, light of different respective colors may be coupled into different waveguides within a stack as a result of the color selective in-coupling optical elements in a manner as described above. For example, if blue light is emitted from the second light source **1112**, the optics **1130** and SLM **1140** will direct the blue light to the second group **1640** of in-coupling optical elements. The light may pass through a first red color in-coupling optical element and a second green color in-coupling optical element in the second group **1640** and be turned by a third blue color in-coupling optical element in the second group **1640** into a third waveguide in the second stack **1610**. The waveguides in the second stack **1610** may include an out-coupling optical element or other optical element that has optical power (e.g., diffractive lens) so as to provide a beam to the eye **210** associated with a particular depth plane or object distance associated with the second stack **1610**.

FIG. **16B** is a top view of the system **1600** in FIG. **16A**. The different groups **1630**, **1640**, **1650** of in-coupling optical elements are shown laterally displaced with respect to each other (e.g., in the x direction). Similarly the light sources **1110**, **1112**, **1114** are shown laterally displaced with respect to each other (e.g., in the x direction).

A wide variety of different variations in the aforementioned systems are possible. For example, the location of the light source **1110** with respect to the waveguide(s) and optics **1130** may be different. FIG. **17**, for example, is a side view

of a system **1700** that has a light source **1110** at a different location with respect to a waveguide **1720** and optics **1130** than shown in FIGS. **11-16B**. Additionally, FIG. **17** shows a design with the waveguide **1720** divided into a first portion **1720a** and a second portion **1720b**. The waveguide **1720** may further include a reflector **1730** configured to couple light that is guided in the first portion **1720a** proximal to the light source **1110** out of the first portion **1720a** and into optics **1130** toward the SLM **1140**. Additionally or in the alternative, the system **1700** may include a diffractive out-coupling optical element to out-couple light in the first portion **1720a** of the waveguide **1720** and into optics **1130** toward the SLM **1140**. This reflector **1730** may be opaque and include an isolator that reduces cross-talk between the first portion **1720a** and the second portion **1720b**. The waveguide **1720** has a first side **1721** and a second side **1723** opposite the first side **1721**, the optics **1130** and the SLM **1140** are disposed on the first side **1721** such that light from the SLM **1140** is directed onto the first side **1721**. In this example, the light source **1110** is disposed on the first side **1721** of the waveguide **1720** such that light from the light source **1110** is incident on the first side **1721** prior to passing through the optics **1130** to the SLM **1140**. The system **1700** may further include in-coupling optical element **1710** disposed on or in the first portion **1720a**. The in-coupling optical element **1710** may be configured to receive light from the light source **1110** and to couple the light into the first portion **1720a**. The in-coupling optical element **1710** may include a diffractive optical element or reflector configured to turn light incident thereon into the first portion **1720a** at an angle to be guided therein by TIR.

The reflector **1730** may be configured to direct light guided in the first portion **1720a** out of the first portion **1720a** and toward the optics **1130** and the SLM **1140**. (As discussed above, in some implementations, a diffractive optical element may in addition or in the alternative be used to direct the light in the first portion **1720a** out of the first portion **1720a** and toward the optics **1130** and the SLM **1140**.) Accordingly, the reflector **1730** may be a mirror, reflective grating, one or more coatings that reflect light of the waveguide **1720** toward the SLM **1140**. The light ejected from the first portion **1720a** by the reflector **1730** passes through the optics **1130**, is incident on the SLM **1140**, and passes through the optics **1130** once again and is incident onto the second portion **1720b**. As described above, light reflected from the SLM **1140** transmitted through the optics **1130** may be incident on an in-coupling optical element **1160** and turn light to be guided in the second portion **1720b**. Light guided in the second portion **1720b** may be outcoupled therefrom by an out-coupling optical element **1180** (not shown) and directed to the eye **210**.

As discussed above, the reflector **1730** may be an isolator that reduces cross-talk between the first portion **1720a** and the second portion **1720b**. The reflector **1730** may include an opaque and/or reflective surface. The reflector **1730** may be disposed within the waveguide **1720** and, in some cases, may define a side of the first portion **1720a** and second portion **1720b**.

Instead of having the first and second portions **1720a**, **1720b** of the waveguide **1720**, separate waveguides may be used. FIG. **18** is a side view of a system **1800** that includes a first waveguide **1822** for receiving light from a light source **1110** and directing light guided therein to the optics **1130** and toward the SLM **1140**. The system **1800** additionally includes a second waveguide **1820** that receives light from the SLM **1140** after the light has again passed through the optics **1130**. The first waveguide **1822** includes in-coupling

and out-coupling optical elements **1730a**, **1730b**, respectively. These in-coupling and out-coupling optical elements **1730a**, **1730b** may include reflective surfaces oriented to in-couple and out-couple light in and out of the waveguide **1822**. The in-coupling optical element **1730a** may, for example, include a reflective surface disposed to receive light from the light source **1110** and oriented (e.g., tilted) to direct the light into the waveguide **1822** at an angle so as to be guided therein by TIR. The out-coupling optical element **1730b** may, for example, include a reflective surface oriented (e.g., tilted) to direct light guided within the waveguide **1822** at an angle so as to be ejected from the waveguide **1822**. The out-coupling optical element **1730b** may be located so light turned out of the waveguide **1822** is directed into the optics **1130**, reflected from the SLM **1140**, passes again through the optics **1130** and is incident on an in-coupling optical element **1730c** of a second waveguide **1820**.

The in-coupling optical element **1730c** in the second waveguide **1820** may include a reflective surface that may be located and oriented (e.g., tilted) so as to receive and turn light incident thereon from the SLM **1140** to be guided in the second waveguide **1820** by TIR. FIG. **18** illustrates the optics **1130** and the light source **1110** disposed on a same side of the waveguides **1820**, **1822**. The system **1800** may further include an isolator to reduce cross-talk between the waveguide **1822** and the waveguide **1820**. The isolator may include an opaque and/or reflective surface. The isolator may be disposed in or on at least one of the waveguides **1820**, **1822**.

A variety of the designs, such as the designs discussed above, can include additional features or components. FIG. **19**, for example, shows a side view of a system **1900** that includes variable focus optical elements (or adaptive optical elements) **1910**, **1920**. The variable focus optical elements **1910**, **1920** may include optical elements that are configured to be altered to provide variable optical power. The variable focus optical elements **1910**, **1920** may include multiple states such as a first state and a second state, wherein in the first state the variable focus optical elements **1910**, **1920** have different optical power than when in the second state. For instance, the variable focus optical elements **1910**, **1920** may have negative optical power in the first state and zero optical power in the second state. In some implementations, the variable focus optical elements **1910**, **1920** have positive optical power in the first state and zero optical power in the second state. In some implementations, the variable focus optical elements **1910**, **1920** have a first negative or positive optical power in the first state and a second different negative or positive optical power in the second state. Some adaptive optical elements or variable focus optical elements **1910**, **1920** may have more than two states and may possibly provide a continuous distribution of optical powers.

The variable focus optical elements **1910**, **1920** may include a lens (e.g., a variable lens) and be transmissive. Transmissive or transparent adaptive optical elements or variable focus optical elements **1910**, **1920** are shown in FIG. **7**. The variable focus optical elements **1910**, **1920** may include liquid lenses (e.g., movable membrane and/or electro-wetting). The variable focus lens may also include liquid crystal lenses such as switchable liquid crystal lenses such as switchable liquid crystal polarization lenses, which may for example comprise diffractive lenses. Alvarez lens may also be used. Other types of variable focus optical elements **1910**, **1920** may possibly be employed. Examples of variable focus optical elements can be found in U.S. Application No. 62/518,539, filed on Jun. 12, 2017, entitled AUGMENTED

REALITY DISPLAY HAVING MULTI-ELEMENT ADAPTIVE LENS FOR CHANGING DEPTH PLANES, which is hereby incorporated by reference in its entirety. The variable focus optical elements **1910**, **1920** may have electrical inputs that receive electrical signals that control the amount of optical power exhibited by the variable focus optical elements **1910**, **1920**. The variable focus optical elements **1910**, **1920** may have positive and/or negative optical power. In addition to variable focus elements (e.g., polarization switches, geometric phase (GP) lenses, fluid lenses, and the like), the variable focus elements **1910**, **1920** may include fixed lenses (e.g., diffractive lenses, refractive lenses, and the like) to generate depth planes desired in a light field.

A first variable focus optical element **1910** may be disposed between a stack **1905** and the eye **210**. The stack **1905** may include different waveguides for different colors as discussed above. The first variable optical element **1910** may be configured to introduce different amounts of optical power, negative and/or positive optical power. The variable optical power may be used to vary the divergence and/or collimation of light coupled out from the stack **1905** to vary the depth at which virtual objects projected into the eye **210** by the system **1900** appear to be located. Accordingly, a 4 dimensional (4D) light field may be created.

A second variable focus optical element **1920** is on the opposite side of the stack **1905** as the first variable focus optical element **1920**. The second variable focus optical element **1920** can thus compensate for the effect of the first optical element **1910** on light received from the world **510** in front of the system **1900** and the eye **210**. Thus, a world view maybe effectively unaltered or altered as desired.

The system **1900** can further include a static or variable prescription or corrective lens **1930**. Such a lens **1930** may provide for refractive correction of the eye **210**. Additionally, if the prescription lens **1930** is a variable lens it may provide different refractive corrections for multiple users. Variable focus lenses are discussed above. The eye **210** may for example have myopia, hyperopia, and/or astigmatism. The lens **1930** may have a prescription (e.g., optical power) to reduce the refractive error of eye **210**. The lens **1930** may be spherical and/or cylindrical and may be positive or negative. The lens **1930** may be disposed between the stack **1905** and the eye **210** such that light from both the world **510** and from the stack **1905** undergoes the correction provided by the lens **1930**. In some implementations, the lens **1930** may be disposed between the eye **210** and the first variable focus optical element **1910**. Other locations for the lens **1930** are possible. In some embodiments, prescriptive lenses may be variable and allow multiple user prescriptions to be implemented.

In some designs, the system **1900** may include an adjustable dimmer **1940**. In some implementations, this adjustable dimmer **1940** may be disposed on a side of the stack of waveguides **1900** opposite to the eye **210** (e.g., world side). Accordingly, this adjustable dimmer **1940** may be disposed between the stack of waveguides **1900** and the world **510**. The adjustable dimmer **1940** may include an optical element that provides variable attenuation of light transmitted there through. The adjustable dimmer **1940** may include electrical inputs to control the level of attenuation. In some cases the adjustable dimmer **1940** is configured to increase attenuation when the eye **210** is exposed to bright light, such as when the user goes outdoors. Accordingly, the system **1900** may include a light sensor to sense the brightness of the ambient light and control electronics to drive the adjustable dimmer **1940** to vary the attenuation based on the light levels sensed by the light sensor.

Different types of adjustable dimmers **1940** may be employed. Such adjustable dimmers **1940** may include variable liquid crystal switches with a polarizer, electrochromic material, photochromic material, and the like. The adjustable dimmer **1940** may be configured to regulate the amount of light entering and/or transmitted through the stack **1905** from the world **510**. The adjustable dimmer **1940** can be used in some cases to reduce the amount of light from the ambient that passes through the waveguide stack **1900** to the eye **210** that may otherwise provide glare and decrease the user's ability to perceive virtual objects/images injected into the eye **210** from the stack **1905**. Such an adjustable dimmer **1940** may reduce the incident bright ambient light from washing out the images that are projected into the eye **210**. The contrast of the virtual object/image presented to the eye **210** may therefore be increased with the adjustable dimmer **1940**. In contrast, if ambient light is low, the adjustable dimmer **1940** may be adjusted to reduce attenuation so that the eye **210** can more readily see objects in the world **510** in front of the user. The dimming or attenuation may be across the system or localized to one or more portion of the system. For example, multiple localized portions may be dimmed or set to attenuate light from the world **510** in front of the user **210**. These localized portions may be separated from each other by portions without such increased dimming or attenuation. In some cases, only one portion is dimmed or caused to provide increased attenuation with respect to other portions of the eyepiece. Other components may be added in different designs. Also the arrangement of the components can be different. Similarly, one or more components may be excluded from the system.

An example of another configuration is shown in FIG. **20A**. FIG. **20A** shows a side view of a system **2000** including laterally displaced in-coupling optical elements **1360**, **1362**, **1364** on different waveguides as well as a color filter array **2030** including laterally displaced color filters **2040**, **2042**, **2044** aligned with respective in-coupling optical elements **1360**, **1362**, **1364**. The color filter array **2030** may be disposed on the side of a stack **2005** proximate the eye **210** and optics **1130**. The color filter array **2030** may be between the stack **2005** and the optics **1130**. The color filter array **2030** may be disposed in or on a coverglass **2050** that is located between the stack **2005** and the optics **1130**. The color filter array **2030** may include one or more different color filters **2040**, **2042**, **2044** such as a red color filter, a green color filter, and a blue color filter, laterally disposed with respect to each other. The system **2000** includes light sources **1110**, **1112**, **1114** laterally displaced with respect to each other. These light sources **1110**, **1112**, **1114** may include different color light sources such as red, green, and blue light sources. The color filters **2040**, **2042**, **2044** may be transmissive or transparent filters. In some implementations, the color filters **2040**, **2042**, **2044** include absorption filters, however, the color filters **2040**, **2042**, **2044** may also include reflective filters. The color filters **2040**, **2042**, **2044** in the color filter array **2030** may be separated and/or surrounded by a mask such as an opaque mask that would reduce propagation of stray light. The filters in the color filter array **2030** may be used to reduce or eliminate undesired reflections within the system such as from the waveguides and/or in-coupling optical elements **1360**, **1362**, **1364** from reentering the waveguides used for different colors through in-coupling optical elements **1360**, **1362**, **1364** for different colors. Examples of color filter arrays can be found in U.S. application Ser. No. 15/683,412, filed on Aug. 22, 2017, entitled "PROJECTOR ARCHITECTURE INCORPORATING ARTIFACT MITIGATION, which is hereby incorpo-

rated by reference in its entirety; and U.S. Application No. 62/592,607 filed on Nov. 30, 2017, entitled PROJECTOR ARCHITECTURE INCORPORATING ARTIFACT MITIGATION, which is hereby incorporated by reference in its entirety. The mask may be a black mask and may include absorbing material to reduce propagation and reflection of stray light. The light sources **1110**, **1112**, **1114** may be disposed with respect to the optics **1130** and SLM **1140** to couple light in to corresponding color filters **2040**, **2042**, **2044** in the color filter array **2030**. For example, the color filter array **2030** may include first, second, and third, (e.g., red, green, and blue) color filters **2040**, **2042**, **2044** that are disposed to receive light from the first, second, and third, light sources **1110**, **1112**, **1114**, respectively. The first, second, and third, (e.g., red, green, and blue) color filters **2040**, **2042**, **2044** may be aligned (e.g., in the x and z direction) with the respective in-coupling optical elements **1360**, **1362**, **1364**. Accordingly, light from the first light source **1110** will be directed through the first color filter **2040** and to a first in-coupling optical element **1360**, light from the second light source **1112** will be directed through the second color filter **2042** and to a second in-coupling optical element **1362**, and light from the third light source **1114** will be directed through the third color filter **2044** and to a third in-coupling optical element **1364**. In some implementations, the in-coupling optical elements **1360**, **1362**, **1364** may be color specific. For example, the first and second in-coupling optical elements **1360**, **1362** may be configured to couple light of respective first and second colors into the first and second waveguides, respectively. Similarly, the first, second, and third in-coupling optical elements **1360**, **1362**, **1364** may be configured to couple light of respective first, second, and third colors into the first, second, and third waveguides, respectively. The first in-coupling optical element **1360** may be configured to couple more light of the first color than the second color (or the third color) into the first waveguide. The second in-coupling optical element **1362** may be configured to couple more light of the second color than the first color (or the third color) into the second waveguide. The third in-coupling optical element **1364** may be configured to couple more light of the third color than the first color or the second color into the second waveguide. In other configurations, the in-coupling optical elements **1360**, **1362**, **1364** may be broad band. For example, the first in-coupling optical element **1360** may be configured to couple light of first, second, and third colors into the first waveguide. The second in-coupling optical element **1362** may be configured to couple light of first, second, and third colors into the second waveguide. The third in-coupling optical element **1364** may be configured to couple light of first, second, and third colors into the third waveguide. The plurality of color filters **2040**, **2042**, **2044**, may, however, be color specific, selectively transmitting light of a particular color. For example, the first color filter **2040** may transmit more of the first color than the second color (and third color). The second color filter **2042**, may transmit more of the second color than the first color (and third color). The third color filter **2044**, may transmit more of the third color than the first color and second color. Likewise, the first, second, and third color filters **2040**, **2042**, **2044** may be color filters that selectively transmit the first, second, and third color, respectively. Accordingly, the first, second, and third color filters **2040**, **2042**, **2044** may be band pass filters that selectively pass the first, second, and third colors, respectively. In some implementations, the first, second, and third light sources **1110**, **1112**, **1114**, may selectively emit the first, second, and third colors, respectively. For example, the first light source **1110**,

may emit more of the first color than the second color (and third color). The second light source **2042**, may emit more of the second color than the first color (and third color). The third light source **2044**, may transmit more of the third color than the first color and second color. The color filters **2040**, **2042**, **2044**, may reduce the amount of stray light that is inadvertently directed to a particular in-coupling optical element. In other implementations, the one or more of the light sources **1110**, **1112**, **1114** are broad band light sources. For example, the first light source **1110** may emit the first and second (and possibly third) colors. The second light source **1112** might also emit the first and second, (and possibly third) colors. The third light source **1114** might also emit the first and second (and possibly third) colors. Although three filters are shown in FIGS. **20A-20G**, more or less filters may be included. For example, in some implementations, two filters (not three) may be used. Accordingly, two colors corresponding to the two color filters may be selectively transmitted into by the filters. In some such implementations, two corresponding in-coupling optical elements may be used and be aligned with the two filters. In some implementations, the two in-coupling optical elements selectively couple the two colors, respectively, into the two respective waveguides. In some implementations, two light sources may be used instead of three. Other variations and other numbers of components may be used. Also, the color filters **2040**, **2042**, **2044** may or may not be integrated together in a single array.

As discussed above, the components and their location and arrangement may vary. For example, although FIG. **20A** shows an analyzer **1150** disposed between the optics **1130** and the stack **1905**, the analyzer **1150** may be located at a different position. FIG. **20B** shows an analyzer **1150** located between the optics **1130** and the SLM **1140**. In some designs, the analyzer (e.g., polarizer) **1150** may attach directly to the SLM **1140**. For instance, the analyzer **1150** may be adhered to or mechanically coupled to the SLM **1140**. For example, the analyzer **1150** may be glued, cemented to the SLM **1140** (e.g., to the SLM window) using adhesive. Accordingly, although FIG. **20B** shows a gap between the analyzer **1150** and the SLM **1140**, in some designs no gap between the analyzer **1150** and SLM **1140** is present. The analyzer **1150** may be affixed to the SLM **1140** mechanically (e.g., using a mechanical fixture), and in such cases may or may not include a gap between the analyzer **1150** and SLM **1140**. Birefringence from the optics **1130** may be cleaned up by positioning a polarizer directly on the SLM **1140** as described above. In some implementations, an analyzer **1150** disposed between the optics **1130** and the in-coupling optical elements **1360**, **1362**, **1364** may also be included to clean up the polarization of light outbound from the optics **1130** (e.g., as illustrated in dashed lines in FIG. **20B**). In addition, a retarder (not shown) such as a quarter waveplate may be included proximal the SLM **1140**, for example, between the optics **1130** and the SLM **1140**. As used herein a quarter waveplate may refer to a quarter wave retarder regardless of if the quarter wave retarder comprises a plate, film, or other structure for providing a quarter wave of retardance. In FIG. **20B**, for example, the retarder (e.g., quarter waveplate) may be disposed between the analyzer **1150** and the SLM **1140**. The retarder (e.g., quarter waveplate) may be used for skew ray management. For example, the retarder (e.g., quarter waveplate) may, for example, compensate for variations caused by differences in wavelength and angle of incidents on the SLM **1140**. As discussed above, a compensator may be included and may provide a more consistent polarization rotation (e.g., of) 90° of the SLM **1140** for different angles

of incidence and different wavelengths. The compensator may be used to increase contrast of the display by providing more consistent orthogonal rotation. The compensator may be attached or affixed to the SLM **1140** such as described above. For example, glue, cement or other adhesive may be used. The compensator may also be attached to the SLM **1140** using a mechanical fixture. A gap or no gap may be included between the compensator or SLM **1140**. Other light conditioning optics may also be included in addition or in the alternative and may be affixed to the SLM **1140** such as described above with respect to the analyzer **1150** and/or compensator.

In some embodiments, large angle spreads (e.g., -70° degrees) may be used. The angle spread may refer to an angle of light entering into the optics **1130**, for example, from the light sources **1110**, **1112**, **1114**, and/or an angle of light exiting the optics **1130** into the in-coupling optical elements **1360**, **1362**, **1364**. In these embodiments, a thinner SLM **1140** may be used. For example, if the SLM **1140** is a liquid crystal (LC) SLM (e.g., a liquid crystal on silicon (LCoS) SLM), the LC layer may be made thinner to accommodate the large angle spread.

A double pass retardance through a polarizer and the analyzer **1150** may need to be a half wave. The polarizer may be between the optics **1130** and the analyzer **1150**. The double pass retardance may be a function of a ratio of a refractive index of the LCoS SLM **1140** and a thickness of the LCoS SLM **1140**. For a given refractive index of the LCoS SLM **1140** and a given thickness of the LCoS SLM **1140**, going in and out of the LCoS SLM **1140** at large angles makes a path length of light longer than going in and out of the LCoS SLM **1140** at small angles. The path length is related to the thickness of the LCoS SLM **1140**. In one example, a LCoS SLM may have a first refractive index and a first thickness. For small angles, a double pass retardance of the LCoS SLM having the first refractive index and the first thickness may be a half wave. For large angles, a double pass retardance of the LCoS SLM having the first refractive index and the first thickness may not be a half wave (e.g., may be greater than a half wave). The thickness of the LCoS SLM may be changed from the first thickness to a second thickness, where the second thickness is less than the first thickness. For small angles, a double pass retardance of the LCoS SLM having the first refractive index and the second thickness may not be a half wave (e.g., may be less than a half wave). For large angles, a double pass retardance of the LCoS SLM having the first refractive index and the second thickness may be a half wave.

Also, although FIGS. **20A** and **20B** illustrate the use of a polarization-based SLM **1140**, other types of SLMs may be utilized. FIG. **20C**, for example, illustrates use of a deflection-based SLM **1140** such as a movable micro-mirror based SLM. As discussed above, such SLM **1140** may include Digital Light Processing (DLP™) and digital micromirror device (DMD) technology. As discussed above, the deflection-based SLM **1140** can couple light from one of the light source **1110**, **1112**, **1114** into the respective in-coupling optical element **1360**, **1362**, **1364**, depending on the state of the pixel of the SLM **1140**. In one state, the light from the light source **1110**, **1112**, **1114** would be directed to the respective in-coupling optical element **1360**, **1362**, **1364** as illustrated in FIG. **20D**. In another state, the light from the light source **1110**, **1112**, **1114** would be directed away from the in-coupling optical element **1360**, **1362**, **1364** as illustrated in FIG. **20E**. In some implementations, while in the off state, the black absorbing mask between color filters **2040**, **2042**, **2044** in the color filter array **2030** may serve as a light

dump. As described above, the color filters **2040**, **2042**, **2044** may be surrounded and/or separated by a mask such as an absorbing mask (e.g., a black mask). This mask may include absorbing material such that of the light incident more is absorbed than reflected therefrom. This mask may also be

Other variations are possible. Although the light sources are shown as emitters **1110**, **1112**, **1114** (e.g., LEDs, laser diodes) coupled to coupling optic **1105** such as nonimaging optical coupling element (e.g., compound parabolic collectors (CPC) or cones), other configurations are possible. For example, the coupling optic **1105** (e.g., CPC) may be tilted with respect to a stack of waveguides. In some cases the projector (i.e., the optics **1130** and the SLM **1140**) may be tilted relative to the eyepiece (e.g., the stack of waveguides). In some implementations, the lens optics **1130** is tilted with respect to the SLM **1140** to reduced distortion such as keystone distortion. A Scheimpflug configuration may be employed to reduce such distortion. Components may be tilted (e.g., optics **1130** and/or spatial light modulator **1140**) as needed, for example, to fit more conformally about a head and/or face. As described above, the light emitter(s) and/or coupling optic **1105** may be tilted. In some configurations, the assembly including the waveguides may be tilted with a side closer to a side of the eye **210** (e.g. temporal side) being closer to the eye **210** to increase perceived field of view of a binocular system as a whole (at a cost of binocular overlap).

As discussed above, components and their location and arrangement may vary. For example, FIG. **20F** is a side view of a system **2000F** including cover glass **2050** disposed between the stack **2005** and the optics **1130**. In some designs, the light sources **1110**, **1112**, **1114** may be disposed on a world side of the cover glass **2050** and configured to propagate light through the cover glass **2050** to the optics **1130** and SLM **1140**. As illustrated, the cover glass **2050** may extend laterally (e.g., parallel to the x-axis) beyond the stack **2005** such that light emitted by the light sources **1110**, **1112**, **1114** enters the optics **1130** without passing through waveguides in the stack **2005**. Although the system **2000F** depicts a deflection-based SLM **1140**, similar configurations of the light source may also be used with a non-deflection-based SLM or in with any other configuration or features disclosed herein.

FIG. **20G** is a side view of a system **2000G** including cover glass **2060** disposed on the world side of the stack **2005** (i.e., opposite the side of the stack **2005** proximal the optics **1130**). In some designs, the light sources **1110**, **1112**, **1114** may be disposed on a world side of the cover glass **2050** and configured to propagate light through the cover glass **2050** to the optics **1130** and SLM **1140**. As illustrated, the cover glass **2060** may extend laterally (e.g., parallel to the x-axis) beyond the stack **2005** such that light emitted by the light sources **1110**, **1112**, **1114** enters the optics **1130** without passing through waveguides in the stack **2005**. Although the system **2000G** depicts a deflection-based SLM **1140**, similar configurations of the light source may also be used with a non-deflection-based SLM or in or with any other configuration or features disclosed herein.

Additionally, as discuss above, a configuration that facilitates light recycling may be employed. FIG. **21**, for example, is a partial side view of a system **2100** outfitted with a configuration that provides light recycling of light from the light source **1110**. The light source **1110** may be disposed with respect to a polarizer **1115** configured to recycle light having an undesired polarization. The polarizer **1115** may include, for example, a wire grid polarizer that transmits

light of a first polarization and retro reflects light of a second opposite polarization. Accordingly, light **2110** may be emitted from the light source **1110** and impinge on the polarizer **1115**. The polarizer **1115** may transmit light of the first polarization, for which a projector (not shown) is configured to use. For example, an SLM may properly operate with light of this first polarization. Light of the second polarization **2120** is reflected back toward the light source **1110** and can be recycled. The polarization of the light **2120** may be altered, for polarization rotated, after reflecting off portions (e.g., sidewalls) of the coupling optic (not shown) such as non-imaging optics like the compound parabolic collector (CPC) at various angles. Some light having suitable polarization (e.g., polarization orientation), that may be passed by the polarizer **1115** may result. Multiple reflections may change polarization of the light and may cause light to exit with a desired polarization. This recycled light **2130** is then emitted back toward the polarizer **1115**. Such a configuration may improve efficiency, e.g., energy efficiency as more of the desired polarization is produced. Also, in addition or in the alternative, a retarder may be used to change a reflected polarization state and reclaim light.

FIG. **22** shows another configuration that includes light sources **1110**, **1112**, **1114** and corresponding light collection optics **2210**, **2212**, **2214**. The light collection optics **2210**, **2212**, **2214** may include lenses or other optics to collect light from the light sources **1110**, **1112**, **1114**. The light sources **1110**, **1112**, **1114** may be laser diodes or other emitters that emit light over a wide range of angles. The light collection optics **2210**, **2212**, **2214** may be used to collect much of that light. The light sources **1110**, **1112**, **1114** may emit light asymmetrically. For example, light may be emitted in a wider range of angles in one direction (e.g., x or z direction) than in the orthogonal directions (e.g., z or x direction). Accordingly, the light collection optics **2210**, **2212**, **2214** may be asymmetric. For example, the light collection optics **2210**, **2212**, **2214** may have different optical power in different possible orthogonal directions. The light collection optics **2210**, **2212**, **2214** may, for example, include lenses such as anamorphic lenses. The light collection optics **2210**, **2212**, **2214** may also possibly include non-imaging optics. Apertures **2220**, **2222**, **2224** may be included. A diffuser **2230** may also be included proximal the apertures **2220**, **2222**, **2224**, for example, when the light sources **1110**, **1112**, **1114** lasers such as laser diode. With the diffuser proximal the apertures **2220**, **2222**, **2224**, the apertures may appear to be the location of the laterally displaced light sources. The apertures **2220**, **2222**, **2224** may be matched with in-coupling optical elements on a waveguide or waveguides via optics and SLM as discussed above. For example, each aperture **2220**, **2222**, **2224** may be matched with a respective in-coupling optical element. Similarly, in certain implementations, such as shown in FIG. **16A**, each aperture **2220**, **2222**, **2224** may be matched with respective groups of (e.g., color selective) in-coupling optical elements.

A wide range of system variations and configurations are possible. For example, although the linearly polarized light is described as being propagated through the optics **1130** to the SLM **1140** and back through the optics to the waveguide stack, in some designs circular polarized light may be used instead. For example, circularly polarized light may be directed into the optics **1130**. A retarder such as a quarter waveplate may be disposed such that this light passes through the retarder prior to being incident on the SLM. The retarder (e.g., quarter waveplate) may be disposed between the optics **1130** and the SLM **1140**. In some cases, such as described above, the retarder (e.g., quarter waveplate) may

be affixed to the SLM 1140, such as for example, using adhesive or a mechanical fixture. The retarder (e.g., quarter waveplate) may transform the linearly polarized light into circularly polarized light after reflection from the SLM 1140. Accordingly, in some implementations, circular polarized light may again pass through the optics 1130 toward the stack. Another retarder (e.g., quarter waveplate), for example, proximal to the analyzer 1150 may transform the circular polarized light into linearly polarized light that may or may not pass through the analyzer depending on the linear polarization (e.g., orientation). Pixels of the SLM 1140 may have states that can be varied to rotate or not rotate the polarization. Still other configurations are possible.

FIG. 23A is a side view of an augmented reality display system 2300 including a light source 2305, a polarization rotator 2307, optics having optical power (e.g., lenses) 2320, polarizers 2312, 2335 such as linear polarizers (e.g. horizontal or vertical polarizers), retarders 2315, 2330, 2340 such as quarter wave retarders (e.g., quarter waveplates), and at least one waveguide 2348 for outputting image information to a user. Such a configuration can be used to illuminate a reflective spatial light modulator (not shown) such that light emitted from light source 2305 is reflected from the spatial light modulator and is coupled into the at least one waveguide 2348 to be directed to a user's eye. The configuration and placement of these elements, particularly the polarizers and retarders, may reduce or eliminate reflections from optical surfaces within the system such as surfaces from the optics 2320, which may otherwise result in ghost images being visible to the user. For example, optical elements that are polarization selective and/or that have retardance (e.g., polarizers 2312, 2335 and retarders 2315, 2330, 2340) can be arranged and configured to convert linearly polarized light into circularly polarized light that changes from left-handed to right-handed or right-handed to left-handed upon reflection from optical surfaces. Similarly, such optical elements that are polarization selective and/or that have retardance (e.g., polarizers 2312, 2335 and retarders 2315, 2330, 2340) can be arranged and configured to convert circularly polarized light into linearly polarized light that can be attenuated or filtered out by the polarizers (e.g., linear polarizers). Circular polarizers that transform linearly polarized light into circularly polarized light and vice versa may be fabricated with such optical elements that are polarization selective and that have retardance (e.g., polarizers 2312, 2335 and retarders 2315, 2330, 2340). For example, a circular polarizer may comprise a linear polarizer and a quarter wave retarder. Circular polarizers can be used to convert linearly polarized light into circularly polarized light having a first state (e.g., handedness) and to filter out circularly polarized light having a second state (e.g., handedness) that is of a different first state. For example, circular polarizers can be used to convert linearly polarized light having a certain orientation into left-handed circular polarized light and to filter out circular polarized light that is right-handed circularly polarized. Circular polarizers can also be used to convert linearly polarized light having a certain orientation into right-handed circular polarized light and to filter out circular polarized light that is left-handed circularly polarized. Circular polarizers or other configurations of optical elements that include retardance that can be used to transform linearly polarized light into circular polarized light and back and that can selectively filter linearly polarized light can be used to reduce back reflection from optical surfaces as discussed below in connection with FIGS. 23A and 23B.

It is noted that left-hand and right-hand circular polarization is illustrated with clockwise and counter-clock-wise arrows, respectively, in FIGS. 23A and 23B. Further, horizontal and vertical linear polarization is depicted using horizontal arrows and circular dots respectively.

As discussed above, FIG. 23A illustrates a configuration of an augmented reality display system 2300 where polarizers 2312, 2335 such as linear polarizers (e.g., horizontal polarizers) and retarders 2315, 2330, 2340 such as quarter wave retarders (e.g., quarter waveplates) are arranged to reduce back reflection from optical surfaces such as the surfaces of optics 2320 in the path of light illuminating and reflecting from a spatial light modulator (not shown). The first polarizer 2312 and first retarder 2315 are disposed between the light source 2305 and the optics 2320. The first polarizer 2312 is disposed between the light source 2305 and the first retarder 2315. Likewise, the first retarder 2315 is disposed between the first polarizer 2312 and the optics 2320.

As illustrated, the light source 2305 emits light as represented by a light ray 2310. In some implementation, the ray 2310 can pass through the polarization rotator 2307. The rotator 2307 is optional and can be used to rotate the polarization of the light from the light source 2305, e.g., ray 2310. In various implementations, the rotator 2307 can rotate the angle of the polarization (e.g., of the linear polarization). For example, the rotator 2307 can rotate the linear polarization of the ray 2310 to an orientation aligned with the first polarizer 2312 so as to be transmitted there-through. In some implementations, the polarization rotation 2307 may comprise a retarder, for example, a half-wave retarder in some cases. The optic axis of the half-wave retarder may be oriented to rotate the polarization of the light from the light source 2305 from vertical to horizontal or vice versa. Alternatively the polarization rotator 2307 may be configured to rotate the angle of polarization of linearly polarized light emitted from the light source 2305 by different amounts. The polarization rotator 2307 need not be included in the system. For example, in implementations where the light source 2305 emits light having the same polarization as the first polarizer 2312, the polarization rotator 2307 may be excluded. As illustrated, the light, for example, the ray 2310, passes through a polarizer 2312, here shown as a horizontal polarizer. In instance where light from the light source 2305 is unpolarized, the light transmitted through the horizontal polarizer 2312, shown as ray 2310, is linearly polarized (e.g., horizontally polarized) after passing through the polarizer 2312. While horizontal linear polarizers are used in this example, it will be understood that the principles taught can be applied using vertical linear polarizers. Alternatively, linear polarizers having different orientations other than vertical or linear may also be used.

The horizontally polarized light ray 2310 travels through the retarder 2315, here shown as a quarter wave retarder. This retarder 2315 may include sufficient retardance to transform the linearly polarized light into circularly polarized light. For example, the horizontally polarized light may be converted into left-handed circularly polarized light as illustrated by the curved (e.g., clockwise directed) arrow. In this example, the combination of the polarizer 2312 and the retarder 2315 (e.g., quarter wave) forms a circular polarizer, referred to here as the first circular polarizer, that can convert light of a particular linear polarization (e.g., horizontal or vertical polarization) into a particular circular polarization (e.g., left- or right-handed circular polarization or vice versa). A circular polarizer may also block light of a par-

ticular circular polarization (e.g., right- or left-handed circular polarization) depending on the configuration.

In some implementations, various optical elements have birefringence. In certain such cases, the retarder **2315** may include an amount of retardance sufficient to convert linearly polarized light into circularly polarized light and need not be a quarter waveplate. More or less than a quarter wave of retardance may be included in the retarder **2315** as retardance may be contributed by other optical elements. Similarly, retardance can be distributed in a number of optical elements. As another example, multiple retarders may be employed to provide the appropriate amount of retardance.

The circularly polarized ray **2310** (here left-handed circularly polarized) then passes through the optics **2320**. Undesirable reflections may occur at any interface in the system with media having dissimilar refractive indices such as, for example, air to material interfaces. These reflections can be problematic if they are allowed to enter the at least one waveguide **2348** as this reflected light may be directed into the user's eye and form "ghost" images visible in the user's eye. For example, in an instance where the display projects a first image into the viewer's eye with the at least one waveguide **2348**, a second faint duplicate image that is displaced (e.g., laterally displaced) with respect to the first image may also be seen by the user. Such "ghost" images, formed by reflections from optical surfaces that are directed into the user's eye, may be distracting or otherwise degrade the viewing experience. For example, as illustrated in FIG. **23A**, light such as a reflected ray **2325** can be reflected from a lens within the optics **2320**. This light may be directed toward the at least one waveguide **2348**, which is configured to direct light into the user's eye for presenting images thereto. However, in this case, the circularly polarized light reverses handedness. For example, upon reflecting off of the lens, the direction of the circular polarization is changed (e.g., from left-handed to right-handed). The right-handed reflected ray **2325** then travels through the retarder **2315** and is transformed into linearly polarized light having a different (e.g., orthogonal) linear polarization than that which is transmitted by the polarizer **2312**. In this case, for example, the light reflected from the optical surface of the lens is converted by the retarder **2315** into vertical linear polarization, which is orthogonal to the polarization transmitted by the horizontal linear polarizer **2312**. The horizontal linear polarizer **2312** selectively passes horizontally polarized light and filters out vertically polarized light. Thus, the reflected ray **2325** is attenuated and/or not transmitted by the horizontal linear polarizer **2312** and is prevented from reaching the at least one waveguide **2348** or at least a reduced amount of such reflected light reaches the at least one waveguide **2348** or is coupled therein, for example, through in-coupling optical elements (e.g., one or more in-coupling gratings). The result would be similar for left-handed circularly polarized rays reflected from different optical surfaces of the optics **2320** or other optical surfaces on different optical elements.

As illustrated, the display system **2300** further includes a second retarder **2330** (e.g., quarter wave retarder or quarter waveplate) as well as second polarizer **2335** (e.g., linear polarizer) disposed between the optics **2320** and the spatial light modulator (not shown). This second retarder **2330** and this second linear polarizer **2335** may form a second circular polarizer in certain implementations. The second retarder **2330** is disposed between the optics **2320** and the second polarizer **2335**. Likewise, the second polarizer **2335** is disposed between the second retarder **2330** and the spatial light modulator. Accordingly, after passing through the

optics **2320**, the ray **2310** may pass through the second retarder **2330** (e.g., quarter wave retarder). The second retarder **2330** is configured (e.g., the optic axis is appropriately oriented) such that the ray **2310** is converted from a left-handed circular polarization to a horizontal linear polarization. Likewise, the second retarder **2330** converts the circularly polarized light back to the original linear polarization state that was output by the first polarizer **2312**. As will be discussed below, this second retarder **2330** and second polarizer **2312** may be useful in reducing "ghost" images caused by light reflected from the spatial light modulator that passes through optical surfaces (e.g., on the powered optics or lenses **2320**) as the light travels to the at least one light guide **2348**.

A third retarder **2340** (e.g., a quarter wave retarder or quarter waveplate) is disposed between the second polarizer **2335** and the spatial light modulator. Accordingly, the third retarder **2340** is disposed between the second retarder **2330** and spatial light modulator. Also, in various implementations such as shown, the second polarizer **2335** is between the second and third retarders **2330**, **2340**. As illustrated, the ray **2310** upon passing through the second polarizer **2335** is linearly polarized and in some implementations, the second retarder **2330**/second polarizer **2335** may convert the light to the original linear polarization of the first polarizer **2312** (e.g., horizontally polarized). This linearly polarized light is incident on the third retarder **2340**. The third retarder **2340** is configured such that the ray is converted back into a circularly polarized light and in some implementations to the same polarization as output by the first retarder **2315** (e.g., left-handed circularly polarized light in this example). In certain implementations, the spatial light modulator is configured to operate on circularly polarized light. In some implementations, the spatial light modulator is a reflective spatial light modulator that reflects the incident circularly polarized light back as circularly polarized light. In some embodiments, the circularly polarized light reflected from the spatial light modulator may have the same handedness (e.g., left-handed circularly polarized) as that incident thereon depending possibly on whether the spatial light modulator pixels are in the "on" or "off" states. In some embodiments, the spatially light modulator may reflect circularly polarized light of the different handedness (e.g., right-handed circularly polarized) as that incident thereon depending possibly on whether the spatial light modulator pixels are in the "on" or "off" states. Other types of spatial light modulators, however, may be used.

FIG. **23A** shows light, illustrated as ray **2342**, reflected from the spatial light modulator and travelling toward the waveguide **2385**. The reflected ray **2342** is depicted as left-hand circularly polarized light. The ray **2342** passes through the third retarder **2340**. The third retarder **2340** converts the circular polarized light into linearly polarized light. In this example, left-handed circularly polarized light is converted into horizontally polarized light. The linearly polarized light is transmitted through the second polarizer **2335**. In this example, the horizontally polarized light passes through the second polarizer **2335**. The linearly polarized light is incident on the second retarder **2330** and is converted into circularly polarized light. In this example, the horizontally polarized light is converted into left-hand polarized light and is transmitted to the optics **2320**. Here again, reflections from optical surfaces such as the surfaces of the optics **2320** having optical power may create ghost images by reflecting back off the spatial light modulator into the at least one waveguide **2348** and to the user's eye. As described above, undesirable reflections may occur at any interface

with media having dissimilar refractive indices such as air to material interfaces. As referenced above, the inclusion of the second retarder and polarizer **2330**, **2335**, may attenuate these reflections and lower the likelihood of ghost reflections. FIG. **23A**, for example, depicts light, illustrated as ray **2346**, reflected from an optical surface of the optics **2320**. The act of being reflected from the surface causes the reflected ray **2346**, which is circularly polarized to switch handedness, in this example, to switch from left-handed circular polarization to right-handed circular polarization. The switched circular polarized light is attenuated by the second circular polarizer formed by the second retarder and polarizer **2330**, **2335**. As illustrated in FIG. **23A**, for example, the reflected circularly polarized light **2346** is incident on the second retarder **2330** and transformed by the second retarder into linearly polarized light having a different, e.g., orthogonal, linear polarization than that which is selectively transmitted by the second linear polarizer **2335**. In this case, for example, the right-handed circularly polarized light reflected from the optical surface of the optics **2320** is converted by the retarder **2330** into vertical linear polarization, which is orthogonal to the polarization selectively transmitted by the polarizer **2335**. The second polarizer **2335** attenuates or prevents transmission of this linearly polarized light. In this example, the light **2346** is vertically polarized while the second polarizer **2335** is a horizontal polarizer that selectively passes horizontally polarized light and filters out vertically polarized light.

In contrast, the light **2342** passing through the optics **2320** and incident on the first retarder **2315** is circularly polarized and has a different handedness than light reflected from optical surfaces of the optics **2320**. This light **2342** directed toward the at least one waveguide **2348** has a polarization (e.g., left-handed polarized) that is converted by the first retarder **2315** into linearly polarization (e.g., horizontal linearly polarized light) that is selectively transmitted by the first polarizer **2312**. In this manner, the light **2342** can reach and be coupled into the at least on one waveguide **2348** and be directed to the user's eye.

In the example shown in FIG. **23A**, first circular polarizer, formed by the first polarizer **2312** and the first retarder **2315**, and second circular polarizer, formed by the second retarder **2330** and the second polarizer **2335**, on opposite sides of the optics **2320**, one closer to the light source **2305** and one closer to the spatial light modulator, are used to reduce reflections that may result in "ghost images". An additional retarder **2340** is included between the second circular polarizer (e.g. the second polarizer **2335**) and the spatial light modulator to convert the light into circularly polarized light. A wide range of variations are possible, however. For example, only one circular polarizer may be included. Alternately, additional circular polarizers or other types of polarization optics may be included.

FIG. **23B** illustrates a third circular polarizer that can be added to an augmented reality system **2300** such as shown in FIG. **23A**. In particular, FIG. **23B** depicts the second circular polarizer including the second polarizer **2335** and second retarder **2330** as well as the third retarder **2340** as introduced above, and further depicts a spatial light modulator **2375**. This spatial light modulator (SLM) **2375** may include a liquid crystal spatial light modulator (e.g., liquid crystal on silicon or LCoS). In some implementations, the SLM **2375** can be covered with a cover glass **2370**.

FIG. **23B** also shows a third circular polarizer including a fourth retarder **2345** such as a quarter wave retarder (e.g. quarter waveplate) and a third polarizer **2355** such as a linear polarizer disposed between the second circular polarizer

including the second polarizer **2335** and second retarder **2330** and the spatial light modulator **2375**. The third polarizer **2355** is between the fourth retarder **2345** and the spatial light modulator **2375**. An additional fifth retarder **2360** such as a quarter wave retarder (e.g., quarter waveplate) as well as a compensator **2365** are disposed between the third circular polarizer including the fourth retarder **2345** and the third polarizer **2355** and the spatial light modulator **2375** or more specifically the cover glass **2370** shown in FIG. **23B**. The fifth retarder **2360** is between the third polarizer **2355** and the compensator **2365**. The compensator **2365** is between the fifth retarder **2360** and spatial light modulator **2375** or specifically the cover glass **2370**.

FIG. **23B** shows how light, for example, ray **2310**, from the light source **2305** (shown in FIG. **23A**) can propagate through the second circular polarizer including the retarder **2330** and second polarizer **2335**, as well as the third retarder **2340** to the third circular polarizer including the fourth retarder **2345** and third polarizer **2355**. The light ray **2310** from the light source **2305** after passing through the second circular polarizer including the second retarder **2330** and second polarizer **2335** is incident on the third circular polarizer and in particular on the fourth retarder **2345**. The fourth retarder **2345** may convert the circular polarizer light of ray **2310** into linearly polarized light. In the example shown in FIG. **23B**, ray **2310** is circularly polarized (e.g., left-hand circularly polarized) and is converted by the fourth retarder **2345** into linearly polarized light (e.g. horizontally polarized light). This linearly polarized light proceeds through the third polarizer **2355**, which in FIG. **23B** includes a horizontal polarizer that selectively transmits horizontally polarized light. This linearly polarized light propagates through the fifth retarder **2360**, which may include a quarter wave retarder that converts the linearly polarized light into circularly polarized light. In the example shown in FIG. **23B**, the horizontally linearly polarized light **2310** incident on the fifth retarder **2360** is transformed into left-handed circularly polarized light. This circularly polarized light is incident on and passes through the compensator **2365**. The compensator **2365** may include a polarization element that adjusts the polarization to the desired polarization. The compensator **2365** may be used to offset birefringence of various optical elements in the system. For example, the light may be slightly elliptically polarized due to retardance contributions of one or more optical elements. In various implementations, the light output from the compensator **2365** is circularly polarized light. In the example shown in FIG. **23B**, the light output from the compensator **2365** is left-handed circularly polarized light. In various implementations, the compensator **2365** may be used to offset residual retardance within the SLM, which may comprise, for example, a liquid crystal (e.g., LCoS) SLM cell. The compensator may introduce in-plane retardance and/or out of plane retardance. In some implementations, the compensator **2365** may include a combination of optical retarders that when combined, produce the retardance that may potentially offset the residual retardance from the SLM (e.g., LCoS panel).

In FIG. **23B**, the light after passing through the compensator **2365** is incident on the cover glass **2370** and the SLM **2375**. This light incident on the cover glass **2370** and the SLM **2375** is depicted as left-hand circularly polarized light. Depending on the type of and the state of the spatial modulator, the SLM **2375** may reflect circularly polarized light of the same handedness. For example, when a pixel of the SLM **2375** is in an "on" state (although this state may be an undriven state in some implementations), the SLM **2375**

may introduce a quarter wave of retardance on each pass through the SLM 2375. Accordingly, on reflection, incident circularly polarized light may remain circular polarized on reflection. In various configurations, the handedness may also remain the same. For example, as shown in FIG. 23B, the incident left-hand circularly polarized light may remain left-handed circularly polarized on reflection. This circularly polarized light reflected from the SLM 2375, represented by ray 2342, may pass through the cover glass 2370 and compensator 2365 and be incident on the fifth retarder 2360, which converts the circularly polarized light into linearly polarized light. In the example shown in FIG. 23B, the circularly polarized light incident on the fifth retarder 2360 is left-handed and the fifth retarder 2360 converts this circularly polarized light into horizontally polarized light. The third polarizer 2355 may be configured to selectively transmit the polarization of light output by the fifth retarder 2360. Accordingly, in the example shown in FIG. 23B where the light output from the fifth retarder 2360 is horizontally polarized, the third polarizer 2355 selectively transmits the horizontally polarized light. This linearly polarized light transmitted by the polarizer 2355 is incident on the fourth retarder 2345 and converted into circularly polarized light. In the example shown in FIG. 23B, this circularly polarized light is left-hand circularly polarized. This light can travel through the second circular polarizer comprising the second retarder 2330 and second polarizer 2335, the optics 2320, as well as the first circular polarizer comprising the first polarizer 2312 and the first retarder 2315 onto the at least one waveguide 2348 and into the eye of the user as discussed above in connection with FIG. 23A.

Light reflected from optical surfaces may, however, be attenuated by the third circular polarizer thereby reducing the likelihood that such reflections will reach the at least one waveguide 2348 and be directed to the user's eye producing ghost images. To illustrate, FIG. 23B shows an example ray 2343 reflected from an optical surface of the third retarder 2340, for example, from the interface between the air and the third retarder 2340. As discussed above, reflections may occur at any interface between media having dissimilar refractive indices such as air to material interfaces or interfaces between different dielectric layers. However, circularly polarized light reverses handedness upon reflection. For example, upon reflecting off of the surface of the third retarder 2340, the direction of the circular polarization is changed (e.g., from left-handed to right-handed). The right-handed reflected ray 2343 then travels through the fourth retarder 2345 and is transformed into linearly polarized light having a different, for example, orthogonal, linear polarization than that which is selectively transmitted by the third polarizer 2355. In this case, for example, the light reflected from the optical surface of the third retarder 2340 is converted by the fourth retarder 2345 into vertical linear polarization, which is orthogonal to the polarization selectively transmitted by the third polarizer 2355. The third polarizer 2355 selectively passes horizontally polarized light and filters out vertically polarized light. Thus, the reflected ray 2343 is attenuated and/or not transmitted by the third polarizer 2355 and is prevented from reaching the at least one waveguide 2348 (e.g., by reflecting off another surface) or at least a reduced amount of such reflected light reaches the at least one waveguide 2348 or is coupled therein.

The result may be the similar for circularly polarized rays reflected from different optical surfaces. FIG. 23B, for example, shows a reflection of incident light ray 2310 off the optical surface of the fourth retarder 2345. The reflection 2350 off of the fourth retarder 2345 switches the handedness

of the polarization. For example, the incident ray 2310 depicted as left-handed circularly polarized is converted upon reflection into a ray 2350 that is shown as having right-handed circular polarization. The reflected ray 2350 passes through the third retarder 2340 and is transformed into vertically polarized light. This vertically polarized light is selectively attenuated or filtered out by the second polarizer 2335.

As described above, a pixel of the SLM 2375 may, for example, be in an "on" state (although an undriven state in some implementations) where light incident on this pixel of the SLM 2375 is reflected therefrom and coupled into the at least one waveguide 2348 and directed to the eye of the user. However, a pixel of the SLM 2375 can be in an "off" state (which may be a driven state in some implementations), in which light incident on the pixel of the SLM 2375 is not coupled into the at least one waveguide 2348 and is not coupled into the user's eye. In this "off" state, for example, various implementations of the SLM 2375 may introduce no retardance upon reflection therefrom. Accordingly, in the example shown in FIG. 23B, circularly polarized light incident on the SLM 2375 may remain circularly polarized on reflection from the SLM 2375. This handedness of the circularly polarized light may, however, change upon reflection from the SLM 2375. For example, the ray 2310 shown in FIG. 23B that is left-handed circularly polarized that is incident on the SLM 2375, may be transformed into right hand circularly polarized light upon reflection from the SLM 2375. This reflected light, however, may be selectively attenuated by the third polarizer 2355. For example, the right circularly polarized light reflected from the SLM 2375 may pass through the cover glass 2370, the compensator 2365, and the fifth retarder 2360. The fifth retarder 2360 may convert the right-handed circularly polarized light into vertically polarized light, which is selectively attenuated by the third polarizer 2355, which may include a horizontal polarizer. Accordingly, in various implementations, the fifth retarder 2360 may convert light reflected from a pixel of the SLM 2375 when the pixel of the SLM is in the "off" state, into a linear polarization that is orthogonal to the linear polarization selectively transmitted by the third polarizer 2355. This third polarizer 2355 may thus selectively attenuate this linearly polarized light thereby reducing or blocking the light from that pixel of the SLM 2375 from reaching the at least one waveguide 2348 and being directed into the eye.

Variations in the configurations, such as variations in the polarization optical elements, are possible. For example, more or less circular polarizers may be included.

In various implementations, for example, the third circular polarizer including the fourth retarder 2345 and third polarizer 2355 is excluded such as shown in FIG. 23C. In this particular implementation, the fourth retarder 2345, third polarizer 2355, and the fifth retarder 2360 are not included in the system. FIG. 23C illustrates a design of the augmented reality system 2300 that includes components illustrated in FIGS. 23A and 23B, with the exception of the fourth retarder 2345, third polarizer 2355, and the fifth retarder 2360. Nevertheless, despite excluding the third circular polarizer, the augmented reality display system is still configured to reduce ghost images. The second circular polarizer, for example, reduces reflection that would otherwise contribute to ghost images. To illustrate, FIG. 23C, depicts light, illustrated as ray 2380, reflected from the third retarder 2340. The act of being reflected from the surface of the third retarder 2340 causes the reflected ray 2380, which is circularly polarized to switch handedness. In this example, the polarization is switched from left-handed circular polar-

ization to right-handed circular polarization. The switched circular polarized light **2380** then passes through the compensator **2365** and is incident on the cover glass **2370** and the SLM **2375**. As discussed above, the SLM **2375** may reflect circularly polarized light of the same handedness. Accordingly, the incident right-hand circularly polarized light may remain right-handed circularly polarized on reflection. This circularly polarized light reflected from the SLM **2375**, represented by ray **2382**, may then pass through the cover glass **2370** and compensator **2365** and be incident on the third retarder **2340**. The switched circular polarized light **2382** is attenuated by the second circular polarizer and in particular by the third retarder **2340** and polarizer **2335**. As illustrated in FIG. 23C, for example, the circularly polarized light **2382** reflected from the SLM **2375** is incident on the third retarder **2340** and transformed by the third retarder **2340** into linearly polarized light having a different, e.g., orthogonal, linear polarization than that which is selectively transmitted by the second linear polarizer **2335**. In this case, for example, the right-handed circularly polarized light **2382** is converted by the third retarder **2340** into vertical linear polarization, which is orthogonal to the polarization selectively transmitted by the second polarizer **2335**. The second polarizer **2335** attenuates or prevents transmission of this linearly polarized light.

Reflections that may contribute to ghost reflections may also potentially be reduced by tilting the optical surfaces in the system. FIG. 24 illustrates an example configuration having a tilted optical surface for reducing reflections that may produce ghost reflections. FIG. 24 shows an augmented reality display system **2400** including a light source **2305** that emits light represented by a ray **2310** that passes through any number of polarizers, retarders, lenses and/or other optical components as the light travels toward a spatial light modulator (SLM) **2375**. A first polarizer **2312** and a first retarder **2315** possibly forming a first circular polarizer as well as lenses **2320** are shown in FIG. 24 for illustrative purposes. However, additional components may be included or components may be excluded or arranged or configured differently. In the example illustrated, the SLM **2375** includes therewith a cover glass **2370**. The cover glass **2370** can be a contributor to reflections that produce ghost images. As such, in some implementations, the cover glass **2370** can be shaped so as to direct reflections that may yield ghost images away from being directed into a user's eye. As illustrated, the cover glass **2370** has a surface that can be tilted such that the surface is not parallel with other components or optical surfaces of the system (e.g., the SLM **2375**, first retarder **2315**, first polarizer **2312**, at least one waveguide **2348**, etc., or optical surfaces thereof). A major surface of the cover glass **2370** may, for example, have a normal that is tilted so as not to be aligned or parallel to the optical axis of the augmented reality display system **2400** or optical components therein such as optics **2320**. By being tilted, reflections from the optical surface of the cover glass **2370** can be directed away from the at least one waveguide **2348** or in-coupling optical elements (e.g., in-coupling gratings or diffractive optical elements) for in-coupling light into the at least one waveguide **2348** and reduce the likelihood that reflections from the cover glass **2370** enter the at least one waveguide **2348**. As depicted, reflected light **2405** is directed back toward the light source **2305** and away from the at least one waveguide **2348** where such light could ultimately reach the eye of a user. In some implementations, the reflected light **2405** can be directed back to the light source and a least a portion recycled at the light source **2305**.

Although FIG. 24 depicts the cover glass **2370** having a surface that is tilted, optical surfaces that are tilted to divert reflections away from being coupled into the at least one waveguide **2348** can be included on any component in the system where undesired reflection is possible. Accordingly, optical surfaces on other components, such as polarizers, retarders, etc., may be tilted to reduce reflection being coupled into the at least one waveguide **2348** and to the eye of the user. Variations in the shape and size of the cover glass **2370** or other optical components are possible. The cover glass **2370** or other optical component may, for example, be thinner. Similarly, the cover glass **2370** or other optical component may have a different aspect ratios (length to thickness) than shown in FIG. 24. In some implementations, the cover glass **2370** or other optical component is wedge shaped. Other shapes, however, are possible.

Still other arrangements are possible. FIG. 25, for example, illustrates an implementation of an augmented reality display system **2500** similar to the system **2400** shown in FIG. 24 but further including a light dump **2505** for absorbing light directed thereto. The system **2500** includes the tilted cover glass **2370** to direct reflections **2510** from the cover glass **2370** to the light dump **2505** instead of being directed back to the light source **2305**. The light dump **2505** may include an absorbing material or structure that is configured to absorb light. The location of the light dump **2505** can change depending on the implementation, for example, depending on the angle of the tilted cover glass **2370**. As discussed above, this approach can be applied to other optical surfaces in the system. In addition, the shapes and sizes of the optical elements may be different.

A wide range of variations in the augmented reality display are possible. Variations in the polarization optical elements are possible. For example, although horizontal polarizers are used, in some implementations, vertical polarizers or a combination of horizontal and vertical polarizers are employed. Additionally, polarizers characterized by polarization other than vertical or horizontal may be used. Likewise, the light shown in the figures need not be horizontally polarized but may be vertically polarized. Similarly, light shown as vertically polarized may be horizontally polarized or vice versa in different implementations. Linearly polarized light having polarizations other than vertical or horizontal may also be used.

Additionally, the retarders may be configured differently. For example, the polarized light in the figures need not be left-hand circularly polarized but may be right-hand circularly polarized light and/or the right-hand polarized light may be left-hand circularly polarized. Still other variations are possible. Different retarder configurations can be employed to produce different combinations of left-handed and/or right-handed polarized light than shown. Also, in some implementations, elliptical polarized light may possibly be used instead of circularly polarized light. Retarders may be employed, for example, to convert elliptically polarized light into linear polarized light and vice versa. Linear polarizers can be used to filter light and may be used to reduce ghost reflections such as described herein.

In some implementations, other types of polarization elements and configurations thereof are employed. For example, the retarders are not limited to quarter wave retarders or quarter waveplates. For example, in some implementations, various optical elements have birefringence. In certain such cases, any one or more of the retarders **2315**, **2330**, **2340** may include an amount of retardance sufficient to convert linearly polarized light into circularly polarized light and need not be a quarter wave retarder. More or less

than a quarter wave of retardance may be included in any one or more of the retarders **2315**, **2330**, **2340** as retardance may be contributed by other optical elements. Similarly, retardance can be distributed in a number of optical elements. As another example, multiple retarders may be employed to provide the appropriate amount of retardance. Also, as described above, in some implementations, elliptical polarized light may possibly be used instead of circularly polarized light. Retarders may be employed, for example, to convert elliptically polarized light into linear polarized light and vice versa. Linear polarizers can be used to filter light and may be used to reduce ghost reflections such as described herein.

Additionally, the optical components may be in the form of optical layers, sheets and/or films as well as stacks or one or more layers, sheets and/or films. Accordingly, different polarization elements, in different amounts, locations, and arrangements may be used. For example, one or more of the retarders and/or polarizers may comprise films.

In some implementations, the spatial light modulator may operate differently. For example the spatial light modulator may operate on light other than circularly polarized light and/or may output light other than circularly polarized light.

Diffraction waveguide image combiners are an attractive approach for conveying computer generated imagery to a user, in AR applications. These waveguides work through repeated splitting, and outcoupling of the image-containing light rays, which spreads the incoupled light to fill the far-field viewing area, as well as acting as a pupil expander. The inventors have determined that the splitting process inherently results in recombining of rays, and when this happens, the opportunity for rays to interfere with each other occurs. This interference effect results in intensity non-uniformities that are addressed by embodiments of the present invention. As described herein, small differences in path length of the interfering rays, such as are introduced via small variations in thickness of the waveguide, lead to random occurrence in these non-uniformities. The addition of an intentionally controlled bias to path length differences has been shown to result in improved and more repeatable luminance uniformity. This bias can be in the form of a controlled variation in waveguide thickness, oriented along certain directions relative to the main propagation paths in the waveguide. Examples of this include a wedge-like taper in optical or physical thickness, parabolic changes in optical or physical thickness, and others. As described herein, embodiments of the present invention result in improvement in a variety of performance characteristics, particularly the modulation transfer function (MTF), image sharpness, and the like.

Total thickness variation (TTV) is one metric for improving performance of an optical waveguide. As used herein, TTV generally refers to the difference between the maximum and minimum values of the thickness of the waveguide or the waveguide display substrate on which the waveguide is formed. As light travels through an optical waveguide, typically by total internal reflection, variations in the thickness a the light propagation path(s). Angular differences in the light propagation path(s) can affect image quality with field distortions, image blurring, and sharpness loss.

In some embodiments described herein, the TTV is achieved by variation in the physical thickness of the optical waveguide. However, embodiments of the present invention are not limited to physical thickness variations and optical thickness variation of the substrate, which is a combination of physical thickness variation and index of refraction variation is included in the scope of the present invention. As

will be evident to one of skill in the art, the optical path length can be represented by the Optical Thickness Variation=Physical Thickness Variation×Index of Refraction Variation. In applications in which beamlet splitting and combining behavior is utilized to control interference and luminance uniformity, optical thickness variation is implemented. In most high quality optical glasses, the index of refraction variation is typically small (ISO NV±2×10⁻⁵ to ±30×10⁻⁵). However, other materials, including polymers and ceramics, do not have such small index of refraction variation. Additionally, optical waveguides can be fabricated using materials with a gradient index material to achieve optical thickness variation in lieu of any physical thickness variation. Thus, discussion of TTV provided herein is understood to include the broader class of optical thickness variation, which can include physical thickness variation with a uniform index of refraction, uniform thickness with a varying (i.e., graded) index of refraction, or combinations of physical thickness variation and varying index of refraction.

Waveguide preparation and processing typically occurs by arranging a number of waveguides to designated areas onto a waveguide display substrate (e.g., a wafer). FIG. **26A** depicts waveguide display substrate **2600** with a radial arrangement of waveguides **2602**.

FIG. **26B** is a plot showing waveguide display substrate thickness as a function of position according to an embodiment of the present invention. In FIG. **26B**, the thickness of the waveguide display substrate corresponds to cross section A-A' illustrated in FIG. **26A**. As illustrated in FIG. **26B**, the thickness is characterized by a dome shape, with half of the dome shown in the figure. The thickness is substantially constant for positions from 0 mm, corresponding to the center of the waveguide display substrate to ~30 mm. The thickness is characterized by a decreasing thickness for positions from ~30 mm to 70 mm, resulting in a variation in thickness from the center to the edge of the waveguide display substrate of approximately 0.7 μm=700 nm. For positions from ~30 mm to 70 mm, the thickness variation is decreasing as a function of the direction aligned with cross-section A-A'. This thickness variation can be defined as a thickness difference gradient in which the gradient is measured along the direction aligned with cross-section A-A'. If the optical path length is considered, similar description is applicable to the gradient of the optical path length difference.

FIG. **26C** is a plot showing waveguide display substrate thickness as a function of position according to an embodiment of the present invention. In FIG. **26C**, the thickness of the waveguide display substrate corresponds to cross section B-B' illustrated in FIG. **26A**. As illustrated in FIG. **26C**, the thickness is substantially uniform as a function of position with a slight increase in thickness observed in the center of the waveguide in comparison to the sides of the waveguide.

In addition to a radial arrangement of waveguides **2602** as illustrated in FIG. **26A**, some embodiments utilize a rectilinear arrangement. FIG. **26D** depicts waveguide display substrate **2610** with a rectilinear arrangement of waveguides **2612**, **2614**, etc. Thus, embodiments include waveguide layouts that have rotational symmetry with four waveguides as illustrated in FIG. **26A**, six waveguides, or another number of waveguides, as well as a rectilinear layout that is typically utilized in semiconductor manufacturing processes.

The rectilinear layout illustrated in FIG. **26D** provides a number of benefits in relation to an optical TTV bias that may not be available using a rotationally symmetric layout.

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FIG. 26E is a simplified schematic diagram illustrating thickness variation for waveguide display substrate with a rectilinear layout according to an embodiment of the present invention. As an example, in a rectilinear layout, a unidirectional wedge can be implemented with the thickness varying linearly from a first thickness t_1 at a first side of the waveguide display substrate (i.e., position A) and a second thickness t_2 at a second side of the waveguide display substrate (i.e., position A'). FIG. 26E illustrates thickness variation for a linear variation by curve 2620. Both waveguide 2612 and waveguide 2614 will thus have a thickness that decreases as the position moves along the line A-A'. Although waveguide 2612 will have a larger nominal thickness than waveguide 2614, the variation in thickness of the waveguide display substrate (e.g., on the order of less than one micron) is small enough in comparison to the overall thickness of the waveguides (i.e., on the order of several hundred microns), that this difference in nominal thickness is acceptable for a variety of applications.

In addition to linear variations, other variations in thickness can be implemented, including quadratic variation, other smoothly varying TTV, or the like. These embodiments can be utilized in order to align the direction of light propagation to the TTV variation. FIG. 26C illustrate thickness variation for a smooth variation by curve 2622. Thus, in some embodiments, a section of a sphere is utilized to provide thickness variation, wherein, in other embodiments, linear variations, quadratic variations, or other suitable variations are utilized to implement TTV, resulting in a bias in thickness across the waveguide.

FIG. 26F is a simplified schematic diagram illustrating a rectangular waveguide display substrate 2630 with a linear arrangement of waveguides 2632. In this exemplary embodiment, a non-circular form, is implemented to take advantage of a tapered waveguide architecture and biased thickness variation that can be characterized by a thickness variation that can depend on the layout and directionality of the waveguide fields. As illustrated in FIG. 26F, the TTV increases linearly from side 2634 to side 2636 as illustrated by the plot included in the figure, characteristic of a wedge thickness variation across substrate 2630. Thus, each waveguide 2632 is characterized by a TTV as measured along the length of the waveguide (i.e., along the y-axis). Although a rectangular geometry is illustrated in FIG. 26F, other non-circular forms, including square forms can be utilized. Moreover, although a linear variation in thickness is illustrated in FIG. 26F, smoothly varying functions can be used to define the TTV. Additionally, although a uniform thickness as a function of waveguide width (i.e., along the x-axis) is illustrated in FIG. 26F, in other embodiments, variation in thickness along both the length and width is utilized.

FIG. 26G is a simplified schematic diagram illustrating a waveguide display substrate with localized TTV according to an embodiment of the present invention. In addition to the overall substrate TTV provided by embodiments of the present invention, for example, as illustrated in FIG. 26F, other embodiments utilize a localized TTV for specific waveguides. As an example, a local wedge or taper specific to an individual waveguide can be implemented by a deterministic polishing process that forms sections with slanted bottom features in a substrate. Thus, FIG. 26G illustrates a substrate with specific "cavities" that correspond to specific waveguides. Thus, in this embodiment, the thickness profile of the waveguides is detached from the thickness of the waveguide display substrate

Referring to FIG. 26G, rectangular waveguide display substrate 2640 includes a linear arrangement of waveguides

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2642. The thickness of waveguides 2642 is uniform as measured in the y-direction. However, as measured in the x-direction, the thickness of each of the waveguides varies, decreasing in a linear manner from side 2634 to side 2636. This wedge shape is characteristic of each waveguide. Thus, using waveguide display substrate 2640 with an overall plano-plano thickness profile, multiple waveguides 2642, each with a localized TTV that has a wedge shape, are provided. Although a linear thickness variation for each of the waveguides is illustrated in FIG. 26G, this is not required and other shapes can be utilized, including smoothly varying functions.

In addition to deterministic polishing to form the localized TTV discussed in relation to FIG. 26G, other methods of implementing localized TTV are included within the scope of the present invention, including molding processes, and the like. Other methods of making localized TTV include patterned (masked) film deposition (organic, inorganic, vapor phase vacuum, atmospheric, or nanoimprint), patterned (masked) etching into the substrate, spatially varying ion implantation or ion diffusion to create changes in index of refraction and therefore optical TTV, creating a mold with the inverse TTV pattern and molding polymer or glass with it, creating the localized areas of desired TTV bias.

TTV can be reduced by fabricating a flat waveguide display substrate (i.e., a waveguide display substrate with zero TTV), for example, by polishing the substrate (e.g., a metal, glass or silicon substrate) or originally molding the substrate (e.g., a polymer substrate) with high-precision. However, polishing can produce a certain amount of curvature upon the waveguide display substrate and the resultant waveguides formed thereon. FIG. 27A depicts polished flat waveguide display substrate 2700. FIGS. 27B and 27C depict polished convex waveguide display substrate 2702 and polished concave waveguide display substrate 2704, respectively. Though polishing can yield convex or concave curvatures, embodiments described herein are explained with reference to a convex curvature, such as that depicted in FIG. 27B.

Since completely flat polishing or molding, such as that depicted in FIG. 27A, requires extensive and costly processing to achieve, a certain degree of TTV is typically tolerated. With most low TTV processes for waveguide display substrates (e.g., $20\text{ nm} < \text{TTV} < 2\text{ }\mu\text{m}$), there exists a thickness shape or profile to the substrates that varies (e.g., randomly) from part to part. Here, "thickness shape" generally refers to the 3D mapping of the height difference between the substrate's top and bottom surfaces. In one example, a typical plano-convex lens has a thickness shape that is a convex spherical or positive "dome". In another example, a meniscus lens with the same radius of curvature on each surface (e.g., non-prescription sunglasses) has a flat near-zero TTV thickness shape but each surface itself is non-flat. As waveguides are stacked to form a multi-layer multi-color waveguide display, the random thickness shape differences can cause each color channel of red, green and blue to have a different luminance uniformity pattern. Differences in luminance uniformity pattern can cause the color to vary over the field of view when the waveguide stack is illuminated with a uniform white light image. These color non-uniformities can result in poorer image quality.

It should be noted that although FIGS. 27A-27C illustrate waveguide display substrates with at least one planar surface, embodiments of the present invention are not limited to plano-convex or plano-concave geometries. FIG. 27D depicts polished meniscus waveguide display substrate 2706 with one convex surface and one concave surface. The

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embodiment illustrated in FIG. 27D is a positive lens, but modification of the curvatures of the convex and concave surfaces can result in a negative lens as well. FIG. 27E depicts polished bi-convex waveguide display substrate 2708. FIG. 27F depicts polished bi-concave waveguide display substrate 2710. Thus, in order to implement a controlled change of path length during splitting and combining of beamlets, a variety of waveguide display substrates can be utilized, including waveguide display substrates that include a planar surface. However, as illustrated in FIGS. 27D-27F, physical thickness changes, refractive index changes, or a combination of both (i.e., optical thickness changes), can be achieved using waveguide display structures in which no planar surfaces are utilized.

Referring to FIG. 28A, waveguide display substrate 300 is shown having a TTV, measured as the highest point to the lowest point of the curved portion of the polished waveguide display substrate. It should be noted that the polished waveguide display substrate is not required to have a planar surface and both surfaces can exhibit curvature. Perfectly flat waveguide display substrate 302, depicted in FIG. 28B has a zero TTV.

FIG. 28C depicts TTV for a set of eyepiece waveguides according to an embodiment of the present invention. Referring to FIG. 28C, a set of three eyepiece waveguides: eyepiece waveguide 2820, second eyepiece waveguide 2830, and third eyepiece waveguide 2840 are included in an eyepiece 2810. In some embodiments, the three eyepiece waveguides are designed for incoupling, propagation, and outcoupling of light in red, green, and blue wavelengths, respectively. Eyepiece waveguide 2820 includes an incoupling grating 2822 and a combined pupil expander 2824 that outcouples light toward a user. Second eyepiece waveguide 2830 includes a second incoupling grating 2832 and a second combined pupil expander 2834 that outcouples light toward the user. Third eyepiece waveguide 2840 includes a third incoupling grating 2842 and a third combined pupil expander 2844 that outcouples light toward the user. In some embodiments, eyepiece waveguide 2820, second eyepiece waveguide 2830, and third eyepiece waveguide 2840 are laminated together into a single structure to form eyepiece 2810. Spacers (not shown) between each of the eyepiece waveguide can be utilized to maintain a fixed relationship between each of the eyepiece waveguides.

As illustrated in FIG. 28C, the thickness, and, therefore, the optical path length measured along the longitudinal axis (i.e., the z-axis) varies as a function of the lateral dimensions (i.e., the x-direction and the y-direction). In the illustrated embodiment, the variation in the optical path length for each of the eyepiece waveguides occurs along the x-direction, but this is not required by the present invention. Thus, as illustrated in FIG. 28C, the varying optical path length difference varies from a first value at a first region of each of the three eyepiece waveguides (e.g., the incoupling grating region adjacent incoupling grating 2822, incoupling grating 2832, and incoupling grating 2842) to a second value at a second region of each of the three eyepiece waveguides (e.g., the combined pupil expander region adjacent combined pupil expander 2824, second combined pupil expander 2834, and third combined pupil expander 2844).

The variation in optical path length for each of the eyepiece waveguides in an eyepiece can be aligned so that the variation in each eyepiece waveguide varies (e.g., decreases) along a single direction and this single direction can be common to all of the eyepiece waveguides in an eyepiece. As illustrated in FIG. 28C, each of the eyepiece waveguides decrease in thickness as a function of the lateral

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direction aligned with the x-axis. That is, the thickness decreases as measurements are made at various points along the x-axis. As a result, the x-axis can be considered as a direction corresponding to the variation. In FIG. 28C, the direction corresponding to the variation is common between each of the three eyepiece waveguides, that is, it is a common direction (i.e., the x-axis) and the direction corresponding to the variation for each eyepiece waveguide is aligned with the direction corresponding to the variation in each eyepiece waveguide being parallel to the direction corresponding to the variation in each of the other eyepiece waveguides. Thus, the combined decrease in optical path length corresponding to the eyepiece 2810 is a sum of the decreases in optical path length corresponding to each eyepiece waveguide.

Referring to FIG. 28C, the portion of each of the eyepiece waveguides adjacent the incoupling grating is thicker than the portion of each of the eyepiece waveguides adjacent the combined pupil expander. Thus, the varying optical path length difference decreases from a first value at a first region of each of the three eyepiece waveguides (i.e., corresponding to the input coupling grating) to a second value at a second region of each of the three eyepiece waveguides (i.e., corresponding to the combined pupil expander). In some implementations, the incoupling grating is disposed near the peripheral region of the eyepiece while the combined pupil expander is disposed near the nasal region of the eyepiece. Thus, in these implementations, the thickness of the eyepiece will be thicker near the peripheral region and thinner near the nasal region.

In other embodiments, the optical path length difference decreases in other manners, for example, thicker near the nasal region and thinner near the peripheral region, thicker in the region of the combined pupil expander and thinner in the region of the incoupling grating, combinations thereof, or the like. Thus, a variety of structures with gradients in their optical path length difference are included within the scope of the present invention and the examples discussed herein are merely exemplary. One of ordinary skill in the art would recognize many variations, modifications, and alternatives.

In binocular applications, an eyepiece associated with the right eye can utilize a variation in optical path length as illustrated in FIG. 28C, for example, with the optical path length decreasing as a function of position with respect to the nasal region of the eyepiece, i.e., shorter optical path length at the periphery than near the nasal region. An eyepiece associated with the left eye can utilize a mirrored structure, with the optical path length also decreasing as a function of position with respect to the nasal region of the eyepiece, but with respect to the left eye rather than the right eye.

FIG. 28D is a simplified schematic diagram illustrating an augmented reality system according to an embodiment of the present invention. Augmented reality system 2850 can be implemented as an augmented reality headset suitable for a user. The augmented reality system 2850 includes a controller and, a first set of elements corresponding to the left eye of the user, and a second set of elements corresponding to the right eye of the user. In the embodiment illustrated in FIG. 28D, the first set of elements includes a left projector 2852, left projector optics 2854, and one or more left eyepiece waveguides 2856. The second set of elements includes a right projector 2852, right projector optics 2854, and one or more right eyepiece waveguides 2856. For each eye, light from the projector is focused using the projector optics onto the eyepiece including the eyepiece

waveguides, for example, onto the incoupling gratings of the eyepiece waveguides. As described herein, due to the optical path length variation associated with the eyepiece, the image quality provided by the augmented reality system **2850** is higher than that available using conventional approaches.

The augmented reality system can be operated to provide virtual content to a user. Accordingly, methods of operating the augmented reality systems described herein, including the augmented reality system **2850** illustrated in FIG. **28D**, are included within the scope of the present invention. These methods can utilize the eyepieces and eyepiece waveguides described herein, which are characterized by optical path length differences as measured across their lateral dimensions, to improve the user experience in comparison with conventional techniques.

FIG. **29** depicts waveguide display substrate **400** having a minimum thickness X . That is, no part of the waveguide display substrate is thinner than X , and any thickness variation is measured as a thickness in addition to X . The greatest thickness is h , and the TTV mathematically is Y , where $Y=h-X$. TTV specifications are typically expressed as a maximum allowed TTV (TTV_{max}), and product specifications target a TTV (Y) below TTV_{max} (i.e., $0 \leq Y < TTV_{max}$). Here, “ultra-low TTV” waveguide display substrates refers to waveguide display substrates for which the target TTV is zero (or as close to zero as practically possible). It should be noted that the polished waveguide display substrate is not required to have a planar surface and both surfaces can exhibit curvature.

In optical products, image quality and uniformity can be sensitive to a particular shape or profile of a polished waveguide display substrate as TTV approaches zero. In one example, the comparative difference of image quality (as measured by uniformity) of waveguides produced on a 20 nm TTV and a 40 nm TTV waveguide display substrate can be much higher than the comparative quality of waveguides produced on a 100 nm TTV substrate and a 120 nm waveguide display substrate, despite both pairs being only 20 nm different. Stated differently, a 100 nm TTV waveguide and a 120 nm waveguide can yield a more similarly uniform image than a 20 nm TTV waveguide and a 40 nm TTV waveguide (the former pair being more dome shaped as compared to the latter pair). The 100 nm TTV waveguide and 120 nm TTV waveguide can thus produce lower image variation and more consistent image uniformity across a product line.

Path length differences in the light propagation path(s) within a waveguide produced on a non-flat waveguide display substrate can also affect image quality with luminance pattern non-uniformities and color non-uniformities. The source of the luminance pattern non-uniformities includes electromagnetic interference patterns produced by numerous pathways through a pupil-replicating waveguide display substrate. A unit-cell of a typical pupil-replicating waveguide resembles a Mach-Zehnder interferometer, where there exist two pathways per unit-cell from the input to an output replicated pupil location. Path length differences between the two pathways are influenced by the path length through the thickness of the waveguide display substrate, which is defined by the TTV metric and thickness profile (whether the thickness changes linearly or quadratically and at an angle with respect to the light in total-internal-reflection within the waveguide display). If the path lengths have equal or opposite phase, there may be constructive or destructive interference, respectively. Hence, the thickness

shape may affect the magnitude within pupil-replicated copies and ultimately the output image outcoupled by the waveguide display.

Thus, it can be advantageous to minimize thickness shape variation as well as TTV. Since perfect replication may not be achieved in polishing or molding processes, certain manufacturing distributions can result. In a hypothetical distribution model, with a circular substrate shape defined by Zernike polynomials, a standard set of shape basis functions can be defined. There are even and odd Zernike polynomials. The even ones are defined as

$$Z_n^{m(p,\psi)} = R_n^m(\rho) \cos(m\psi)$$

and the odd ones are defined as

$$Z_n^{-m(p,\psi)} = R_n^m(\rho) \sin(m\psi),$$

where m and n are nonnegative integers with $n \geq m$, is the azimuthal angle, ρ is the radial distance, and R_n^m are the radial polynomials defined below. Zernike polynomials have the property of being limited to a range of -1 to $+1$, i.e. The radial polynomials R_n^m are defined as

$$R_n^m(\rho) = \sum_{k=0}^{n-m} \frac{(-1)^k (n-k)!}{k! \left(\frac{n+m}{2} - k\right)! \left(\frac{n-m}{2} - k\right)!} \rho^{n-2k}$$

for $n-m$ even, and are identically 0 for $n-m$ odd.

Image quality, in particular luminance uniformity, is non-linearly sensitive to thickness shape as TTV approaches zero. In other words, as TTV decreases below certain thresholds, image quality becomes increasingly varied among waveguides exhibiting even minor changes in thickness shape. To compensate for this anomaly, a biased TTV with a consistent thickness shape can be incorporated into substrate processing. As used herein, a “biased” TTV generally refers to a TTV having a nonzero target. More specifically, a “biased” TTV generally refers to a substrate thickness shape having one or more coefficients of Zernike fit polynomials with nonzero targets and all remaining coefficients of Zernike fit polynomials having zero targets. Consistency of thickness shape within a plurality of waveguide display substrates generally refers to a plurality of waveguide display substrates having a low variation of coefficients of Zernike fit polynomials. In the case of “biased” TTV, consistency of thickness shape refers to a set of waveguide display substrates having (i) all nonzero-targeted Zernike coefficients having minimal variation from their target magnitudes, for example the nonzero-targeted coefficients of all waveguide display substrates being about 70% to about 130% of their target magnitude, and (ii) all zero-targeted Zernike coefficients having an absolute magnitude substantially less than the nonzero-targeted Zernike coefficients, for example the zero-targeted coefficients being 0 to about 30% of the nonzero-targeted coefficients.

An example of biased TTV and consistent substrate thickness shape is a substrate that is substantially polished (or molded) to a spherical shape with a TTV many times larger than the typical minimum TTV range of a substrate polishing (or molding) process. This substrate shape can be described as having wedge (linear change in thickness) and dome (quadratic change in thickness) components. FIG. **30** depicts waveguide display substrate **3000** with wedge component **3002** having height W , dome component **504** having height D , and cylindrical component **3006** with planar surface **3008** and having a thickness t and a diameter d .

Using Zernike fit polynomials, an average thickness of the waveguide display substrate is Z_0^0 , the height of the “wedge” component of TTV can be defined as $\sqrt{Z_1^{-1} + Z_1^1}$, and the height of the convex “dome” component of TTV can be defined as $-2 \times Z_2^0$. An average thickness of the waveguide display substrate is typically between 200 μm and 2000 μm . In a substrate polishing or molding process that produces a “dome”-biased TTV, the waveguide display substrates may have a target “dome” height D_t in the range of 10^{-7} to 10^{-6} of the substrate diameter, and each substrate may have a “dome” height less than 30% of the target “dome” height and a “wedge” height less than 30% of the target “dome” height. In one example, a plurality of waveguide display substrates having a diameter of 150 nm, a mean “dome” (D_{mean}) height of 1000 nm, a range of “dome” height (D_{min} to D_{max}) of 700 nm to 1300 nm, and a maximum “wedge” height (W_{max}) of less than 300 nm will have increased efficiency, luminance uniformity and color uniformity and a reduced part-to-part variation in efficiency, luminance uniformity, and color uniformity compared to a plurality of waveguide display substrates having close-to-zero targeted TTV with “dome” and “wedge” height in the range of 0 to 300 nm. In the radial display layout of FIG. 26 and consistent-shape radially symmetric substrate thickness profiles biased to a dome, a consistent thickness change from a waveguide display input coupler to an output coupler can be achieved between the numerous parts arranged on a single substrate as well as from substrate to substrate. Such an arrangement of consistent thickness shape shows improved color uniformity and image quality compared to the typical ultra-low TTV with random thickness shapes used in conventional waveguide displays.

The biased TTV with consistent shape can be applied to the waveguide display substrate in numerous methods. For glass or crystalline substrates, it may be applied by polishing to a biased shape or by applying a coating with non-uniform thickness (of consistent shape and magnitude) to an ultra-low TTV waveguide display substrate with small but random thickness shape variation. For moldable polymer materials, the biased thickness profile can be designed into the mold that produces the waveguide display substrate.

FIGS. 31A-31C show waveguide display substrate yield versus TTV, dome height, and wedge height for ultra-low TTV waveguide display substrates. In these ultra-low TTV waveguide display substrates, the magnitude of dome height and wedge height are typically similar to each other. Also, the range of variation of dome height is similar in magnitude to the mean of the dome height. FIGS. 31D-31F show waveguide display substrate yield versus TTV, dome height, and wedge height for biased TTV waveguide display substrates. In these biased TTV waveguide display substrates, the magnitude of the dome height is larger than the magnitude of the wedge height, and the variation of the dome height is smaller than the mean dome height. FIGS. 31D-31F represent depict a biased relationship without reference to specific values. A specification for a biased TTV tolerance can be expressed as:

$$W_{\text{max}}/D_{\text{mean}} < X$$

$$(D_{\text{mean}} - D_{\text{min}})/D_{\text{mean}} < Y$$

$$(D_{\text{max}} - D_{\text{mean}})/D_{\text{mean}} < Z$$

With these relationships, X, Y, and Z typically range from 0 to 10 among different substrate polishing or molding processes. FIGS. 31A-31C show a set of waveguide display substrates that result in higher values of X, Y, and Z than the

set of waveguide display substrates shown in FIGS. 31D-31F. As X, Y, and Z approach zero, as opposed to having TTV itself approach zero, overall efficiency of a plurality of waveguides produced on numerous subsections of numerous waveguide display substrates both increases and has fewer variations. Also, as X, Y, and Z approach zero, luminance uniformity and color uniformity among the resultant waveguides also increases and has fewer variations. In a typical substrate polishing or molding process, X, Y, and Z increase as the targeted TTV_{max} approaches zero. Ultra-low TTV waveguide display substrates typically have X, Y and Z in the range of 1-10, and biased TTV waveguide display substrates have X, Y and Z in the range of 0 to 0.3, so a biased (or nonzero target) TTV can contribute toward improved waveguide display image quality.

FIG. 32A shows waveguide eyepiece efficiency versus “dome” TTV (nm) of a spherical waveguide display substrate shape on a 6 inch wafer for a typical diffractive waveguide display with 300 μm average thickness and 0 nm “wedge” TTV. As seen in FIG. 32A, eyepiece efficiency is a maximum between a TTV of 400 nm and a TTV of 600 nm. Waveguide eyepiece efficiency here refers to the sum of light entering a $15 \times 20 \text{ mm}^2$ rectangular area 15 mm^2 away from the eye-side of a diffractive waveguide display’s output grating relative to the sum of light incident on the waveguide display’s input coupling grating as calculated by a typical diffractive waveguide simulation.

FIG. 32B shows waveguide eyepiece efficiency versus “dome” TTV (in μm) of a spherical waveguide display substrate shape on a 6 inch wafer for a typical diffractive waveguide display with 300 μm average thickness. The TTV is measured from the center of the wafer to the edge of the wafer. In FIG. 32B, the pupil efficiency increases as the TTV increases from 0 μm to $\sim 0.6 \mu\text{m}$. Thus, if the waveguide is flat, it would have a pupil efficiency of ~ 0.0023 corresponding to a TTV of 0 μm . Thus, at a TTV of $\sim 0.6 \mu\text{m}$, the pupil efficiency is maximum. As illustrated in FIG. 32B, the range over which the pupil efficiency is acceptable (e.g., greater than 0.003) is a function of the design of the eyepiece waveguide. Accordingly, a large range of TTV values can be utilized according to embodiments of the present invention.

FIG. 32C shows waveguide eyepiece efficiency versus “dome” TTV (in μm) of a spherical waveguide display substrate shape on a 6 inch wafer for another diffractive waveguide display with 300 μm average thickness. In FIG. 32C, the diffractive waveguide display utilizes a two dimensional grating design. For this case, the pupil efficiency increases to 0.001 at a TTV of $\sim 0.4 \mu\text{m}$ and remains at this efficiency for a large range of TTV values. Thus, the desired TTV value can be a function of the grating design, with embodiments of the present invention providing a range of acceptable values for TTV.

The inventors have determined that color uniformity of an eyepiece stack made from multiple eyepiece waveguide layers (e.g., three distinct layers) can be improved if the multiple (e.g., three) layers have similar TTV profiles. In a manner similar to that illustrated in FIGS. 32A-32C, in which the pupil efficiency increased with increasing TTV, the color uniformity also improves as the TTV variation increases, up to a certain value. Without limiting the scope of the present invention, the inventors believe that this increase in color uniformity is because the more similar the uniformity pattern of each layer is to each other layer, the more similar the color of the combined image made from all three layers. The inventors have demonstrated that the color uniformity RMS error, when plotted as a function of the spherical component of TTV for the wafers used to fabricate

the eyepieces, decreases with increasing TTV. This improvement in color uniformity is likely due to each of the eyepiece waveguide layers having similar luminous uniformity pattern. Thus, as the TTV increases, and is consistent between the multiple eyepiece waveguide layers, the color uniformity error is reduced.

Additionally, in addition to color uniformity, the inventors have also determined that the use of TTV can improve the image uniformity. Using a measure of low spatial frequency variation in the image intensity, the inventors determined that introduction of TTV, compared to flat eyepiece waveguide layers, resulted in significant reductions in low spatial frequency variation. This decrease in variation results in improved image uniformity. Thus, eyepiece waveguides with low values of TTV were characterized by higher pixel-to-pixel uniformity compared to eyepiece waveguides with higher values of TTV (e.g., TTV in the range of 0.5 μm).

According to embodiments of the present invention, methods and systems related to eyepiece waveguides with optical thickness variation are provided. The total thickness variation can result from variations in index of refraction and/or variations in physical thickness and can be implemented as total thickness variation (TTV) across a substrate or as local thickness variation (LTV) in multiple areas in a substrate. As described herein, either TTV or LTV, which can improve the image and color uniformity of an eyepiece waveguide, can be implemented by using at least one or more thin film coatings that can be disposed on at least one or both sides of the substrate. The thin film coating, referred to as a film herein, can have a similar index of refraction to that of the substrate, for example, silicon oxynitride ($\text{Si}_x\text{O}_y\text{N}_z$) with an index of refraction in the range of ~ 1.42 to ~ 2.0 . Additionally, films of ZnO , ZrO_2 , TiO_2 , SiC , and the like can be utilized to achieve index of refraction values in the range of ~ 2.0 to ~ 2.7 . Such inorganic films can be deposited using Physical Vapor Deposition processes (PVD) such as Evaporation, Sputter, etc. and/or Chemical Vapor Deposition processes (CVD) such as Low Pressure Plasma Enhanced CVD (LPPECVD), Atmospheric Pressure Plasma Enhanced CVD (APPECVD), Atomic Layer Deposition (ALD), etc. Thin film coatings can also consist of organic polymer which may contain higher index inorganic nanoparticles (ZrO_2 , TiO_2) which can give a modified index of this polymeric base film a range of index from 1.7–2.0. In some implementations, the base polymer material can include a resin material, such as an epoxy vinyl ester. The resin can include a vinyl monomer (e.g., methyl methacrylate) and/or difunctional or trifunctional vinyl monomers (e.g., diacrylates, triacrylates, dimethacrylates, etc.), with or without aromatic molecules in the monomer in addition to Sulphur. In some implementations, the base polymer material can have a refractive index ranging from approximately 1.5 to 1.75. In some implementations, the base material can include a cyclic aliphatic epoxy containing resin can be cured using ultraviolet light and/or heat. Further, the base polymer material can include an ultraviolet cationic photoinitiator and a co-reactant to facilitate efficient ultraviolet curing in ambient conditions. The prepolymer composite resin can have nanoparticles (e.g. ZrO_2 , TiO_2) with a functional surface to prevent particle agglomeration and maintain uniform particle dispersity in the composite resin solution, which once coated and cured helps improve the scatter loss of light undergoing total internal reflection within the now modified TTV of the based substrate. These polymeric resin can be deposited using inkjet dispense, spin coating, slot-die coating, gravure coating, etc. These thin films can be

used in conjunction with substrates made of SiO_2 ($n=1.42$), barium flint (e.g. BaF) ($n=1.58$), lanthanum flint (e.g. LaF) ($n=1.75$), tantalum flint (e.g. TaF) ($n=1.77$), dense tantalum flint (e.g., TAFD55) ($n=2.0$), LiNbO_3 ($n=2.25$), or SiC ($n=2.65$). In some embodiments, the films can be shaped in a manner similar to those shown in FIG. 33 using, for example, dry etching methods like reactive ion etching (RIE), Inductively Coupled Plasma enhanced RIE (RIE-ICP), or ion beam etching (IBE).

FIG. 33 is a simplified schematic diagram illustrating process flows used in forming an inverse dome thickness variation according to an embodiment of the present invention. In addition to methods that form an inverse dome thickness variation via polishing, FIG. 33 illustrates a set of methods that utilize at least one thin film coating, which can be formed by various methods described herein, on at least one side of a waveguide substrate of a similar index to enhance light propagation internally such that the virtual image that is outcoupled towards the user's eye is more uniform and fills the intended field of view, as well as being characterized by improved color uniformity.

Referring to FIG. 33, substrate 3305, for example a glass substrate with an index of refraction of 1.9, is coated with a thin film 3307, typically a deposited film of silicon nitride (Si_3N_4), that has an index of refraction that is substantially matched to the index of refraction of substrate 3305. Thin film 3307 and substrate 3305, both of which have uniform thickness profiles in this example, thus form a substrate structure 3311 with a uniform thickness profile. The substrate structure 3311 can be etched using a preferential etch to form varied thickness film 3308, resulting in a substrate structure 3313 having TTV. Substrate structure 3313 has an inverse dome profile that is thicker at the periphery than in the center of the substrate structure. In embodiments in which the thin film is index matched to the substrate, the substrate structure is optically equivalent to a substrate with TTV. In a typical implementation, substrate 3305 has a thickness on the order or 350 μm and varied thickness film 3308 has a thickness on the order of 1 μm , with the TTV being in the range of 300 nm–600 nm.

In optional process 3330, the substrate structure 3313 can be processed on second side 3315 to form a double sided structure, for example, with the TTV being defined by the TTV of varied thickness film 3308 combined with the TTV of an additional varied thickness film (not shown) that is fabricated on second side 3315. After optional double sided processing in optional process 3330, process 3332 can be utilized to imprint diffraction patterns on varied thickness film 3308, second side 3315, and/or the additional varied thickness film (not shown) that can be fabricated on second side 3315. Either contact printing or photolithography processes can be utilized. After imprinting, etching and/or deposition processes can be utilized to produce gradations in thickness defined by the imprint, the residual layer thickness (RLT), combinations there, or the like.

As an alternative, varied thickness film 3320 can be deposited on substrate 3305 to produce a substrate structure 3322 with a TTV. Varied thickness film 3320 can be uniformly etched to produce varied thickness film 3308 or proceed to optional double sided processing in optional process 3330. Thus, embodiments that utilize an etching process to form a varied thickness film, for example, as illustrated by varied thickness film 3308, as well as embodiments that utilize a deposition process to form a varied thickness film, for example, as illustrated by varied thickness film 3320, are included within the scope of the present invention.

The inventors have determined that the inverse dome profile illustrated in FIG. 33 is difficult to achieve using wafer polishing. Accordingly, embodiments of the present invention enable the use of thin film deposition/etching to fabricate structures that are not readily achieved using conventional wafer polishing techniques. It should be noted that although substrate 3305 is illustrated as having a uniform thickness, a substrate having a first TTV can be utilized with a film having a second TTV to produce a coated substrate with a combined TTV equal to a predetermined thickness profile. Thus, embodiments of the present invention are not limited to substrates with a predetermined TTV, but expand the concept of TTV to the use of films having TTV, thereby enabling implementations independent of wafer polishing techniques.

FIG. 34 is a simplified schematic diagram illustrating process flows used in forming an inverse dome thickness variation according to an embodiment of the present invention. The processes and structures illustrated in FIG. 34 share common elements with the processes and structures illustrated in FIG. 33, but producing a dome thickness variation rather than the inverse dome thickness variation illustrated in FIG. 33.

Referring to FIG. 34, substrate 3405, for example a glass substrate with an index of refraction of 1.9, is coated with a thin film 3407, typically a deposited film of silicon nitride (Si_3N_4), that has an index of refraction that is substantially matched to the index of refraction of substrate 3405. Thin film 3407 and substrate 3405, both of which have uniform thickness profiles in this example, thus form a substrate structure 3411 with a uniform thickness profile. The substrate structure 3411 can be etched using a preferential etch to form varied thickness film 3408, resulting in a substrate structure 3413 having TTV. Substrate structure 3413 has a dome profile that is thicker at the center of the substrate structure than at the periphery. In embodiments in which the thin film is index matched to the substrate, the substrate structure is optically equivalent to a substrate with TTV. In a typical implementation, substrate 3405 has a thickness on the order of 350 μm and varied thickness film 3408 has a thickness on the order of 1 μm , with the TTV being in the range of 300 nm-600 nm.

In optional process 3430, the substrate structure 3413 can be processed on second side 3415 to form a double sided structure, for example, with the TTV being defined by the TTV of varied thickness film 3408 combined with the TTV of an additional varied thickness film (not shown) that is fabricated on second side 3415. After optional double sided processing in optional process 3430, process 3432 can be utilized to imprint diffraction patterns on varied thickness film 3408, second side 3415, and/or the additional varied thickness film (not shown) that can be fabricated on second side 3415. Either contact printing or photolithography processes can be utilized. After imprinting, etching and/or deposition processes can be utilized to produce gradations in thickness defined by the imprint, the residual layer thickness (RLT), combinations there, or the like.

As an alternative, varied thickness film 3420 can be deposited on substrate 3405 to produce a substrate structure 3422 with a TTV. Varied thickness film 3420 can be uniformly etched to produce varied thickness film 3408 or proceed to optional double sided processing in optional process 3430. Thus, embodiments that utilize an etching process to form a varied thickness film, for example, as illustrated by varied thickness film 3408, as well as embodiments that utilize a deposition process to form a varied

thickness film, for example, as illustrated by varied thickness film 3420, are included within the scope of the present invention.

The LTV and TTV discussed herein can be implemented using stencils that control the plasma density during deposition or etching. A plasma enhanced deposition technique such as plasma enhanced chemical vapor deposition (PECVD) can be used to fabricate films with TTV. Additionally, uniform film coating techniques that rely on evaporation, sputtering, or the like (e.g., using physical vapor deposition (PVD), chemical vapor deposition (CVD), PECVD, or the like) can be utilized to form a uniform film and then followed by an optional stencil controlled etching process, for example, a plasma etch, that defines a specific etched pattern and results in a desired LTV/TTV profile on the substrate. The coating or the etched profile can then be patterned using imprint lithography or photolithography to form a diffraction pattern, which may or may not be followed by an etch or deposition step to define a diffraction pattern in the coating, an additional coating, or in the coating underneath the pattern.

FIGS. 35A-35B are simplified cross-sectional diagrams illustrating showerhead designs according to various embodiments of the present invention. Referring to FIG. 35A, a showerhead 3510 with a plurality of apertures 3512 is provided that can be utilized during a deposition process. The plurality of apertures 3512 are disposed adjacent the periphery of showerhead 3510, but not at the center of showerhead 3510. The spacing of the plurality of apertures 3512 can vary as a function of lateral position. Using a showerhead such as the one illustrated in FIG. 35A, which reduces the plasma density and the resulting deposition rates near the center of the showerhead compared to the deposition rates near the periphery, an inverse dome profile can be achieved similar to the profile associated with varied thickness film 3320 illustrated in FIG. 33.

Referring to FIG. 35B, a showerhead 3520 with a plurality of apertures 3522 is provided that can be utilized during a deposition process. The plurality of apertures 3522 are disposed adjacent the center of showerhead 3520, but not at the periphery of showerhead 3520. The spacing of the plurality of apertures 3522 can vary as a function of lateral position. Using a showerhead such as the one illustrated in FIG. 35B, which reduces the plasma density and the resulting deposition rates near the periphery of the showerhead compared to the deposition rates near the center, a dome profile can be achieved similar to the profile associated with varied thickness film 3420 illustrated in FIG. 34.

FIG. 35C is a simplified schematic diagram illustrating an etch mask according to an embodiment of the present invention. Referring to FIG. 35C, an etch mask 3530 with a plurality of apertures 3532 having varying size is provided that can be utilized during an etching process. The plurality of apertures 3532 are disposed across the face of etch mask 3530, with the size of the aperture varying from a small dimension adjacent the periphery of etch mask 3530 to a large dimension adjacent the center of etch mask 3530. The spacing of the plurality of apertures 3532 can also vary as a function of lateral position. Using an etch mask such as the one illustrated in FIG. 35C, which reduces the etching rates near the periphery of the etch mask compared to the etching rates near the center, an inverse dome profile can be achieved similar to the profile associated with varied thickness film 3320 illustrated in FIG. 33. In some embodiments, portions of the etch mask are free of apertures, reducing the etch rates in these portions, for example, to zero. Additional description related to shadow masks and variable density

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plasma deposition and etching is provided in U.S. Pat. No. 10,527,865, issued on Jan. 7, 2020, the disclosure of which is hereby incorporated by reference in its entirety for all purposes.

Thus, using embodiments of the present invention, one or more thin films can be deposited, for example, using PECVD and a modified plasma showerhead to either increase the density of reaction and reactive species in certain zones or decrease the density of reaction/reactive species. Accordingly, the ability to fabricate thin films (e.g., a Si_3N_4 film) with a dome shape or an inverse dome shape on a substrate is provided by embodiments of the present invention. During the deposition process, the precursors can be selected to form films of a predetermined index of refraction. As an example, by varying the oxygen content, silicon oxynitrides can be grown, with indices of refraction ranging from 1.45 (i.e., SiO_2) to ~ 2.0 (Si_3N_4).

FIG. 36A is a simplified plan view of a substrate illustrating thickness variation according to an embodiment of the present invention. As illustrated in FIG. 36A, the deposition and/or etching process is controlled to produce a TTV covering the entire substrate area. As discussed more fully in relation to FIG. 36F, the thickness variation can only cover portions or areas of the substrate, resulting in LTV that can be repeated a number of times in specific areas of the substrate. Referring to FIG. 36A, the thickness variation across substrate 3610 is illustrated by the legend, which shows a thickness variation ranging from ~ 0 nm near the periphery of the substrate to ~ 500 nm at the center of the substrate. Thus, in this figure, a dome shaped profile is illustrated.

FIG. 36B is a simplified cross-sectional diagram illustrating a single sided thickness variation according to an embodiment of the present invention. This single sided thickness variation can be implemented to achieve the dome shaped profile illustrated in FIG. 36A, with a variation of ~ 500 nm from periphery to center. In FIG. 36B, substrate 3620 is a uniform thickness substrate and film 3622 varies in thickness as a function of lateral position.

FIG. 36C is a simplified cross-sectional diagram illustrating a double sided thickness variation according to an embodiment of the present invention. This double sided thickness variation can be implemented to achieve the dome shaped profile illustrated in FIG. 36A, with a variation of ~ 250 nm from periphery to center on the first side of the substrate and a variation of ~ 250 nm from periphery to center on the second side of the substrate. In FIG. 36C, substrate 3630 is a uniform thickness substrate and film 3632 and film 3634 vary in thickness as a function of lateral position. Although film 3632 and film 3634 are illustrated as matching in FIG. 36C, this is not required and they could have differing thickness profiles, for example film 3632 varying by ~ 100 nm from periphery to center and film 3634 varying by ~ 400 nm from periphery to center. One of ordinary skill in the art would recognize many variations, modifications, and alternatives.

FIG. 36D is a simplified schematic diagram illustrating varying height nanofeatures according to an embodiment of the present invention. As illustrated in FIG. 36D, the thickness of film 3622 varies as a function of lateral dimension. As discussed in relation to FIGS. 36A and 36B, the thickness variation can be on the order of 350 nm. Referring to FIG. 36D, the height of the nanofeatures 3640 that are formed in the film 3622 can vary as a function of lateral dimension, for example, on the order of 100 nm (i.e., illustrated by the left nanofeature) down to ~ 0 nm (i.e., illustrated by the right nanofeature) across the varying height film or the substrate.

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These varying height nanofeatures can be imprinted, for example, using contact printing, formed using photolithography, etched, or otherwise formed in film 3622, which also varies in height as discussed above.

FIG. 36E is a simplified schematic diagram illustrating single height nanofeatures according to an embodiment of the present invention. In the embodiment illustrated in FIG. 36E, film 3622 includes nanofeatures 3650 that have a uniform height as a function of lateral position. These uniform height nanofeatures can be imprinted, etched, or otherwise formed in film 3622, which varies in height as discussed above.

FIG. 36F is a simplified plan view of a substrate illustrating thickness variation according to another embodiment of the present invention. In FIG. 36F, six regions 3661, 3662, 3663, 3664, 3665, and 3666 of substrate 3660 are characterized by local thickness variation (LTV). Rather than a single height variation across the entire substrate, each of the six regions is characterized by the height variation characteristic of substrate 3610 shown in FIG. 36A. Stencils or other masking patterns can be used to define the six regions, each of which can correspond to an eyepiece element. Thus, tailoring of the height variation can be implemented as illustrated in FIG. 36F. Although six regions with LTV are illustrated, it will be appreciated that a fewer number of regions or a greater number of regions can be utilized. One of ordinary skill in the art would recognize many variations, modifications, and alternatives.

FIG. 36G is a simplified cross-sectional diagram illustrating a single sided thickness variation according to an embodiment of the present invention. FIG. 36G, which shows a cross section corresponding to cross section A-A' in FIG. 36F, illustrates region 3662 and 3665, which are characterized by a single sided thickness variation implemented to achieve a tapered profile, with a variation of ~ 500 nm from an outer edge to an inner edge of each region. In FIG. 36G, substrate 3660 is a uniform thickness substrate and film 3668 and film 3669 vary in thickness as a function of lateral position.

FIG. 36H is a simplified cross-sectional diagram illustrating a double sided thickness variation according to an embodiment of the present invention. FIG. 36H, which also shows a cross section corresponding to cross section A-A' in FIG. 36F, illustrates region 3662 and 3665, now characterized by a double sided thickness variation implemented to achieve a tapered profile, with a linear variation of ~ 250 nm from an outer edge to an inner edge of each region on each side of the substrate.

FIG. 37A is a simplified plan view of a substrate illustrating thickness variation according to an embodiment of the present invention. In FIG. 37A, substrate 3710 is rectangular in form rather than circular as illustrated with respect to substrate 3610. Referring to FIG. 37A, the thickness variation across substrate 3710 is defined in four regions 3711, 3712, 3713, and 3714, each having a thickness variation ranging from ~ 0 nm near the left side of each region periphery of the substrate to ~ 500 nm at the right edge of the region. Thus, in this figure, a series of tapered profiles with a linear increase in height across the region are illustrated. In addition to circular and rectangular substrates, other substrate form factors can be utilized, including square substrates or substrates in the form of a web.

FIG. 37B is a simplified schematic diagram illustrating a single sided thickness variation according to an embodiment of the present invention. This single sided thickness varia-

tion can be implemented to achieve the set of linear, tapered profiles illustrated in FIG. 37A, with a variation of ~500 nm across each region.

FIG. 37C is a simplified schematic diagram illustrating a double sided thickness variation according to an embodiment of the present invention. This double sided thickness variation can be implemented to achieve the linear, tapered profiles illustrated in FIG. 37A, with a variation of ~250 nm from one edge of each region to the opposing edge of each region and a corresponding variation of ~250 nm from edge to edge on the second side of the substrate.

FIG. 37D is a simplified plan view of a substrate illustrating thickness variation according to another embodiment of the present invention. In FIG. 37D, in a manner similar to that discussed in relation to FIG. 37A, nine regions 3721, 3722, 3723, 3724, 3725, 3726, 3727, 3728, and 3729 of substrate 3720 are characterized by LTV. Rather than a single height variation across the entire substrate, each of the nine regions is characterized by the height variation characteristic of four regions shown in FIG. 37A. Stencils or other masking patterns can be used to define the nine regions, each of which can correspond to an eyepiece element. In the example, illustrated in FIG. 37D, a rectangular substrate is utilized and the thickness varies along the length of the rectangle, providing an alternative to the polar-coordinate based layout illustrated in FIG. 36F. In other embodiments, rather than the thickness variation increasing in the same direction for all eyepiece waveguides, the thickness variation can be reversed for some eyepiece waveguides, i.e., decreasing along the length, increasing/decreasing along the width, or the like. Thus, embodiments of the present invention can be implemented on a variety of substrate geometries and a variety of eyepiece waveguide geometries.

FIG. 37E is a simplified schematic diagram illustrating a single sided thickness variation according to an embodiment of the present invention. FIG. 37E, which shows a cross section corresponding to cross section B-B' in FIG. 37D, illustrates regions 3727, 3728, and 3729, characterized by a single sided thickness variation implemented to achieve a tapered profile, with a linear variation of ~500 nm from a left edge to a right edge of each region.

FIG. 37F is a simplified schematic diagram illustrating a double sided thickness variation according to an embodiment of the present invention. This double sided thickness variation can be implemented to achieve the linear, tapered profiles illustrated in FIG. 37D, with a variation of ~250 nm from one edge of each region to the opposing edge of each region and a corresponding variation of ~250 nm from edge to edge on the second side of the substrate.

FIGS. 38A-41F are simplified cross-sectional diagrams illustrating eyepiece waveguides with a variety of thickness variations based on thin film TTV. As described more fully below, in examples discussed in relation to FIGS. 38A-38F, the thin film thickness decreases from the region corresponding to the incoupling grating to the region corresponding to the combined pupil expander. The examples described in relation to FIGS. 38A-41F are merely exemplary and are not intended to limit the scope of the present invention or the architectures that can be utilized in accordance with embodiments of the present invention. Referring to FIG. 28C, the eyepiece waveguides illustrated in FIGS. 38A-41F can be utilized in place of eyepiece waveguide 2820, second eyepiece waveguide 2830, and third eyepiece waveguide 2840 as elements of an eyepiece. Thus, embodiments in which the substrate varies in optical path length (e.g., thickness) and

embodiments in which the substrate structure varies in optical path length (e.g., thickness) are interchangeable in the systems described herein.

FIG. 38A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation according to an embodiment of the present invention. As illustrated in FIG. 38A, a variable thickness film 3812 is formed, for example, via deposition, on a substrate 3810 that is characterized by a uniform thickness. Although substrate 3810 is illustrated as having a uniform thickness and the thickness variation for the substrate structure 3813, which includes variable thickness film 3812 and substrate 3810, is only due to the variation in thickness of variable thickness film 3812, it will be appreciated that substrate 3810 can also be characterized by some thickness variation, which can be taken into account during the deposition of variable thickness film 3812 to produce a substrate structure 3813 with the desired total thickness variation.

Diffraction structures, including incoupling grating 3814 and combined pupil expander 3816, which includes variable height nanofeatures in this embodiment, are formed on variable thickness film 3812. As will be evident to one of skill in the art, the representation of incoupling grating 3814 and combined pupil expander 3816 is merely for purposes of illustration. For the eyepiece waveguide illustrated in FIG. 38A, input light is received below incoupling grating 3814, propagates through anti-reflection coating 3818 and substrate 3810, is coupled into variable thickness film 3812 and substrate 3810 by incoupling grating 3814, and propagates in substrate 3810 and variable thickness film 3812 prior to outcoupling by combined pupil expander 3816. Thus, the eyepiece waveguide decreases in thickness as light propagates from incoupling grating 3814 to combined pupil expander 3816. In an alternative embodiment of a single sided eyepiece waveguide, incoupling grating 3814 and combined pupil expander 3816 are fabricated on substrate 3810 and variable thickness film 3812 is positioned between anti-reflection coating 3818 and substrate 3810.

FIG. 38B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 38B shares common elements with the eyepiece waveguide illustrated in FIG. 38A, with the addition of second variable thickness film 3820 and second combined pupil expander 3826. Thus, substrate structure 3823 includes variable thickness film 3812, substrate 3810, and second variable thickness film 3820 and is characterized by a total variable thickness due to the thicknesses of these elements varying in the lateral dimension. Although a double sided structure is illustrated in FIG. 38B, an alternative embodiment removes second variable thickness film 3820 and defines second combined pupil expander 3826 on substrate 3810. Thus, in this, and other embodiments, double side patterning for diffractive structures can be implemented in the context of a single sided variable thickness film.

FIG. 38C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 38C shares common elements with the eyepiece waveguide illustrated in FIG. 38B, with the removal of the interconnecting layer between diffractive nanofeatures.

FIG. 38D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to yet another embodiment of the present invention. The eyepiece waveguide illustrated in FIG.

38D shares common elements with the eyepiece waveguide illustrated in FIG. 38C, with the addition of etching of portions of variable thickness film 3812 and second variable thickness film 3820 to increase the diffraction efficiency of the diffractive structures, which are defined by incoupling grating 3814 and combined pupil expander 3816, along with remaining portions of variable thickness film 3812, and second combined pupil expander 3826, along with remaining portions of variable thickness film 3820.

FIG. 38E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an alternative embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 38E shares common elements with the eyepiece waveguide illustrated in FIG. 38D, with the removal of combined pupil expander 3816 and second combined pupil expander 3826. In this embodiment, the diffractive properties associated with combined pupil expander 3816 and second combined pupil expander 3826 are implemented via the remaining portions of variable thickness film 3812 and the remaining portions of second variable thickness film 3820. In some embodiments, incoupling grating 3814 is also partially or fully removed in favor of a diffractive structure etched or otherwise formed in variable thickness film 3812.

FIG. 38F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and an overcoat according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 38F shares common elements with the eyepiece waveguide illustrated in FIG. 38B, with the addition of an overcoat 3830. The overcoat 3830, which may be a high index of refraction material compared to the index of refraction of variable thickness film 3812 and second variable thickness film 3820. Overcoat 3830 can be deposited in the desired pattern or deposited and then etched to form the desired pattern.

The eyepiece waveguides illustrated in FIGS. 38A-38F can be utilized in an augmented reality device including a projector, projector optics optically coupled to the projector, and an eyepiece optically coupled to the projector optics and including one or more eyepiece waveguides. As illustrated in FIGS. 38A-38F, each of the eyepiece waveguides can be implemented a substrate structure characterized by lateral dimensions and one or more layers, with at least one of the one or more layers being characterized by an optical path length difference (e.g., a thickness variation) as a function of one or more of the lateral dimensions. As illustrated in FIG. 38A, the substrate structure can include a substrate and a variable thickness layer coupled to the substrate, with the indices of refraction of the substrate and variable thickness layer being matched or within a given value. Multiple eyepiece waveguides can be laminated, with the gradient of the thickness variations for each of the eyepiece waveguides being aligned, to form an eyepiece. Thus, in some embodiments using three eyepiece waveguides, the variable thickness layer in each of the eyepiece waveguides is aligned such that the thickness of each of the variable thickness layers decreases along a same direction. Thus, for an eyepiece using three eyepiece waveguides similar to the eyepiece waveguide illustrated in FIG. 38A, the thickness of each variable thickness layer (i.e., variable thickness film 3812 in FIG. 38A) would decrease along a direction pointing from the incoupling grating to the combined pupil expander. Thus, the gradient of each of the variable thickness layers would be aligned, pointing from the incoupling grating of each eyepiece waveguide to the combined pupil expander of each eyepiece waveguide.

FIGS. 39A-39F are simplified cross-sectional diagrams illustrating eyepiece waveguides with a variety of thickness variations based on thin film TTV. These examples correspond to the examples discussed in relation to FIGS. 38A-38F, but with the thin film thickness increasing from the region corresponding to the incoupling grating to the region corresponding to the combined pupil expander.

FIG. 39A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 39A shares common elements with the eyepiece waveguide illustrated in FIG. 38A, with the thickness of variable thickness film 3912 increasing in a manner opposite to that shown in FIG. 38A, namely an increase in thickness as light propagates in substrate 3910 and variable thickness film 3912 from incoupling grating 3914 toward combined pupil expander 3916.

FIG. 39B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 39B shares common elements with the eyepiece waveguide illustrated in FIG. 38B, with the thickness of variable thickness film 3912 and second variable thickness film 3920 increasing in a manner opposite to that shown in FIG. 38B.

FIG. 39C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 39C shares common elements with the eyepiece waveguide illustrated in FIG. 39B, with the removal of the interconnecting layer between diffractive nanofeatures.

FIG. 39D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to yet another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 39D shares common elements with the eyepiece waveguide illustrated in FIG. 39C, with the addition of etching of portions of variable thickness film 3912 and second variable thickness film 3920 to increase the diffraction efficiency of the diffractive structures, which are defined by incoupling grating 3914 and combined pupil expander 3916, along with remaining portions of variable thickness film 3912, and second combined pupil expander 3926, along with remaining portions of second variable thickness film 3920.

FIG. 39E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an alternative embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 39E shares common elements with the eyepiece waveguide illustrated in FIG. 39D, with the removal of combined pupil expander 3916 and second combined pupil expander 3926. In this embodiment, the diffractive properties associated with combined pupil expander 3916 and second combined pupil expander 3926 are implemented via the remaining portions of variable thickness film 3912 and the remaining portions of second variable thickness film 3920.

FIG. 39F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and an overcoat according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 39F shares common elements with the eyepiece waveguide illustrated in FIG. 39B, with the addition of an overcoat 3930.

Thus, as illustrated in the embodiments illustrated in FIGS. 38A-39F, the variable thickness films formed on the substrate, including films that are index matched to the

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substrate, can utilize an imprint with thin RLT, imprint with no index matched photoresist, photoresist plus an etched film pattern, an etched film pattern or the film deposited onto the imprint or photoresist patterned polymer, or the like. The variable thickness film(s) can be deposited using CVD, PVD, or resin dispense method like ink jetting, slot-die coating for high index polymeric film coatings or the like to produce a film that is conformal, bi-directional, or formed using a glancing angle deposition. The etched or deposited pattern architectures that are disposed on the substrate structure with predetermined TTV may be planarized with a polymer, for example, a lower index polymer, or an inorganic material such as a flowable SiO₂ using CVD processes so as to reduce rainbow artifacts as well as increase the see through transmission as discussed in more detail in relation to FIGS. 42A-42C.

In the different architectures illustrated in FIGS. 38A-41F, the film(s) with tailored TTV are formed on a uniform thickness substrate. It should be noted that the film(s) can be a combination of two or more variable thickness films in which the first film adjacent the substrate can be a film with a lower index of refraction, for example, SiO₂, MgF₂, etc., and the additional film(s) distal from the substrate can be film(s) with a higher index of refraction, for example, Si₃N₄, ZnO, ZrO₂, TiO₂, SiC, ZnTe, BP, or the like. The first, lower index film can be formed as a uniform film, for example, between approximately 10 nm and 50 nm in thickness, while the additional, higher index film(s) can be incorporate the thickness variation, for example with a thickness in the range of 1 μm and a TTV of 300 nm-600 nm.

FIG. 40A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation and a low index film according to an embodiment of the present invention. As illustrated in FIG. 40A, a variable thickness film 4012 is formed, for example, via deposition, on an intermediate film 4011, which is formed on a substrate 4010 that is characterized by a uniform thickness. Intermediate film 4011, which is referred to as an intermediate film because it is disposed between substrate 4010 and variable thickness film 4012, can be characterized by a uniform thickness and can have an index of refraction less than the index of refraction of substrate 4010 and variable thickness film 4012. Although substrate 4010 is illustrated as having a uniform thickness and the thickness variation for the substrate structure 4013, which includes variable thickness film 4012, intermediate film 4011, and substrate 4010, is only due to the variation in thickness of variable thickness film 4012, it will be appreciated that substrate 4010 and/or intermediate film 4011 can also be characterized by some thickness variation, which can be taken into account during the deposition of variable thickness film 4012 to produce a substrate structure 4013 with the desired total thickness variation.

The inventors have determined that the addition of intermediate film 4011 serves to optically divide light rays that are propagating via TIR in the substrate structure. This multiplication of the light rays produces image filling in the virtual content and an improved user experience.

Diffraction structures, including incoupling grating 4014 and combined pupil expander 4016, which include variable height nanofeatures in this embodiment, are formed on variable thickness film 4012. As will be evident to one of skill in the art, the representation of incoupling grating 4014 and combined pupil expander 4016 is merely for purposes of illustration. For the eyepiece waveguide illustrated in FIG. 40A, input light is received below incoupling grating 4014, propagates through anti-reflection coating 4018 and sub-

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strate 4010, is coupled into variable thickness film 4012, intermediate film 4011, and substrate 4010 by incoupling grating 4014, and propagates in substrate 4010, intermediate film 4011, and variable thickness film 4012 prior to outcoupling by combined pupil expander 4016. Thus, the eyepiece waveguide decreases in thickness as light propagates from incoupling grating 4014 to combined pupil expander 4016. In an alternative embodiment of a single sided eyepiece waveguide, incoupling grating 4014 and combined pupil expander 4016 are fabricated on substrate 4010 and intermediate film 4011 and variable thickness film 4012 are positioned between anti-reflection coating 4018 and substrate 4010.

FIG. 40B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 40B shares common elements with the eyepiece waveguide illustrated in FIG. 40A, with the addition of second intermediate film 4021, second variable thickness film 4020, and second combined pupil expander 4026. Thus, substrate structure 4023 includes variable thickness film 4012, intermediate film 4011, substrate 4010, second intermediate film 4021, and second variable thickness film 4020 and is characterized by a total variable thickness due to the thicknesses of these elements varying in the lateral dimension. Although a double sided structure is illustrated in FIG. 40B, an alternative embodiment removes second intermediate film 4021 and second variable thickness film 4020 and defines second combined pupil expander 4026 on substrate 4010. Thus, in this, and other embodiments, double side patterning for diffractive structures can be implemented in the context of a single sided variable thickness film.

FIG. 40C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 40C shares common elements with the eyepiece waveguide illustrated in FIG. 40B, with the removal of the interconnecting layer between diffractive nanofeatures.

FIG. 40D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to yet another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 40D shares common elements with the eyepiece waveguide illustrated in FIG. 40C, with the addition of etching of portions of variable thickness film 4012 and second variable thickness film 4020 to increase the diffraction efficiency of the diffractive structures, which are defined by incoupling grating 4014 and combined pupil expander 4016, along with remaining portions of variable thickness film 4012, and second combined pupil expander 4026, along with remaining portions of second variable thickness film 4020.

FIG. 40E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an alternative embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 40E shares common elements with the eyepiece waveguide illustrated in FIG. 40D, with the removal of combined pupil expander 4016 and second combined pupil expander 4026. In this embodiment, the diffractive properties associated with combined pupil expander 4016 and second combined pupil expander 4026 are implemented via the remaining portions of variable thickness film 4012 and the remaining portions of second

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variable thickness film 4020. In some embodiments, incoupling grating 4014 is also partially or fully removed in favor of a diffractive structure etched or otherwise formed in variable thickness film 4012.

FIG. 40F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film and an overcoat according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 40F shares common elements with the eyepiece waveguide illustrated in FIG. 40B, with the addition of an overcoat 4030. The overcoat 4030, which may be a high index of refraction material compared to the index of refraction of variable thickness film 4012 and second variable thickness film 4020. Overcoat 4030 can be deposited in the desired pattern or deposited and then etched to form the desired pattern.

FIG. 41A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with single sided total thickness variation and a low index film according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 41A shares common elements with the eyepiece waveguide illustrated in FIG. 40A, with the thickness of variable thickness film 4112 increasing in a manner opposite to that shown in FIG. 40A, namely an increase in thickness as light propagates in substrate 4110, intermediate film 4111, and variable thickness film 4112 from incoupling grating 4114 toward combined pupil expander 4116.

FIG. 41B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 41B shares common elements with the eyepiece waveguide illustrated in FIG. 40B, with the thickness of variable thickness film 4112, intermediate film 4111, and second variable thickness film 4120 increasing in a manner opposite to that shown in FIG. 40B. Second combined pupil expander 4126 is also illustrated.

FIG. 41C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 41C shares common elements with the eyepiece waveguide illustrated in FIG. 41B, with the removal of the interconnecting layer between diffractive nanostructures.

FIG. 41D is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to yet another embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 41D shares common elements with the eyepiece waveguide illustrated in FIG. 41C, with the addition of etching of portions of variable thickness film 4112 and second variable thickness film 4120 to increase the diffraction efficiency of the diffractive structures, which are defined by incoupling grating 4114 and combined pupil expander 4116, along with remaining portions of variable thickness film 4112, and second combined pupil expander 4126, along with remaining portions of second variable thickness film 4120.

FIG. 41E is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film according to an alternative embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 41E shares common elements with the eyepiece waveguide illustrated in FIG. 41D, with the removal of combined pupil expander 4116 and second combined pupil expander 4126. In this embodiment, the

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diffractive properties associated with combined pupil expander 4116 and second combined pupil expander 4126 are implemented via the remaining portions of variable thickness film 4112 and the remaining portions of second variable thickness film 4120.

FIG. 41F is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a low index film and an overcoat according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 41F shares common elements with the eyepiece waveguide illustrated in FIG. 41B, with the addition of an overcoat 4130.

FIG. 42A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 42A corresponds to a portion of the eyepiece waveguide illustrated in FIG. 38D, with substrate 3810, variable thickness film 3812, and second variable thickness film 3820 illustrated. Combined pupil expander 3816 and second combined pupil expander 3826 are also illustrated, along with remaining portions of variable thickness film 3812 and remaining portions of second variable thickness film 3820.

In some embodiments, the presence of the diffractive structures associated with combined pupil expander 3816, second combined pupil expander 3826, and the remaining portions of variable thickness film 3812 and second variable thickness film 3820, can result in artifacts, for example, rainbow artifacts, due to coupling of undesired world light into the eyepiece waveguide. The planarization process may either full encapsulate the etched or deposited patterns or may partially encapsulate the etched or deposited patterns.

FIG. 42B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a first level of planarization according to an embodiment of the present invention. As illustrated in FIG. 42B, combined pupil expander 3816 has been partially encapsulated in layer 4210 and combined pupil expander 3826 has been partially encapsulated in layer 4212. The extent of the partial encapsulation illustrated in FIG. 42B can be selected to provide a desired eyepiece waveguide efficiency.

FIG. 42C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and a second level of planarization according to an embodiment of the present invention. As illustrated in FIG. 42C, combined pupil expander 3816 has been fully encapsulated in layer 4220 and combined pupil expander 3826 has been fully encapsulated in layer 4222.

FIG. 43A is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation and an overcoat according to an embodiment of the present invention. The eyepiece waveguide illustrated in FIG. 43A corresponds to a portion of the eyepiece waveguide illustrated in FIG. 38F, with substrate 3810, variable thickness film 3812, and second variable thickness film 3820 illustrated. Combined pupil expander 3816 and second combined pupil expander 3826 are also illustrated. Overcoat 3830 is also illustrated.

FIG. 43B is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation, an overcoat, and a first level of planarization according to an embodiment of the present invention. As illustrated in FIG. 43B, combined pupil expander 3816 has been partially encapsulated in layer 4310 and combined pupil expander 3826 has been partially encapsulated in layer

4312. The extent of the partial encapsulation illustrated in FIG. 43B can be selected to provide a desired eyepiece waveguide efficiency.

FIG. 43C is a simplified cross-sectional diagram illustrating an eyepiece waveguide with double sided total thickness variation, an overcoat, and a second level of planarization according to an embodiment of the present invention. As illustrated in FIG. 43C, combined pupil expander **3816** has been fully encapsulated in layer **4320** and combined pupil expander **3826** has been fully encapsulated in layer **4322**.

In the foregoing specification, the disclosure has been described with reference to specific embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the disclosure. The specification and drawings are, accordingly, to be regarded in an illustrative rather than restrictive sense.

Indeed, it will be appreciated that the systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combinations and subcombinations are intended to fall within the scope of this disclosure.

Certain features that are described in this specification in the context of separate embodiments also may be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment also may be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination. No single feature or group of features is necessary or indispensable to each and every embodiment.

It will be appreciated that conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise. Similarly, while operations may be depicted in the drawings in a particular order, it is to be recognized that such operations need not be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable

results. Further, the drawings may schematically depict one more example processes in the form of a flowchart. However, other operations that are not depicted may be incorporated in the example methods and processes that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously, or between any of the illustrated operations. Additionally, the operations may be rearranged or reordered in other embodiments. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems may generally be integrated together in a single software product or packaged into multiple software products. Additionally, other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results.

Although this disclosure contains many specific embodiment details, these should not be construed as limitations on the scope of the subject matter or on the scope of what may be claimed, but rather as descriptions of features that may be claimed, but rather as descriptions of features that may be specific to particular embodiments. Certain features that are described in this disclosure in the context of separate embodiments can also be implemented, in combination, in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments, separately, or in any suitable sub-combination. Moreover, although previously described features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

Particular embodiments of the subject matter have been described. Other embodiments, alterations, and permutations of the described embodiments are within the scope of the following claims as will be apparent to those skilled in the art. While operations are depicted in the drawings or claims in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed (some operations may be considered optional), to achieve desirable results.

Accordingly, the previously described example embodiments do not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure.

Accordingly, the claims are not intended to be limited to the embodiments shown herein, but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features disclosed herein.

What is claimed is:

1. An augmented reality device comprising:

a projector;

projector optics optically coupled to the projector; and an eyepiece optically coupled to the projector optics, wherein the eyepiece comprises an eyepiece waveguide including a uniform thickness portion, a tapered portion, a thinner portion, a diffractive incoupling element, and a diffractive outcoupling element, wherein the eyepiece waveguide is characterized by lateral dimensions and an eyepiece waveguide optical thickness that

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decreases from the uniform thickness portion of the eyepiece waveguide to the thinner portion of the eyepiece waveguide as a function of one or more of the lateral dimensions, wherein the diffractive incoupling element is located adjacent the uniform thickness portion of the eyepiece waveguide and the diffractive outcoupling element is located at the thinner portion of the eyepiece waveguide.

2. The augmented reality device of claim 1 wherein the eyepiece further comprises:

- a second eyepiece waveguide characterized by second lateral dimensions and a second optical path length difference as a function of one or more of the second lateral dimensions; and
- a third eyepiece waveguide characterized by third lateral dimensions and a third optical path length difference as a function of one or more of the third lateral dimensions.

3. The augmented reality device of claim 2 wherein the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide form the eyepiece.

4. The augmented reality device of claim 3 wherein the eyepiece is a laminated structure including the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide.

5. The augmented reality device of claim 2 further comprising:

- a second projector;
- second projector optics optically coupled to the second projector; and
- a second eyepiece optically coupled to the second projector optics, wherein the second eyepiece comprises a fourth eyepiece waveguide characterized by fourth lateral dimensions and a fourth optical path length difference as a function of one or more of the fourth lateral dimensions.

6. The augmented reality device of claim 5 wherein the second eyepiece further comprising:

- a fifth eyepiece waveguide characterized by fifth lateral dimensions and a fifth optical path length difference as a function of one or more of the fifth lateral dimensions; and
- a sixth eyepiece waveguide characterized by sixth lateral dimensions and a sixth optical path length difference as a function of one or more of the sixth lateral dimensions.

7. The augmented reality device of claim 2 wherein: the diffractive outcoupling element of the eyepiece waveguide comprises a combined pupil expander; the second eyepiece waveguide comprises a second combined pupil expander; and the third eyepiece waveguide comprises a third combined pupil expander, wherein a thickness of the eyepiece waveguide varies across the combined pupil expander, a thickness of the second eyepiece waveguide varies across the second combined pupil expander, and a thickness of the third eyepiece waveguide varies across the third combined pupil expander.

8. The augmented reality device of claim 1 wherein the projector, the projector optics, and the eyepiece are mounted in an augmented reality headset.

9. The augmented reality device of claim 1 wherein the eyepiece waveguide is characterized by an index of refraction variation as a function of the one more of the lateral dimensions.

10. The augmented reality device of claim 1, wherein the projector, the projector optics, and the eyepiece are config-

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ured such that light from the projector passes through the projector optics and enters the eyepiece waveguide at the diffractive incoupling element located adjacent the uniform thickness portion of the eyepiece waveguide.

11. The augmented reality device of claim 10, wherein the diffractive outcoupling element located at the thinner portion of the eyepiece waveguide is configured such that the light that enters the eyepiece waveguide at the diffractive incoupling element traverses the eyepiece waveguide from the tapered portion to the thinner portion and exits the eyepiece waveguide through the diffractive outcoupling element located at the thinner portion of the eyepiece waveguide.

12. The augmented reality device of claim 1 wherein the tapered portion is positioned between the uniform thickness portion and the thinner portion.

13. The augmented reality device of claim 1 wherein the tapered portion of the eyepiece waveguide is characterized by a nonlinear cross section.

14. The augmented reality device of claim 13 wherein the nonlinear cross section comprises a domed shape.

15. The augmented reality device of claim 1 wherein the eyepiece waveguide comprises:

- a substrate; and
- a variable thickness layer coupled to the substrate.

16. An augmented reality device comprising:

- a projector;
- projector optics optically coupled to the projector; and
- an eyepiece optically coupled to the projector optics, wherein the eyepiece comprises an eyepiece waveguide characterized by lateral dimensions and one or more layers, wherein at least one of the one or more layers includes a uniform thickness portion, a tapered portion, a thinner portion, a diffractive incoupling element, and a diffractive outcoupling element, wherein the eyepiece waveguide is characterized by an eyepiece waveguide optical thickness that varies decreases from the uniform thickness portion of the eyepiece waveguide to the thinner portion of the eyepiece waveguide as a function of one or more of the lateral dimensions, wherein the diffractive incoupling element is adjacent the uniform thickness portion of the eyepiece waveguide and the diffractive outcoupling element is at the thinner portion of the eyepiece waveguide.

17. The augmented reality device of claim 16 wherein the eyepiece comprises:

- a substrate; and
- a variable thickness layer coupled to the substrate.

18. The augmented reality device of claim 17 wherein an index of refraction of the substrate is substantially equal to an index of refraction of the variable thickness layer.

19. The augmented reality device of claim 16 wherein the eyepiece further comprises:

- a second eyepiece waveguide characterized by second lateral dimensions and one or more second layers, wherein at least one of the one or more second layers is characterized by a second optical path length difference as a function of one or more of the second lateral dimensions; and
- a third eyepiece waveguide characterized by third lateral dimensions and one or more third layers, wherein at least one of the one or more third layers is characterized by a third optical path length difference as a function of one or more of the third lateral dimensions.

20. The augmented reality device of claim 19 wherein the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide form the eyepiece.

21. The augmented reality device of claim 20 wherein the eyepiece is a laminated structure including the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide.

22. The augmented reality device of claim 19 wherein:
the one or more layers, the one or more second layers, and the one or more third layers are characterized by a thickness variation as a function of the one or more lateral dimensions; and
the thickness variation of each of the one or more layers, the one or more second layers, and the one or more third layers decreases along a lateral dimension of each of the eyepiece waveguide, the second eyepiece waveguide, and the third eyepiece waveguide, wherein the lateral dimensions are parallel.

23. The augmented reality device of claim 19 further comprising:

a second projector;
second projector optics optically coupled to the second projector; and
a second eyepiece optically coupled to the projector optics, wherein the second eyepiece comprises a fourth eyepiece waveguide characterized by fourth lateral dimensions and one or more fourth layers, wherein at least one of the one or more fourth layers is characterized by a fourth optical path length difference as a function of one or more of the fourth lateral dimensions.

24. The augmented reality device of claim 23 wherein the second eyepiece further comprising:

a fifth eyepiece waveguide characterized by fifth lateral dimensions and one or more fifth layers, wherein at least one of the one or more fifth layers is characterized by a fifth optical path length difference as a function of one or more of the fifth lateral dimensions; and
a sixth eyepiece waveguide characterized by sixth lateral dimensions and one or more sixth layers, wherein at least one of the one or more sixth layers is characterized by a sixth optical path length difference as a function of one or more of the sixth lateral dimensions.

25. The augmented reality device of claim 16, wherein the projector, the projector optics, and the eyepiece are configured such that light from the projector passes through the projector optics and enters the eyepiece waveguide at the diffractive incoupling element located adjacent the uniform thickness portion of the eyepiece waveguide.

26. The augmented reality device of claim 25, wherein the diffractive outcoupling element located at the thinner portion of the eyepiece waveguide is configured such that the light that enters the eyepiece waveguide at the diffractive incoupling element traverses the eyepiece waveguide from the tapered portion to the thinner portion and exits the eyepiece waveguide through the diffractive outcoupling element located at the thinner portion of the eyepiece waveguide.

27. The augmented reality device of claim 16 wherein the projector, the projector optics, and the eyepiece are mounted in an augmented reality headset.

28. An augmented reality device comprising:
a projector;
projector optics optically coupled to the projector; and
an eyepiece waveguide optically coupled to the projector optics and including:
an incoupling element; and
an outcoupling element, wherein:
the eyepiece waveguide is:
characterized by lateral dimensions;

configured to propagate light by total internal reflection (TIR) in a TIR region positioned between the incoupling element and the outcoupling element; and
has an optical thickness that decreases continuously from the incoupling element, over an entirety of the TIR region, to the outcoupling element, as a function of one or more of the lateral dimensions.

29. The augmented reality device of claim 28 wherein the incoupling element comprises a diffractive incoupling element.

30. The augmented reality device of claim 29 wherein the diffractive incoupling element further comprises a reflector.

31. The augmented reality device of claim 28 wherein the incoupling element comprises a reflective incoupling element.

32. The augmented reality device of claim 28 wherein the outcoupling element comprises a diffractive outcoupling element.

33. The augmented reality device of claim 28 wherein the outcoupling element comprises a combined pupil expander.

34. The augmented reality device of claim 28 wherein the projector, the projector optics, and the eyepiece waveguide are mounted in an augmented reality headset.

35. The augmented reality device of claim 28 wherein the eyepiece waveguide is characterized by an index of refraction variation as a function of the one more of the lateral dimensions.

36. An augmented reality device comprising:

a projector;
projector optics optically coupled to the projector; and
an eyepiece optically coupled to the projector optics, wherein the eyepiece comprises an eyepiece waveguide including an incoupling element and an outcoupling element and characterized by lateral dimensions and an eyepiece waveguide optical thickness that varies from a thicker portion of the eyepiece waveguide to a thinner portion of the eyepiece waveguide as a function of one or more of the lateral dimensions, wherein the incoupling element is located at a tapered portion of the eyepiece waveguide disposed between the thicker portion and the thinner portion and the outcoupling element is located at the thinner portion of the eyepiece waveguide.

37. The augmented reality device of claim 36 wherein the entire eyepiece waveguide is tapered.

38. The augmented reality device of claim 36 wherein the outcoupling element of the eyepiece waveguide comprises a combined pupil expander.

39. The augmented reality device of claim 36 wherein the projector, the projector optics, and the eyepiece are mounted in an augmented reality headset.

40. The augmented reality device of claim 36 wherein the eyepiece waveguide comprises:

a substrate; and
a variable thickness layer coupled to the substrate.

41. The augmented reality device of claim 36 wherein the incoupling element comprises a diffractive incoupling element.

42. The augmented reality device of claim 41 wherein the diffractive incoupling element further comprises a reflector.

43. The augmented reality device of claim 36 wherein the incoupling element comprises a reflective incoupling element.

44. The augmented reality device of claim 36 wherein the outcoupling element comprises a diffractive outcoupling element.